

**DEVELOPMENT OF A DISTRIBUTED IOT MECHATRONIC SYSTEM FOR
HVAC OPTIMIZATION USING HYBRID AI LOGIC**

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APPROVAL SHEET



VISAYAS
STATE UNIVERSITY

DEPARTMENT OF
**MECHANICAL
ENGINEERING**

THESIS

APPROVAL SHEET

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CERTIFICATION

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ABSTRACT

This study developed and evaluated a distributed Internet of Things (IoT) mechatronic framework integrating hybrid artificial intelligence (AI), a Building Energy Modeling (BEM)-based Digital Twin, and edge-computing control architecture for Heating, Ventilation, and Air Conditioning (HVAC) optimization. The system was designed to reduce energy consumption while maintaining thermal comfort in accordance with ASHRAE Standard 55. A Raspberry Pi 5 served as the supervisory controller, coordinating ESP32-based sensor nodes through the MQTT protocol to acquire environmental, occupancy, and energy-consumption data. Occupancy information was obtained using a headless YOLOv8n computer-vision pipeline, while all control and processing functions were performed locally through edge computing.

For HVAC optimization, a Bayesian Optimized Random Forest (BORF) model was developed to learn the nonlinear relationship between thermal comfort and energy consumption using experimentally collected operational data.

To address the limitations of data-driven models under dynamic real-world conditions, the machine-learning model was integrated with deterministic physics-based guardrails that supervised and corrected control actions when necessary. Although the standalone BORF model achieved modest predictive performance ($R^2 = 0.204$ for Predicted Mean Vote and $R^2 = 0.338$ for energy prediction), the hybrid architecture successfully compensated for these limitations during live deployment. Experimental results demonstrated weather-normalized energy savings of

approximately 20% relative to conventional manual operation while maintaining acceptable occupant comfort conditions.

A physics-based Digital Twin of the target facility was developed using OpenStudio and EnergyPlus to provide virtual parallel assessment of building performance. The model underwent automated Bayesian calibration through Gaussian Process optimization using measured operational data and was evaluated in accordance with ASHRAE Guideline 14. Following calibration, the Digital Twin successfully reproduced the thermal and energy behavior of the physical facility and satisfied established validation criteria. Its deployment enabled continuous comparison between measured and simulated building conditions, demonstrating its capability as a decision-support and performance-monitoring platform for HVAC operation.

The distributed IoT-mechatronic controller was implemented using a Raspberry Pi 5 supervisory hub and ESP32-based sensing and monitoring nodes. Technical evaluation demonstrated stable computational performance, reliable communication, minimal packet loss, and low power consumption suitable for long-term deployment. Economic analysis indicated favorable implementation feasibility through measurable annual energy savings and positive investment returns. Collectively, the findings demonstrate that the integration of hybrid AI control, calibrated Digital Twin technology, and distributed IoT-mechatronic architecture provide a practical, scalable, and techno-economically viable framework for intelligent HVAC management.

Keywords: *HVAC Optimization; Hybrid Artificial Intelligence; Digital Twin; Building Energy Modeling (BEM); Internet of Things (IoT); Distributed Mechatronic Controller; Thermal Comfort (PMV); Bayesian Optimized Random Forest (BORF); Edge Computing; Technoeconomic Analysis.*

CHAPTER 1

INTRODUCTION

This chapter presents the foundation of the study by discussing the need for intelligent and energy-efficient HVAC control systems in modern buildings. It introduces the integration of hybrid artificial intelligence, distributed Internet of Things (IoT) mechatronic architecture, and Digital Twin technology as a unified framework for optimizing thermal comfort and energy consumption.

Nature and Importance of the Study

Heating, Ventilation, and Air Conditioning (HVAC) systems account for approximately 30% to 40% of the total energy consumption in residential and commercial buildings, representing a primary driver of operational expenditures and greenhouse gas emissions.

Conventional HVAC control frameworks typically rely on static thermal setpoints and rigid, time-based schedules. Consequently, these systems fail to account for the dynamic influences of transient occupancy patterns, varying metabolic activity levels, and fluctuating ambient environmental conditions. This lack of responsiveness leads to systemic energy inflation, prolonged thermal discomfort, and operational inefficiencies.

As global demand for energy-efficient, occupant-centric building management intensifies, a critical requirement has emerged for intelligent control strategies capable of transitioning from reactive adjustments to predictive and adaptive decision-making.

Recent advancements in Artificial Intelligence (AI), the Internet of Things (IoT), and cyber-physical modeling present a clear opportunity to transform HVAC operations through integrated, data-driven automation. Machine learning algorithms such as Random Forest regression and Predicted Mean Vote (PMV)-based thermal equilibrium models have demonstrated significant efficacy in forecasting thermal indoor states and optimizing energy expenditure based on multi-dimensional feature vectors including temperature, relative humidity, CO₂ concentration, and occupancy patterns. Concurrently, IoT-enabled distributed control architectures facilitate real-time coordination among edge sensors, actuators, and localized computational nodes, thereby enhancing scalability and fault tolerance compared to legacy centralized systems.

Complementing these technologies is the emergence of digital twin frameworks, which create real-time virtual replicas of physical structures for continuous monitoring, simulation, and operational optimization. Unlike static Building Information Modeling (BIM) architectures, digital twins coupled with Building Performance Simulation (BPS) platforms such as EnergyPlus and OpenStudio enable predictive validation of HVAC strategies prior to physical deployment, thereby mitigating trial-and-error energy losses and operational risks.

At the local hardware level, modern split-type inverter air-conditioning units, such as the Samsung and General Royal systems utilized in this study, rely on internal Proportional-Integral-Derivative (PID) control loops embedded within their factory printed circuit boards (PCBs) to regulate compressor frequency and thermal output. Because these proprietary internal feedback loops are closed and governed via

consumer infrared (IR) remote protocols, implementing advanced supervisory optimization requires an external mechatronic interface capable of translating high-level algorithmic decisions into compatible IR control vectors.

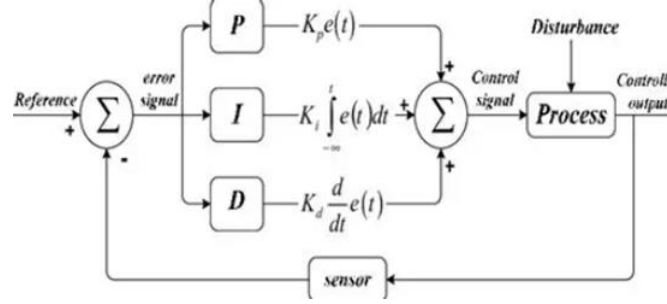


Figure 1. PID Logic of ACU Systems (Sahu et al., 2025)

Historically, residential cooling systems utilized a binary (On-Off) control strategy, also known as hysteresis or "*bang-bang*" control. In this conventional architecture, the compressor operates at a fixed, maximum angular velocity (ω_{max}) until the temperature setpoint is reached, at which point the system undergoes a complete shutdown.

The compressor remains inactive until the ambient temperature drifts beyond a predefined threshold, triggering a restart. This cyclic behavior characterized by frequent inrush current spikes and thermal oscillations results in significant energy inefficiency and mechanical fatigue due to compressor short cycling.

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \frac{de(t)}{dt} \quad (1)$$

In contrast, modern inverter-based systems employ a Proportional-Integral-Derivative (PID) control logic to achieve variable speed regulation. Rather than binary states, the PID controller continuously evaluates the system error ($e(t)$), defined as the

instantaneous difference between the user-defined reference (setpoint, T_s) and the measured process variable (ambient temperature, T_o). By modulating the compressor's RPM in proportion to the magnitude, duration, and rate of change of this error, the system maintains a state of dynamic equilibrium.

The advantages of this modulated approach are two-fold:

1. **Energy Conservation.** The inverter unit significantly lowers cumulative power consumption by avoiding the high-torque startup phases associated with On-Off systems and operating at reduced partial-load frequencies.
2. **Mechanical Longevity.** The reduction in thermal and mechanical shock loads extends the operational lifespan of the compressor assembly.

By utilizing this efficient internal PID loop as the foundational layer, this study seeks to layer a high-level Machine Learning supervisory controller to further optimize the setpoint inputs, thereby maximizing both energy savings and the comfort of the occupants.

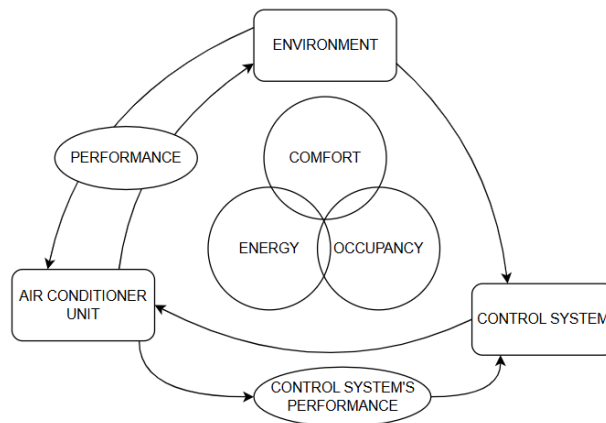


Figure 2. Multi-Dimensional System of HVAC Optimization

While ‘Refrigeration Cycle Theory’ is fundamentally grounded in the second law of thermodynamics, achieving human thermal comfort introduces a stochastic and multi-variable layer of complexity. As defined by the ASHRAE Standard 55, comfort is not a function of dry-bulb temperature alone; it is an equilibrium state influenced by a synergy of environmental variables including relative humidity, air velocity, and mean radiant temperature, as well as physiological factors like metabolic rate (*met*) and clothing insulation (*clo*).

The challenge in modern applied physics lies in the integration of these disparate domains: the fluid dynamics of airflow, the latent heat transfer of humidity, and the homeostatic regulation of human physiology. Current inverter technology provides a highly efficient engine for cooling; however, the overall performance of the system is often throttled by the human-in-the-loop.

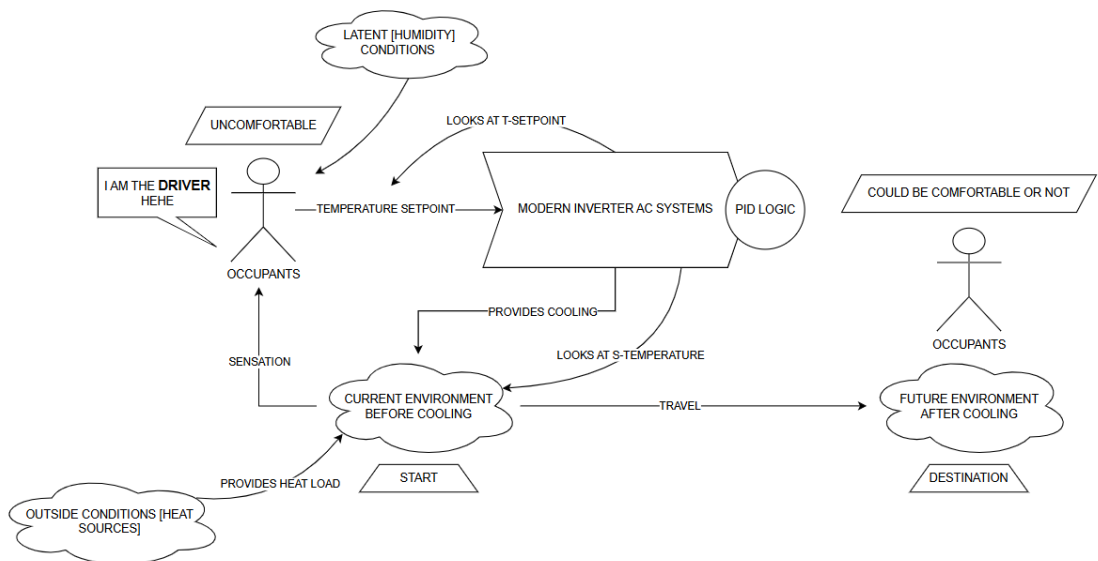


Figure 3. Conventional Inverter ACU Control Logic

To illustrate this systemic limitation, an analogy can be drawn to a high-performance vehicle optimized for fuel efficiency and precision dynamics. Regardless of the inherent mechanical and thermodynamic efficiency of the machine, its aggregate energy consumption and trajectory precision remain fundamentally dictated by the human operator.

If the driver lacks the high-fidelity technical feedback required to leverage the advanced powertrain features of the vehicle, the coupled human-machine system operates in a highly suboptimal state. Analogously, a modern inverter-driven air-conditioning unit represents a highly efficient thermodynamic tool; however, when governed by manual human inputs relying on subjective thermal perception rather than deterministic environmental data, the system fails to achieve its theoretical peak operating efficiency.

Modern inverter air-conditioning systems are equipped with sophisticated operational modes and features designed for efficiency; however, there is a significant discrepancy between system capability and actual user utilization. Empirical observation suggests that most users bypass advanced settings such as dehumidification (Dry), sleep curves, or programmable timers and ultra-saving modes, opting instead for a simplistic interaction model restricted to basic temperature setpoint adjustments. This manual intervention often results in suboptimal performance, as users lack the real-time technical data required to align AC settings with dynamic environmental changes.

Furthermore, conventional AC units operate on a closed-loop feedback system localized primarily to air temperature. This "single-variable" approach is fundamentally limited, as it ignores critical parameters of human thermal comfort, such as relative

humidity, mean radiant temperature, and external ambient fluctuations. By relying solely on a thermostat, the system fails to achieve true thermal neutrality as defined by ASHRAE standards. Consequently, the burden of regulation is placed on the user, who must constantly and inaccurately adjust the unit to compensate for a lack of comprehensive environmental awareness within the native control logic of the AC unit.

This research proposes the implementation of a Machine Learning supervisory layer to act as the professional driver. By synthesizing complex environmental inputs that the average user ignores, the proposed controller can navigate the operational states of the inverter to achieve a precise destination which is the optimal thermal comfort with minimal energy expenditure.

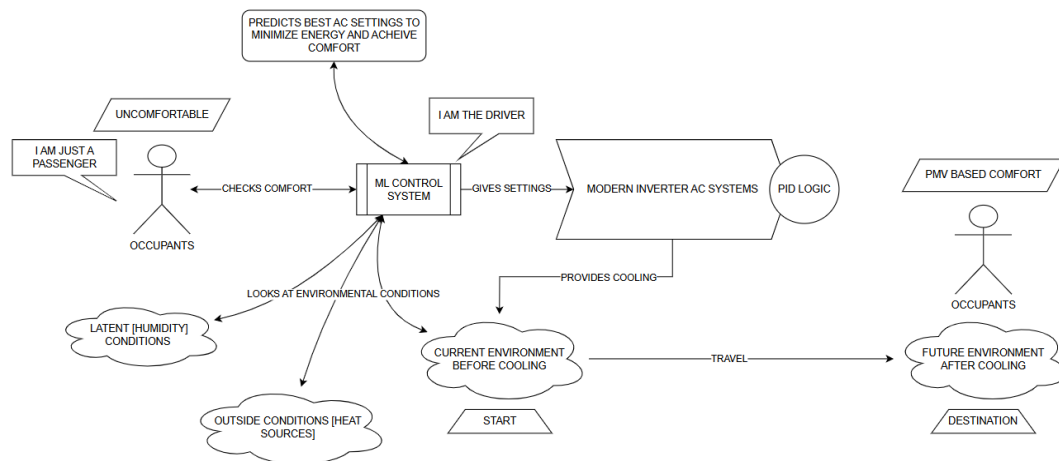


Figure 4. Applied Hybrid AI Predictive Control Logic

While existing literature extensively explores integrated control strategies for centralized HVAC systems, such implementations typically necessitate direct modification of the unit's internal firmware or proprietary hardware. This research diverges from traditional centralized approaches by proposing a non-invasive, exogenous control architecture.

As illustrated in Figure 4, the system utilizes an external sensor suite to monitor multi-dimensional environmental factors influencing thermal comfort. These data points are processed via a Machine Learning (ML) optimizer, which dynamically determines the ideal operational parameters. These setpoints are then transmitted to the air conditioning (AC) unit via an Infrared (IR) interface. This method ensures that the external controller acts as a high-level supervisor analogous to a pilot providing navigation inputs, while the unit's internal PID (Proportional-Integral-Derivative) logic remains responsible for low-level compressor modulation and frequency control.

By decoupling the comfort-seeking logic (ML) from the hardware-level efficiency logic (PID), the system achieves a dual-optimization state:

1. **Environmental Optimization.** The ML layer ensures the space adheres to ASHRAE-based thermal comfort standards.
2. **Operational Efficiency.** The native PID controller continues to manage the inverter's duty cycle without interference, maintaining high energy-efficiency ratios (EER).

Ultimately, this design transitions the human occupant from an active controller to a passive recipient, automating the pursuit of thermal comfort while leveraging the inherent efficiency of modern inverter technology.

Statement of the Problem

Despite the individual progress in AI-based control, IoT automation, and digital twin modeling, few studies attempt to unify these three domains under a single techno-

economically validated framework. Existing implementations often focus solely on either control accuracy or simulation fidelity without addressing long-term feasibility in terms of hardware cost, energy savings, or return on investment.

To address this gap, this study introduces an integrated smart HVAC control framework combining (1) machine learning-driven adaptive thermal regulation, (2) an IoT-enabled hierarchical distributed mechatronic controller, and (3) a digital twin-based simulation environment for pre-deployment validation. The system learns comfort-energy trade-offs through real-time data acquisition, execute control logic through a modular distributed network, and be benchmarked through both experimental testing and techno-economic analysis.

The expected outcome is a practical yet scalable blueprint for intelligent HVAC automation that achieves measurable energy reduction without compromising occupant comfort. By utilizing low-cost microcontrollers, open-source AI libraries, and accessible simulation platforms, the study aims to demonstrate that advanced smart building technologies can be realistically implemented even in resource-constrained settings. Ultimately, the research intends to contribute to the progression of sustainable building management practices by bridging the gap between academic innovation and real-world applicability.

Objectives of the Study

To design, develop, and implement a distributed mechatronic controller compatible with IR supported AC Systems with the goal to reduce energy consumption while maintaining thermal comfort acceptable by ASHRAE Standards.

Specifically, the study aims to:

1. Develop a machine learning algorithm capable of learning non-linear relationships between energy and comfort;
2. Develop a digital twin model of the case study building using an open-source simulation platform to validate energy and comfort of the target building through virtual parallel assessment; and
3. Design and implement an IoT distributed mechatronic controller, compatible with IR-supported air-conditioning units and perform techno-economic analysis.

Scope and Limitations of the Study

This section delineates the operational scope and inherent technical limitations of the study, establishing the boundaries that guided the researchers throughout the execution of this study.

Scope of the Study

To establish clear operational parameters, the scope of this study was explicitly defined. The following boundaries were strictly observed and enforced throughout the execution of the research:

1. The machine learning framework utilized Random Forest Regression optimized via Bayesian Optimization to characterize the non-linear trade-offs between indoor thermal comfort and HVAC energy consumption within the target thermal zone.

2. The model incorporated the following variables: outside temperature, outside humidity, inside temperature, inside humidity, occupant count, electrical energy consumption, and time of day.
3. A distributed mechatronic controller prototype was engineered and developed for this study, utilizing hardware including, but not limited to, the Raspberry Pi 5 and ESP32 microcontrollers.
4. The physical deployment encompassed a hybrid control architecture designed to support the machine learning model and ensure operational safety. A physics-informed, rule-based supervisory logic was integrated alongside the BORF model to govern final HVAC operational settings during live edge computing execution.
5. The development of a digital twin capable of reproducing the operational behavior of the target facility through virtual parallel assessment, utilizing an open-source building performance simulation platform.
6. The research prioritized the representation of measured building performance and operational responses within the digital twin environment to facilitate continuous comparison between the physical asset and its virtual counterpart.

Limitations of the Study

To establish the boundary conditions of this research, the operational and methodological limitations were explicitly defined. The framework was governed by

the following constraints throughout its development, deployment, and evaluation phases:

1. **Control Interface and Integration.** The physical execution of the control system was limited to interfacing with split-type or window-mounted air-conditioning (AC) units equipped with infrared (IR) remote-control capabilities. The prototype operated strictly as an external supervisory overlay without modifying the internal circuitry, hardware components, or the native proportional-integral-derivative (PID) control loops embedded by the manufacturer.
2. **Spatial and Thermal Zone Configuration.** The applicability of the control system was restricted to single-zone office or classroom environments characterized by open, unpartitioned layouts. This study excluded spaces serviced by centralized air-handling units (AHUs) or variable air volume (VAV) air-distribution cooling networks.
3. **Dataset Volumetric Constraints.** Due to physical operational constraints, the machine learning model was trained on a bounded dataset of approximately 3,000 steady-state data points. Consequently, the standalone mathematical extrapolation of the predictive model outside this baseline training envelope is constrained, necessitating the integration of a deterministic supervisory layer to ensure operational safety under real-world conditions.
4. **Digital Twin Boundary Conditions.** The developed digital twin was engineered to replicate the operational response of the target facility based strictly on empirical sensor data. The analytical validity of the virtual model

was measured by its capacity to reproduce observed building performance under specific, localized ambient conditions experienced during the testing window.

5. **Calibration Target Limitations.** In accordance with ASHRAE Guideline 14, calibration and validation protocols focused primarily on whole-building HVAC electrical energy consumption. While thermal comfort indicators such as indoor temperature, relative humidity, and Predicted Mean Vote (PMV) were calculated and monitored via virtual parallel assessment, they did not serve as primary objective functions during the automated mathematical calibration phase.
6. **Digital Twin calibration method.** Model calibration was performed using a Bayesian optimization framework based on Gaussian Process Regression. Consequently, the resulting calibrated parameter set represents the optimal solution identified within the calibration objectives, parameter ranges, and search space defined in this study and should not be interpreted as the definitive physical characteristics of the target building.

Selected Room of Study and Technical Specifications

The physical deployment and empirical data collection for this study were conducted inside the Department of Mechanical Engineering (DME) Conference Room. This specific facility was selected due to its representative single-zone thermal configuration, the presence of a target split-type inverter air-conditioning unit (ACU), and its high variability in dynamic occupant density. A comprehensive architectural overview and geometric modeling of the space are detailed in Appendix A.

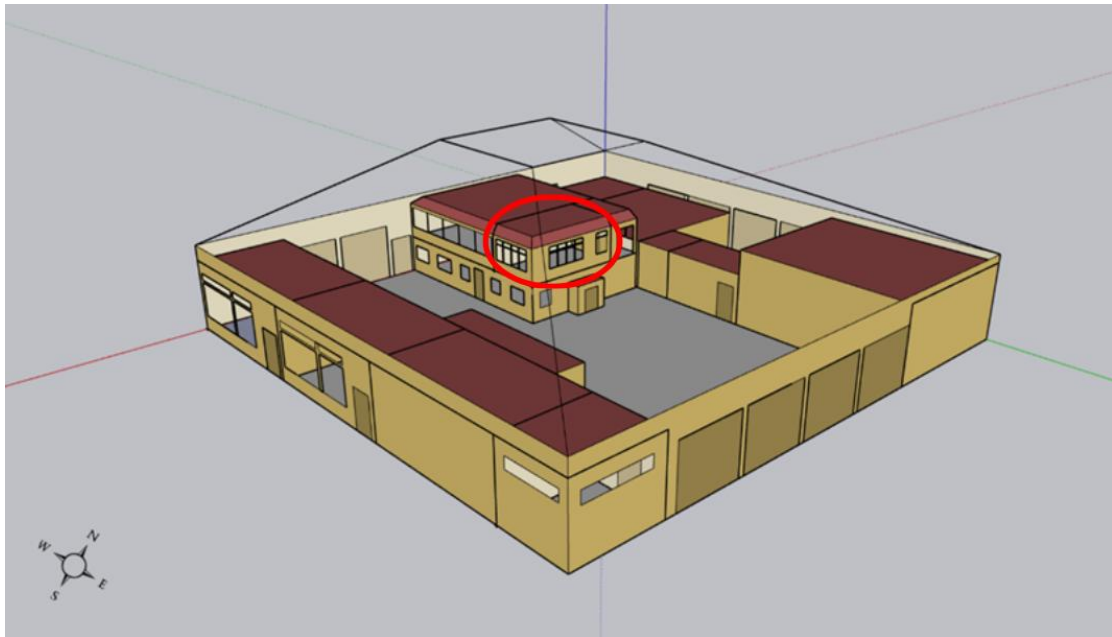


Figure 5. Overview of the DME Workshop

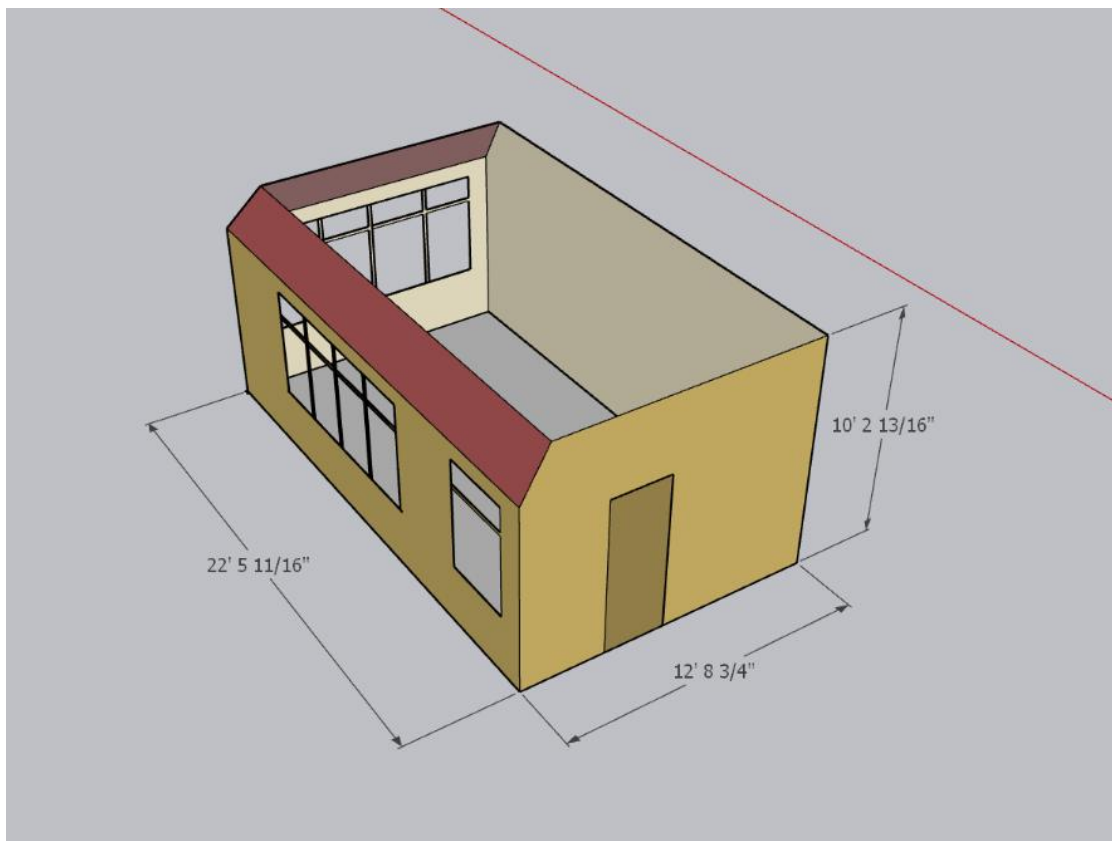


Figure 6. DME Conference Room (Room of Study)

The DME Conference Room is situated within the Department of Mechanical Engineering Workshop, as demarcated in Figure 1. Serving primarily as a dual-purpose lecture and academic assembly facility, the zone's thermal environment is regulated by two independent, wall-mounted split-type inverter air-conditioning units (ACUs): a Samsung 1.5 HP (1 TR) unit and a General Royal 1.5 HP (1 TR) unit. Comprehensive technical specifications and electrical profiles for these cooling assets are documented in Appendix C.

SAMSUNG	
AIR CONDITIONER	
MODEL NAME	AR12MVFHWWKN
CAPACITY (COOL / HEAT)	12 600 kJ/h / -
RATED VOLTAGE & FREQUENCY	230 V~ 60 Hz
RATED CURRENT (COOL / HEAT)	5.5 A / -
RATED POWER INPUT (COOL / HEAT)	1240 W / -
CLIMATE CLASS	T1

SAMSUNG ELECTRONICS CO., LTD.
MADE IN THAILAND

SPLIT TYPE AIR CONDITIONER			
INDOOR MODEL	GRI-15ASI		
OUTDOOR MODEL	GRI-15ASO		
COOLING CAPACITY	12000Btu/h		
HEATING CAPACITY	—		
REFRIGERANT	R410A/0.7kg		
EXCESSIVE OPERATING PRESSURE	DISCHARGE	4.2MPa	
	SUCTION	1.5MPa	
POWER SOURCE	230V~ 60Hz, 1Ph		
STANDARD RATING CONDITIONS	COOLING	CURRENT	4.5A
		INPUT	1035W
	HEATING	CURRENT	—
		INPUT	—
RATED CURRENT	8.5A		
RATED INPUT	2000W		
OUTDOOR UNIT RESISTANCE CLASS	IP 24		

GENERAL[®]
Royal

⚠ CAUTION (OUTDOOR UNIT)

1. Ensure to evacuate the air inside the indoor unit and pipes with vacuum pump.
2. Make sure the additional amount of refrigerant to be charged is based on the pipe size and length, please refer to the INSTALLATION INSTRUCTIONS for details.
3. Incorrect installation due to ignoring of the instruction will cause serious problem to the machine.

Figure 7. Air-conditioning Unit Specifications



Figure 8. Samsung 1.5 HP 1 Ton Inverter



Figure 9. General Royal 1.5 HP 1 Ton Inverter

CHAPTER 2

REVIEW OF RELATED LITERATURE

This chapter presents the body of literature and related studies relevant to intelligent HVAC control, thermal comfort optimization, Digital Twin technology, and distributed IoT-based automation systems. It establishes the theoretical and technical foundations of the study by examining existing methodologies, frameworks, standards, and research gaps that support the development of the proposed system.

Intelligent HVAC Control and Optimization Systems

The optimization of heating, ventilation, and air conditioning (HVAC) systems has become an important research domain in response to the need for energy efficiency and occupant comfort in modern buildings. Various approaches have been developed, ranging from model-based control to machine learning and reinforcement learning techniques. This section presents selected works relevant to the proposed study, highlighting their methods, formulations, outcomes, contributions, recommendations, and research gaps.

IoT-Based and Intelligent HVAC Control Systems

Carli et al. (2020) introduced an Internet of Things (IoT)-based architecture that integrates Model Predictive Control (MPC) for HVAC systems in smart buildings. Their method formulated HVAC optimization as a quadratic optimization problem, where the objective function minimized both energy consumption and deviations from the desired Predicted Mean Vote (PMV) comfort levels.

To address the nonlinearity of PMV, they applied linearized approximations, enabling real-time computations. The architecture included a network of environmental sensors, actuators, a control unit executing the MPC algorithm, and a user interface connected through the Internet.

Outcomes showed that the system achieved significant energy savings while maintaining thermal comfort, outperforming classical thermostat control. Their main contribution lies in demonstrating that IoT integration makes advanced control strategies feasible in real environments, not just simulations. They recommended further work in scaling MPC for multi-zone buildings and incorporating renewable energy integration. However, the gap remains in the computational intensity of MPC and its dependence on accurate thermal models, limiting its application in resource-constrained or smaller-scale settings.

Javed et al. (2017) proposed a smart controller for HVAC systems using Random Neural Networks (RNNs) combined with wireless sensor networks (WSNs) and cloud computing. Their formulation involved two neural models, with one estimating the number of occupants and the other estimating the PMV-based comfort setpoints

The models were trained using a hybrid Particle Swarm Optimization–Sequential Quadratic Programming (PSO-SQP) algorithm, which optimized network weights for better convergence. They compared three architectures: (1) cloud-only computation, (2) base station-only processing, and (3) distributed computation between nodes and base stations.

Results revealed that the distributed approach reduced power consumption by 4.4% compared to the cloud-only method and by 19.23% compared to base-station-only processing. Their contribution lies in proving that intelligent controllers embedded in local infrastructure can save energy while maintaining comfort. They recommended designing future Building Energy Management Systems (BEMS) to distribute intelligence across nodes for scalability and robustness.

Namatēvs (2018) explored HVAC optimization using Deep Reinforcement Learning (DRL), combining Q-learning with deep neural networks (DNN). The HVAC system was modeled as a Markov Decision Process (MDP), where the state space included indoor and outdoor parameters (temperature, airflow, humidity, occupancy), the action space represented control signals (heating or cooling intensity, airflow levels), and the reward function balanced energy consumption against occupant comfort indices such as PMV.

The DNN was trained to approximate the Q-function, allowing the agent to learn optimal control policies through trial and error. Outcomes showed that DRL can dynamically adapt to uncertain and nonlinear environments, offering potential for long-term energy cost minimization.

The contribution of this study lies in presenting the DRL as an adaptive, model-free method that avoids reliance on explicit thermal models. However, the research was limited to simulations, and the computational requirements of the DRL remain a barrier to real-time embedded implementation. Namatēvs recommended exploring hybrid approaches that combine predictive models with reinforcement learning to reduce computational overhead.

Trivedi et al. (2017) developed iSchedule, a campus-scale HVAC scheduling system that leverages Wi-Fi connectivity logs as a proxy for occupancy detection. Their method applied an ensemble gradient boosting regressor to learn occupancy patterns for different zones and generate adaptive HVAC schedules.

The formulation used Wi-Fi association and dissociation events from thousands of access points across 112 campus buildings, treating device counts as soft sensors for occupancy. The outcomes indicated a 37% reduction in energy waste and a 3% improvement in comfort levels compared to static scheduling. Their main contribution is showing that existing infrastructure can be repurposed for scalable HVAC scheduling without requiring additional hardware. They recommended applying such learning-based schedules to other large-scale facilities with dense network infrastructures. However, gaps exist in terms of granularity, as Wi-Fi cannot precisely track occupancy counts or occupant comfort, and the approach is limited to institutions with pervasive Wi-Fi coverage.

Fuzzy Logic-Based HVAC Optimization

A study by Abuhussain et al. (2023) proposed an adaptive HVAC system employing a fuzzy controller approach, utilizing a Mamdani Fuzzy Inference System with automatic rule generation to optimize energy consumption based on inputs such as occupancy status, ambient temperature, humidity, and real-time electricity tariffs. Simulations conducted via EnergyPlus software, incorporating local weather data from Saudi Arabia and evaluated through the Fanger thermal comfort model, demonstrated up to 50% energy savings compared to the baseline IoFClima system when prioritizing

energy over comfort, with the controller maintaining satisfactory predicted mean vote (PMV) ranges across varying scenarios.

The authors discussed the limitations of traditional inputs like ambient temperature and humidity for precise comfort management, emphasizing the benefits of occupancy-based controls and demand response for handling thermal inertia, though the approach relies on simulation-only validation and a narrow input set, potentially overlooking real-world sensor inaccuracies and broader parameters such as CO₂ levels.

Complementing this fuzzy logic emphasis, Acquaah et al. (2020) explore occupancy detection for smart HVAC efficiency using a deep learning neural network framework with thermal imagery, comparing a pre-trained ResNet-50 model for feature extraction combined with multi-class Support Vector Machine classification against a pre-trained AlexNet with transfer learning.

Data acquisition involved thermal cameras and sensors in a laboratory setup, achieving accuracies of approximately 96% for ResNet-50 + SVM and over 98% for AlexNet, with the latter performing better on larger datasets despite increased computational time. The discussion highlights the privacy advantages of thermal imaging over RGB cameras and its potential to reduce HVAC energy loads by conditioning occupied zones only, outperforming prior methods like ThermoSense, yet the study is limited to small-scale detection (up to three occupants) without full HVAC integration or energy quantification, restricting its applicability to dynamic, multi-room environments.

Further advancing fuzzy-based alternatives to conventional controls, Chojecki et al. (2023) advocate replacing proportional-integral-derivative (PID) controllers in air handling units with fuzzy logic controllers structured to mimic PID functions without requiring tuning, validated through Simulink simulations using real building data and implemented on a custom hardware control board. The fuzzy controller improved integral control quality indicators by at least 27.4%, reduced supply air temperature variations, and achieved 12.7% energy savings compared to untuned PIDs, with discussions underscoring the prevalence of sub-optimal PID tuning in over 50% of installations leading to oscillations and waste. However, the method neglects advanced inputs like occupancy or PMV, focusing solely on temperature, and assumes no initial fuzzy rule adjustments, which may limit its plug-and-play feasibility in diverse building contexts.

PMV and Thermal Comfort-Oriented HVAC Control

In a practical experimental context, Kim et al. (2020) introduced an optimal control method for HVAC systems in offices using algorithms based on thermal environment factors, regulating temperature and humidity ranges (18-21°C and 40-60% RH) alongside PMV control (-0.5 to +0.5) in a real South Korean office setting over three days.

Surveys indicated enhanced occupant comfort and productivity, with PMV control effectively addressing temperature-humidity coupling and energy regulations, though energy savings were implied rather than precisely quantified. The authors emphasize balancing energy conservation with comfort to mitigate productivity losses

(thermal environment impacting 44.1% of efficiency), but the single-case, short-duration experiment is vulnerable to weather dependencies and lacks automated sensors for real-time PMV, highlighting a need for broader scalability.

Addressing more complex system dynamics, Homod (2018) presents a feedforward hybrid layers control for HVAC optimization in smart buildings, hybridizing Takagi-Sugeno-Kang fuzzy inference with neural networks trained offline via Gauss-Newton regression and tuned online with gradient descent, controlling five parameters (valves, dampers, and fan speed) in variable water and air volume systems while leveraging PMV sensors to handle nonlinearity, inertia, and coupling. Compared to hybrid PID-cascade benchmarks, the approach yielded superior precision, disturbance rejection, and over 30% energy savings versus on/off controls, with discussions noting its ability to decouple temperature and humidity without additional hardware. Nonetheless, the simulation-based validation assumes PMV sensor availability, overlooks occupancy integration, and may be computationally intensive for legacy systems, underscoring opportunities for hardware simplification.

Data-Driven and Machine Learning-Based Optimization

Building on black-box optimization techniques, Lin et al. (2023) propose a Bayesian optimization framework for HVAC system control, modeling the problem as a contextual optimization task with Gaussian process regression as the surrogate model for energy efficiency (negative total power) and employing an upper confidence bound acquisition function enhanced with noise-based exploration and dual constraint handling (feedforward optimization via genetic algorithms and feedback correction).

The framework automates sample construction from time series trending data, excluding transition periods through algorithmic identification, and was evaluated on a simulated data center system calibrated with real data, achieving over 10% energy savings within weeks compared to baseline building automation controls.

The discussions emphasized on the efficiency and stability of the framework for large-scale deployment, surpassing data-intensive methods like model predictive control and reinforcement learning in practical applications, although it relies on quasistatic setpoints for near-optimal rather than fully dynamic performance, with manual feedback rules and simulation-based validation potentially limiting generalization to diverse HVAC systems.

The study by Manjarres et al. (2017) aimed to automate energy-efficient HVAC strategies in tertiary buildings through the Next24h-Energy framework, targeting a reduction in HVAC energy consumption, which typically accounts for 30-50% of building energy use, while maintaining indoor temperature margins for user comfort. Their methodology involved Random Forest regression for predictive modeling of indoor and outdoor conditions, enabling daily adaptive optimization of HVAC ON/OFF schedules and mechanical ventilation to address dynamic variability in occupancy and environmental factors, utilizing a black-box approach that decreases computational complexity and engineering knowledge requirements compared to physical or grey-box models.

The framework included integration with Building Energy Management Systems (BEMS) for two-way communication, a dedicated database for storing sensor data on temperature, humidity, and occupancy, and simulation software likely based on

MATLAB or Simulink, validated using real sensors deployed in the Mikeletegi 1 office building in Donostia-San Sebastian, Spain. The results demonstrated substantial savings of 48% in heating energy and 39% in cooling energy over the test period, with no compromise to comfort as indoor temperatures remained consistently within predefined margins.

The discussions emphasized the superiority of Random Forest in handling nonlinear dynamics over static physical or grey-box models, reducing reliance on human intervention through automated daily parameter adaptation and facilitating generalization across varying building conditions.

The authors concluded that this framework offers a practical solution for bridging the gap between sensor infrastructures and facility management actions, yielding median energy savings of 15% from building commissioning. Recommendations include extending the approach to incorporate additional environmental variables and testing in diverse climatic zones to enhance robustness. However, gaps in their study are notable, including the absence of Predictive Mean Vote (PMV) integration for a more comprehensive thermal comfort assessment beyond simple temperature margins, which overlooks factors like humidity, air velocity, and metabolic rates; a heavy reliance on simulations that limits empirical validation in varied real-world scenarios, potentially underestimating practical implementation challenges such as sensor inaccuracies or network delays; assumptions of detailed building structural knowledge, which restrict scalability to buildings without extensive prior data and may not account for retrofitting older structures; and a lack of cost-

benefit analysis for hardware deployment, ignoring economic feasibility in resource-constrained environments.

Ait Tchakoucht et al. (2024) focused on predicting electrical and thermal efficiencies in air-based photovoltaic-thermal (PV/T) systems to optimize energy performance and reduce dependency on traditional HVAC components, emphasizing the simultaneous generation of electricity and heat for cost-effective and environmentally friendly solar energy solutions.

Their method employed Random Forest with feature selection, such as temperature and humidity, combined with 10-fold cross-validation to model nonlinear thermal exchanges, facilitating pre-cooling strategies for HVAC integration through an automatic fuzzy rule generation method that simplifies inference system design.

The tools include EnergyPlus for conducting simulations based on local weather data, Python with Scikit-learn libraries for implementing Random Forest and cross-validation processes, and implicit basic sensors for integrating weather data like ambient temperature and solar irradiance.

The results yield exceptional accuracy with R-squared values of 99.9996% for electrical efficiency and 81.2% for thermal efficiency, alongside a mean absolute error of 0.0034 for electrical and 2.64 for thermal predictions, and a 50% reduction in processing time through feature selection.

The discussions highlight the outperformance of Random Forest over artificial neural networks, support vector machines, and linear regression in predictive accuracy and stability, enabling scalable solutions with potential 50% energy savings when

comfort is not prioritized, though the focus remains PV/T-specific and demonstrates robust generalization across validation folds.

The authors concluded that this approach serves as a viable alternative to traditional physical modeling for electrical and thermal performance estimation, offering impressive metrics that underscore its utility in renewable energy systems.

Recommendations involve exploring hybrid models with additional machine learning techniques and validating in real-world PV/T installations to bridge simulation-reality gaps.

The limitations encompass the lack of direct HVAC control mechanisms, as the predictions do not extend to real-time system adjustments like setpoint modulation; exclusive reliance on simulations without incorporation of physical sensors, which may overlook practical issues such as sensor drift or environmental noise; high data requirements for effective cross-validation, potentially straining computational resources and limiting applicability in scenarios with sparse datasets; and a narrow PV/T focus that neglects broader HVAC integration, such as occupancy-driven adaptations or PMV-based comfort evaluations, thereby restricting its relevance to full building energy management.

Razzano et al. (2025) sought to optimize centralized Air Handling Units (AHU) in multi-zone buildings for energy efficiency and indoor temperature control by leveraging deep reinforcement learning (DRL) and rule extraction techniques, addressing the substantial energy demand of HVAC systems while ensuring occupant comfort and health.

The methodology involved DRL for learning multi-action policies, such as adjusting supply water temperature, chiller valve positions, and economizer dampers, followed by decision tree-based rule extraction using regression trees for interpretability, compared against ASHRAE 2006 and Guideline 36 standards via high-fidelity co-simulation that models building envelope, loads, and HVAC components.

The tools comprised Spawn of EnergyPlus integrated with Modelica for equation-based dynamic modeling, Python libraries like Stable Baselines for DRL implementation, and virtual sensors for real-time data on occupancy, thermal loads, and environmental conditions.

The results indicated that the extracted rules achieve only 3% higher energy consumption than pure DRL but surpass ASHRAE sequences in reducing chiller and fan energy while minimizing temperature violations, with the co-simulation validating the framework's ability to approximate DRL policies effectively.

The discussions underscored the transparency of rule extraction in making complex DRL decision-making actionable for building automation systems, with a mechanical emphasis on Variable Air Volume (VAV) efficiency in multi-zone setups and the potential for integrating AI-driven methods with interpretable rules to facilitate adoption in real-world scenarios.

The authors concluded that this hybrid approach bridges the gap between advanced control technologies and practical implementation, demonstrating superior energy management over conventional rule-based systems.

The recommendations included extending the framework to incorporate real-time occupant feedback and testing in physical buildings to validate simulation assumptions.

The gaps included high computational demands for DRL training and execution, which require significant processing power and specialized programming expertise, making it inaccessible for low-resource applications; simulation exclusivity without hardware validation, potentially failing to account for real-world variabilities like sensor inaccuracies, equipment wear, or unforeseen environmental disturbances; omission of occupant behavior inference, such as activity levels or preferences, which limits the model's ability to adapt to dynamic human factors beyond basic occupancy; and a lack of economic analysis, including implementation costs and payback periods, overlooking feasibility in budget-constrained contexts.

Xu et al. (2024) aimed to predict daily heating energy consumption in residential buildings to enhance efficiency through a hybrid Random Forest model optimized by meta-heuristic algorithms, addressing key variables like architectural characteristics, occupancy, and ambient temperature.

The approach integrated Random Forest with optimizers like the Arithmetic Optimization Algorithm (AOA) and Ant Lion Optimizer (ALO) for hyperparameter tuning, employing K-fold cross-validation on features such as building envelope properties, occupant density, and external weather conditions to ensure model robustness and prevent overfitting.

The tools involved were Python with Scikit-learn for Random Forest implementation and optimizer libraries for meta-heuristics, alongside simulated datasets implying standard sensors for temperature, occupancy, and energy monitoring. Results show superior performance with an R-squared of 0.977, root mean squared error of 0.118, mean absolute error of 0.057, and AOA-RF as the optimal variant, demonstrating convergence after 101 iterations for ALO with a Theil Inequality Coefficient of 0.015.

The discussions noted the superiority of the hybrid over the standalone models, like artificial neural networks or support vector machines, mechanically enabling reductions in over-heating by 15-25% through accurate load forecasts that inform heating system operations.

The authors concluded that the hybrid framework provides a robust tool for energy consumption modeling, outperforming traditional methods in accuracy and generalization. Recommendations included applying the model to commercial buildings and incorporating real-time data streams for enhanced predictive capabilities.

The limitations included a prediction-only focus without a closed-loop control mechanism, restricting its utility to forecasting rather than active system modulation; residential specificity that does not extend to office or multi-zone environments, potentially overlooking complexities like varying occupancy patterns; computational intensity for meta-heuristics, which demands significant processing time and expertise, posing challenges for real-time applications; and reliance on simulated datasets without physical sensor validation, which may not capture nuances like sensor noise or environmental interferences.

Personalized Thermal Comfort and Occupant-Centric Approaches

Qi et al. (2023) investigated factors influencing thermal sensation vote (TSV) to develop a personalized regression model for HVAC setpoint optimization, using local skin temperature as a key indicator in moderate summer conditions.

The methodology correlated ambient temperature, multiple local skin temperatures (with emphasis on the forehead due to its strong correlation with TSV and ease of measurement), and subjective TSV from a survey of 40 subjects aged 20-59, incorporating individual characteristics like body fat percentage and setpoints to compensate for personal differences.

The tools included were infrared skin temperature sensors for non-contact measurements, survey instruments for collecting TSV data, and analytical software like Python or MATLAB for regression modeling and correlation analysis.

The results revealed a root mean squared error 15.8%, lower than support vector regression and comparative models by Zhang and Zhou, with the lowest mean absolute error, demonstrating the ability of the model to capture individual variations.

The discussions affirmed the forehead temperature's strong correlation for non-intrusive prediction, mechanically supporting adaptive cooling strategies that could yield 10-20% energy savings by fine-tuning setpoints without compromising satisfaction.

The authors concluded that this approach offers a practical, scalable solution for international ergonomic standards, outperforming existing models in accuracy and

minimal input requirements. Recommendations involved expanding the study to diverse climatic conditions and larger sample sizes for broader applicability.

The gaps comprised applicability limited to moderate summer conditions, failing to address extreme climates or seasonal variations that could alter skin temperature correlations; neglect of dynamic occupancy factors, such as real-time changes in occupant numbers or activities, which influence overall thermal loads; reliance on potentially intrusive surveys for initial data collection, which may introduce bias or limit real-time deployment; a small sample size of 40 subjects, reducing generalizability across demographics like age or gender; and absence of energy consumption metrics, overlooking the direct link between comfort predictions and HVAC operational efficiency.

Ermis et al. (2025) developed a practical personalized thermal comfort model by minimizing hard-to-measure features through meta-learning and feature importance analysis, addressing challenges in traditional models like Predicted Mean Vote that require complex inputs such as metabolic rate and clothing insulation.

The method used model-agnostic meta-learning (MAML) with a novel importance technique that outperforms SHAP in handling feature correlations, leveraging ensembles like Random Forest and support vector machines on minimal thermal sensation votes pretrained on subsets of the ASHRAE RP-884 database. Tools encompass Python with SHAP and meta-learning libraries for analysis, and basic environmental sensors for collecting temperature and humidity data in daily life environments.

The results exhibited higher prediction accuracy than non-personalized general models, effectively substituting complex personal factors with thermal sensation votes and providing robust insights into feature importance.

The discussions emphasized reduced measurement complexity for adaptive HVAC predictions, potentially yielding 10-15% energy savings via personalization while leveraging techniques like transfer learning to improve efficiency with limited data. The authors conclude that this methodology offers a parsimonious framework for thermal comfort modeling, outperforming aggregate models in real-world scenarios. Recommendations included further exploration of ensemble methods and integration with additional datasets for enhanced personalization.

Limitations include dependence on user-provided thermal sensation votes, which can be time-consuming and intrusive, potentially limiting scalability in automated systems; absence of direct energy integration, focusing solely on comfort prediction without linking to HVAC consumption metrics; handling of feature correlations that, while improved, may still overlook interdependent variables in highly variable environments; reliance on limited personal data, which could introduce bias in underrepresented demographics; and a lack of validation in extreme conditions, restricting applicability beyond controlled settings.

Woo et al. (2025) proposed an occupant-centric HVAC control using multi-modal infrared array sensing for real-time occupancy, counting, and operative temperature detection in open offices, aiming to enhance energy efficiency while considering adaptive comfort.

The methodology integrated convolutional neural networks for processing infrared data with two-position control strategies, simulated to evaluate energy-comfort trade-offs and incorporating multiple heat transfer mechanisms for accuracy.

The tools included infrared arrays with 110-degree vision angles for multi-modal functionality, EnergyPlus for modeling a 100-square-meter office space using 2023 Michigan weather data, and Python with Keras or TensorFlow libraries for neural network implementation.

The results indicated 33.7% heating energy savings compared to baseline controls, an 8.1-year payback period for the infrared sensor, and 95% detection accuracy, though with a 13.6% adaptive comfort penalized percentage during heating seasons.

The discussions highlighted the multi-modal robustness for reducing fan and chiller loads, emphasizing the trade-off between energy efficiency and comfort, and the potential of edge computing for real-time deployment.

The authors concluded that this framework advances intelligent HVAC systems, outperforming presence-based controls in energy performance. The recommendations involved real-world deployments and integration of dynamic occupant behavior to refine the approach.

The gaps involved inherent comfort trade-offs, where energy savings compromise thermal satisfaction, potentially unsuitable for high-occupancy spaces; simulation limitations that may not capture real-world variabilities like sensor degradation or environmental interferences; lack of behavior inference beyond basic

counting, ignoring activity levels that affect metabolic rates and loads; high initial sensor costs without detailed economic breakdowns for varying building scales; and focus on open offices, limiting generalizability to enclosed or residential settings.

Moreover, Challa et al. (2025) introduced a multi-modal system for real-time classroom occupancy detection to enhance HVAC efficiency, combining video analysis and gas sensors to provide precise data for dynamic control.

The method utilized YOLOv4 for face detection and counting from video feeds, fused with MS1100 gas sensor data for robustness in low-visibility or obstructed conditions, trained on a dataset of over 20,000 labeled human face images.

The tools comprised a Logitech C20 webcam for video capture, the MS1100 module for CO₂ and volatile organic compound sensing, OpenCV for stream processing, and integration frameworks for seamless HVAC optimization.

The results achieved over 98% accuracy in occupancy detection, demonstrating significant potential for reducing energy consumption through scalable building management.

The discussions highlighted the reliability of the fusion in addressing environmental challenges, offering a practical solution for smart buildings with enhanced responsiveness.

The authors concluded that the dual-modal approach provided a robust framework for energy-efficient infrastructure. The recommendations included scaling to diverse building types and incorporating additional modalities for further reliability.

The limitations included specificity to classroom environments, which may not translate to offices or varying layouts; omission of PMV or occupant behavior assessments, focusing solely on count without activity or comfort metrics; unaddressed hardware costs and deployment feasibility, potentially overlooking budget constraints; reliance on large labeled datasets, which could be resource-intensive to acquire; and lack of real-world energy savings quantification beyond potential estimates.

Amiri et al. (2025), as a review, synthesized occupancy detection techniques for HVAC control from over 190 studies between 2017 and 2024, categorizing methods as direct, indirect, and fusion-based while analyzing machine learning roles in real-time estimation. Notable studies include Kong et al. (2018) on CO₂-motion fusion achieving 20% office energy savings through sensor integration and Dong et al. (2019) on convolutional neural networks for 18% counting accuracy in building simulations.

The discussions emphasized the multimodal fusion's prominence for 95% accuracy and edge computing's potential, concluding that these advances enable resource-efficient HVAC but highlighted gaps in the standardized benchmarks and real-world data utilization.

The recommendations involved developing privacy-preserving frameworks and integrating occupant behavior for validation. The gaps across reviewed studies include dataset scarcity and bias, limiting model generalization; privacy concerns in camera-based methods without adequate edge processing; high costs and suboptimal sensor placement affecting accuracy by up to 20%; algorithmic adaptability issues in varying buildings; and underutilization of real deployments, often relying on simulations that overlook practical challenges like data resolution and preprocessing needs.

Multi-Objective Optimization and Digital Twin Applications

Latifah et al. (2025) presented a multi-objective genetic algorithm within a digital twin model to balance thermal comfort and energy efficiency in shared workspaces, addressing the trade-off between occupant well-being and HVAC operational costs.

The methodology enhanced genetic algorithms with artificial neural network surrogates for surrogate-assisted optimization, focusing on real-time setpoint adjustments and benchmarked against NSGA-II and multi-objective particle swarm optimization using ASHRAE 55 standards for comfort (20-24°C, 30-60% humidity).

The tools involved were the EnergyPlus integrated with Python for simulations, feedforward neural networks trained with Adam optimizers on sensor data for temperature, humidity, and occupancy, and IoT sensors for real-time inputs.

The results showed a 25% reduction in HVAC energy consumption and 78% occupant comfort satisfaction, outperforming NSGA-II (22% savings, 75% comfort) and MOPSO (20% savings, 74% comfort) by 3-5%.

The discussions highlighted the Digital Twin's role in predictive mechanical control for fan speeds, damper positions, and airflow optimization, concluding that this framework provides actionable insights for building engineers in energy management.

The recommendations included real-world deployments, integration of dynamic occupant behavior, and generalization across climatic conditions for future validation. The gaps comprised simulation exclusivity without physical building tests, potentially

missing real-world discrepancies like equipment inefficiencies or sensor failures; neglect of diverse climates, limiting applicability beyond moderate conditions; omission of occupant behavior details, such as activity variations that influence thermal loads; high computational demands for genetic algorithms, requiring substantial resources for population iterations; and lack of cost analyses or scalability assessments for different building types.

Collectively, these studies underscore the shift toward fuzzy logic, PMV-based strategies, and deep learning for HVAC optimization, achieving energy reductions of 12-50% through enhanced comfort management, yet common gaps include simulation-dominant validations, limited input scopes (e.g., excluding multi-modal data like CO₂ or weather forecasts), and insufficient quantification of health or productivity impacts.

For the present thesis, integrating deep learning occupancy detection from Acquaaah et al. (2020) as dynamic inputs to fuzzy/PMV controllers from Abuhussain et al. (2023), Chojecki et al. (2023), and Kim et al. (2020), while adopting the hybrid fuzzy-neural structure of Homod (2018) for nonlinearity and the efficient Bayesian framework of Lin et al. (2023) for handling uncertainties, could yield a more robust framework. Improvements may involve multi-zone real-world testing, adaptive online learning for uncertainties, and comprehensive metrics encompassing energy, comfort, and occupant well-being, potentially realizing 20-40% additional savings in variable-load buildings.

The Role of BEM in Digital Twin Architecture

The decarbonization of the built environment has accelerated the need for sophisticated predictive modeling to optimize facility operations as shown in Figure 10. While traditional Building Energy Modeling (BEM) has been extensively utilized during the design phase to forecast thermal dynamics, an acknowledged performance gap routinely emerges between "as-designed" simulations and "as-operated" energy consumption.

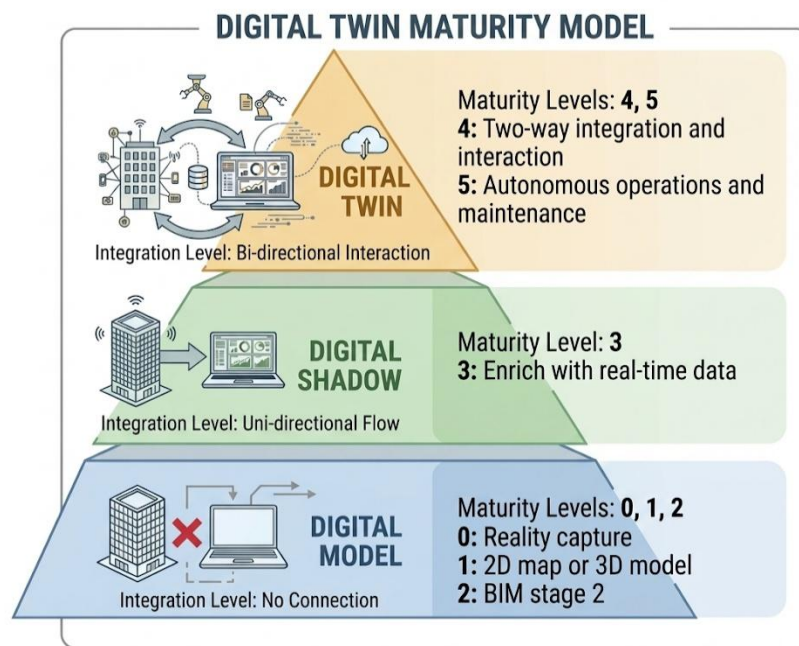


Figure 10. Digital Twin Maturity Model

To reconcile this operational discrepancy, the transition toward BEM Digital Twins (DT) has become a focal point in contemporary energy research. By establishing a bidirectional data flow between physical assets and virtual models, BEM DTs facilitate real-time monitoring, model predictive control, and dynamic energy optimization.

This review explores the paradigm shift from static modeling to real-time integration, examines edge-computing architectures, and emphasizes the critical programmatic validation protocols necessary to ensure the predictive robustness of these digital environments.

The development of a BEM-based Digital Twin represents a transformative approach to architectural and systems engineering. By utilizing simulation engines such as EnergyPlus, IESVE, or eQUEST, stakeholders can generate high-fidelity virtual models that account for complex geometric and thermodynamic variables. Alpin (2026) emphasizes that the core strength of BEM lies in its ability to mitigate the "performance gap" by anticipating energy demands, carbon footprints, and peak loads before a facility is physically realized. This digital representation serves as a controlled environment where designers can iterate through various building envelope configurations and HVAC strategies, thereby identifying energy waste at the design stage rather than through costly retrospective interventions.

Constructing a precise digital twin requires a multi-stage integration of architectural and environmental data. Current literature outlines a rigorous methodological sequence: importing CAD-based geometries into the simulation environment, defining climate-specific boundary conditions, and inputting granular material and system properties. Crucially, the validity of these modeling rests on the comparison of simulated outputs against historical performance benchmarks. Discrepancies between the virtual model and the actual building performance necessitate systematic adjustments to the internal variables of the model, such as internal heat gains and occupancy-dependent schedules. This cyclical process of

simulation and calibration is essential for ensuring that the digital twin functions as an accurate, predictive representation of the physical asset.

Edge-Hosted Digital Twins

Historically, BEM has operated under static, schedule-based assumptions regarding occupancy, microclimates, and boundary conditions. This inherent rigidity renders traditional models highly susceptible to performance drift as physical building usage diverges from design-phase estimates.

The integration of Internet of Things (IoT) sensors enables the continuous streaming of operational data such as temperature, CO₂ concentrations, and spatial occupancy into virtual replicas. Sayed et al. (2025) demonstrate that deploying a DT framework dynamically adapts to environmental changes and occupancy patterns, effectively shifting energy management from reactive to proactive see in Figure 10.

Notably, the authors addressed significant limitations in conventional cloud-dependent DTs by deploying the virtual replica on edge devices using user-centric platforms like Home-Assistant. This edge implementation significantly reduces latency, secures data privacy through localized processing, and maintains the continuous synchronization necessary to support high-fidelity, real-time energy efficiency recommendations.

Machine Learning and Data-Driven Optimization

A central component of the recent BEM DT architectures is the integration of Machine Learning (ML) to process stochastic operational inputs. Accurate modeling of

building thermodynamics requires precise knowledge of internal heat gains, which are profoundly influenced by occupant behavior.

Sayed et al. (2025) utilized ML algorithms to develop a data-driven occupancy detection mechanism, achieving a testing accuracy of 95.12%. This deterministic understanding of spatial utilization allows the DT to systematically optimize energy-intensive systems, such as HVAC and lighting, strictly based on real-time usage patterns rather than hypothetical schedules. While data-driven models demonstrate high efficacy in empirical interpolation, the literature indicates that pure black-box models frequently lack fundamental thermodynamic interpretability and struggle to generalize when exposed to unseen operational states outside of their initial training datasets. Consequently, integrating these dynamic inputs into physics-based BEM engines remains a critical requirement for scalable reliability.

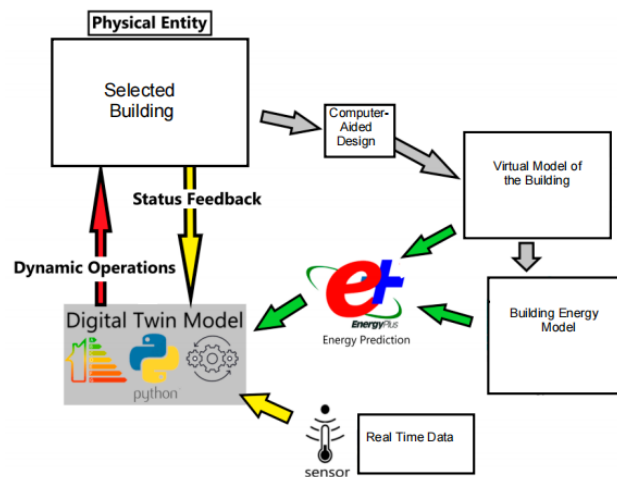


Figure 11. Real-time IoT Telemetry and Physics-Based BEM Engines

The successful convergence of real-time IoT telemetry and physics-based BEM engines hinges upon sophisticated, automated interoperability as shown in Figure 11.

Manually updating complex simulation input files to reflect shifting operational states is computationally prohibitive. Consequently, the development of automated programmatic workflows constitutes a significant methodological advancement in the field. Utilizing scripting libraries, such as Python's `eppy`, enables researchers to programmatically parse and manipulate complex thermodynamic simulation objects. This programmatic approach allows a script to systematically overwrite initial training boundary conditions with newly aggregated, dynamic schedules as shown in Figure 12. By circumventing manual data entry, these environments enable the high-frequency execution of building physics simulations required to maintain genuine parity between the physical asset and its digital twin.

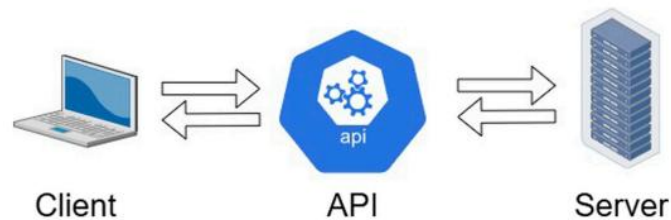


Figure 12. EnergyPlus Application Programming Interface (API)

Implementing Digital Twin to Quantify Energy Consumption

Guerrero Ramírez et al. (2025) developed a Digital Twin framework based on calibrated white-box Building Energy Models to reproduce the thermal behavior of a real building.

The Digital Twin was calibrated using measured indoor environmental conditions and demonstrated the capability of representing the thermal response of the building under operational conditions. Through calibration and validation procedures,

the study showed that Digital Twins can provide reliable representations of physical buildings and can support building operation and control strategies through continuous comparison between simulated and measured data.

A significant contribution of the study is the distinction made between thermal behavior calibration and energy consumption quantification. Although the developed Digital Twin successfully reproduced indoor thermal conditions using calibrated Building Energy Models and Ideal Air Loads systems, the authors acknowledged that HVAC system calibration was not incorporated into the modeling framework. Consequently, the study identified HVAC calibration and accurate energy consumption quantification as a critical direction for future research. The authors specifically stated that future work should focus on extending the calibrated models to accurately quantify energy consumption, emphasizing that this step is essential for achieving whole-system optimization and unlocking the full operational energy management capabilities of Digital Twins.

This observation highlights an important consideration in Digital Twin development. A model may successfully reproduce indoor environmental conditions while still requiring additional HVAC-specific calibration to accurately represent energy use. Therefore, the ability of a Digital Twin to predict thermal conditions should not automatically be interpreted as evidence of accurate energy quantification. Instead, dedicated calibration of HVAC-related parameters is necessary when energy consumption becomes the primary performance variable.

The distinction identified by Guerrero Ramírez et al. (2025) reinforces the importance of aligning calibration objectives with the intended application of the

Digital Twin, particularly when the objective extends beyond thermal behavior representation toward operational energy assessment and management.

Open-Source Building Energy Modeling

Open-source Building Energy Modeling (BEM) platforms have become essential tools for analyzing building energy performance, thermal behavior, and HVAC system operation. These tools provide researchers and engineers with accessible and flexible environments for developing virtual building models and evaluating energy efficiency strategies. Among the most widely used open-source platforms are OpenStudio and EnergyPlus, which enable detailed simulation of building systems and support advanced applications such as optimization and digital twin development. Therefore, the following sections discuss their fundamental concepts, methodologies, and roles in modern building energy analysis.

OpenStudio Software

The evolution of the built environment toward achieving net-zero energy consumption relies heavily on the capacity to accurately simulate and predict building thermal dynamics.

Before the advent of dynamic Building Energy Modeling (BEM) Digital Twins, the industry required accessible, high-fidelity spatial and thermodynamic modeling platforms to establish the virtual geometry of physical assets. The integration of 3D modeling interfaces with advanced simulation engines, such as the coupling of CAD software with the U.S. Department of Energy's EnergyPlus via OpenStudio, marked a

pivotal democratization of these capabilities as shown in Figure 13. This review analyzes the foundational methodology of static BEM design, discusses the inherent limitations of graphical-interface modeling in a continuous-operation context, and conceptually bridges these early simulation frameworks to the programmatic, automated architectures required for modern BEM Digital Twins.

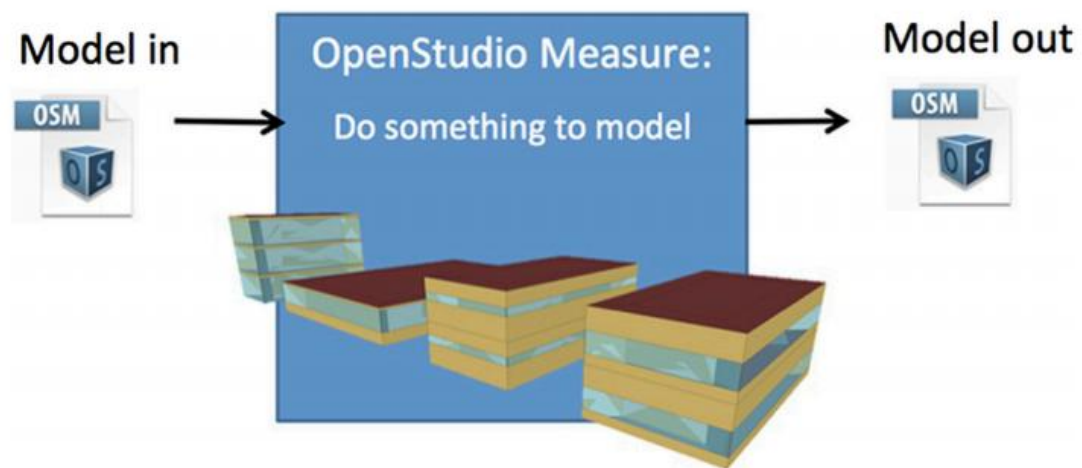


Figure 13. OpenStudio Software

OpenStudio Plugin

The accurate representation of the physical and thermodynamic properties of the building is the mandatory first step in constructing any virtual replica.

Al-Zubaydi (2013) detailed a comprehensive methodological workflow for conceptualizing energy-efficient buildings using the OpenStudio plug-in within CAD software as shown in Figure 14. This approach facilitates the creation of a building's spatial geometry by sequentially defining core and perimeter thermal zones, establishing surface matching for heat transfer calculations, and configuring critical

architectural elements such as the Window-to-Wall Ratio (WWR) and exterior fenestrations. By assigning specific material properties (e.g., U-factors, Solar Heat Gain Coefficients) and occupancy schedules based on established benchmarks like ASHRAE Standard 90.1, engineers can execute EnergyPlus simulations to estimate HVAC loads and overall consumption. This methodology effectively translates physical structural intentions into a computable mathematical model, establishing the static baseline necessary for subsequent digital twin integration.

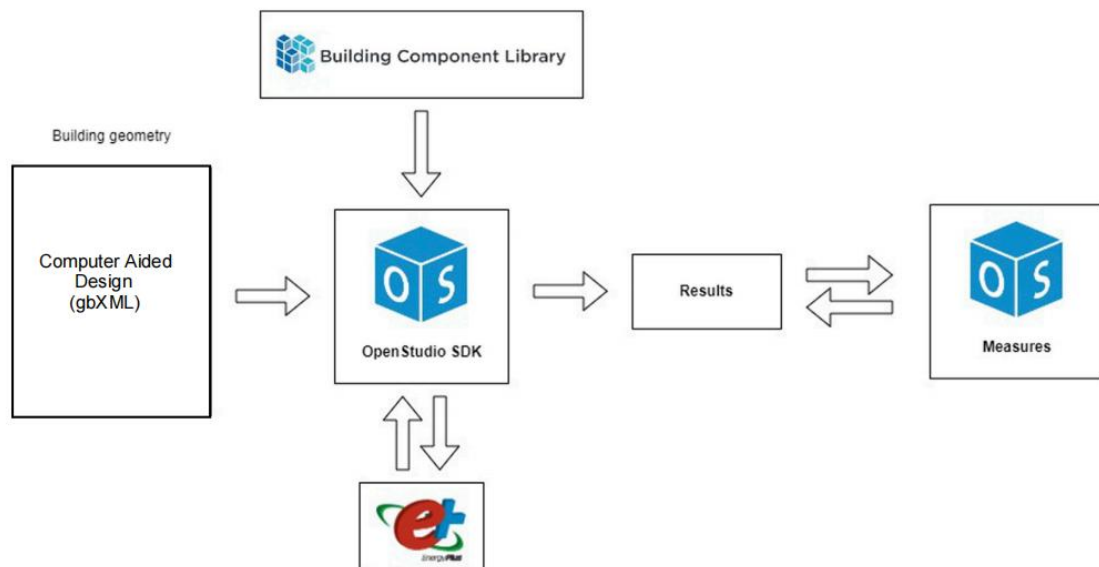


Figure 14. Building Energy Modeling (BEM) Digital Twin

While graphical user interfaces (GUIs) like SketchUp and OpenStudio are instrumental during the architectural design phase, the literature highlights their operational limitations when transitioning toward BEM Digital Twins. Al-Zubaydi (2013) illustrates a predominantly manual process: adjusting glazing properties, inserting overhangs, and configuring thermal zones require direct user interaction within the software's inspector windows. In a true Digital Twin ecosystem, which

demands the continuous ingestion of real-time Internet of Things (IoT) telemetry and stochastic environmental data, relying on manual GUI interventions is computationally unscalable and methodologically flawed. Static models inherently assume idealized schedules that quickly diverge from actual building utilization, resulting in a pronounced performance gap between simulated estimates and empirical energy usage.

EnergyPlus Engine

In the domain of whole-building energy simulation (WBES), the precision of virtual models is intrinsically tied to the underlying computational engine. EnergyPlus, developed by the United States Department of Energy (DOE), has established itself as the preeminent open-source simulation program for modeling building heating, cooling, lighting, ventilation, and other energy flows. Designed to replace legacy programs such as DOE-2 and BLAST, EnergyPlus integrates advanced thermodynamic algorithms to provide sub-hourly, high-fidelity energy analysis. This review focuses exclusively on the architectural and computational framework of EnergyPlus, examining its core heat balance methodology, its text-based interoperability, its inherent limitations in dynamic operational contexts, and its critical role in advanced predictive modeling and digital twin research.

The foundational strength of EnergyPlus lies in its rigorous application of the fundamental principles of thermodynamics and building physics. Unlike earlier simulation tools that utilized sequential calculation methods in solving space loads first, followed by system response, and finally plant equipment in EnergyPlus employs an integrated, simultaneous solution architecture. At its core is the Heat Balance Method,

which meticulously calculates the heat transfer across all building surfaces, accounting for conduction, convection, and both short-wave and long-wave radiation. This simultaneous solving mechanism ensures that the HVAC system accurately responds to the instantaneous thermal conditions of the building zones. By allowing for sub-hourly time steps, EnergyPlus can capture the transient thermodynamic behavior of complex building envelopes, internal thermal mass, and rapidly shifting environmental variables, rendering it indispensable for high-accuracy performance benchmarking.

While EnergyPlus is a highly sophisticated computational engine, it operates without a native graphical user interface, relying instead on ASCII text-based Input Data Files (IDF) and specific weather data formats (EPW). This text-based architecture, while initially presenting a steep learning curve, provides unparalleled flexibility and programmatic interoperability. Literature surrounding advanced building simulation emphasizes that the plain-text nature of the IDF allows researchers to interface with the simulation engine using high-level programming languages. Consequently, EnergyPlus is frequently utilized not just as a standalone tool, but as a modular physics engine embedded within larger computational frameworks. This interoperability is crucial for performing large-scale parametric analyses, multi-objective optimization, and sensitivity testing, where thousands of building permutations must be generated and simulated algorithmically without manual intervention.

Despite its computational rigor, the literature identifies significant limitations when EnergyPlus is deployed in continuous, real-time operational environments. As a deterministic simulation engine, EnergyPlus relies on static boundary conditions, idealized occupancy schedules, and fixed equipment performance curves.

In reality, building operations are highly stochastic. When the physical asset's usage deviates from the static parameters hardcoded into the IDF, a pronounced performance gap emerges between simulated outputs and empirical energy consumption. Traditional EnergyPlus simulations lack the native capacity to dynamically ingest real-time data from Internet of Things (IoT) sensors or Building Management Systems (BMS) without the aid of external co-simulation middleware (such as the Building Controls Virtual Test Bed) or custom scripting environments. Therefore, while EnergyPlus provides an exceptional design-phase baseline, its standalone configuration is insufficient for the dynamic calibration required by modern digital twin architectures.

HVAC System in EnergyPlus Engine

The Packaged Terminal Air Conditioner (PTAC) is one of zone-level HVAC systems commonly used to provide heating and cooling to individual spaces as shown in Figure 15. In EnergyPlus, the PTAC is represented through the HVACTemplate:Zone:PTAC object, which simplifies the modeling of direct-expansion air-conditioning systems serving a single thermal zone.

According to the EnergyPlus Input Output Reference, the PTAC template automatically generates the detailed HVAC components required for simulation, allowing users to model cooling, heating, ventilation, and fan operation without manually defining each component (EnergyPlus, 2015).

According to EnergyPlus (2015), a PTAC system consists of several key components, including an outdoor air mixer, a supply fan, a direct expansion (DX)

cooling coil, and a heating coil. The outdoor air mixer combines return air from the conditioned zone with outdoor ventilation air before the air passes through the cooling or heating coil. The conditioned air is then delivered back to the thermal zone through the supply fan. This configuration enables the PTAC system to simultaneously address zone cooling and ventilation requirements while maintaining indoor thermal comfort.

The airflow sequence within a PTAC system begins with the collection of return air from the conditioned space. This return air is mixed with outdoor air through the outdoor air mixer and then directed through the cooling or heating coil depending on the zone load requirements. During cooling operation, the DX cooling coil removes sensible and latent heat from the mixed air stream before the supply fan distributes the conditioned air back into the zone. Conversely, when heating is required, the heating coil raises the supply air temperature before delivery to the occupied space (EnergyPlus, 2015).

Based on the EnergyPlus Engineering Reference, PTAC systems operate under thermostat control, where system performance is continuously adjusted according to the zone temperature setpoint. The system cycles the cooling coil, heating coil, and fan operation to meet the required thermal demand while calculating associated energy consumption. EnergyPlus also allows the specification of parameters such as cooling capacity, airflow rate, fan characteristics, and outdoor air ventilation rate, enabling the PTAC model to represent a wide range of packaged terminal air-conditioning applications (U.S. Department of Energy, 2016).

The PTAC model is particularly suitable for building energy simulation because it integrates thermal conditioning, ventilation, and energy consumption calculations

within a single HVAC object. By incorporating component-level representations of fans, coils, and airflow pathways, the PTAC system provides a simplified yet comprehensive approach for simulating zone-level HVAC performance in building energy models.

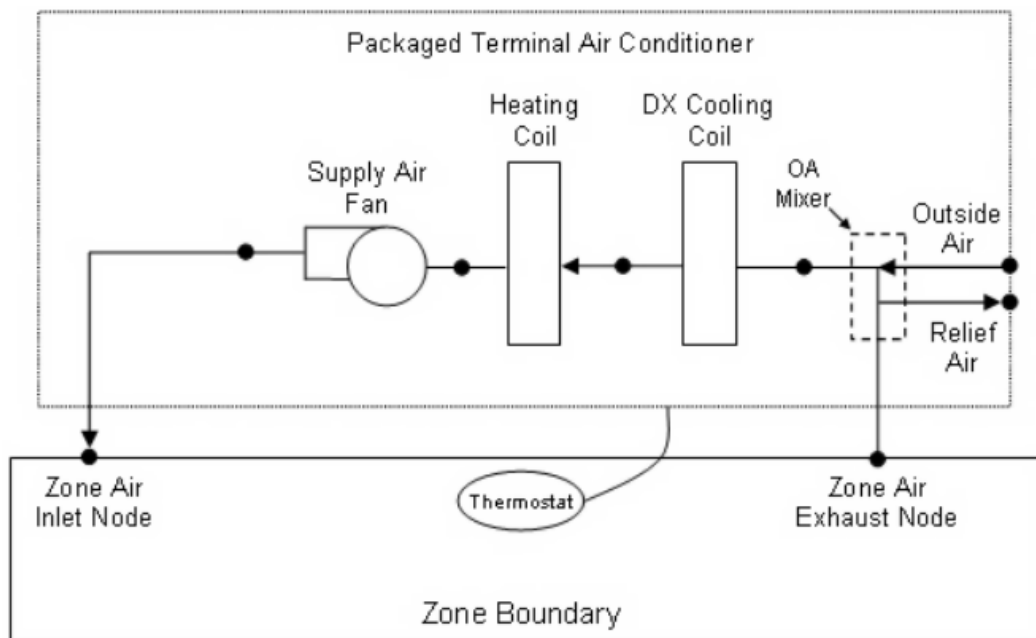


Figure 15. EnergyPlus Packaged Terminal Air Conditioning Design

Calibration Frameworks

Calibration frameworks play a critical role in Building Energy Modeling (BEM) by reducing discrepancies between simulated and actual building performance. As building systems become more complex, researchers have increasingly adopted automated calibration techniques to improve model accuracy, reliability, and computational efficiency.

Modern calibration approaches commonly employ Bayesian optimization, surrogate modeling, and uncertainty quantification methods to systematically identify optimal model parameters while minimizing computational cost. The following sections discuss the principles, methodologies, and applications of calibration frameworks in enhancing the predictive capability of building energy models.

Automated Parametric Calibration in Building Energy Modeling

The deployment of Building Energy Models (BEMs) is a cornerstone of modern sustainability analysis and HVAC performance evaluation; however, substantial discrepancies frequently persist between simulated predictions and actual field measurements (Jiang et al., 2024). To bridge this performance gap, rigorous model calibration is required to align the simulation with historical operational data.

Historically, BEM calibration relied heavily on manual "trial and error" approaches, which are highly subjective, time-intensive, and heavily dependent on an analyst's intuitive tuning of inputs (Fabrizio & Monetti, 2015). To overcome the limitations of these ad-hoc procedures, the field has transitioned toward automated parametric calibration. This approach utilizes algorithmic search techniques to mathematically infer unknown or uncertain building parameters such as equipment efficiencies, lighting densities, and thermal mass properties until the error between metered consumption, and the simulation output satisfies stringent statistical thresholds, such as those defined by ASHRAE Guideline 14 (Herbinger et al., 2023). Despite these advancements, achieving high-resolution (e.g., hourly or sub-hourly) automated calibration remains a profound computational hurdle due to the complex,

coupled non-linear heat and mass transfer dynamics inherent in whole-building simulation engines like EnergyPlus (Jiang et al., 2024).

Why Bayesian Calibration Method?

Building Energy Simulation (BES) models are frequently calibrated to measured building performance data to reduce discrepancies between simulated and observed results. According to Coakley et al. (2014), calibration is necessary because it is often difficult to achieve an accurate representation of real-world building operation due to the complexity of the built environment and the large number of interacting variables affecting building performance. Calibration is therefore defined as the process of reconciling model outputs with measured data to improve the accuracy and reliability of simulation results.

Coakley et al. (2014) classified calibration approaches into manual and automated methods. Automated approaches employ mathematical and statistical techniques to assist the calibration process and reduce dependence on user-driven trial-and-error adjustments. Among the optimization techniques identified in the review is Bayesian calibration, which is described as an alternative statistical approach to model calibration. The authors noted that Bayesian calibration offers the advantage of naturally accounting for uncertainty in model predictions using prior input distributions.

The review further emphasizes the importance of uncertainty in building simulation and calibration. Building energy models are affected by specification uncertainty, modeling uncertainty, numerical uncertainty, and scenario uncertainty. As

a result, approaches that explicitly account for uncertainty are valuable for improving confidence in simulation results. Bayesian calibration is recognized as one such approach because uncertainty is incorporated directly into the calibration framework through probabilistic parameter representations.

When deploying highly parameterized mathematical models to approximate the behavior of complex physical systems, it is scientifically imperative to conduct rigorous predictive and uncertainty analyses. Because simulation models are fundamentally idealized abstractions of stochastic physical environments, deterministic approaches frequently fail to capture the true predictive boundaries of the system. To address this, Bayesian calibration frameworks offer a robust methodological solution by intrinsically embedding uncertainty directly into the parameter estimation process. Unlike traditional deterministic calibration which yields a singular, absolute point-estimate for unknown variables.

Bayesian techniques quantify the posterior probability distributions of the inputs. This explicitly captures the residual uncertainty remaining across the calibrated parameter space. Once these probabilistic distributions are established, the uncertainties can be systematically propagated throughout the entire simulation architecture via probabilistic sensitivity analysis, ensuring that the final predictive outputs reflect a mathematically sound confidence interval rather than a falsely precise absolute value.

Furthermore, a critical advantage of the Bayesian approach is its advanced capacity to address and mitigate structural "model inadequacy." This inadequacy characterizes the systematic discrepancies between empirical observations and virtual predictions in divergence that inevitably persists even when simulation parameters are

optimally tuned. By introducing a mathematically rigorous discrepancy term, Bayesian calibration actively corrects for these inherent structural limitations, effectively bridging the gap between simplified theoretical equations and complex operational realities.

Finally, Bayesian frameworks possess the unique statistical flexibility to fuse highly heterogeneous data streams. They provide a resilient architecture capable of synthesizing multiple sources of empirical evidence, seamlessly accommodating datasets that inherently differ in spatial scale, temporal resolution, and measurement reliability.

Derivative-Free Search via Bayesian Optimization

Building simulation engines operate as complex "black box" functions where exact mathematical derivatives cannot be analytically computed; consequently, derivative-free global optimization strategies are strictly required to tune BEM parameters. Among these strategies, Bayesian Optimization has emerged as a highly robust framework uniquely suited to address the computational bottlenecks associated with high-resolution calibration. Unlike traditional deterministic approaches or heuristic methods, Bayesian frameworks effectively utilize prior knowledge to iteratively close the discrepancy between model predictions and observed data without relying on rigid functional assumptions (Jiang et al., 2024).

By probabilistically mapping the error space, the algorithm intelligently navigates the search for optimal parameter combinations by mathematically balancing two competing imperatives: exploiting areas of known low error and exploring regions

of high uncertainty. This intelligent, sample-efficient strategy allows researchers to optimize computationally heavy BEMs rapidly, yielding accurate parameter identification in significantly fewer iterations than standard population-based methods like Genetic Algorithms (Kubwimana & Najafi, 2023).

Furthermore, Bayesian frameworks provide a rigorous mathematical basis for incorporating bias-correction and uncertainty quantification, ensuring robust parameter estimation even when systematic discrepancies or sensor noise exist within the metered boundary conditions (Jeong & Byon, 2024).

Coakley et al. (2014) classified building energy modeling approaches into black-box, grey-box, and detailed simulation approaches. Black-box models are based on mathematical or statistical relationships that relate influential input variables to measured outputs. In these models, the coefficients used to generate predictions have no direct connection to physical building parameters.

Grey-box approaches differ from black-box models by incorporating key or aggregated system parameters derived from a physical representation of the building. According to Coakley et al. (2014), grey-box models combine engineering-based system descriptions with measured data, allowing the model to remain linked to physical building, system, and environmental parameters. Compared with detailed simulation models, grey-box approaches require less development effort while maintaining a connection to the physical characteristics of the building.

Building energy simulation models are subject to several sources of uncertainty, including specification uncertainty and modeling uncertainty. Coakley et al. (2014)

defined specification uncertainty as uncertainty arising from incomplete or inaccurate descriptions of building characteristics and HVAC systems. Modeling uncertainty, on the other hand, results from simplifications and assumptions used to represent complex physical processes within simulation tools.

The authors emphasized that simplification is an inherent challenge in building energy simulation because building models represent abstractions of complex real-world systems. They further noted that calibration of building energy models is an over-specified and under-determined problem, as simulation models may contain thousands of input parameters while only a limited number of measurable outputs are available for comparison.

To address this challenge, various model simplification techniques have been reported in the literature. These include parameter reduction, zone-typing, day-typing, and simplified energy analysis procedures that reduce model complexity while maintaining acceptable predictive performance. Coakley et al. (2014) described these approaches as methods for reducing the number of simulation variables and simplifying building representations to support calibration and analysis.

Surrogate Modeling Using Gaussian Processes

The primary mechanism driving the exceptional sample efficiency of Bayesian Optimization is the implementation of a surrogate model, which acts as a fast-computing mathematical substitute for the computationally expensive physical simulation engine (Herbinger et al., 2023). In advanced Bayesian BEM calibrations, a Gaussian Process (GP) is predominantly utilized as this surrogate emulator (Kristensen

et al., 2017). A Gaussian Process regression model effectively maps the non-linear relationship between the building's input parameters and the simulated outputs. Crucially, it accounts for both the deterministic predictions (the expected error score) and the inherent uncertainty (the variance) surrounding unexplored areas of the parameter space (Kristensen et al., 2017).

By utilizing a GP to represent these complex, multidimensional relationships, the optimization framework avoids executing the physical BEM at every single step of the search (Jiang et al., 2024). Instead, the algorithm relies on an acquisition function to evaluate the surrogate's probabilistic map, only triggering the actual EnergyPlus simulation when the Gaussian Process identifies a highly promising parameter combination. This methodological decoupling of the physical simulation from the parametric search space drastically reduces the computational overhead, allowing the digital twin to achieve strict calibration standards within a fraction of the time required by traditional optimization methodologies.

ASHRAE Guideline 14

The transition of Building Energy Modeling (BEM) from a static, design-phase estimating tool to a continuous, operationally dynamic Digital Twin (DT) necessitates rigorous, standardized validation protocols. While the theoretical architectures of DTs such as IoT integration, edge computing, and bidirectional data synchronization are well-documented, the empirical utility of these virtual models is strictly governed by their predictive accuracy. Within the built environment, the internationally recognized benchmark for this accuracy is ASHRAE Guideline 14: Measurement of Energy,

Demand, and Water Savings. This review focuses exclusively on the application of ASHRAE Guideline 14 compliance criteria to BEM Digital Twins. It examines the transition of these metrics from static calibration to continuous verification, the methodological challenges inherent in dynamic thermodynamic alignment, and the necessity of automated frameworks to ensure enduring predictive robustness.

ASHRAE Guideline 14 establishes the definitive quantitative thresholds for model fidelity, primarily utilizing two statistical indices: the Normalized Mean Bias Error (NMBE) and the Coefficient of Variation of the Root Mean Square Error (CVRMSE).

For a building energy model to be deemed calibrated using hourly data the resolution required for DTs the guideline mandates an NMBE of within plus or minus 10% and a CVRMSE of less than 30%. Historically, these thresholds were applied retrospectively; an engineer would manually tune a model until its historical outputs fell within these margins. However, recent literature entitled Physics-informed machine learning for ASHRAE-compliant heat transfer coefficient estimation from short-term IoT monitoring by Sami and Sierra (2026) emphasizes a critical paradigm shift: for a Digital Twin, ASHRAE compliance cannot be a one-time, static achievement. Because DTs operate continuously and must respond to stochastic variations in occupancy, weather, and system degradation, compliance must be viewed as a dynamic, continuous state. The literature asserts that a DT must autonomously maintain these error margins across rolling operational horizons, requiring the virtual replica to continuously recalibrate against incoming telemetry without diverging from fundamental thermodynamic laws.

Achieving initial ASHRAE compliance during the training phase of a digital twin is relatively straightforward, particularly with the advent of advanced optimization algorithms. However, maintaining compliance under continuous operational drift presents substantial methodological hurdles. Real-world building operations introduce extensive stochastic noise, unmeasured internal heat gains, and sensor degradation. When a DT, calibrated strictly to historical data, is exposed to these transient, out-of-distribution events, its CVRMSE frequently spikes beyond the 30% threshold, invalidating its utility for model predictive control or automated fault detection. Contemporary research highlights that maintaining strict ASHRAE Guideline 14 compliance requires the continuous, algorithmic calculation of these error indices. Furthermore, it necessitates the isolation of simulation outputs from empirical sensor data to prevent mathematical conflation during the verification process.

Automated Frameworks for Quantitative Evaluation

To resolve the bottleneck of continuous compliance verification, researchers are increasingly reliant on automated, programmatic workflows. Computational scripting is deployed to interface directly with the DT's simulation engine because evaluating hourly NMBE and CVRMSE across thousands of data points cannot be executed manually.

The literature demonstrates that frameworks utilizing Python libraries can systematically extract the necessary thermodynamic outputs such as zone air temperatures, HVAC energy consumption, and thermal loads at high frequencies. To

rigorously prove ASHRAE compliance, these data arrays must be meticulously processed.

Advanced methodologies advocate for the programmatic isolation of these validation arrays, systematically exporting them as clean, headless, single-column CSV files. This rigorous data structuring is not merely an administrative preference; it is a vital prerequisite for feeding the data into strict, automated error-calculation algorithms that definitively verify whether the DT remains within the mandated 10% NMBE and 30% CVRMSE thresholds.

Standardization of Evaluation Metrics for Environmental Variables

While international protocols, such as ASHRAE Guideline 14, originally developed the Normalized Mean Bias Error (NMBE) and the Coefficient of Variation of the Root Mean Square Error (CVRMSE) to standardize the calibration of building energy consumption, contemporary literature frequently extends these statistical thresholds to evaluate indoor thermodynamic states. Researchers have increasingly applied these same indices to validate simulated indoor environmental parameters, specifically air temperature and relative humidity because digital twin models must accurately predict complex internal heat balances prior to calculating equipment energy demands. For instance, in their development of an operationally responsive building energy model, Tanasa et al. (2019) calibrated their simulation based on monitored interior conditions, actively assessing the performance of the environmental outputs by verifying that the resulting NMBE and CVRMSE remained within acceptable tolerances.

Recently, Palaić et al. (2023) formally validated this methodological approach by demonstrating that a building energy simulation could successfully predict room temperatures with a CVRMSE well below prescribed limits (3.58%). Their research explicitly noted that while these metrics were formulated for energy, they are now widely and reliably used to assess the calibration performance of indoor temperature prediction models (Palaić et al., 2023). Consequently, the present study adopts this validated academic precedent, utilizing NMBE and CVRMSE not only as the primary validation limits for HVAC electrical consumption but also to rigorously evaluate the thermodynamic fidelity of the simulated indoor temperature and relative humidity.

International Organization for Standardization (ISO) 13786:2017

International standards provide a consistent framework for evaluating the thermal performance of building components and ensuring the reliability of building energy analyses. Among these standards, ISO 13786:2017 establishes procedures for determining the dynamic thermal characteristics of building elements under periodic thermal conditions. The standard is widely applied in building energy modeling and thermal performance assessment because it accounts for the transient behavior of building envelopes beyond conventional steady-state metrics. The following sections discuss the theoretical basis, computational implementation, and significance of ISO 13786:2017 in advanced building simulation and digital twin applications.

Dynamic Thermal Characterization of Building Components

The accurate prediction of building energy performance heavily relies on characterizing the thermodynamic behavior of the building envelope. Historically, energy efficiency regulations and preliminary design phases have prioritized static thermal transmittance, commonly expressed as the U-value. However, static metrics are fundamentally insufficient for capturing the transient nature of real-world environmental conditions, such as diurnal solar cycles and fluctuating ambient temperatures. To address this deficiency, the international standard ISO 13786 provides a rigorous mathematical framework for calculating the dynamic thermal characteristics of complete building components. This review critically examines the theoretical principles of ISO 13786, its computational implementation in modern modeling tools, its inherent methodological limitations, and its crucial role in establishing baseline parameters for advanced building energy simulations and digital twin architectures.

Theoretical Framework of ISO 13786:2017

The core objective of ISO 13786 is to quantify the thermal mass and dynamic behavior of multi-layer building assemblies under steady periodic boundary conditions. The standard establishes the calculation of heat transfer matrices to model how thermal waves propagate through diverse material layers. HTFlux (n.d.) emphasizes that understanding this dynamic behavior requires evaluating three primary metrics derived from the standard: thermal admittance, the decrement factor, and the time shift (or phase lag).

Thermal admittance characterizes the ability of a surface to absorb and release heat in response to periodic temperature variations, serving as a critical indicator of a room's thermal inertia and its susceptibility to overheating. The decrement factor represents the attenuation of the temperature wave amplitude as it travels from the exterior environment to the interior space, while the time shift quantifies the delay in this thermal transmission. Together, these metrics dictate how a building envelope mitigates peak thermal loads, shifting cooling demands to off-peak hours and maintaining indoor environmental stability.

Calculating the dynamic thermal properties of homogeneous, single-layer materials is mathematically straightforward; however, modern building envelopes comprise complex, multi-layer assemblies where the sequence of materials drastically alters dynamic performance. ISO 13786 addresses this through a complex matrix calculation method that accounts for the specific arrangement of resistive and capacitive layers.

Given the mathematical complexity of this matrix multiplication, computational tools are essential for practical application. Software solutions, such as the calculation tools provided by HTFlux, programmatically execute the ISO 13786 algorithms, allowing engineers to input material thicknesses, thermal conductivities, densities, and specific heat capacities to automatically derive the dynamic parameters. This automation is a vital precursor for advanced energy modeling, as it translates raw material data into the standardized transient indices required by whole-building simulation engines.

International Organization for Standardization (ISO) 15099:2003

Fenestration systems, including windows, doors, and shading devices, play a significant role in building energy performance by influencing heat transfer, solar gains, and indoor thermal comfort. To ensure consistent evaluation of these components, ISO 15099:2003 provides standardized procedures for determining their thermal and optical properties under various operating conditions.

The standard serves as a fundamental reference for building energy modeling and HVAC load calculations, supporting the development of accurate and reliable simulation models. The following sections discuss the theoretical principles and computational methodologies established by ISO 15099:2003 for fenestration performance analysis.

Thermodynamic Characterization of Fenestration Systems

Within the thermodynamics of the built environment, fenestration systems consistently represent the most vulnerable interface for unintended heat transfer, profoundly influencing a facility's overall energy consumption and peak thermal loads. To accurately simulate the thermodynamic behavior of these complex transparent assemblies, modern architectural engineering relies on universally standardized computational algorithms.

ISO 15099:2003, titled Thermal performance of windows, doors and shading devices, stands as the definitive international standard for quantifying the optical and thermal transmission properties of fenestration. This review critically examines the

theoretical and methodological frameworks embedded within ISO 15099:2003, explores its integration into advanced Building Energy Modeling (BEM) simulation engines, discusses its inherent limitations regarding transient environmental states, and establishes its foundational role in the development and validation of high-fidelity BEM Digital Twins.

Thermodynamic Algorithms

The core utility of ISO 15099:2003 lies in its rigorous, component-based methodology for calculating the overall thermal transmittance (U-value) and the Solar Heat Gain Coefficient (SHGC) of complex window assemblies. Rather than treating a window as a homogeneous surface, the standard mathematically partitions the fenestration system into three distinct interactive domains: the center-of-glass, the edge-of-glass, and the framing elements.

Hoang (2018) noted that the widespread adoption of this standard evidenced by its direct translation into national frameworks such as Vietnam's TCVN 11857 stems from its comprehensive algorithmic approach to one-dimensional non-linear heat balances. The standard dictates precise calculations for radiative and convective heat transfer across multi-pane glazing cavities, utilizing specific gas fill properties and wavelength-dependent optical data to determine how solar radiation is transmitted, absorbed, and subsequently re-radiated into the interior zone.

Furthermore, ISO 15099 provides the foundational algorithms for assessing the impact of complex shading devices, such as venetian blinds and woven shades, evaluating their angle-dependent effect on luminous and thermal transmittance.

The theoretical principles of ISO 15099:2003 are not merely academic; they form the operational bedrock of the world's most advanced whole-building energy simulation programs, notably EnergyPlus. When energy modelers define complex fenestration geometries and material properties within a text-based Input Data File (IDF), the simulation engine utilizes the algorithms prescribed by ISO 15099 to perform sub-hourly calculations of solar gains and conductive heat losses. This algorithmic integration is crucial for digital twin architectures. Accurate prediction of localized indoor environmental quality, HVAC zone loads, and perimeter thermal stratification is fundamentally impossible without the detailed, layer-by-layer optical and thermal characterizations dictated by this standard. The mathematical rigor of ISO 15099 ensures that the digital twin's physical geometry and envelope baseline are grounded in verified, standardized building physics before any empirical operational data is introduced.

International Organization for Standardization (ISO) 6946:2017

The thermal performance of building envelopes is influenced not only by solid construction materials but also by the presence of air layers and cavities within wall, roof, and floor assemblies.

ISO 6946:2017 provides standardized procedures for determining the thermal resistance and thermal transmittance of building components, including the effects of enclosed and ventilated air spaces. The standard serves as a fundamental reference in building energy modeling by establishing consistent methods for evaluating heat transfer through opaque building elements. The following sections discuss the

thermodynamic principles and calculation procedures defined by ISO 6946:2017 and their significance in building performance assessment.

Thermodynamic Characterization of Interstitial Air Cavities

The energetic performance of a multi-layer building envelope is not governed solely by the conductive resistance of solid materials but heavily relies on the thermodynamic behavior of internal air cavities. Within the domain of steady-state thermal calculations, EN ISO 6946:2017 serves as the definitive international framework for quantifying the thermal resistance of opaque assemblies. While the standard establishes general rules for homogenous layers, its most mathematically intricate components address the modeling of unventilated, weakly ventilated, and well-ventilated air spaces.

This review critically examines the specific algorithms embedded within ISO 6946:2017 for calculating heat transfer across interstitial air gaps, discusses the inherent limitations of these steady-state assumptions in dynamic environments, and establishes the necessity of programmatic digital twin validation to overcome the rigidity of these static baseline calculations.

Thermodynamic Mechanics

Within the framework of ISO 6946:2017, an unventilated air cavity is treated not as a simple conductive layer, but as a complex multidimensional heat transfer domain. The standard dictates that the total thermal resistance of such a space is governed by the simultaneous action of conduction, internal convection, and surface-

to-surface radiation. The mathematical derivation relies heavily on the physical dimensions of the airspace (thickness and orientation) and the radiative properties of the bounding surfaces.

According to the standard, the radiative heat transfer coefficient is directly proportional to the effective emissivity of the facing materials. Consequently, the standard provides precise algorithms to calculate how low-emissivity barriers (such as reflective aluminum foils) drastically increase the thermal resistance of the air gap by neutralizing the radiative heat exchange.

Further, Alshuraiaan (2022) highlighted the importance of these calculations in lightweight composite structures, where the strategic configuration of internal air gaps and foil facings constitutes the primary defense against extreme thermal loads.

Real-world building envelopes rarely maintain perfectly sealed air cavities; therefore, ISO 6946 establishes strict operational thresholds to mathematically categorize the degree of ventilation. The standard classifies air layers into three regimes: unventilated, weakly ventilated, and well-ventilated. This classification fundamentally alters the mathematical treatment of the wall assembly.

For a weakly ventilated airspace, the standard requires an interpolation between the resistance of an unventilated cavity and that of a well-ventilated one, acknowledging a partial loss of insulative capacity due to external air infiltration. Conversely, for a well-ventilated airspace such as the cavity behind a rainscreen cladding system, ISO 6946 dictates a radical simplification: the thermal resistance of the airspace itself, as well as all material layers between the airspace and the external environment, must be

entirely neglected. In this scenario, the calculation assumes the temperature of the air gap is equal to the external ambient temperature, relying solely on the internal surface resistance to model the boundary condition.

IoT Architecture in HVAC Systems

The successful implementation of smart HVAC systems depends not only on advanced control algorithms but also on the underlying Internet of Things (IoT) architecture and the methods used to evaluate system performance. IoT technologies enable communication among sensors, controllers, and HVAC equipment, facilitating real-time monitoring, data exchange, and automated decision-making. At the same time, rigorous validation methods are necessary to assess controller reliability, accuracy, energy consumption, and economic feasibility. The following sections review the architectures, communication frameworks, performance metrics, and evaluation approaches commonly used in IoT-based HVAC automation systems.

HVAC Automation and Control System Architectures

Much of the existing research on smart home HVAC automation has focused on software-based optimization through artificial intelligence, machine learning and predictive scheduling algorithms. However, the performance of these solutions ultimately depends on the responsiveness and scalability of the underlying control hardware, which acts as the interface between intelligent algorithms and physical HVAC units.

A study by Elkholy et al., (2022) tested their control system on a scaled prototype house, with 3 rooms and a reception (physical dimensions: 1m x 1.5 m x 0.5). Data from this model were integrated into a graphical user interface (GUI) to support decision-making. These results suggested that the system was easy to use, highly reliable, and has a low cost compared to other systems, with a simulated consumption reduction of 18.161 kWh, and operating cost savings of 23.13%, all without sacrificing comfort. However, such values were derived through LabVIEW simulations based on prototype data rather than validation in a real-scale residential setting. This presents a persistent gap in literature; while scaled models and simulations can demonstrate feasibility, the real-world effectiveness of such systems depends on validated hardware performance, particularly in terms of controller responsiveness and scalability.

Optimal control approaches in building automation can generally be classified into centralized and distributed systems. In centralized approaches, a single controller collects all data and determines the optimal set-points for the entire system. Distributed approaches, on the other hand, assign decision-making to multiple local controllers, which may either coordinate through a central agent (hierarchical distributed) or operate through direct peer-to-peer communication (fully distributed) (Li & Wang, 2022).

Fully distributed control has been widely recognized for its strengths in robustness and scalability. Without a single point of failure, these systems are more resilient to node or communication failures, since each controller can continue operating independently (Bonawitz et al., 2017). Their peer-to-peer structure also makes them inherently scalable, as new nodes can be added with minimal reconfiguration of the system (Su et al., 2021). Several HVAC-oriented studies further

highlight these advantages. An example would be a study by Eimi and Abdalwahed (2019), a distributed MPC (model predictive control) testbed work, which achieved ~48% less computation time, and ~25% lower energy use compared to centralized MPC, with less tracking error. Another would be the work of Yang et al. (2020), which indicates that if there are less than 20 zones in a single building, there is only a performance gap of around ~2% to 3% and will only increase the more zones are present. These results show that distributed control can achieve near-centralized performance in small systems, and even outperform it as scale grows

Despite the fully distributed HVAC control systems promising robustness and flexibility, they also introduced several implementation challenges. Communication delays in distributed optimal control can drastically degrade performance, due to it requiring many more iterations for convergence thus sacrificing control accuracy (Su et al., 2021). Similarly, decentralized energy management becomes practically difficult because computational load, data transfer, and coordination overhead increase significantly as more agents are added (Asad et al., 2023).

Multi-agent systems literature further warns that hardware constraints at the node level, plus inter-agent communication and decision conflicts, can limit how “distributed” a system can realistically be (Astudillo & Koch, 2024). These factors suggest that fully distributed designs are harder to implement – especially in IoT / smart home environments with limited hardware, constrained networks, and need for real-time response. In contrast, hierarchical distributed designs are more practical for the study’s setting.

In the study of Wu et al. (2018)), a proposed hierarchical control framework that utilizes HVAC loads for primary frequency regulation (PFR) was simulated. This approach can notably reduce communication requirements and may enable more efficient dispatch of HVAC units at the control level, while ensuring rapid response at the local level. This was adopted by a simulated hierarchical distributed model predictive control scheme, as detailed in studies by Long et al. (2016) and by Wang and Wang (2021). The scheme is divided into two layers: the upper, which collects all data and information to generate predictive reference trajectories, and the lower, which only uses local information to track the reference and optimize energy efficiency and thermal comfort. Both studies, however, are only validated in a simulated setup, suggesting the need for practical testing for the hierarchical distributed approach.

Internet of Things (IoT) Integration in HVAC Systems

The Internet of Things (IoT) refers to the integration of physical devices (sensors, actuators, and controllers) that collect, exchange, and process data to enable extensive monitoring and automated decision-making (Atzori et al., 2010).

Within the context of hierarchical distributed control, IoT provides the foundation for enabling individual nodes to function with greater independence, reducing the need for constant coordination between higher-level managers (Long et al., 2016)).

This autonomy is especially valuable in HVAC systems, where localized decisions can improve efficiency and comfort while minimizing network load. In practice, IoT implementations in HVAC and smart-home settings typically use

lightweight publish/subscribe protocols such as MQTT over Wi-Fi, which provide low-overhead, scalable messaging suitable for resource-constrained devices (Masdani & Darlis, 2018). For instance, Froiz-Miguez et al. (2018) demonstrated an IoT-based home automation testbed using MQTT and Wi-Fi/ZigBee nodes, reporting comparative and practical performance metrics such as response time, packet delivery success rate, and consumption.

Performance Evaluation of IoT-Based HVAC Controllers

Beyond control architectures and IoT integration, it is also necessary to review how the effectiveness of HVAC automation systems is evaluated. Much of the existing literature approaches this through techno-economic analysis, which combines technical performance metrics with economic assessments.

On the technical side, studies measure factors such as power consumption, response time, control accuracy, and scalability to establish the operational value of the controller. On the economic side, analyses typically focus on cost-savings, investment requirements, and payback periods to assess feasibility. Positioning this study within such a framework ensures that both the practical functionality and the financial viability of hierarchical distributed IoT-based HVAC control are adequately considered.

Due to the lack of research on performance testing of HVAC IoT controllers, papers that did the analysis on non-HVAC territory will be studied instead. As long as the premise of the paper is about IoT-performance testing, the technical metrics and method of measurement will be indispensable for building this paper's foundation.

Energy Consumption Measurement Methods

The study conducted by Masdani and Darlis (2018) compared three protocols in terms of energy consumption: MQTT, RESTful HTTP, and CoAP. Among the three, MQTT consumes more power due to it requiring a communication broker, like Mosquitto, with average amperage of 70.87 mA, followed by RESTful HTTP, 62.66 mA, and the least consuming protocol is CoAP, with 61.58 mA. While the study focused solely on energy consumption, other literature suggests that protocol choice also influences communication latency and reliability, which indicate power consumption is only one part of the tradeoff. The method of measurement uses the shunt resistor technique, wherein a one-ohm resistor is placed as a shunt so that the voltage and current would be equal as per Ohm's Law. The schematic diagram used is shown in the Figure 16.

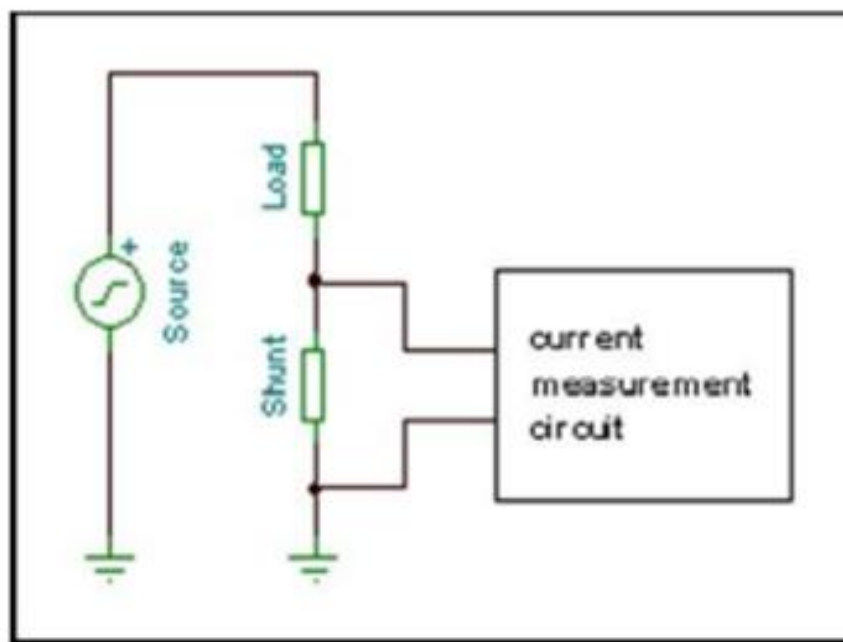


Figure 16. DC Power Monitoring via a Shunt Resistor (Masdani, et al., 2018)

Reliability and Communication Performance Metrics

The study Qiu et al. (2021) did a performance analysis of a fail-safe IoT-based fire alarm control panel, which aimed to quantify its reliability by inspecting packet loss rate (PLR), packet failure rate (PFR), and the probability of failure per hour (PFH). The formulas for the three data are given below.

$$PLR = \frac{N_L}{N_T} \quad (2)$$

Where:

N_L = lost packets

N_T = sent packets

$$PFR = \frac{N_F}{N_T} \quad (3)$$

Where:

N_F = failed packets

$$PFH = \frac{N_F}{N_h} \quad (4)$$

Where:

N_h = testing hours

Packets are used to deliver broken down data from a sender to a receiver both subscribed to the same MQTT broker. Integration with other platforms can also be done, as what Moch et al. (2024) did: Prometheus to collect data from the broker, and Grafana

to visualize it into graphs, which included the CPU and RAM usages. The latency (Kusuma & Oktiawati, 2022), packet loss (Hakim & Setiawan, 2024), and network load (Trubák et al., 2025) can also be shown in this way.

Control Accuracy Assessment

Control accuracy is also important in determining whether the proposed controller is doing its intended purpose. This can be done via the root mean square error method, which is also done by Palaić et al. (2025), wherein a physical model was built, which also aims to balance out indoor temperature and energy consumption, with a computed RMSE of 0.85 °C. RMSE is preferred over average error metrics because it penalizes larger deviations more strongly, which is important in thermal comfort control. The formula for RMSE is given below.

$$T_{RMSE} = \sqrt{\frac{\sum_{l=1}^n (P_l - O_l)^2}{n}} \quad (5)$$

Where:

P_l = predicted temperature

O_l = observed temperature

n = number of data points

Scalability Analysis of Distributed IoT Systems

Scalability is widely regarded as a key advantage of distributed systems, but performance ultimately degrades once communication, or computation demand

exceeds capacity. A recent study (Hamad et al., 2024) investigated this effect in an SDN-enabled testbed by proportionally scaling both end devices and access points, maintaining a constant IoT-to-AP ratio of 4:1 across four trials. Under this balanced configuration, latency remained below 1 ms even at 64 devices, while packet loss rose to 0.78%, and throughput increased from 27.4 Mbit/s at 8 devices to 33 Mbit/s at 64 devices. However, because the infrastructure was scaled alongside the device count, the system was effectively expanding, rather than being subjected to true congestion. As a result, the experiment did not reflect the conditions where many devices compete over a single communication bottleneck.

Techno-Economic Evaluation of HVAC IoT Controllers

Aside from measuring system performance, the economic implications of the controller are also important in figuring out whether the system is sustainable long-term. This implies the need to conduct several economic analyses. However, due to the lack of IoT-enabled HVAC economic analysis studies, concepts and ideas will be taken from the neighboring topic: economic analysis of electromechanical systems.

This gap is also addressed in a review by Hossain Sikdar et al., (2024), wherein 125 published articles between 2015 and 2023, focusing on IoT, AI or renewable energy in HVAC systems, and energy efficiency, predictive maintenance, or data privacy, were reviewed. It has been found out that only 15 focused on the economic feasibility of their proposed method.

Generally, approximately 80% of the articles recorded an ROI (return of investment) of between five to seven years, in commercial applications. They also noted

that, driven by technological advancements and supportive policy measures, there is now a positive shift in perspectives regarding the economic viability of the smart HVAC systems. For residential users however, affordability concerns are still in need of further research.

Economic Performance Metrics

A study (Belgard et al., 2024) which focused on techno-economic analyses, stated that CapEX (capital costs), OpEX (operating costs), and profitability measures are to be included in the economic analyses. Profitability measures are grouped into two, approximate (net cash flow, payback period (PBP), and return of investment (ROI)) and rigorous (net present value). However, since the present study's scope does not include marketing the controller, net cash flow and net present value will not be calculated and considered. The formulas for the necessary economic metrics are also given by the literature, as written below.

$$CapEX = C_E + C_L \quad (6)$$

Where:

C_E = Equipment costs

C_L = Labor costs

$$OpEX = C_R + C_{LO} + C_{EL} + C_A \quad (7)$$

Where:

C_R = Cost of raw materials

C_{LO} = Cost of labor costs per year

C_{EL} = Electricity costs

C_A = Additional costs

$$PBP = \frac{CapEX}{Annual\ Net\ Income} \quad (8)$$

Where:

Annual Net Income = Energy costs per year saved by the ML model

$$ROI = \frac{Annual\ Net\ Income}{CapEX} \quad (9)$$

CHAPTER 3

METHODOLOGY

This chapter presents the methodology employed in the design, development, implementation, and evaluation of the proposed distributed IoT mechatronic controller integrated with a hybrid AI model and a Building Energy Modeling (BEM) Digital Twin. The study follows a systematic workflow consisting of hardware and software development, system integration, prototype deployment, real-world data acquisition, machine learning model development, digital twin calibration and validation, and performance assessment. Real-world operational data are utilized to train and optimize the predictive algorithms while simultaneously supporting the calibration of the digital twin model to satisfy established validation criteria. Finally, the developed system is evaluated through technical, energy, comfort, and techno-economic analyses to determine its effectiveness, reliability, and practical feasibility for HVAC automation applications.

Research Workflow

To maintain structural organization and methodological rigor, the study was executed through a sequential workflow, with each operational stage refined to satisfy the specific research objectives within the defined scope and technical limitations. Detailed technical specifications and procedural breakdowns for each constituent phase are presented in the subsequent subsections.

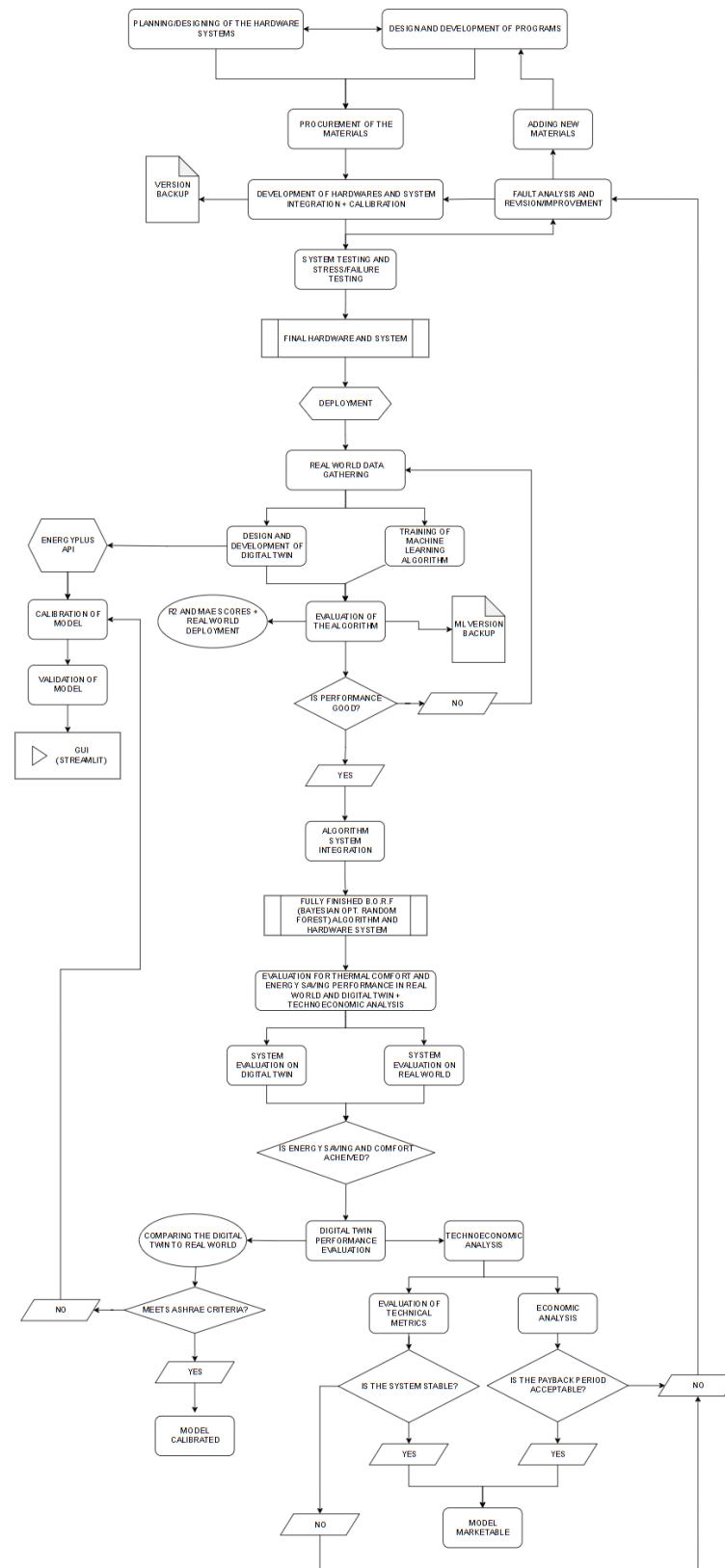


Figure 17. Research Workflow

Planning and Design of the System

Figure 19 illustrates the comprehensive system architecture, detailing both the physical hardware topology and the logical data streams within the control network. The developed distributed control system operates on a localized edge-computing framework, utilizing a Raspberry Pi 5 as the primary supervisory controller. This central unit serves as both the core processing engine and the local MQTT broker, interfacing via a wireless star network with five ESP32 microcontrollers configured as specialized peripheral nodes.

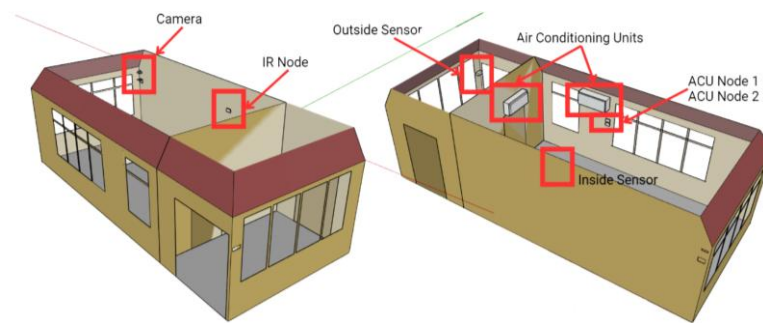


Figure 18. Locations of Each Node Relative to Test Room

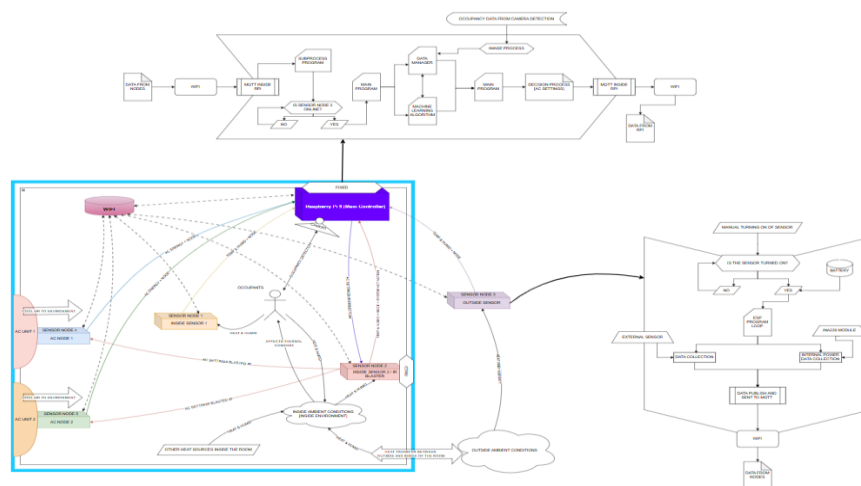


Figure 19. System Architecture

To ensure comprehensive environmental monitoring and precise actuation, the peripheral sensor and actuator network was strategically categorized into four distinct functional nodes:

1. **Dynamic Inside Sensor Node:** This is engineered for spatial flexibility to monitor localized ambient conditions within the occupied zone. This node was dynamically positioned within high-density occupancy areas to capture localized ambient temperature and relative humidity telemetry.
2. **Static Inside Sensor Node:** This is fixed permanently to a vertical partition at a height of 2.5m above the finished floor line to capture vertical thermal stratification and boundary condition variances. This node was equipped with an Infrared (IR) transmitter, serving as the non-invasive primary control interface to actuate optimal thermal setpoints on the air-conditioning unit without modifying its internal circuitry.
3. **Outside Sensor Node:** This is deployed externally to log macroclimatic ambient data, thereby providing critical environmental boundary condition variables required by the predictive machine learning algorithm.
4. **Air Conditioning Unit (ACU) Power Monitoring Nodes 1 and 2:** These are interfaced directly with the two independent split-type inverter units to track real-time operational state transitions and aggregate electrical power consumption metrics.

The specific hardware configurations, localized processing loops, and wireless communication protocols (*via Wi-Fi*) governing each of these subsystems are delineated in the subsequent sections.

internal microclimatic conditions. As demonstrated in the system architecture, the Bayesian Optimized Random Forest (BORF) supervisory controller must dynamically compensate for these environmental variables to maintain a stabilized indoor thermal state against external and internal disturbances:

Exogenous Thermal Loads (Building Envelope Heat Transfer)

This external load is driven by the transient temperature differential established across the boundary separating the unconditioned outdoor environment from the conditioned indoor space. Thermal energy transport penetrates the building envelope through four primary coupled thermodynamic mechanisms:

- **Convection:** Convective heat exchange occurring between the boundary-layer ambient outdoor air and the exterior vertical and horizontal structural surfaces.
- **Conduction:** Conductive thermal transmission migrating through the solid composite structural materials (walls, roof, and floor slabs) into the interior envelope core.
- **Radiation:** Direct and diffuse solar heat gain resulting from shortwave solar irradiance striking the building exterior and penetrating transparent fenestrations.
- **Mass Transfer (Infiltration):** Combined sensible and latent heat ingress via unconditioned outdoor air mass flow seeping through structural micro-fissures, doors, and window seals.

Metabolic Heat Generation (Occupancy Load)

Occupants act as dynamic, biological heat sources within the controlled boundary. The human body continuously rejects both sensible heat (via convection and radiation from the skin) and latent heat (via respiration and perspiration). This specific thermal load scales proportionally with localized occupancy density and fluctuates based on the occupants' physical activity levels (metabolic rate, measured in met). Consequently, higher occupancy or increased activity necessitates a proportional adjustment in the cooling capacity to offset the biological heat rejection.

Internal Equipment Loads (Parasitic Heat Gains)

The third heat source comprises parasitic sensible heat generated by electrical and electronic devices operating within the space. This includes localized heat dissipation from lighting fixtures, computing equipment (e.g., laptops, monitors), mobile devices, and other operational appliances that convert electrical energy into waste heat.

Developing an explicit, deterministic mathematical model to calculate these transient thermal loads in real-time introduces severe computational overhead, rendering it impractical for low-latency HVAC control. To circumvent the limitations inherent in traditional analytical modeling, this study adopts an empirical, data-driven approach. A distributed network of peripheral sensor nodes continuously captures the real-time environmental boundary conditions across the interior and exterior zones. When coupled with the dynamic occupancy telemetry generated by the YOLOv8 computer vision subsystem, this hardware architecture aggregates high-density, multi-

variable data streams. A Bayesian-Optimized Random Forest (BORF) regression model is trained on this empirical dataset, enabling the algorithm to implicitly map and predict the transient thermodynamic behavior of the single-zone space. This ensemble architecture allows the system to execute predictive thermal regulation without requiring the manual determination of complex building envelope properties or the explicit integration of nonlinear heat transfer equations.

Thermal Comfort Optimization Method Using ASHRAE 55

The Predicted Mean Vote (PMV) index, originally developed by P.O. Fanger, is the definitive analytical framework for evaluating steady-state indoor thermal comfort and serves as the core calculation methodology within ANSI/ASHRAE Standard 55-2023. Rather than prescribing static, deterministic temperature setpoints which fail to account for physiological variations and transient spatial dynamics, the PMV model evaluates a dynamic thermal balance. It quantifies human thermal sensation by solving a mathematical heat balance equation that integrates six primary parameters integrating two personal factors which is the occupant metabolic rate and clothing insulation and four measurable environmental parameters such as air temperature, radiant temperature, air speed, and humidity.

The output of the PMV equation is mapped across a standardized 7-point psychophysical scale ranging:

- +3 Hot
- +2 Warm
- +1 Slightly Warm

- 0 Neutral
- -1 Slightly Cool
- -2 Cool
- -3 Cold

To achieve strict compliance with ASHRAE Standard 55 comfort zones, the control system must maintain the environment within an acceptable PMV range of -0.5 to +0.5. The boundaries of the comfort zone are defined by equal contours of PMV values. The boundary value is set at ± 0.5 PMV in thermal sensation scale units. In field studies of actual buildings, environmental conditions within comfort zones bounded by this ± 0.5 value were found satisfactory by roughly 80% of occupants. This percentage varies depending on additional confounding circumstances beyond the PMV model of an occupant's heat balance.

Figure 21 delineates the systematic logic tree utilized to compute the Predicted Mean Vote (PMV) index. The algorithm executes a series of conditional evaluations to dynamically adjust human physiological variables before final computation. Initially, the system evaluates the presence of direct solar radiation to calibrate the Mean Radiant Temperature T_r . Following this, the core input vector is ingested, comprising air temperature T_a , mean radiant temperature T_r , relative humidity RH , air velocity V_a , metabolic rate met , and clothing insulation clo .

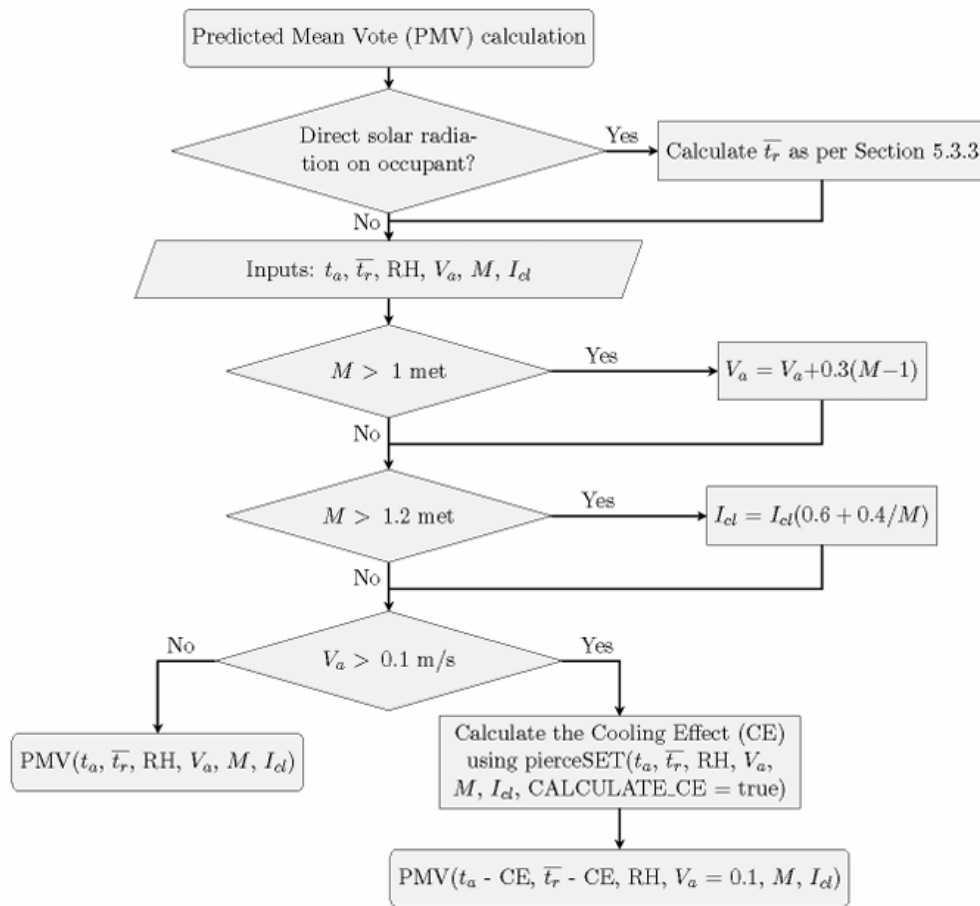


Figure 21. PMV Adjustment Workflow

To account for the physiological impacts of physical exertion, the algorithm applies two sequential corrections. If the metabolic rate exceeds 1.0 met, the effective air velocity is mathematically increased to simulate the relative air movement generated by the occupant's own motion. Furthermore, if the metabolic rate exceeds 1.2 met, the clothing insulation value *clo* is proportionally reduced to account for the "pumping effect" of air circulating through garments during strenuous activity.

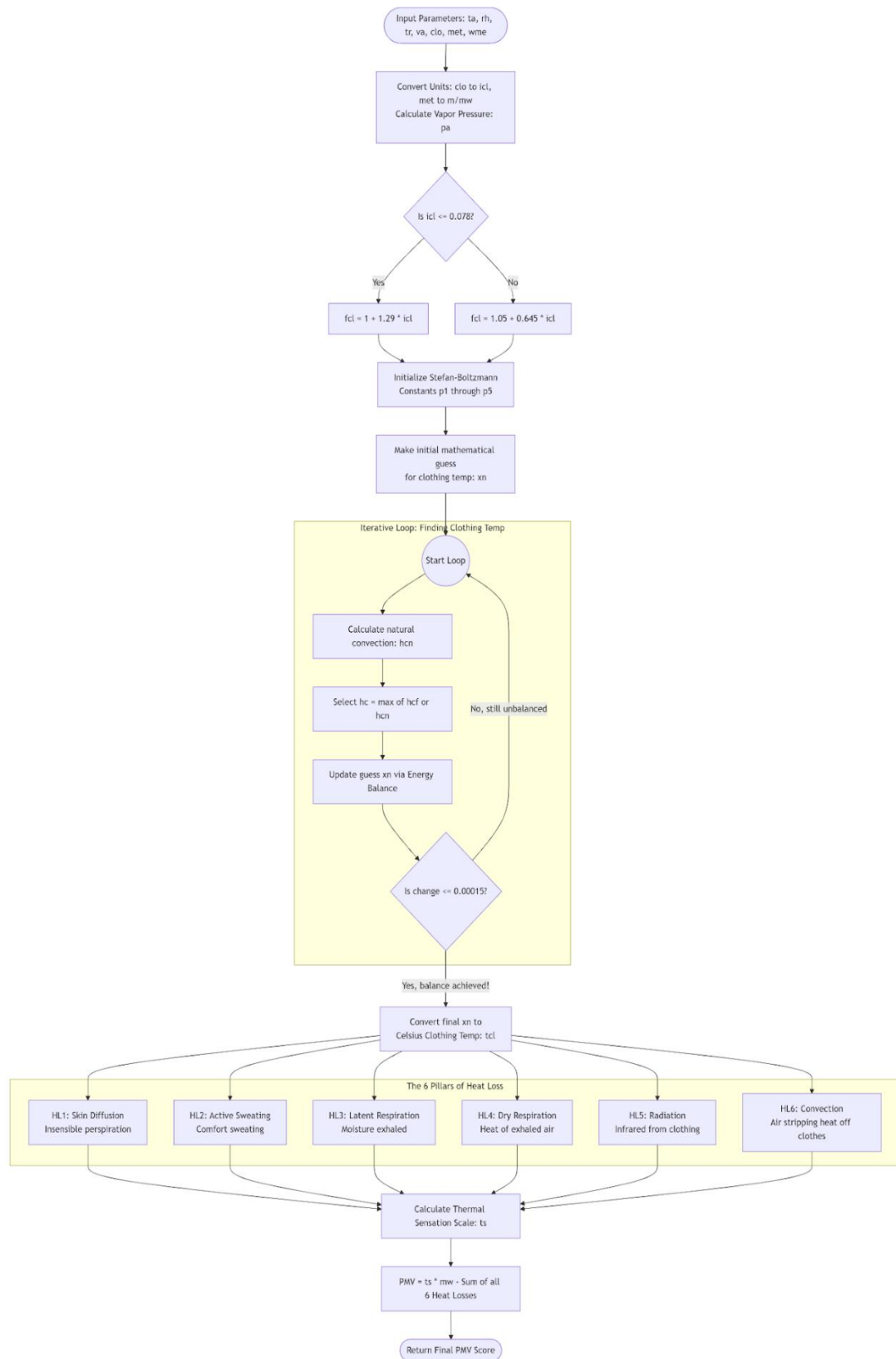


Figure 22. Thermal Comfort Index – PMV

Finally, the algorithm assesses the elevated air speed threshold. If the ambient air velocity is strictly 0.1 m/s, the standard PMV heat balance equation is executed. However, if air velocity exceeds 0.1 m/s, the system integrates the Standard Effective Temperature (SET) model to calculate a distinct Cooling Effect (CE). This CE factor mathematically depresses both T_a and T_r before the final PMV integer is computed, ensuring accurate thermal modeling in draft-prone environments.

The computational program for the PMV index presented in Figure 22 and detailed program found in GitHub Link in Appendix A was derived directly from the reference algorithms published in ASHRAE Standard 55. Originally provided as Java source code, the algorithm was translated into Python to ensure native compatibility with the system's edge computing environment.

```
def calculate_pmv(ta, rh, tr=None, va=0.1, clo=0.5, met=1.1, wme=0):
    """
    ISO 7730 PMV calculation (Fanger model)
    ta : air temperature (°C)
    tr : mean radiant temperature (°C) (if None, tr = ta)
    rh : relative humidity (%)
    va : air velocity (m/s)
    clo : clothing insulation (clo)
    met : metabolic rate (met)
    wme : external work (met), normally 0
    """
```

Figure 23. Derived PMV Parameters

The variables T_a and RH represent the dry-bulb air temperature and relative humidity, respectively. Both parameters are continuously monitored and acquired directly from the environment via the temperature and humidity sensors deployed across the room.

The T_r represents the mean radiant temperature and depicts the radiant heat from the walls. It is typically measured by Black Globe Thermometer along with formulas from ASHRAE 55. While ASHRAE Standard 55 dictates the inclusion of long-wave radiation from exterior building envelopes, the hardware constraints of the distributed sensor nodes necessitated a mathematical approximation. Short-wave solar radiation was nullified because the room is situated within a covered structure. For the scope of this study, the Mean Radiant Temperature (T_r) was approximated as equivalent to the Ambient Air Temperature (T_a).

$$T_{radiant} = T_{ambient} \quad (10)$$

It is acknowledged that during peak external thermal loads, this assumption may slightly underestimate the true PMV index due to unmeasured long-wave radiation from the exterior-facing walls.

Due to the hardware limitations of the distributed sensor network, real-time omnidirectional air velocity (V_a) could not be empirically measured. In accordance with standard ASHRAE 55 practices for typical indoor office environments, the air velocity was treated as a constant and mathematically initialized at 0.1 m/s.

$$V_{air} = 0.1 \text{ m/s} \quad (11)$$

This value represents the standardized baseline for 'still air,' accounting for the natural buoyancy-driven convective boundary layer generated by human metabolic heat, while remaining below the physiological threshold for draft perception. Consequently, this constant was integrated into the Python-based PMV calculation module prior to its execution by the Machine Learning algorithm.

The variable ***CLO*** quantifies the thermal insulation provided by an occupant's clothing ensemble. Given the limitations of the computer vision subsystem, conducting real-time, individualized visual extraction of clothing insulation for dynamic occupant loads is computationally impractical. To resolve this, a static, homogenized clothing insulation value was established based on the designated function of the space and the prevailing regional climate.

In the tropical climate of the Philippines, standard classroom attire consists of lightweight garments, typically represented by short-sleeved school uniforms, thin trousers, or skirts. This ensemble perfectly aligns with the standardized 'typical summer indoor clothing' classification.

$$clo = 0.5 \quad (12)$$

Consequently, a constant value of 0.5 *clo* was integrated into the model, sourced directly from the standardized ensemble metrics provided in ASHRAE Standard 55, Table 5-2 (shown in Figure 24).

Which ensemble best matches what you are wearing right now?

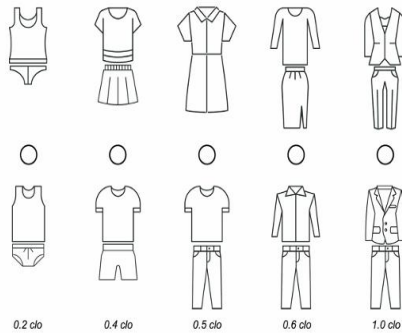


Table 5-2 Clothing Insulation I_{cl} Values for Typical Ensembles

Clothing Description	Garments Included ^a	I_{cl} , clo
Trousers	(1) Trousers, short-sleeve shirt	0.57
	(2) Trousers, long-sleeve shirt	0.61
	(3) #2 plus suit jacket	0.96
	(4) #2 plus suit jacket, vest, t-shirt	1.14
	(5) #2 plus long-sleeve sweater, t-shirt	1.01
	(6) #5 plus suit jacket, long underwear bottoms	1.30
Skirts/dresses	(7) Knee-length skirt, short-sleeve shirt (sandals)	0.54
	(8) Knee-length skirt, long-sleeve shirt, full slip	0.67
	(9) Knee-length skirt, long-sleeve shirt, half slip, long-sleeve sweater	1.10
	(10) Knee-length skirt, long-sleeve shirt, half slip, suit jacket	1.04
	(11) Ankle-length skirt, long-sleeve shirt, suit jacket	1.10
	(12) Walking shorts, short-sleeve shirt	0.36
Overalls/coveralls	(13) Long-sleeve coveralls, t-shirt	0.72
	(14) Overalls, long-sleeve shirt, t-shirt	0.89
	(15) Insulated coveralls, long-sleeve thermal underwear tops and bottoms	1.37
Athletic	(16) Sweat pants, long-sleeve sweatshirt	0.74
Sleepwear	(17) Long-sleeve pajama tops, long pajama trousers, short 3/4 length robe (slippers, no socks)	0.96

^a All clothing ensembles, except where otherwise indicated in parentheses, include shoes, socks, and briefs or panties. All skirt/dress clothing ensembles include pantyhose and no additional socks.

Figure 24. Clothing Factors by ASHRAE 55 Standards

The metabolic rate (*met*) quantifies the anthropogenic heat generation of the occupants. While computer vision systems can theoretically estimate individualized metabolic variations based on micro-movements or posture, extracting this data for every occupant imposes immense computational overhead on the edge device. Furthermore, the physical limitations of a single-zone HVAC architecture render individualized thermal tracking impractical, as the actuator can only apply a uniform cooling load across the entire space. Therefore, the system utilizes a homogenized spatial average for the occupant metabolic rate.

Table 1. Metabolic rates for typical tasks by ASHRAE 55

Activity	Metabolic Rate		
	met	W/m ²	Btu/h·ft ²
Resting			
Sleeping	0.7	40	13
Reclining	0.8	45	15
Seated, quiet	1.0	60	18
Standing, relaxed	1.2	70	22
Walking (on level surface)			
0.9 m/s, 3.2 km/h, 2.0 mph	2.0	115	37
1.2 m/s, 4.3 km/h, 2.7 mph	2.6	150	48
1.8 m/s, 6.8 km/h, 4.2 mph	3.8	220	70
Office Activities			
Reading, seated	1.0	55	18
Writing	1.0	60	18
Typing	1.1	65	20
Filing, seated	1.2	70	22
Filing, standing	1.4	80	26
Walking about	1.7	100	31
Lifting/packing	2.1	120	39

Based on empirical observations of standard academic environments, where occupants are predominantly engaged in seated, light-activity tasks (e.g., listening, writing, or typing), a constant value of $MET = 1.1$ was adopted (Found on Table 1). This value strictly adheres to the standardized activity levels defined in ASHRAE Standard 55 for educational and office environments.

The external mechanical work variable (WME) within the PMV calculation represents the physical energy an individual expends performing labor. For typical classroom environments, student activity is restricted to seated, non-strenuous tasks. Because no significant mechanical work is being performed against external physical forces, the mechanical efficiency is practically zero. Therefore, following standard ASHRAE 55 calculation protocols for light indoor activities, it initialized as $wme = 0$. This assumption ensures that the entirety of the 1.1 met metabolic rate is correctly calculated as sensible heat dissipated into the conditioned space.

Headless Occupancy Detection and Counting (YOLOv8n)

To dynamically quantify the anthropogenic heat load within the controlled space, the system utilized a computer vision-based occupancy detection pipeline. The core inference engine is powered by YOLOv8n (*You Only Look Once*, version 8, Nano architecture) developed by *Ultralytics*. The "Nano" variant was specifically selected for its highly optimized, lightweight neural network structure, which allows real-time object detection to run efficiently on Raspberry Pi 5 without requiring a dedicated external Graphics Processing Unit. Visual data is continuously acquired via a localized

Raspberry Pi Camera 3 Wide module configured to a resolution of 1280x720 pixels.

Specifications of this module are presented below.

Specification	
	Back-illuminated, stacked CMOS 12-megapixel Sony IMX708 image sensor
	High signal-to-noise ratio (SNR)
	Built-in 2D Dynamic Defect Pixel Correction (DPC)
	Phase Detection Autofocus (PDAF) for rapid autofocus
	QBC Re-mosaic function
	HDR mode (up to 3 megapixel output)
	CSI-2 serial data output
	2-wire serial communication (supports I2C fast mode and fast mode plus)
	2-wire serial control of focus mechanism
	Resolution: 11.9 megapixels
	Sensor size: 7.4mm sensor diagonal
	Pixel size: 1.4µm × 1.4µm
	Horizontal/vertical: 4608 × 2592 pixels
	Diagonal field of view: 75 degrees (Camera Module 3, Camera Module 3 NoIR), 120 degrees (Camera Module 3 Wide, Camera Module 3 NoIR Wide)
	Common video modes: 1080p50, 720p120
	Output: RAW10
	IR cut filter: Integrated in standard variants; not present in NoIR variants
	Dimensions: 25 × 24 × 11.5mm (12.4mm height for Wide variants)
	Ribbon cable length: 200mm
	Cable connector: 15 × 1mm FPC

Figure 25. Raspberry Pi Cam 3 Wide Specifications

The raw RGB matrices are captured and translated into standard BGR color space arrays using the OpenCV (*Open-Source Computer Vision*) library before being passed to the machine learning model.



Figure 26. Image Processing Flowchart

Rather than allocating extensive computational resources to train a custom computer vision model from scratch, the system utilized a pre-trained model “yolov8.pt” that has been comprehensively trained on the Microsoft COCO (*Common*

Objects in Context) dataset. The COCO dataset contains over 250,000 annotated human instances, making the pre-trained weights highly robust and proficient at identifying human figures across various lighting conditions and angles. Leveraging this existing, state-of-the-art model eliminates redundant training phases while guaranteeing high-speed, accurate inference suitable for edge-computing environments. The model is specifically filtered to isolate the "person" class (*Class 0*), ensuring the system only counts human occupants and ignores other objects within the frame.

To manage the real-time visual data pipeline, the Open-Source Computer Vision Library (OpenCV) was integrated into the software architecture. OpenCV functions as the primary bridge between the camera hardware and the AI model. Its role is twofold: first, it actively captures the continuous live video stream and preprocesses the individual image frames (handling resizing and color-space formatting) before feeding them into the YOLOv8 inference engine. Second, OpenCV acts as the graphical rendering engine; once the YOLOv8 model returns the coordinate data of a detected person, OpenCV is utilized to draw bounding boxes around the occupants and continuously update the real-time occupancy count text on the visual display monitor.

The critical hyperparameters governing the YOLOv8n inference engine specifically input image size (*imgsz*), confidence threshold (*conf*), and Intersection over Union (*iou*)—were empirically optimized through a rigorous iterative calibration process. The objective of this calibration was to minimize the delta between the algorithm's raw detection output and the actual occupant load (the ground truth), thereby ensuring a highly stable and accurate spatial representation.

The empirical tuning procedure was conducted through the following experimental phases:

- A verified number of human subjects representing the ground truth were deployed within the target room. To accurately simulate the unpredictable physical dynamics of a real-world environment, subjects were instructed to assume various morphological poses such as sitting, standing, walking, and resting across diverse spatial coordinates and angular orientations relative to the camera lens.
- The YOLOv8n model was executed via a live inference diagnostic feed directly on the Raspberry Pi edge computing device. The network hyperparameters were systematically adjusted in real-time, allowing the system architecture to instantly calculate and display the resulting variance between the "raw model count" and the established ground truth.
- This iterative validation loop was repeated across a comprehensive matrix of environmental scenarios. Testing parameters ranged from low to maximum room capacities (density variance) and included both static (seated/stationary) and highly dynamic (constant movement) occupant behaviors.



Figure 27. Sample Live View of Calibrated YOLOv8n

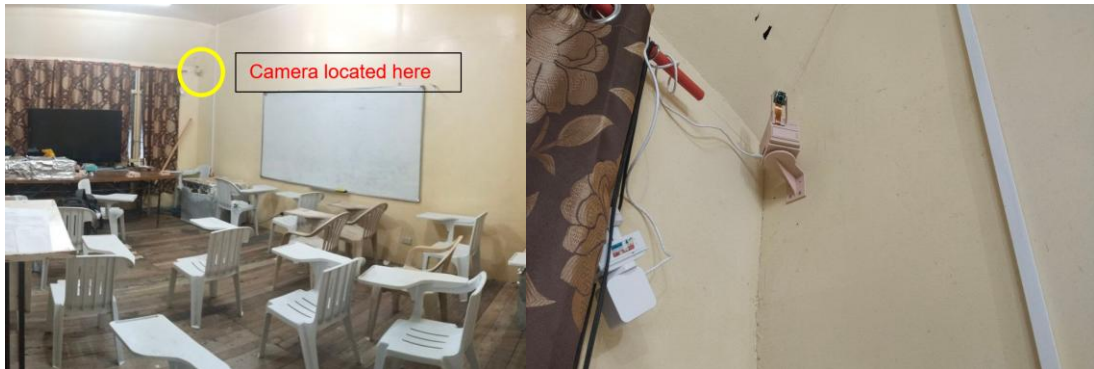


Figure 28. Detailed Camera Location

The final deployed configuration ($imgsz=832$, $conf=0.12$, $iou=0.45$) was explicitly selected based on these empirical trials (Shown in Table 2). It represents the optimal algorithmic equilibrium, yielding the highest detection stability, maximum occlusion resilience, and the lowest mean error rate across all tested scenarios.

Table 2. YOLOv8n configuration values

Factors	Values	Reasons
Image Size ($imgsz$)	832	Lower image size loses occupancy reading capability; higher values can cause longer analysis time.
Confidence Threshold ($conf$)	0.12	Lower values like 0.11 cause hallucination, and any values higher than 0.12 decrease accuracy at crowded settings.
Intersection Over Union (iou)	0.45	Higher values produce multiple bounding boxes on 1 person; lower values make it hard to read a partially visible person behind another person.

Data Acquisition and Machine Learning Training

The empirical data acquisition phase was executed within the controlled boundary of the DME Conference Room. The primary objective of this phase was to

generate a good dataset to serve as the foundational training matrix for the Bayesian Optimized Random Forest (*BORF*) machine learning model.

Variable Classification and Measurement Parameters

To quantify the thermal equilibrium of the space, the localized data management script was configured to monitor, calculate, and log eleven specific systemic variables. These variables are classified into four distinct operational categories:

1. Environmental Boundary Conditions:

- Inside Dry-Bulb Temperature T_{in} , measured in degrees Celsius (°C).
- Inside Relative Humidity RH_{in} , measured as a percentage (%).
- Outside Dry-Bulb Temperature T_{out} , measured in °C.
- Outside Relative Humidity RH_{out} , measured as a percentage (%).

2. Anthropogenic and Temporal Variables:

- Occupancy Count N measured as a discrete integer via the YOLOv8n computer vision subsystem.
- Time of Day recorded in a 24-hour (HH:MM) temporal format to track time variable.

3. HVAC Actuator States:

- AC Temperature Setpoint T_s , logged in °C.
- AC Operating Mode (categorical string such as cool, fan, dry, eco).
- AC Fan Speed (categorical string).
- AC Energy Consumption measured in kilowatts(W) via PZEM-004T.

4. Target Function:

- Predicted Mean Vote (PMV), computed as a dimensionless floating-point index utilizing the real-time environmental inputs parsed through the ASHRAE 55 mathematical formulation.

Telemetry Architecture and Data Aggregation

The localized ESP32 sensor nodes transmitted environmental and electrical power telemetry to the Raspberry Pi 5 supervisor via the Message Queuing Telemetry Transport (*MQTT*) protocol. To mitigate spatial gradients and localized sensor bias, variables utilizing redundant physical hardware deployment specifically indoor temperature and relative humidity were mathematically averaged within the central Python orchestration script to establish a generalized, single-zone environmental metric.

Simultaneously, the YOLOv8n occupancy pipeline delivered real-time occupant counts into the system's data gathering program. The main Python aggregation script synchronized these disparate hardware inputs such as environmental data, AC hardware states, and camera telemetry and appended the unified array to a localized Comma-Separated Values (CSV) database. The system enforced a strict, non-blocking sampling interval of exactly 60 seconds (1 minute) for every dataset appended.

Experimental Boundary Scenarios

To prevent machine learning model overfitting and ensure the BORF algorithm could map the nonlinear thermodynamic behavior of the environment, data was

extracted across a controlled matrix of varying operational states. Over 4 weeks of tracking period, the system captured data under the following defined scenarios:

- The space was logged during periods of zero occupancy, low-density academic sessions, and maximum-capacity scenarios to measure the response of the HVAC system to sudden biological heat injection.
- Data logging encompassed multiple shifting weather conditions, including high solar radiation (sunny), negligible solar load (cloudy), and high ambient moisture (rainy) to map the exterior envelope's thermal bridging effects.

Dataset Finalization

Following about 4 weeks of empirical logging phase, the continuous 60-second sampling cycle yielded a primary dataset containing over 7,000 discrete operational instances. From an initial collection of over 7,000 data points, a systematic filtering process was conducted to isolate steady-state environmental conditions. Any transient data where the PMV values were actively fluctuating was discarded.

The filtering criteria strictly retained data where the PMV stabilized, indicating that the room had reached thermal equilibrium. This process refined the raw telemetry down to approximately 3,000 high-quality data points, which were subsequently utilized as the final training dataset for the machine learning model. Each instance correlates an exact timestamp with the corresponding physical room state, actuator setting, and calculated PMV value, forming the explicit multi-feature matrix required for training the BORF predictive architecture.

Machine Learning Framework and Model Training (BORF)

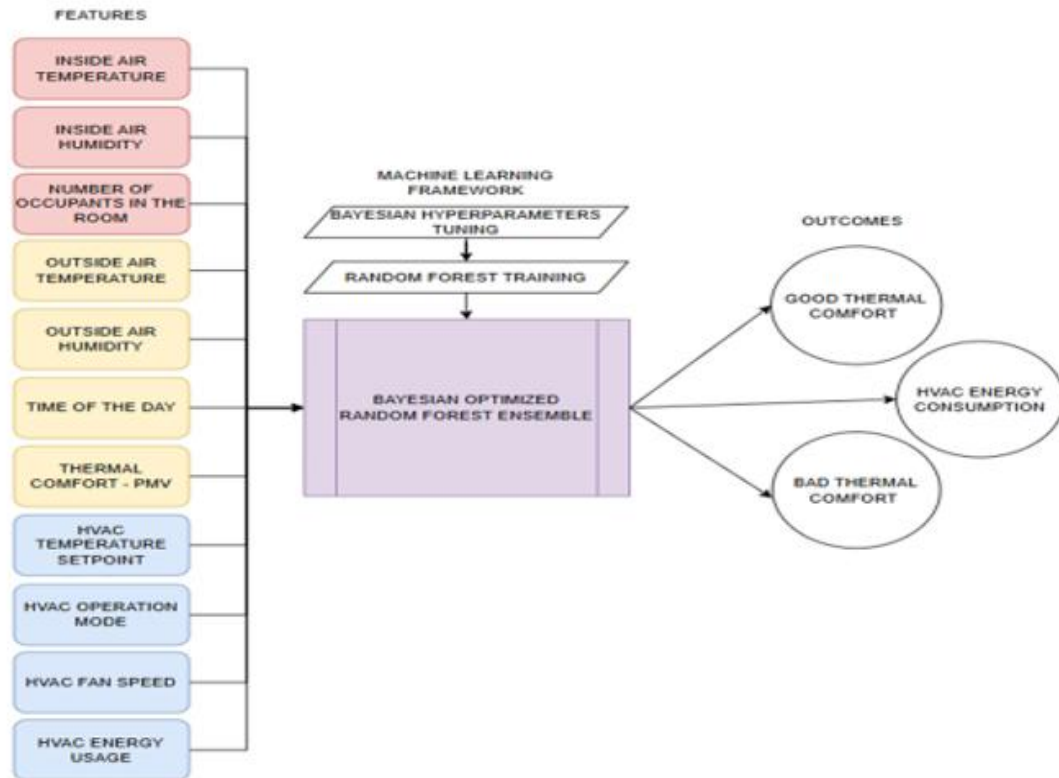


Figure 29. Bayesian Optimized Random Forest Model

Figure 29 illustrates the architectural schematic of the Bayesian Optimized Random Forest (*BORF*) training pipeline. The empirical dataset, comprising the 3,000 discrete operational instances gathered during the data acquisition phase, served as the foundational training matrix for the algorithm.

As shown in the schematic, the machine learning framework is structured to ingest a multi-dimensional feature vector. These input features encompass environmental parameters such as inside/outside temperature and humidity, occupancy metrics, temporal data, and dynamic HVAC actuator states such as setpoint, operation mode, fan speed, and baseline energy usage. Within the framework, the Random Forest

algorithm is subjected to Bayesian Hyperparameter Tuning. This optimization process iteratively adjusts the model's parameters to maximize predictive accuracy. The finalized BORF ensemble maps the input vectors to defined outcome nodes, specifically predicting categorical thermal comfort states alongside projected HVAC Energy Consumption.

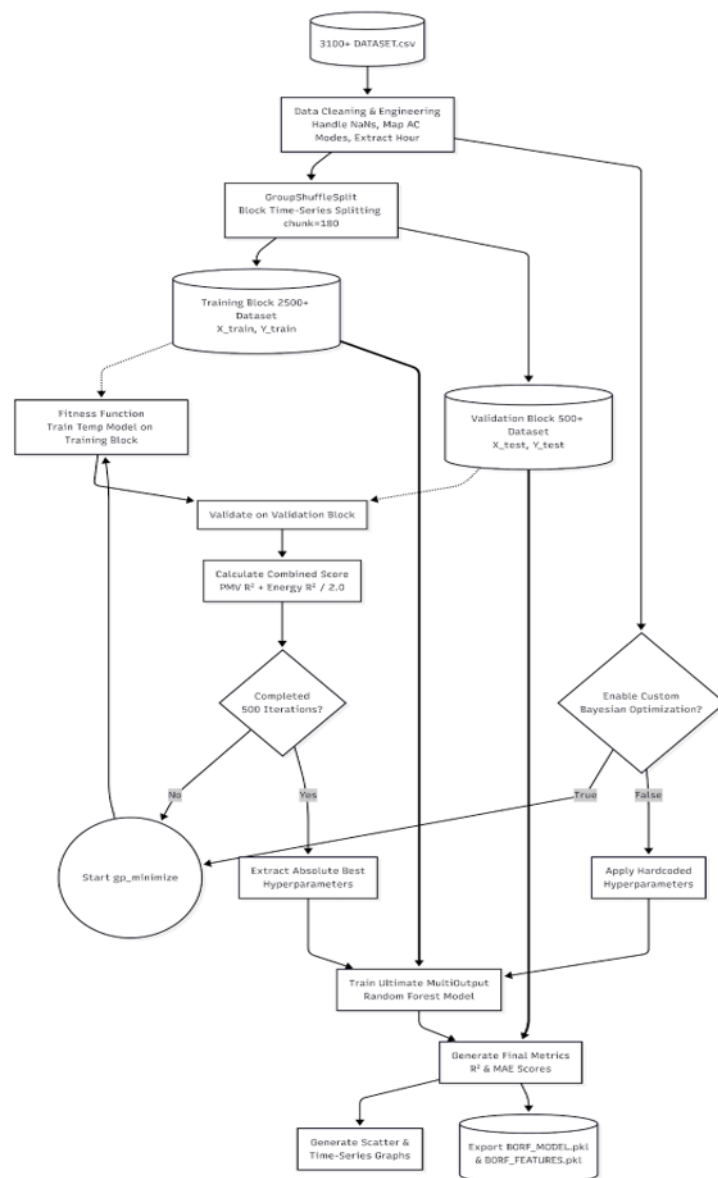


Figure 30. BORF Training Chart

The BORF model was trained utilizing a subset of 2,500 distinct data instances that represented various operational states within the controlled environment. Following this primary training phase, the model's predictive accuracy was rigorously evaluated using a separate testing set of 500 instances. This testing set was withheld during training, ensuring the model's performance was validated on entirely unseen data.

To systematically navigate the hyperparameter search space, a Bayesian optimization framework was implemented. The specific configuration parameters including the minimum and maximum search boundaries, the total number of evaluation iterations, and the k-fold cross-validation architecture is documented in Appendix A (*BORF* Training Program uploaded in GitHub link).

The continuous and categorical bounds for these hyperparameters were derived from a comprehensive synthesis of relevant academic literature, notably the optimization methodologies established by Yang et al., and supplemented by recognized computational best practices for Random Forest ensembles.

The parameters used to train and develop the machine learning model are shown in Table 3.

Table 3. Bayesian training search range and optimization factors

Parameter Category	Variable Name	Value / Search Space	Description
Model Search Space	<i>n_estimators</i>	Integer(50, 300)	The number of decision trees in the random forest ensemble.

Parameter Category	Variable Name	Value / Search Space	Description
	<i>max_depth</i>	Integer(3, 30)	The maximum number of hierarchical splits allowed in a single tree.
	<i>min_samples_split</i>	Integer(2, 20)	The minimum number of data points required to execute a node split.
	<i>min_samples_leaf</i>	Integer(1, 10)	The minimum number of data points required to form a final leaf (output) node.
	<i>max_features</i>	Real(0.5, 1.0)	The fraction of total features evaluated when determining the optimal split.
Optimizer Settings	<i>n_iter</i>	500	The total number of hyperparameter combinations tested by the optimizer.
	<i>cv</i>	3	The number of cross-validation folds used to evaluate each combination.
	<i>scoring</i>	'r2'	The statistical metric (R-squared) used to define optimization success.
	<i>n_jobs</i>	-1	Hardware utilization flag; instructs the system to use all available CPU cores.
	<i>random_state</i>	42	The fixed seed value ensuring mathematical reproducibility.
	<i>verbose</i>	2	The detail level of console outputs printed during the training process.

The computational development and subsequent training of the model were executed utilizing the Python programming language within a Visual Studio Code (*VS Code*) Integrated Development Environment (*IDE*). The core machine learning architecture was constructed leveraging the scikit-learn library. Prior to model compilation, all requisite computational packages and algorithmic dependencies were configured within the local environment. The physical processing and training workload were executed on a dedicated local workstation (*Acer Aspire 5*).

The complete Python algorithmic script utilized for the BORF training sequence documentation uploaded in GitHub link found in Appendix A.

The Supervisory Controller [Raspberry Pi 5]

Designated as a fixed node within the conditioned space, the Raspberry Pi 5 operates as the central computational hub and localized edge server. It is tasked with hosting the MQTT broker to facilitate low-latency network communication and executing the core Machine Learning (ML) optimization algorithms.

Furthermore, the controller processes real-time video feeds via an integrated wide-angle optical peripheral (Raspberry Pi Camera Module 3 Wide). This visual data stream is utilized to perform computer vision-based object detection and spatial counting, thereby supplying the control logic with highly accurate, dynamic metabolic load variables. By aggregating this real-time occupancy data with the environmental telemetry gathered from the distributed sensor network, the primary controller synthesizes the optimal HVAC setpoints and transmits the corresponding actuation protocols to the designated infrared interface (Sensor Node 2).

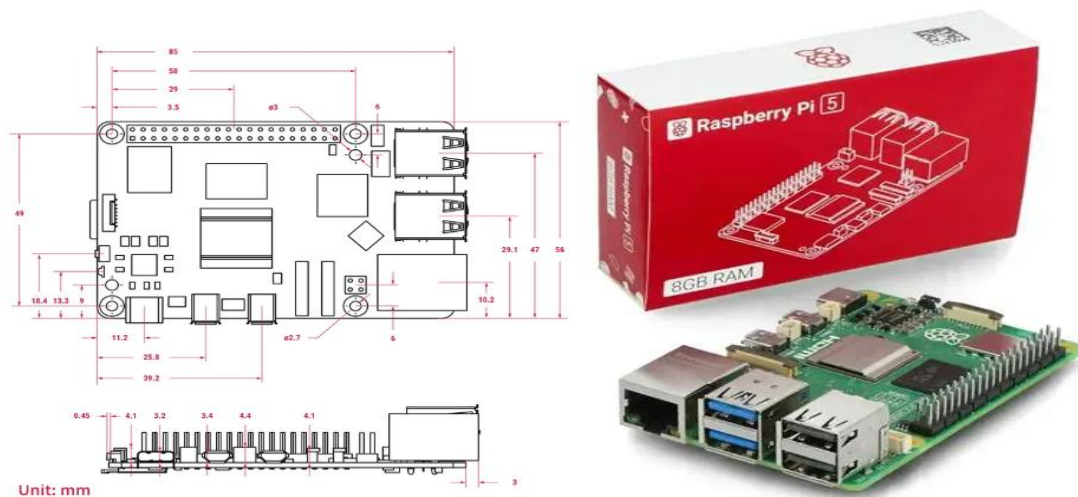


Figure 31. Raspberry Pi 5 8Gb

This study utilized the 8GB variant of the Raspberry Pi 5 to serve as the system's central edge server. The detailed hardware capabilities and technical specifications of the microcomputer are shown in Figure 32.

Specification	
	• Broadcom BCM2712 2.4GHz quad-core 64-bit Arm Cortex-A76 CPU, with cryptography extensions, 512KB per-core L2 caches and a 2MB shared L3 cache • VideoCore VII GPU, supporting OpenGL ES 3.1, Vulkan 1.2 • Dual 4Kp60 HDMI® display output with HDR support • 4Kp60 HEVC decoder • LPDDR4X-4267 SDRAM (4GB and 8GB SKUs available at launch) • Dual-band 802.11ac Wi-Fi® • Bluetooth 5.0 / Bluetooth Low Energy (BLE) • microSD card slot, with support for high-speed SDR104 mode • 2 × USB 3.0 ports, supporting simultaneous 5Gbps operation • 2 × USB 2.0 ports • Gigabit Ethernet, with PoE+ support (requires separate PoE+ HAT) • 2 × 4-lane MIPI camera/display transceivers • PCIe 2.0 x1 interface for fast peripherals (requires separate M.2 HAT or other adapter) • 5V/5A DC power via USB-C, with Power Delivery support • Raspberry Pi standard 40-pin header • Real-time clock (RTC), powered from external battery • Power button

Figure 32. Raspberry Pi 5 Specifications

To ensure the Raspberry Pi 5 could function as the autonomous Master Controller for the HVAC system, a highly structured and isolated software environment was established. The setup procedure was divided into operating system configuration, environment isolation, and communication protocol integration to ensure maximum stability.

Operating System Installation and Configuration

Raspberry Pi 5 required a foundational operating system before it could be deployed. This was achieved using the official Raspberry Pi Imager software. A 64-bit Raspberry Pi OS based on the Debian Bookworm architecture was flashed onto the primary storage drive. This specific OS version was selected because it was the latest and recommended during the time of installation.

During the flashing process, foundational settings such as Wi-Fi credentials, localization, and Secure Shell (SSH) access were pre-configured based on the recommended settings and the community suggestions found on the internet. By system requirements, the local network hostname of the Master Controller was strictly configured as “*cvmpi*”, and the default administrative user account was secured with the password “*cvmpiloin123*”.

The Wi-Fi interface of the Raspberry Pi 5 was specifically configured at the operating system level (via the network manager credentials) to bind exclusively to a designated Service Set Identifier (SSID): “*TP-Link_95CC*”. Auto-connection features to any other previously known networks or available public access points were strictly disabled. This hardcoded network lock ensures that in the event of a power outage or

system reboot, the Master Controller will autonomously seek out and connect only to the system's dedicated router, preventing accidental network drift.

Python Environment and Version Control

Once the operating system was updated, the programming environment was prepared. Python 3.11 was specifically selected and verified as the target interpreter. This version was chosen because it represents the most stable release for the required data science and machine learning libraries used in the system's architecture, preventing compatibility errors during prolonged inference loops.

Furthermore, to facilitate on-device debugging and software maintenance, Integrated Development Environments (IDEs) such as Visual Studio Code (VS Code) are configured within the operating system.

Virtual Environment (venv) Initialization

To maintain a clean system architecture and prevent dependency conflicts, a Python virtual environment (*venv*) was established. This isolated directory ensured that all project-specific libraries were kept entirely separate from the core operating system's Python packages, resulting in a tidy, modular, and easily replicable setup.

Dependency and Library Installation

With the virtual environment activated, all necessary third-party libraries were downloaded and installed using Python's package installer (*pip*). The core dependencies included:

NumPy. This is installed to handle complex, high-performance numerical computations and standard deviation calculations for room stability tracking.

Pandas. This is installed to structure, manipulate, and analyze the live CSV datasets efficiently.

Paho-MQTT. This is installed as the dedicated Python client library, enabling the Master Controller scripts to publish and subscribe to communication topics across the network.

MQTT Broker System Integration

While *paho-mqtt* enabled the Python code to communicate, the core message broker itself required deep integration into the Raspberry Pi's operating system. The Eclipse Mosquitto MQTT broker was installed via the system's package manager.

To ensure the communication infrastructure was perpetually ready for full functionality, the broker was configured as a background system daemon. Utilizing system commands, the MQTT broker was integrated directly into the OS boot sequence. This guaranteed that the moment the Raspberry Pi 5 receives power, the local MQTT server automatically initializes, allowing instantaneous communication between the master controller, environmental sensors, and the dual AC infrared blasters without requiring manual intervention.

Following the initial hardware configuration, the Raspberry Pi 5 will be ready to function as the primary deployment environment for all system operations. To ensure

software stability and streamline the debugging process, a host-target development methodology was employed.

The core software components specifically the machine learning algorithms and the supervisory Main Program were initially authored, simulated, and rigorously tested on a separate, dedicated development workstation. Upon successful pre-deployment validation, the finalized source code was systematically transferred and integrated into the localized edge computing environment of the Raspberry Pi for live execution.

The comprehensive process flow diagram governing the internal software execution of the Raspberry Pi is depicted in Figure 33 below.

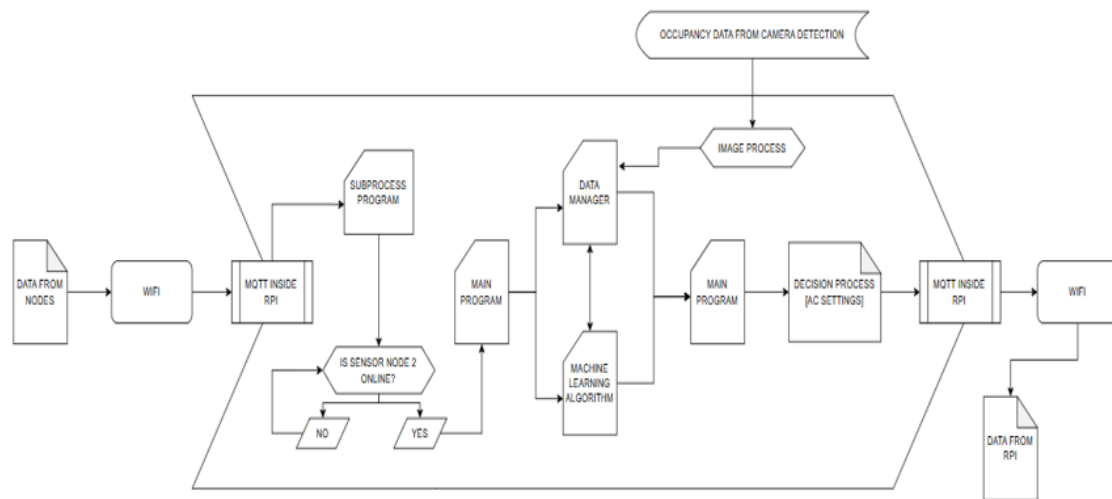


Figure 33. Main Program Process Flowchart

Figure 33 illustrates the software execution flow and data processing pipeline operating within the primary supervisory controller. Upon power cycling, the Raspberry Pi 5 initiates an automated boot sequence, immediately establishing a connection to a localized IEEE 802.11 (Wi-Fi) access point. This localized network serves as the primary communication backbone for the entire distributed control system.

Following successful network acquisition, the operating system initializes all required software dependencies, most notably the local MQTT (*Message Queuing Telemetry Transport*) broker. The broker is configured to facilitate asynchronous publish-subscribe messaging across the network using the following local parameters:

- MQTT Hostname: *cvmpi*
- Port ID: 1883 (Standard MQTT transmission)

Maintaining this active Wireless Local Area Network (WLAN) connection is a strict prerequisite. Without the local access point acting as the routing medium, the MQTT broker cannot establish telemetry synchronization or transmit control payloads to the peripheral edge nodes.

Once the MQTT communication protocol is fully initialized, a background daemon, designated as the Subprocess Program, executes automatically. This subprocess functions as a continuous system interlock and health monitor. Its primary directive is to query the MQTT topics to verify the active "heartbeat" status of Sensor Node 2 (the Fixed Inside Sensor and IR Actuator).

Since the system relies on Sensor Node 2 to physically execute the calculated setpoints, the control architecture dictates that actuation capability must be confirmed before processing logic begins. If the subprocess verifies that Sensor Node 2 is online and responsive, it subsequently triggers the execution of the Main Program.

As the primary software orchestrator, the Main Program simultaneously initializes and bridges two critical operational modules: The Data Manager responsible for aggregating raw sensor telemetry from the MQTT broker and parsed occupancy

data from the camera's image processing pipeline and The Machine Learning Algorithm receives the aggregated arrays from the Data Manager to compute the optimal HVAC operational parameters. This architecture design specifically separating the data aggregation logic from the machine learning inference engine into distinct executable files is a deliberate design choice based on the principle of modularity. By compartmentalizing these processes, the system facilitates independent diagnostics and streamlines maintenance.

In the event of a runtime error, software bug, or sensor failure, this separation enables rapid fault isolation. It allows the researcher to trace anomalies and deploy localized code modifications to a specific subsystem without risking unintended alterations or cascading failures within the broader control logic.

The Main Program

The Main Program functions as the primary supervisory orchestrator for all data aggregation and decision-making workflows. Its core responsibilities include the concurrent management of the decoupled Machine Learning module and the enforcement of deterministic safety guardrails. These hard-coded guardrails act as a critical fail-safe mechanism, designed to intercept and override the ML algorithm in the event of a computational anomaly or the generation of setpoints that violate predefined thermal safety thresholds. A simplified logic flowchart detailing the process flow of this Main Program is presented in Figure 34.

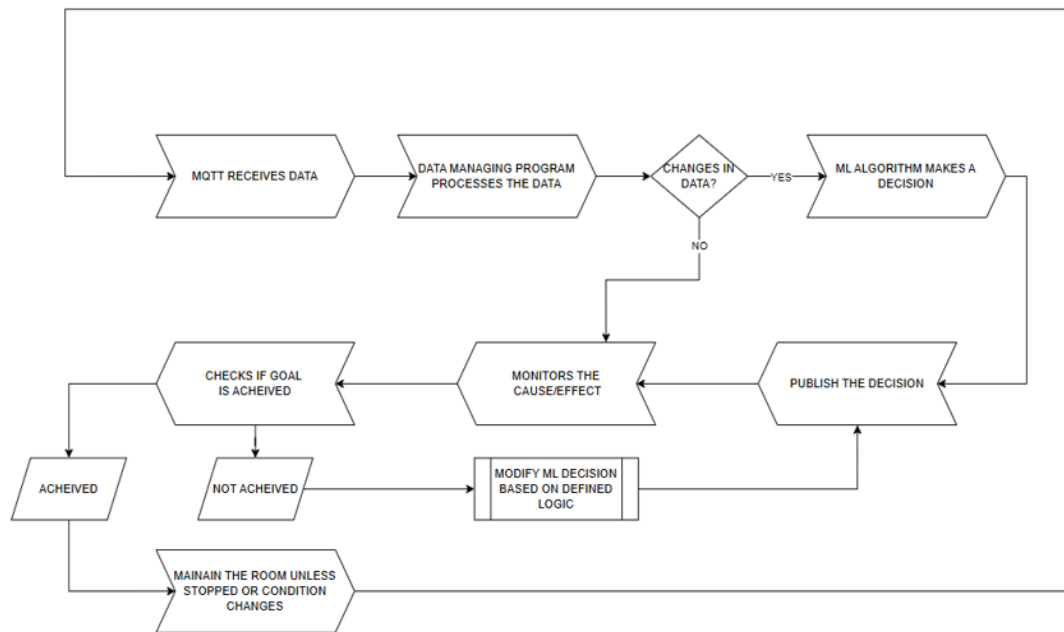


Figure 34. Hybrid AI Model Decision Process

The sequential logic of the Main Program operates as a continuous, closed-loop control cycle, as illustrated in Figure 34. Upon initialization, the system enters a listening state, awaiting incoming telemetry from the distributed sensor network via the MQTT broker. Upon data reception, the Data Management Module is triggered to preprocess the variables for the Machine Learning algorithm while simultaneously archiving the raw telemetry into CSV files for subsequent empirical analysis. If the processed data indicates a change in the environmental conditions, the Main Program invokes the ML inference engine. The algorithm analyzes the multi-variable data array to compute the optimal AC setpoints. These parameters are then packaged as an instruction payload and published to the IR Actuator node to physically adjust the AC unit.

Following actuation, the system transitions into an observation phase to monitor the environment's transient thermal response (cause and effect). During this phase, a parallel function continuously evaluates the trend of the Predicted Mean Vote (PMV) index. By calculating the standard deviation of the PMV over a rolling time window, the system mathematically determines when the environmental metrics have "flatlined." This confirms that the room has reached a steady-state condition, ensuring the system does not prematurely react to temporary thermal fluctuations.

Once steady state is confirmed, the system evaluates whether the ultimate thermal comfort goal has been achieved. The control flow bifurcates based on this evaluation:

- **Goal Achieved.** If the PMV reflects thermal neutrality, the system enters a maintenance state, holding the current AC settings until a new environmental disturbance (data change) is detected.
- **Goal Not Achieved.** If the PMV indicates persistent discomfort, the system's deterministic, hard-coded logic intervenes. This fail-safe overrides the previous ML decision, applies a predefined corrective adjustment to the AC settings, and republishes the new instruction. The system then returns to the monitoring phase, iterating this corrective loop until absolute thermal comfort is established.

A more detailed process flow is presented in Figure 35.

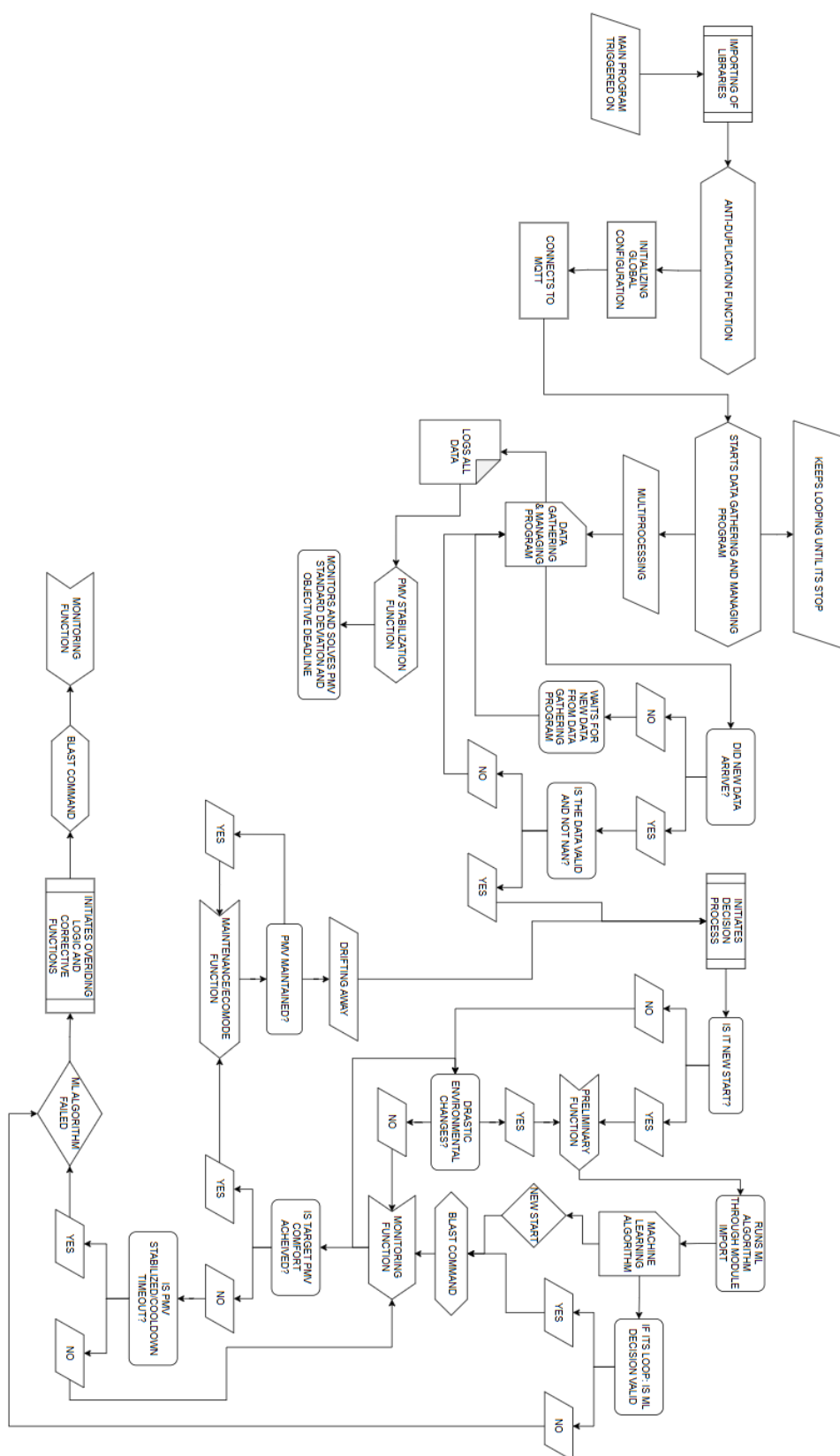


Figure 35. Detailed Logic Flow for Main Program

The Data Managing Program

Figure 36 illustrates the algorithmic workflow and operational logic of the central Data Management Program. Acting as the core computational hub running on the edge device, this Python-based script coordinates the asynchronous ingestion, validation, computation, and storage of all system telemetry. The programmatic workflow is structured into four primary sequential operations.

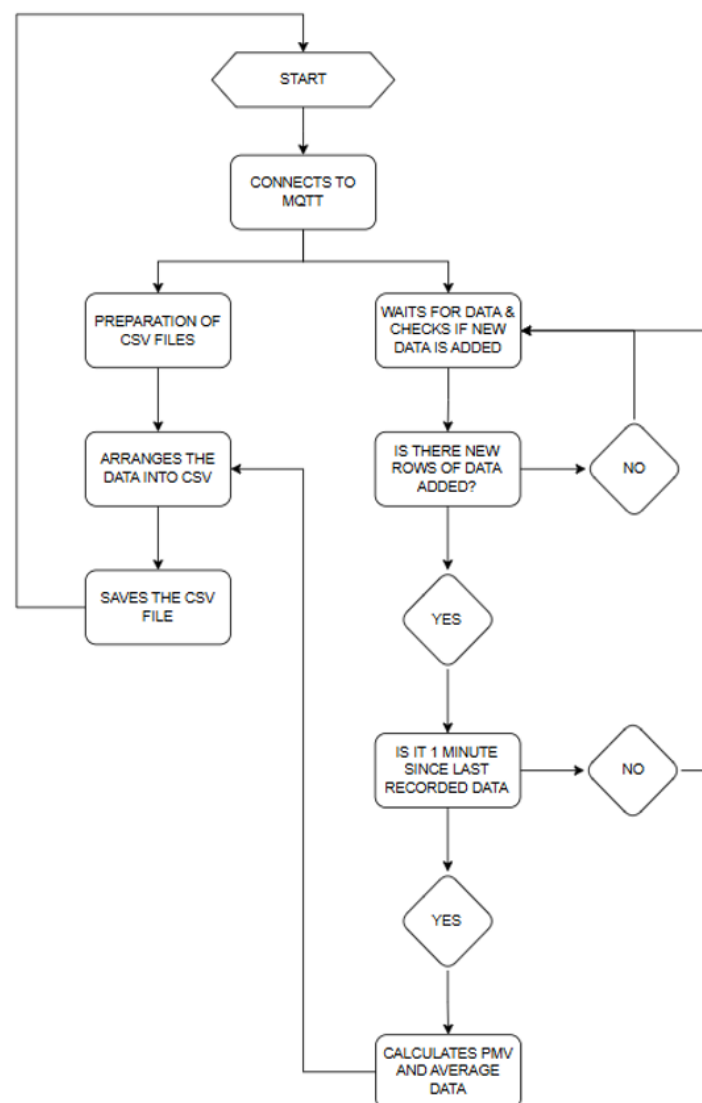


Figure 36. Data Managing Program Logic Flow

The program continuously monitors the MQTT broker for incoming sensor payloads while simultaneously polling the YOLOv8n computer vision thread. To ensure hardware reliability, the system conducts real-time health diagnostics on the distributed sensor nodes. By tracking the exact timestamp of the last received message from each ESP32 microcontroller, the program algorithmically identifies network timeouts and flags unresponsive nodes as "*Offline*."

To prevent systemic crashes caused by corrupted data packets, the script employs a strict validation protocol. Incoming payloads are parsed through exception-handling routines to verify they contain valid numerical floats; anomalous string characters or corrupted values are automatically rejected. Furthermore, for environmental metrics utilizing redundant physical sensors (such as the dual indoor temperature nodes), the program executes a spatial averaging function to establish a singular, generalized room metric.

Once the synchronized data array containing environmental boundaries, HVAC states, and occupancy count are sanitized, it is fed into the thermodynamic calculation module. The script executes the complex, non-linear mathematical operations required to calculate the Predicted Mean Vote (PMV) index based on Fanger's heat balance equations.

In the final step of the pipeline, the validated raw data, calculated averages, and the resultant PMV index are aggregated into a unified temporal array. This complete feature set is then systematically appended to the centralized Comma-Separated Values (CSV) database at strict one-minute intervals. The complete algorithmic

implementation of the Data Management Program is documented and can be downloaded in GitHub link found in Appendix A.

The Decision Maker Program

A separate Python script serves as the main decision-making program for the system. Its primary task is to receive the organized data from the data management program and use the trained machine learning model to simulate all possible combinations of AC settings. The program evaluates these combinations to find the specific AC instructions that achieve the acceptable ASHRAE thermal comfort range while consuming the minimum amount of energy. Once the best combination is identified, the system outputs these settings to control the AC units. The schematic logic flowchart for this decision-making process is shown in Figure 37.

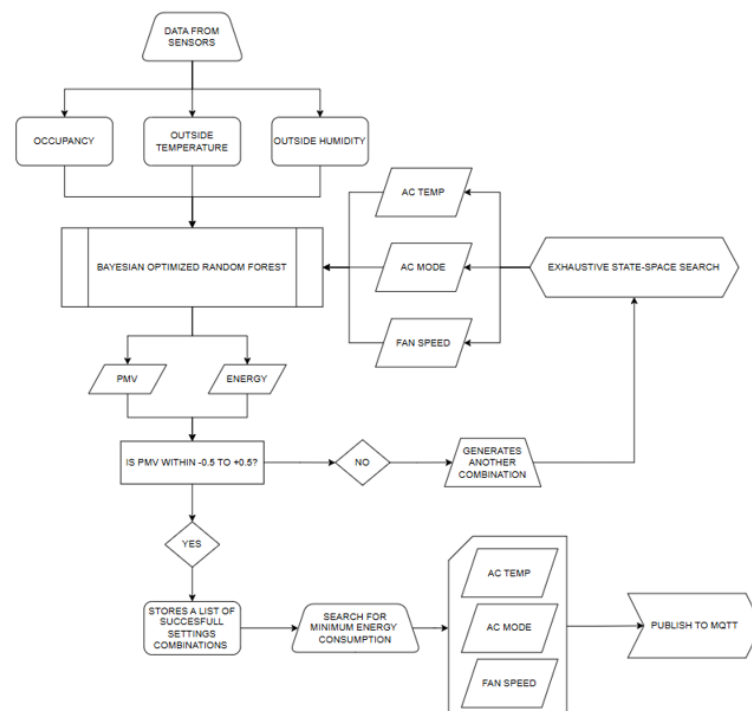


Figure 37. Machine Learning Algorithm

The process begins with the central Data Management Program, which continuously collects and organizes real-time environmental data, specifically occupancy, outside temperature, and outside humidity from the physical sensors. Rather than the sensors communicating directly with the AI, the Data Manager acts as an intermediary, processing the raw telemetry and passing only the required information to the machine learning module. Simultaneously, the decision-making program uses an exhaustive search method to systematically generate every possible combination of AC setting, including temperature, operating mode, and fan speed. Both the live room data and these generated AC combinations are then fed into the trained Bayesian Optimized Random Forest model.

The model simulates each setup to predict the resulting thermal comfort (PMV) and expected energy consumption. If a predicted PMV falls within the acceptable comfort range of -0.5 to +0.5, that specific AC combination is saved to a list of successful options; if it fails the comfort check, the program discards it and moves on to test the next combination. Once all possible settings have been simulated, the system reviews the list of successful options, selects the single combination that requires the minimum amount of energy, and publishes these final instructions via MQTT to physically adjust the AC units.

Hybrid Predictive Control and Deterministic Guardrails

While the Bayesian Optimized Random Forest (*BORF*) model serves as the primary decision-making engine, Random Forest architectures possess an inherent mathematical limitation: as ensemble decision trees, they are incapable of extrapolating

beyond their training domain. If the system encounters environmental boundary conditions exceeding the training dataset (e.g., an extreme ambient heatwave of 36°C when the maximum trained temperature was 33°C), the model cannot accurately calculate new thermodynamic curves and may generate suboptimal setpoints. Furthermore, unforeseen physical disturbances such as abrupt transient heat loads from a door left open or sudden, massive surges in occupancy can induce rapid thermal shifts that confuse the predictive logic.

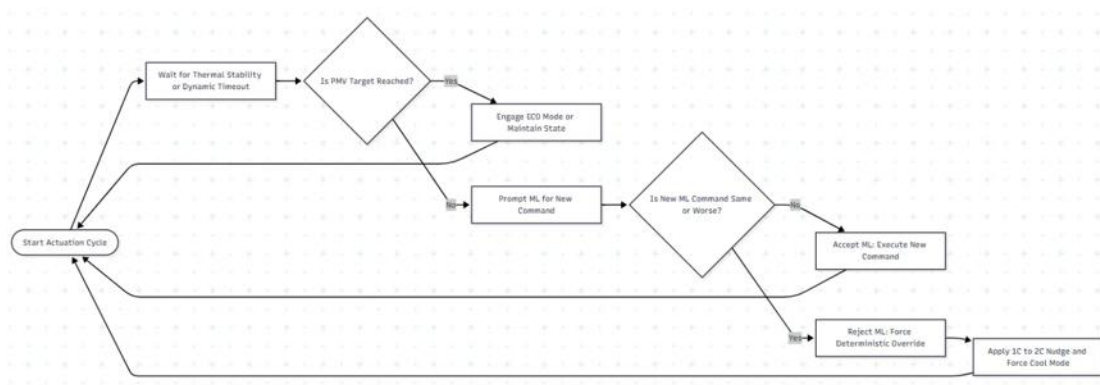


Figure 38. Deterministic Guardrails for AI Model

To ensure operational safety, continuous energy optimization, and strict adherence to ASHRAE 55 occupant comfort standards, a deterministic, rule-based supervisory logic was embedded within the Main Program. The program structure is documented in GitHub link found in Appendix A. This hybrid approach acts as a fail-safe, intervening only when the primary ML algorithm fails to reach thermal equilibrium. The deterministic logic operates across four primary stages:

Transient Response and Steady-State Monitoring

Following the execution of an AC command, the system must allow the thermodynamic environment time to react. A background monitoring function continuously samples the Predicted Mean Vote (*PMV*) index at 60-second intervals, aggregating these data points into a rolling 10-minute sliding window (evaluating the 0th to the 10th latest reading).

To mathematically determine if the room has reached thermal equilibrium, the algorithm calculates the standard deviation σ of this temporal array. A standard deviation of $\sigma = 0.015$ serves as the strict threshold for steady-state conditions. If the variance is equal to or less than this value, the room is considered stabilized. If $\sigma > 0.015$, the environment is still in a transient state, and the system continues to monitor the sliding window.

Dynamic Actuation Timeout Windows

To prevent the system from remaining in an inadequate cooling state indefinitely, a dynamic timeout deadline is assigned to every published AC command. This evaluation window scales proportionally based on the initial distance between the room's current PMV and the target PMV boundary, ranging from a minimum of 15 minutes to a maximum of 40 minutes. Larger thermal deviations require longer thermodynamic "*pull-down*" times, granting the AC unit a wider temporal window to cool the space. However, if the PMV standard deviation logic (detailed above) indicates the room has stabilized before this deadline expires, the countdown is immediately bypassed to evaluate the final comfort goal.

ML Re-Evaluation and Directional Guardrails

If the timeout deadline is reached or if the room stabilizes early and the PMV has still not reached the acceptable comfort zone, the system initiates its override protocol. The system does not immediately discard the ML model; first, it prompts the algorithm to re-evaluate the current environmental data and issue a new command. However, the system applies strict "directional guardrails" to this new prediction. If the ML algorithm issues the exact same failing command, or issues a theoretically worse command (e.g., suggesting a 28°C setpoint when a 27°C setpoint already failed to cool the room), the deterministic safeguard actively rejects the ML prediction and forces a hard-coded override.

Hard-Coded Corrective Actuation

Upon overriding a rejected ML command, the system applies predefined mathematical corrections to the AC settings to force the room into compliance:

- **Temperature Micro-Nudges:** If the distance from the current PMV to the target PMV boundary is ≤ 0.3 , the system deducts 1°C from the previous setpoint. If the thermal distance is > 0.3 , a more aggressive 2°C deduction is applied.
- **Mode Switching (Latent vs. Sensible Cooling):** If the AC is operating in 'Dry' (dehumidification) mode and fails to reach the target PMV, the logic forces a transition to 'Cool' mode. This prioritizes rapid sensible heat removal over latent heat removal. The system will only revert to 'Dry' mode

once the PMV approaches the target goal, and only if the ML algorithm explicitly dictates it.

Steady-State Maintenance and Energy Optimization

In the event that the PMV goal is successfully achieved, the system transitions into an active maintenance state. If the PMV stabilizes deeply within a safe, optimal neutral zone (≈ 0 PMV), the system autonomously switches the AC unit into ECO MODE. This mode physically limits the inverter compressor's rotational speed (RPM) to a partial-load state, minimizing aggregate energy consumption while producing just enough cooling capacity to offset ongoing heat infiltration

To prevent boundary drift, this extreme energy-saving condition is strictly prohibited if the PMV is hovering near the outer limits of the comfort zone (e.g., ± 0.4), ensuring the room does not accidentally drift into an uncomfortably hot (>0.5) or cold (<-0.5) state.

Weather-Normalized Energy Evaluation (ASHRAE Guideline 14 - CDH)

To accurately quantify the energy conservation achieved by the machine-learning-driven system (*BORF*) compared to standard manual human operation, it is necessary to mathematically normalize the energy data against daily outdoor weather variations. In accordance with ASHRAE Guideline 14 protocols, this study utilizes the Cooling Degree Hour (*CDH*) metric. CDH is an established industry-standard evaluation metric designed to normalize HVAC energy consumption across varying

environmental conditions by quantifying the exact external thermal load imposed on the building envelope.

For tropical climates such as the Philippines, the standard thermal baseline T_{base} is established at 24°C. Under ASHRAE assumptions, this represents the ambient threshold at which no mechanical cooling is required to maintain indoor comfort. Consequently, any outdoor temperature exceeding this baseline constitutes an active thermal load that the AC unit must counteract.

To evaluate the comparative energy efficiency of both operational modes (System-Operated vs. Human-Operated), the weather-normalized energy consumption is calculated at strict one-hour intervals using the following procedural sequence:

- **Temperature Aggregation.** The average outdoor dry-bulb temperature $T_{avg}(avg, outdoor)$ is calculated for the specific one-hour timeframe.
- **CDH Calculation.** The hourly thermal load is calculated by subtracting the baseline temperature:

$$CDH = \max(0, T_{avg, outdoor} - T_{base}) \quad (13)$$

- **Energy Logging.** The total electrical energy consumed by the AC units during that exact hour is recorded in kilowatt-hours (kWh).
- **Normalization Ratio.** The Normalized Energy Consumption metric is derived by dividing the energy used by the thermal load:

$$Normalized\ Energy\ Cons. = \frac{Energy\ Used\ (kWh)}{CDH} \quad (14)$$

Experimental Control of Exogenous Heat Loads

While the CDH methodology provides a mathematically rigorous framework for comparing weather-dependent energy use, its standard formulation exclusively tracks ambient outdoor temperature. It does not inherently account for direct solar radiation or transient internal heat gains. To isolate the performance of the BORF controller and ensure an equivalent, unbiased comparison between the automated and human-operated scenarios, the following exogenous variables were strictly controlled within the physical testbed:

- **Occupant Heat Load.** The biological heat generated by occupants significantly impacts the AC cooling load. Therefore, occupancy levels within the controlled space were deliberately monitored, and comparative energy analyses were only conducted across timeframes possessing matching occupancy densities.
- **Solar Radiation.** Since the target DME Conference Room is an interior space situated beneath a protective roof structure, the envelope is physically shielded from direct solar insolation. This effectively nullifies disparities in radiant heat gain between testing days.
- **Internal Equipment Loads.** The introduction of any unmonitored anthropogenic heat sources or high-draw electrical equipment was strictly prohibited across all testing scenarios to prevent the artificial inflation of the room's sensible heat load.

Digital Twin Architecture

The development of the operational digital twin follows a systematic, data-driven computational framework divided into initial geometric construction and a two-phase optimization process as illustrated in Figure 39. \

The methodology initiates with the 3D geometric modeling of the physical asset, utilizing the OpenStudio plug-in to define the spatial boundaries and thermal zoning. This architectural foundation transitions directly into Phase A: an automated, closed-loop calibration process. Within this phase, a baseline Building Energy Model (BEM) is established in the EnergyPlus simulation environment in accordance with ASHRAE Guideline 14 protocols.

The simulation generates critical environmental and energy outputs in specifically zone air temperature, relative humidity, and AC electricity consumption driven by defined boundary conditions and internal control parameters. These simulated outputs are continuously benchmarked against actual measured telemetry. If the performance criteria are not met, the system triggers a Bayesian optimization sequence via Gaussian Process Regression to iteratively tune the model. This loop repeats until the virtual outputs successfully align with the physical measurements, yielding a calibrated model.

Upon achieving successful calibration, the optimized parameters are locked to initiate Phase B: the independent validation cycle. This phase rigorously tests the predictive robustness of the calibrated digital twin by exposing it to novel, unseen boundary conditions. A secondary validation run is executed in EnergyPlus, and the

new simulation outputs are evaluated against a distinct dataset of physical measurements. If the model fails to satisfy the accuracy criteria under these new conditions, the workflow mandates a reassessment of the Phase A calibration parameters. Conversely, successfully meeting the criteria confirms the structural and mathematical integrity of the model, officially establishing a fully validated digital twin capable of reliably mirroring the thermodynamic and electrical behaviors of the physical space.

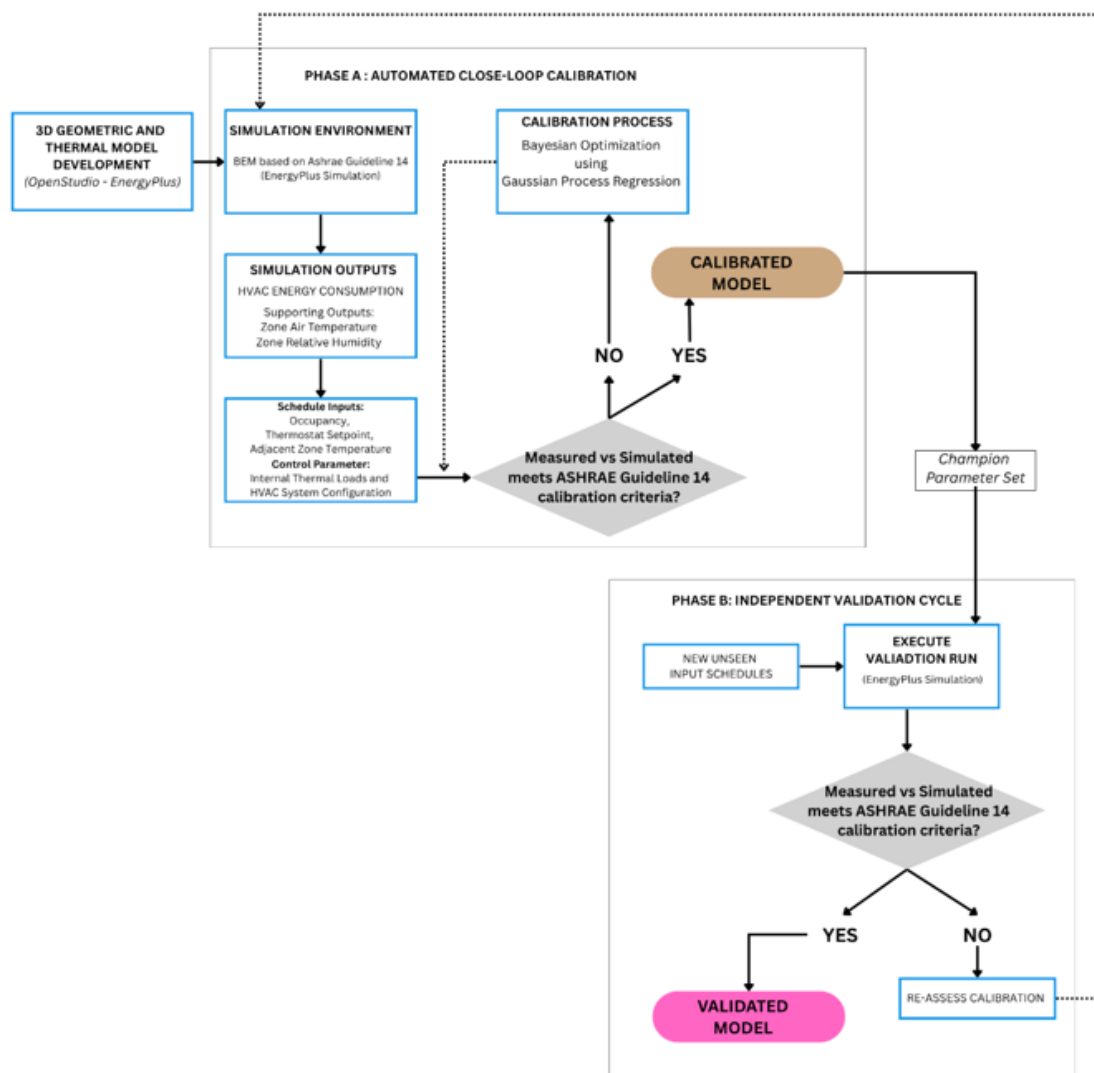


Figure 39. Data-driven Computational Framework

Site Survey and Architectural Data Acquisition

The geometric foundation of the digital twin was established by obtaining official architectural floor plans and as-built drawings of the DME Conference Room and its surrounding spaces, followed by an on-site audit to verify and supplement missing dimensions and architectural details. The validated spatial data were then modeled in OpenStudio and converted into an EnergyPlus IDF, while the building envelope properties were defined using thermo-physical parameters calculated in accordance with relevant ISO standards to accurately represent heat transfer behavior.

Acquisition of Archival Blueprints

The development of a highly calibrated digital twin fundamentally relies on the precise mathematical replication of the physical facility's spatial and geometric characteristics. The initial phase of this research involved a comprehensive architectural audit of the Department of Mechanical Engineering (DME) Conference Room and its immediate structural adjacencies within the larger College of Engineering workshop building. To establish the geometric baseline, official architectural floor plans and as-built drawings were formally procured from the university's engineering and facility management office.

These two-dimensional schematics provided the foundational blueprint, detailing the building's primary footprint, structural grid, and the macroscopic spatial relationships between the nested conference room, the adjacent drawing classroom, and the overarching workshop envelope as shown in Appendix B.

On-Site Geometric Verification and Manual Auditing

While archival blueprints provide essential baseline topography, they frequently lack the high-fidelity spatial data required for rigorous thermodynamic modeling. Standard floor plans often omit critical vertical dimensions, post-construction structural modifications, interior wall thicknesses, and precise fenestration coordinates. To resolve these data gaps and ensure absolute spatial accuracy, a meticulous on-site physical audit was conducted.

Dimensions and architectural features not explicitly covered in the archival plans were manually mapped and measured using physical surveying tools (such as laser distance meters and calibrated measuring tapes). This manual verification protocol captured the exact ceiling heights, window-to-wall ratios, door dimensions, and the precise physical placement of the split-system air conditioning units. The synthesis of the official architectural blueprints with the empirical manual measurements yielded a highly accurate, verified three-dimensional geometric profile of the facility. This verified spatial dataset served as the definitive structural ground truth, allowing for the subsequent and accurate transition into the digital modeling environment using the CAD Software and OpenStudio platforms.

Geometrical and Envelope Modeling

The geometric representation of the digital twin was developed using the OpenStudio Plug-in for CAD software. The resulting spatial model was subsequently translated into an EnergyPlus Input Data File (IDF) to facilitate rigorous dynamic thermal simulation as shown in Figures 40 and 41.

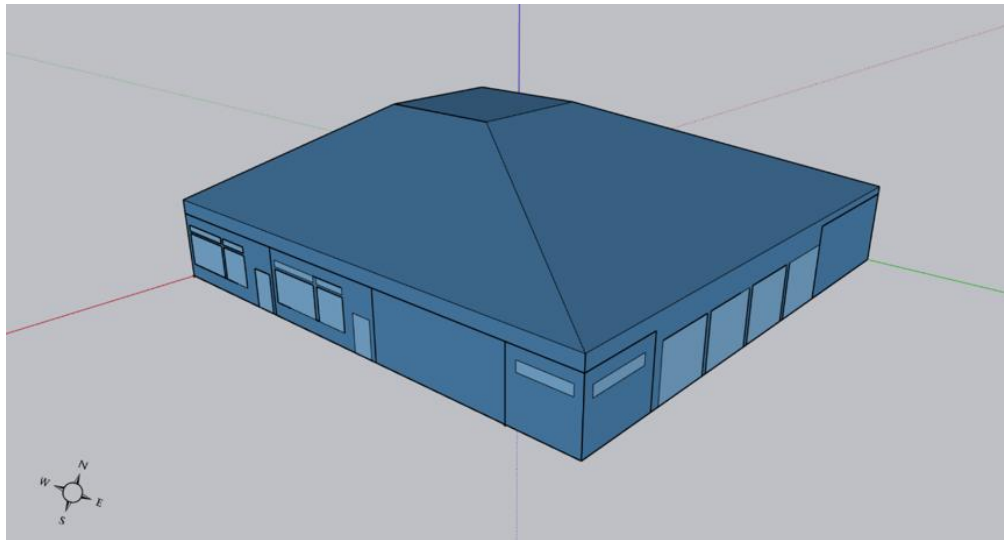


Figure 40. Boundary Condition of the Model in OpenStudio Plug-in

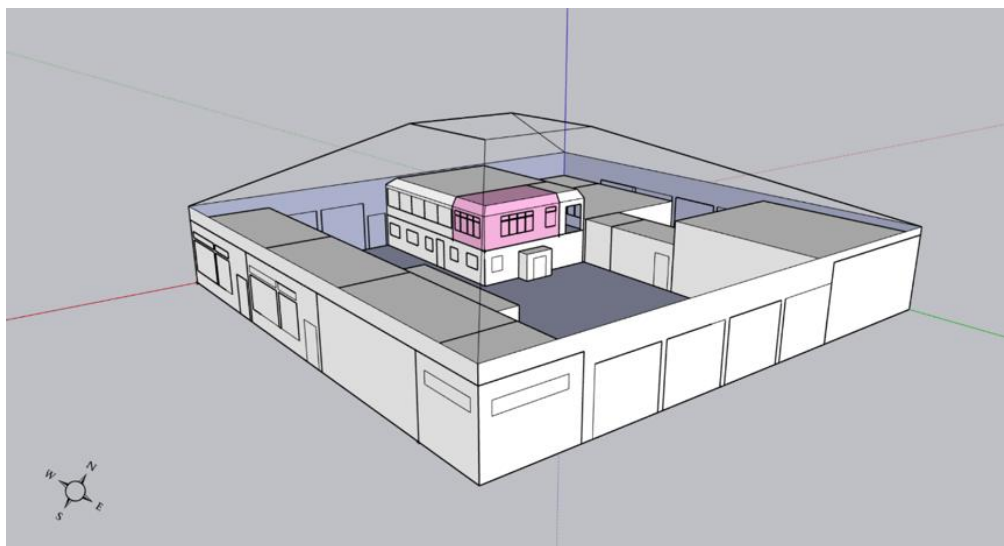


Figure 41. Thermal Zoning of the Model in Open Studio Plug-in

The building envelope constructions were defined by establishing the precise thermo-physical properties of each constituent material. To accurately capture the transient heat transfer behavior, the dynamic thermal characteristics of the building components such as conductivity, density, specific heat, thermal resistance were

calculated in accordance with the methodologies outlined in ISO 13786:2017, ISO 15099:2003, ISO 6946:2017 as shown in Table 4.

Specific interior assemblies were parameterized based on their structural composition as shown in Table 5. Interior partitions were modeled as composite assemblies comprising wood planks and an encapsulated air cavity (designated as *Confe-Int Wall*), incorporating the dedicated thermal resistance of the air space. Similarly, the ceiling structure was defined as a composite assembly of wood plank and plywood (*Confe-Int Ceiling*).

Table 4. Thermo-physical properties

Horizontal Elements	
Floor Inner surface Wood Plank_20mm Coco Lumber_105mm Air Gap Gypsum plaster_15mm Inner surface	Roof Ceiling Inner surface Wood Plank_20mm Coco Lumber_105mm Air Gap Plywood_12mm Inner surface
Vertical Elements	
Interior Window Inner surface Glass louver_6mm Airflow gap Aluminium operating rod Inner surface	Interior Wall Inner surface Air gap Plywood_12mm Coco lumber_70mm Wood plank_19mm Inner surface
	Interior Door Inner surface Fixed window glass_6mm Solid Wood_50mm Inner surface

Table 5. Structural composition of the materials

Materials	Conductivity [W/m.K]	Specific Heat [J/kg.K]	Density [kg/m³]	Resistance [m².K/W]
Plywood	0.170	1600	700	--
Gypsum	0.250	1000	900	--
Wood Plank	0.130	1600	500	--
Coco Lumber	0.180	1600	700	--
Jalousie Window	--	--	--	0.3205
Air Gap (30 cm)	--	--	--	0.1620
Air Gap (10 cm)	--	--	--	0.5490

Given that the target conference room is nested entirely within a larger enclosing facility, its physical boundaries do not interface with the exterior ambient environment. To accurately represent the steady-state and dynamic thermal interactions between the conference room and the adjacent spaces (specifically, the workshop and drawing classroom), the boundary conditions were configured utilizing the OtherSideCoefficients algorithm within EnergyPlus. Consequently, the relevant envelope surfaces were assigned NoSun and NoWind exposure parameters, strictly enforcing their status as interior boundaries and preventing the miscalculation of solar heat gain or convective wind losses.

Simulation Environment

The digital twin simulation utilized the nearest available EnergyPlus weather dataset from Ormoc City to define outdoor boundary conditions, providing essential environmental variables such as temperature, humidity, wind speed, and solar radiation

for cooling load calculations. Any discrepancies between the weather data and actual site conditions were addressed during the calibration process by tuning the model against measured indoor environmental data.

Following each EnergyPlus simulation, AC energy consumption, indoor temperature, and relative humidity outputs were extracted and temporally aligned with the measured datasets for performance evaluation. The simulated AC energy was converted to kilowatt-hours and scaled to account for differences between the physical monitoring setup and the digital twin configuration, while all datasets were synchronized to ensure accurate statistical comparison during calibration and validation.

Meteorological Data and Environmental Boundary Conditions

To accurately simulate the dynamic thermal behavior of the digital twin, site-specific meteorological data are required to establish the exterior boundary conditions. Due to the unavailability of a standardized EnergyPlus Weather (EPW) file specific to the Baybay City site, the simulation utilizes the dataset for Ormoc City (Latitude 11.067° N, Longitude 124.567° E, Elevation 25 m).

As the most geographically and climatically proximate available dataset, the Ormoc EPW file provides the essential hourly environmental variables including outdoor dry-bulb temperature, relative humidity, wind speed, and solar irradiance. These continuous transient parameters directly drive the exterior convection algorithms, configured utilizing the DOE-2 methodology, and are fundamental for the

accurate computation of dynamic cooling loads during the designated design sizing periods.

The application of proximate meteorological data is a recognized and robust practice in building performance simulation when highly localized files are absent. Furthermore, any minor microclimatic discrepancies between the Ormoc dataset and the localized Baybay site conditions was inherently mitigated during the subsequent model calibration phase. Through this process, the simulation parameters were statistically tuned against empirically measured indoor environmental data, effectively neutralizing the impact of external weather data variance.

Extraction and Temporal Alignment of Simulation Outputs

Upon the successful headless execution of each EnergyPlus simulation iteration, the resultant output data are systematically extracted and evaluated against the empirical baseline to quantify model fidelity. A dedicated Python scoring script (`12b_scoring.py`) interfaces with the EnergyPlus SQL output database (`eplusout.sql`). From this repository, the script queries 1 primary target and 2 supporting variables specific to the conditioned spatial envelope (designated as the "DME-CONFETHERMAL ZONE"): Packaged Terminal Air Conditioner (PTAC) electrical energy consumption, Zone Air Temperature, and Zone Air Relative Humidity (RH),

To ensure strict chronological parity with the measured field data, the extracted SQL temporal indices are mapped to standardized datetime objects corresponding to the empirical measurement period. The simulated timestep temperature and RH values are exported at a 5-minute resolution to match the high-frequency empirical sensor data.

Conversely, the PTAC electrical consumption is extracted, mathematically converted from Joules to Kilowatt-hours (kWh), and aggregated into hourly sums, aligning with the temporal resolution mandated by industry-standard energy calibration protocols.

Notably, a scaling factor of 0.5 is applied to the simulated AC energy array. This adjustment accounts for the physical measurement protocol, wherein the physical PZEM sensor is monitored only one of the two identical AC units, whereas the digital twin simulates the combined consumption of both. The simulated and empirical arrays are subsequently merged via a temporal inner join on the timestamp index. A secondary sequential alignment protocol serves as a programmatic fallback to guarantee data parity in the event of minor temporal index desynchronization.

Control Parameters

The digital twin utilized the nearest available EnergyPlus weather dataset from Ormoc City to define outdoor environmental boundary conditions, providing essential meteorological parameters for thermal simulation. Any differences between the weather data and actual site conditions were addressed during model calibration by tuning simulation parameters against measured indoor data.

Following each simulation run, EnergyPlus outputs for AC energy consumption, indoor temperature, and relative humidity were extracted, temporally aligned with measured data, and converted to matching time resolutions for comparison. A scaling factor was applied to simulated AC energy consumption to account for differences between the physical monitoring setup and the digital twin representation, ensuring accurate model validation.

Internal Thermal Loads

Table 6. Control Parameters

Control Parameter	Sub-Parameter	Unit
Internal Thermal Loads	Lighting	W/m ²
	Equipment	W
	Occupancy	--
HVAC System	Cooling Capacity	W
	COP	--
	Rated Air Flow Rate	m ³ /s

It is important to emphasize that the specific numerical inputs assigned to these internal loads serve primarily as baseline calibration parameters as shown in Table 6. While grounded in standard operational assumptions, they function as an adjustable starting point for the uncalibrated digital twin. The preliminary boundary conditions and heat gain parameters were established as follows:

- **Occupancy.** The spatial capacity was initially configured for a baseline maximum occupancy of 24 individuals. Sensible and latent heat dissipation rates per person were defined based on standardized metabolic rates corresponding to medium office activity levels.
- **Lighting Loads.** The initial thermal contribution from artificial illumination was modeled by assigning a baseline Lighting Power Density (LPD) of 3 W/m².
- **Equipment Loads.** To account for localized electronic devices, computers, and miscellaneous internal heat gains, a preliminary peak design electrical load of 200 W was assigned to the space.

During the subsequent model validation phase, these deterministic baseline values was iteratively tuned. Adjustments to the peak thermal capacities and their corresponding fractional schedules was systematically applied to minimize the discrepancy between the simulated internal heat gains and the empirical usage data observed within the physical space.

HVAC System Configuration

Transitioning from the standard Packaged Terminal Air Conditioner (PTAC) schematic provided by EnergyPlus as illustrated in Figure 42, the digital twin's virtual HVAC architecture was explicitly reconfigured to mimic the actual thermodynamic behavior of the physical ductless split-system.

As detailed in the system diagram, the baseline PTAC parent object was adapted to restrict fluid mass transfer to a strictly 100% closed-loop air recirculation cycle. Simulated indoor air is drawn from the zone exhaust node, conditioned by the single-speed DX Cooling Coil, and discharged directly back into the room, with all outdoor air intake parameters constrained to zero to prevent artificial mass exchange.

To simultaneously resolve the required heat rejection without over-parameterizing the model with physical outdoor fluid nodes, the DX coil is governed by an AirCooled condenser parameter. This precise reconfiguration enables the simulation engine to mathematically reject the total absorbed thermal energy inclusive of the compressor's mechanical heat penalty directly to the outdoor weather file environment, successfully isolating the indoor fluid dynamics from the outdoor heat

transfer to provide a highly accurate mechanical baseline for the subsequent Bayesian calibration phase.

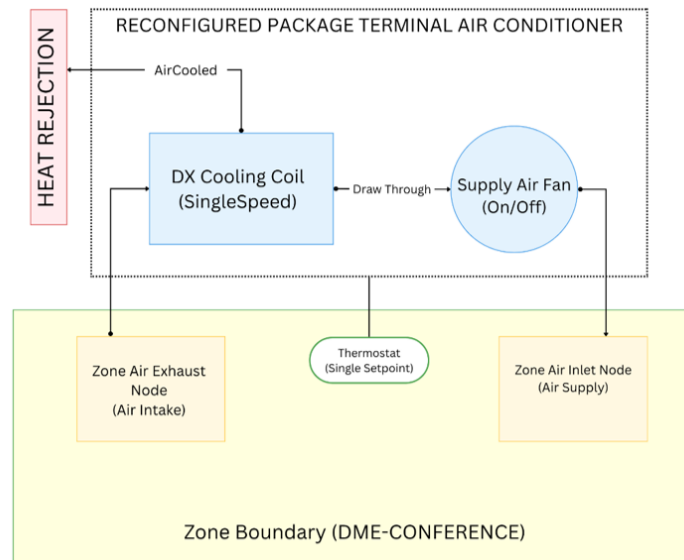


Figure 42. Reconfigured Package Terminal Air Conditioner (PTAC) in OpenStudio

Active thermal conditioning for the conference room is provided by two simulated split-system air conditioning units. Within the OpenStudio and EnergyPlus simulation architecture, these systems were represented using Packaged Terminal Air Conditioner (PTAC) objects (specifically designated as Cycling PTAC DX Clg Elec Htg and Cycling PTAC DX Clg Elec Htg 1).

Each localized HVAC node incorporates a single-speed direct expansion (DX) cooling coil. Initial simulation inputs for the gross rated cooling capacity, air flow rate and the rated Coefficient of Performance (COP) were established at 2,500 W, Auto-sizing and 3.2, respectively. It is critical to note that these specific figures do not represent static, empirical manufacturer specifications derived from the physical units. Rather, they function as primary calibration parameters.

Real-world HVAC performance frequently deviates from nominal factory specifications due to factors such as equipment age, maintenance history, and localized environmental conditions. Consequently, these initial assigned values serve strictly as an adjustable baseline. During the model validation phase, the capacity, air flow rate and COP were iteratively tuned to minimize the discrepancy between the simulated thermal behavior of the digital twin and the empirical data recorded within the actual conference room.

Operational control of the thermal environment is simulated via a cycling supply air fan mechanism, which dynamically responds to the zone cooling demands as dictated by a setpoint thermostatic control logic.

Computational Environment Setup and Empirical Data Processing

The calibration workflow was implemented in Python using Visual Studio Code, where custom scripts were developed to process empirical data, interface with EnergyPlus, and prepare datasets for model calibration and validation. The raw measurements of indoor conditions, occupancy, setpoints, workshop temperature, and AC energy consumption were standardized, synchronized, and converted into high-resolution datasets to ensure accurate comparison between simulated and measured performance.

To comply with ASHRAE Guideline 14, the processed data were aggregated into hourly intervals for statistical evaluation using RMSE, CV(RMSE), NMBE, and R^2 metrics. Additionally, the measured data were expanded into continuous year-long schedules by converting occupancy into fractional values, filling unmonitored periods

with baseline assumptions, and exporting the resulting schedules into EnergyPlus-compatible formats for simulation.

Software Environment and Dependency Configuration

To facilitate the dynamic calibration of the building energy model, a custom computational workflow was developed using Python as the core programming language. This environment, managed within Visual Studio Code (VS Code) as the primary Integrated Development Environment (IDE), was utilized to interface with the EnergyPlus simulation engine and to process the large temporal datasets generated during the study. The data processing and calibration pipelines rely fundamentally on pandas, a robust Python library specialized for high-performance time-series analysis, structural data manipulation, and automated formatting.

Empirical Data Acquisition and Pre-processing Pipeline

A critical prerequisite for calibrating the digital twin was the acquisition and quality assurance of empirical data collected from the physical Department of Mechanical Engineering (DME) Conference Room and the adjacent unconditioned workshop. Raw longitudinal data encompassing occupancy schedules, thermostat setpoints, air conditioner (AC) energy consumption, and zone environmental metrics were logged via field sensors and compiled into initial datasets.

To systematically ingest and prepare these data for algorithmic comparison against the simulation outputs, a dedicated data pre-processing script (`1_prep_act.py`)

was developed. The execution methodology involves the following sequential operations:

- **Data Ingestion and Standardization.** Raw sensory data capturing the internal conference room conditions (`actual_data_1.xlsx`) and the thermal boundary conditions within the unconditioned workshop (`workshoptemp.xlsx`) are programmatically imported. To ensure computational consistency across the calibration pipeline, target variables are isolated and their nomenclature is strictly standardized (e.g., mapping arrays to `temp_C`, `RH`, `ac_kWh`, and `workshop_temp`).
- **Time-Series Synchronization.** Because dynamic building energy simulations are inherently time-dependent, the internal conference room dataset and the external workshop thermal dataset are merged into a single, cohesive dataframe. This dataframe is synchronously keyed to a common, continuous timestamp index to eliminate any temporal misalignment between the disparate sensors.
- **Chronological Sorting and High-Resolution Output.** The merged dataset is cast into a formal datetime object and sorted chronologically to guarantee sequential integrity. The processed data is subsequently exported as a high-resolution timestep dataset (`measured_timestep.csv`). This file establishes the precise empirical boundary conditions and comparative baselines required by the EnergyPlus simulation at each individual computational time step.

Temporal Aggregation for Statistical Calibration Validation

While high-resolution timestep data are vital for driving the physical simulation equations, industry-standard calibration protocols such as ASHRAE Guideline 14 require model accuracy to be evaluated based on hourly or monthly data aggregations.

To align with this validation framework, the pre-processing script executes a temporal aggregation protocol. The timestep temporal data are resampled to the nearest hour, and the sub-hourly AC electrical consumption measurements (ac_kWh) are mathematically integrated to yield total hourly energy consumption. This aggregated array is extracted and exported as a discrete dataset (measured_hourly_ac.csv). This specific dataset serves as the empirical ground truth against which the digital twin's hourly energy predictions are statistically evaluated, utilizing globally recognized performance indices: the Root Mean Square Error (RMSE), Coefficient of Variation of the Root Mean Square Error (CV(RMSE)), Normalized Mean Bias Error (NMBE) and the Coefficient of Determination (R^2).

Annualization and Formatting of Empirical Boundary Schedules

While the empirical data collection captures the high-resolution dynamic behavior of the facility during the specific field monitoring period, the EnergyPlus simulation engine requires continuous, unbroken temporal arrays for the entire simulation year (8,760 hours). To bridge the gap between finite field measurements and the continuous annual simulation requirements, a dedicated Python script (5b_expand_data.py) was developed to expand, normalize, and format the boundary condition schedules.

The methodology for generating these annualized schedules involves the following systematic computational steps:

- **Data Alignment and Resolution Standardization.** The script ingests the previously merged environmental master dataset alongside the manually logged occupancy data. To ensure strict synchronization with the EnergyPlus simulation timesteps, all timestamps are uniformly downsampled to 5-minute intervals. The script includes localized parsing logic to accurately interpret the Filipino date format (DD/MM/YYYY) utilized by the field logging personnel.
- **Occupancy Fractionation.** EnergyPlus fundamentally calculates internal human heat gains using a fractional multiplier (ranging from 0.0 to 1.0) applied to the zone's design occupancy. Consequently, the raw headcount data is programmatically converted into a fractional schedule by dividing the measured headcount by the established design capacity of 24 persons. To ensure peak thermal loads are not underestimated during highly dynamic periods, the script applies a maximization function, retaining the highest observed occupancy count within any given 5-minute window.
- **Full-Year Temporal Expansion and Gap Imputation.** A continuous, 5-minute temporal index is mathematically generated for the entire simulation year (2026), yielding an array of 105,120 discrete calculation nodes. Because the field sensors only capture data during the actual monitoring campaign, programmatic imputation is required for the unmeasured periods. The base matrix is initialized with the following physical and operational defaults:

- **Occupancy** is set to 0.0, representing an empty, unpopulated zone during unmonitored hours.
- **Workshop Temperature Boundary** is set to a baseline of 32°C to simulate the typical daytime thermal conditions of the surrounding unconditioned envelope.
- **Thermostat Setpoint Override** is an artificially high setpoint of 40°C is assigned as the default value. This serves as a programmatic forcing function; by raising the setpoint far above normal ambient conditions, the digital twin's HVAC cooling system is effectively commanded to remain inactive during periods when the actual physical system was known to be off.
- **Empirical Data Injection and Export.** The simulated baseline matrix is subsequently merged with the actual field measurements. Wherever empirical data exists, it overrides the baseline defaults, creating a continuous, year-long schedule that perfectly mimics the actual conditions during the measurement period while maintaining logical baseline behavior throughout the remainder of the year. Finally, the isolated arrays (occupancy fractions, setpoints, and boundary temperatures) are exported as headerless, single-column CSV files. This specific structural formatting is an absolute requirement for direct integration with the EnergyPlus Schedule:File object, allowing the simulation engine to read the arrays efficiently during runtime.

Automated Parametric Mutation and Simulation Execution

The calibration framework employed automated Python scripts to dynamically modify the EnergyPlus model, inject empirical boundary conditions, and execute simulations in a headless environment, enabling thousands of iterative runs without manual intervention. Bayesian optimization was then used to calibrate six key parameters related to internal loads, occupancy, and HVAC performance within physically realistic bounds derived from field observations and manufacturer specifications.

Model performance was evaluated using ASHRAE Guideline 14 statistical metrics, including CV(RMSE), NMBE, and R^2 , for temperature, relative humidity, and AC energy consumption. A weighted composite score was utilized to identify the best-performing model, while an automated ranking system archived the champion model and its associated outputs for subsequent validation and analysis.

Programmatic IDF Manipulation via eppy

To facilitate the highly iterative nature of the Bayesian optimization process, the static EnergyPlus Input Data File (IDF) initially exported from OpenStudio is treated as a dynamic computational template. A dedicated Python execution script (`11b_run_sim.py`) leverages the eppy library robust scripting interface for EnergyPlus to programmatically parse, mutate, and overwrite the IDF for each successive simulation pass as shown in Figure 43. This programmatic approach circumvents the need for manual graphical interface interactions, thereby enabling high-throughput batch processing. During each iteration, the script retrieves a newly proposed vector of

calibration parameters (current_params.csv) generated by the Bayesian optimizer and systematically injects these dynamic values into the architectural and HVAC configuration of the building model.

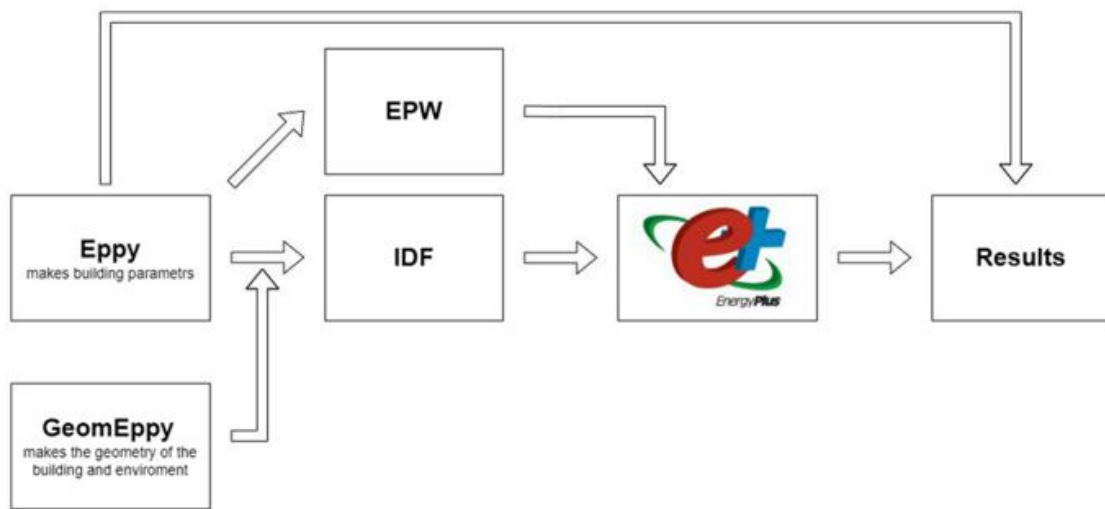


Figure 43. Programmatic IDF Manipulation via Eppy

Dynamic Boundary Condition Mapping

A critical function of the execution script is the programmatic assignment of empirical boundary conditions, ensuring the digital twin responds dynamically to the localized, pre-processed temporal data. This mapping encompasses two primary operations:

Schedule Injection. The script dynamically constructs Schedule:File objects, directly linking the high-resolution, annualized CSV arrays in specifically occupancy profiles, thermostat setpoints, and empirical workshop temperatures to the simulation engine. To guarantee that the modeled HVAC systems strictly adhere to these injected, data-driven schedules, the internal availability schedules for all PTAC units and

associated supply fans are programmatically overridden to a discrete "Always On" state.

Surface Property Allocation. The external boundaries of the simulated thermal zone are dynamically re-routed utilizing the SurfaceProperty:OtherSideCoefficients (OSC) algorithm. The primary envelope surfaces are assigned to a designated "Workshop_OSC," which forces these boundaries to experience the dynamically fluctuating, empirically measured temperatures of the adjacent physical workshop. Conversely, specific internal boundaries (e.g., the floor slab and designated interior partitions) are isolated and routed to a static boundary coefficient ("CooledRoom_OSC") to emulate their adjacency to consistently conditioned spaces. Consistent with the physical topology, all mutated envelope surfaces are restricted to NoSun and NoWind exposure parameters, accurately reflecting their nested, interior architecture.

Headless Simulation Execution

Upon completion of the parametric mutations and boundary condition mappings, the modified IDF is saved to an isolated execution directory (`current_run`). The execution script subsequently utilizes Python's subprocess module to initialize the `energyplus.exe` simulation engine in a "headless" batch mode. To optimize system memory utilization and prevent console buffer overflows across thousands of optimization iterations, the standard terminal output generated by EnergyPlus is programmatically suppressed (`capture_output=True`). However, standard error streams are selectively retained to facilitate diagnostic debugging in the event of a computational or simulation failure.

Parameterization and Definition of the Optimization Search Space

A Bayesian optimization framework was employed to calibrate the digital twin by minimizing the discrepancy between simulated and measured performance while maintaining physically realistic model behavior. To prevent unrealistic parameter combinations, the optimization process was constrained within predefined bounds established from field observations, manufacturer specifications, and expected HVAC operational characteristics.

The calibration focused on six key parameters related to internal loads, occupancy, and HVAC performance, including lighting load, equipment load, occupancy level, coefficient of performance (COP), cooling capacity, and airflow rate. These parameters were assigned realistic operating ranges to ensure that the optimized model accurately represented the physical building while capturing the effects of equipment performance, internal heat gains, and system degradation.

Algorithmic Constraints and Physical Realism

A Bayesian optimization framework was implemented to systematically minimize the error between the digital twin's simulated outputs and the empirical datasets. However, unconstrained mathematical optimization algorithms are highly susceptible to converging on local minima that, while mathematically optimal, violate the laws of thermodynamics or the operational realities of the physical building environment.

To guarantee that the final calibrated model retains strict physical authenticity, a rigidly constrained search space was established. A dedicated Python script (`script6a_para_bounds.py`), leveraging the `scikit-optimize` (`skopt`) library, was utilized to define the continuous real-number boundaries for each target calibration parameter.

Selection and Justification of Calibration Variables

Six critical parameters across the building envelope, internal loads, and HVAC system were targeted for algorithmic mutation. The upper and lower bounding constraints for each variable were established a priori, grounded in on-site empirical observations, verified manufacturer specifications, and expected mechanical degradation factors. The specific parameters and their operational justifications are defined as follows:

- **Internal Lighting and Equipment Loads (`lighting_multiplier`, `equipment_multiplier`).** Fractional scaling multipliers for artificial illumination and plug loads were strictly bounded between 0.08 and 1.0. These tight constraints restrict the optimizer to a highly specific utilization profile, ensuring the precise calculation of latent and sensible heat emissions from localized electronics (e.g., laptops, monitors) without permitting the algorithm to artificially inflate general zone loads to compensate for other thermal discrepancies.
- **Occupancy Limit (`occupancy_multiplier`).** Tightly bounded between 0.999 and 1.000. This near-static constraint functions as a rigid algorithmic ceiling,

ensuring the simulated anthropogenic heat gains never mathematically exceed the empirically logged physical headcount of the conference space.

- **HVAC Coefficient of Performance (cop).** Bounded between 2.7 and 3.2. This range intentionally restricts the cooling efficiency below idealized factory standards. It serves to mathematically represent real-world operational degradation, accounting for the elevated electricity draw of aged mechanical units operating within a challenging, high-heat external environment.
- **HVAC Cooling Capacity (cooling_capacity_multiplier).** Constrained to an exceptionally narrow variance (0.85 to 1.45). Because the foundational OpenStudio model was already populated with verified, real-world manufacturer specifications for these specific split-system units, this constraint prevents the optimizer from grossly inflating total capacity. Instead, it restricts the algorithm to micro-adjustments that account for minor localized inefficiencies, such as evaporator coil fouling.
- **HVAC Airflow Rate (fan_flow_multiplier).** Bounded between 0.50 and 0.65. This parameter scales the supply air volumetric flow rate downward to reflect realistic operational airflow restrictions, encompassing systemic pressure drops caused by particulate accumulation on air filters and general mechanical wear on the fan motors.

Search Space Initialization

Upon execution, the parameterization script compiles these defined boundaries into a structured dataframe, exporting them as a centralized configuration matrix (parameter_bounds.csv). This file serves as the definitive reference matrix for the

Bayesian optimizer, dictating the absolute limits of mathematical exploration for every subsequent iterative simulation pass throughout the calibration campaign.

Statistical Indices for Calibration and Validation

The performance of the digital twin at each iteration is quantitatively assessed using statistical indices prescribed by ASHRAE Guideline 14 for the calibration of building energy models. Specifically, the evaluation framework computes the Coefficient of Variation of the Root Mean Square Error (CV(RMSE)) and the Normalized Mean Bias Error (NMBE).

The CV(RMSE) provides a normalized measure of the model's ability to predict the overall shape and variance of the measured data across the evaluation period. It is mathematically defined as:

$$CV(RMSE) = \frac{1}{\underline{y}} \times \left[\frac{\sum (y_i - \hat{y}_i)^2}{(n-p)} \right]^{\frac{1}{2}} \times 100\% \quad (15)$$

The NMBE quantifies the overarching systematic bias, indicating the tendency of the model to overpredict or underpredict the target variable. It is defined as:

$$NMBE = \frac{1}{\underline{y}} \times \frac{\sum_{i=1}^n (y_i - \hat{y}_i)}{(n-p)} \times 100\% \quad (16)$$

Where y_i represents the empirically measured value, $\hat{y}_i$ is the corresponding simulated value, n denotes the total number of data points, $\underline{y}$ is the mean of the measured values, and p represents the number of adjustable model parameters (conservatively set to 1, per standard convention). These statistical metrics are

calculated independently for zone temperature, relative humidity, and hourly AC electrical consumption.

The Coefficient of Determination (R^2) describes how well the prediction values (i.e., the regression line) fit the reference values and is defined according to equation:

$$R^2 = 1 - \frac{\sum_{k=1}^n (r-p)^2}{\sum_{k=1}^n (r-\bar{r})^2} \quad (17)$$

The R^2 value ranges from 0 to 1. A value of 0 indicates no correlation between the predicted and reference values, while a value of 1 signifies a perfect agreement between them. In general, the closer the R^2 value is to 1, the higher the predictive accuracy. When R^2 approaches 1, the data points tend to cluster near the $y = x$ line, which represents a perfect fit. However, it is important to note that an R^2 value of 1 may still occur even when the predicted values contain a systematic bias or error.

Composite Scoring and Model Selection

To guide the subsequent Bayesian optimization algorithm toward a holistic thermal and energetic calibration rather than optimizing solely for energy consumption at the expense of indoor environmental quality accuracy a weighted composite score is calculated for each optimization iteration:

$$Score = (0.45 \times CV(RMSE)_{Temperature}) + (0.25 \times CV(RMSE)_{RH}) + (0.30 \times CV(RMSE)_{AC}) \quad (18)$$

The scoring script maintains an automated, persistent performance ledger (top10_models.csv) that tracks the historical trajectory of the optimization campaign.

For every iteration, the input parameter vector and the resulting CV(RMSE) values are logged.

If the current iteration achieves a new global minimum for the AC energy CV(RMSE) or the composite score, the script designates it as the optimal "Champion Model." Upon this designation, the script automatically isolates and archives the specific mutated IDF file (Best_Model_Top1.idf), alongside its associated high-resolution comparative diagnostic plots (Temperature, RH, and AC consumption) and raw data arrays. This automated archival mechanism ensures that the single most accurate iteration of the digital twin is definitively preserved at the conclusion of the automated optimization campaign.

Bayesian Optimization Framework and Automated Calibration Loop

The digital twin was calibrated using a Bayesian optimization framework based on Gaussian Process Regression, which efficiently explored the parameter space by learning from previous simulation results and proposing improved calibration parameters. An automated workflow was developed to iteratively modify the EnergyPlus model, execute simulations, evaluate performance metrics, and update the optimization process until the most accurate parameter set was identified.

Model acceptance was governed by the validation criteria of ASHRAE Guideline 14, requiring hourly AC energy predictions to satisfy specified CV(RMSE) and NMBE thresholds. An early stopping mechanism was implemented to automatically terminate the optimization process once these criteria were achieved,

ensuring computational efficiency while preserving a scientifically validated and calibrated digital twin.

Algorithmic Selection: Gaussian Process Minimization

Building energy models function as computationally expensive, non-linear "black-box" systems, rendering traditional brute-force grid-search or purely stochastic random-search calibration methods computationally prohibitive. To address this inefficiency, a Bayesian optimization framework was deployed utilizing the Gaussian Process Regression (gp_minimize) algorithm from the scikit-optimize Python library. This probabilistic, model-based approach is exceptionally efficient for optimizing continuous parameter spaces. It sequentially constructs a surrogate model of the objective function, dynamically balancing the mathematical exploration of uncharted parameter domains with the exploitation of empirically verified, high-performing parameter regions.

The Iterative Execution Pipeline

A central orchestration script (10c_bayesian.py) was developed to govern the automated calibration loop. The optimization campaign was constrained to a maximum of 60 iterations (n_calls), initialized with 6 randomized exploratory evaluations (n_initial_points) to successfully seed the baseline surrogate model. During each iteration, the framework executes the following autonomous sequence:

- **Parameter Proposal.** The Bayesian optimizer generates a specific parameter vector strictly bounded by the predefined continuous search space constraints.

- **Model Mutation and Execution.** The orchestration script triggers the execution module (11b_run_sim.py), which programmatically injects the proposed parameter vector into the EnergyPlus Input Data File (IDF) and initializes the headless simulation engine.
- **Data Extraction and Scoring.** Upon simulation termination, the scoring module (12b_scoring.py) extracts the resultant output data, synchronizes it temporally with the empirical baseline, and computes the defining statistical performance metrics.
- **Feedback and Learning.** The calculated composite score is returned to the Bayesian optimizer, which iteratively updates its internal Gaussian Process model to propose a theoretically superior parameter vector for the subsequent simulation pass.

ASHRAE Guideline 14 Validation and Early Stopping Criteria

The primary objective of the optimization protocol extends beyond absolute mathematical minimization; it necessitates the derivation of a scientifically rigorous model compliant with established industry standards. Consequently, the orchestration script continuously evaluates the simulated hourly Packaged Terminal Air Conditioner (PTAC) energy consumption against the stringent validation thresholds defined by ASHRAE Guideline 14: a maximum Coefficient of Variation of the Root Mean Square Error (CV(RMSE)) of $\leq \pm 30\%$, and a Normalized Mean Bias Error (NMBE) constraint of $\leq \pm 10\%$,

To optimize computational resource allocation, an algorithmic "early stopping" mechanism was integrated into the pipeline. The script persistently tracks the statistical trajectory of the highest-performing iteration. If a specific simulation pass successfully satisfies both ASHRAE statistical thresholds concurrently, the framework formally designates the model as "ASHRAE-Compliant." Upon this designation, the script immediately isolates the optimal parameter vector, archives the calibrated IDF into a finalized directory, and proactively terminates the optimization loop entirely. This mechanism ensures that computational bandwidth is not expended seeking marginal, mathematically imperceptible improvements once the digital twin has been proven to accurately mimic physical reality within recognized scientific tolerances.

Preparation of Unseen Empirical Data for Model Validation

The calibrated digital twin was validated using an independent dataset that was not included during the Bayesian optimization process, allowing the assessment of its predictive accuracy and generalization capability under unseen operating conditions.

A dedicated preprocessing workflow was developed to ingest, synchronize, and standardize measurements of AC energy consumption, occupancy, thermostat setpoints, workshop temperature, indoor temperature, and relative humidity for direct comparison with simulation outputs.

To support EnergyPlus validation simulations, the measured data were expanded into continuous annual schedules using the same methodology applied during calibration, while unmonitored periods were filled with baseline assumptions. The processed datasets were then exported as simulation boundary conditions and ground

truth reference files, enabling the computation of validation metrics including RMSE, CV(RMSE), NMBE, and R^2 for both energy and indoor environmental performance.

Purpose of Independent Data Validation

To rigorously verify the predictive robustness and generalization capability of the calibrated digital twin, the model must be evaluated against a discrete, independent dataset entirely excluded from the Bayesian optimization process. Accurate performance against this "unseen" temporal dataset empirically demonstrates that the algorithm successfully captured the underlying building physics, rather than merely overfitting to the idiosyncratic conditions of the training period. A dedicated Python pre-processing module (`15_prep_data.py`) was developed to systematically ingest, clean, and format this secondary validation dataset.

Ingestion and Temporal Synchronization of Unseen Data

The validation dataset comprises continuous field measurements encompassing Packaged Terminal Air Conditioner (PTAC) energy consumption, occupancy headcounts, thermostatic setpoints, bounding workshop temperatures, and internal conference room environmental metrics (temperature and relative humidity).

- **Data Parsing.** The raw data arrays (`actual_data_new`) are programmatically ingested, and localized timestamp strings (`DD/MM/YYYY HH:MM`) are computationally parsed into standardized datetime objects to ensure temporal consistency.

- **Resolution Alignment.** To guarantee computational synchronicity with the EnergyPlus simulation engine, all disparate sensor logs are mathematically normalized to strict 5-minute temporal intervals utilizing a mathematical floor function. In instances where multiple high-frequency readings fall within a single 5-minute epoch, specific aggregation logic is applied based on the variable's physical nature: occupancy data are maximized to capture peak anthropogenic loads, while continuous environmental parameters (setpoints, interior/exterior temperatures, and RH) are averaged to attenuate sensor noise and capture the mean thermodynamic state.

Annualization of Validation Boundary Conditions

Consistent with the initialization protocol established during the calibration phase, the EnergyPlus simulation engine necessitates continuous 8,760-hour schedules to execute the validation simulation. Consequently, the script extrapolates the finite field measurements across a complete annual 5-minute temporal index (yielding 105,120 discrete rows).

- **Imputation of Physical Defaults.** Unmonitored temporal gaps within the annual index are populated with programmatic baseline assumptions designed to enforce a thermodynamically neutral state. Occupancy fractions are defaulted to 0.0, the external workshop boundary is held at a static 32°C, and the AC thermostat setpoint is overridden to an artificially high 40°C, thereby forcing the simulated HVAC system into an inactive state when physical operational data are absent.

- **Schedule Extraction.** The empirical validation data systematically overwrites these baseline defaults during the specifically monitored periods. The finalized chronological arrays are subsequently isolated and exported as headerless, single-column CSV files (val_occupancy.csv, val_setpoint.csv, val_workshop_temp.csv) for direct programmatic injection into the model's Schedule:File objects.

Formatting Target Ground Truth Data for Scoring

Concurrently, the pre-processing script isolates and formats the empirical target variables, establishing the definitive "ground truth" against which the validation model was statistically scored.

- **Hourly AC Energy.** The high-resolution energy consumption timestamps are resampled to the nearest hour, and the sub-hourly Kilowatt-hour (kWh) values are mathematically integrated. This aggregation yields the discrete hourly baseline (val_measured_hourly_ac.csv) strictly required for the computation of the validation, RMSE, CV(RMSE), NMBE and R2 metrics.
- **Timestep Thermal Conditions.** The internal zone temperature and relative humidity arrays are cleansed of null values and exported at their native 5-minute temporal resolution (val_measured_temp_rh.csv). This high-frequency baseline facilitates the rigorous validation of the digital twin's dynamic thermodynamic and psychrometric responses.

Execution of Independent Model Validation and Performance Verification

The final validation phase assessed the calibrated digital twin using independent boundary conditions and unseen operational data to verify its predictive accuracy. The optimized model was updated with validation occupancy, setpoint, and workshop temperature schedules, executed in EnergyPlus, and configured to simulate only the target validation period to ensure direct comparability with the measured dataset.

Simulation outputs for indoor temperature, relative humidity, and AC energy consumption were extracted, processed, and statistically compared against the measured data using RMSE, CV(RMSE), NMBE, and R^2 metrics. The model was considered validated only if it satisfied the ASHRAE Guideline 14 acceptance criteria, while automated reports, comparative plots, and archived datasets were generated to document and preserve the validated performance of the digital twin.

Integration of Validation Boundaries into the Calibrated Model

The culminating phase of the methodology involves evaluating the optimized digital twin against the independent validation dataset to empirically confirm its predictive accuracy and generalization capability. A dedicated execution module (15_validation.py) was developed to automate this verification framework. The script initializes by importing the finalized, optimized Input Data File (calibrated_model.idf) derived from the Bayesian calibration loop. Leveraging the eppy library, the script programmatically supplants the training boundary conditions with the newly synthesized, unseen validation schedules (specifically occupancy profiles, thermostat setpoints, and bounding workshop temperatures).

Furthermore, the simulation RunPeriod object is dynamically restricted to simulate exclusively the target validation month of March, ensuring computational parity with the specific chronological window of the independent field measurements.

Simulation Execution and Output Extraction

Following the programmatic injection of the validation boundary conditions, the EnergyPlus simulation is executed in a headless batch environment. Upon termination, the script interfaces with the generated SQLite output database (eplusout.sql) to extract the simulated performance metrics. To maintain strict methodological consistency, the data extraction protocol explicitly mirrors the logic employed during the calibration phase:

- **Thermodynamic Data.** Zone Air Temperature and Zone Air Relative Humidity (RH) are extracted and mapped to a continuous 5-minute temporal index.
- **Energy Data.** The simulated Packaged Terminal Air Conditioner (PTAC) electrical consumption is extracted in Joules, mathematically converted to Kilowatt-hours (kWh), and programmatically scaled by a factor of 0.5. This fractional scaling is dictated by the simulation architecture, which models the aggregate spatial load across two identical PTAC units, whereas the physical PZEM validation sensor by monitoring the discrete consumption of a single unit. The scaled energy data are subsequently aggregated into hourly sums.

Statistical Verification Against Industry Standards

The extracted simulation outputs are merged with the unseen empirical ground truth datasets via their synchronized timestamp indices. The framework subsequently quantifies the model's predictive fidelity by calculating the Root Mean Square (RMSE), Normalized Mean Bias Error (NMBE), Coefficient of Variation of the Root Mean Square Error (CV(RMSE)), and the Coefficient of Determination (R2) for the hourly AC electricity consumption.

To formally validate the digital twin, these metrics are strictly evaluated against the performance thresholds established by ASHRAE Guideline 14: an NMBE within $\leq \pm 10\%$, and a CV(RMSE) not exceeding $\leq \pm 30\%$. An automated logic gate is incorporated into the algorithm, definitively flagging the model as "VALIDATED" only if it concurrently satisfies both criteria on this unseen dataset. This rigorous independent evaluation demonstrates that the model has accurately captured the underlying thermodynamic physics of the facility, avoiding mathematical overfitting to the initial training data.

Automated Reporting and Graphical Analysis

To formally document the validation outcomes, the script generates a comprehensive, automated performance report. In addition to the console-level statistical readout, the module constructs and exports high-resolution comparative time-series plots. These graphical outputs visualize the simulated versus empirical behavior for indoor temperature, relative humidity, and hourly electrical consumption. Finally, the raw comparative data arrays are permanently archived as discrete CSV files

(final_val_comparison_ac.csv and final_val_comparison_temp_rh.csv). This archival process provides a transparent, immutable record of the digital twin's verified predictive performance, intended for formal inclusion in the research documentation.

Deployment of the Operational Digital Twin for Real-Time Simulation

The live operational architecture of the calibrated and validated Digital Twin functions as a continuous closed-loop cyber-physical system, as illustrated in Figure 44. The framework originates at the physical asset, specifically the DME-CONFERENCE room and its corresponding HVAC system, where live environmental, occupancy, thermal-comfort, and operational data are continuously acquired. These measurements are transmitted through the Real-Time Data Processing Pipeline via MQTT telemetry and temporarily buffered within a thread-safe memory structure to ensure stable concurrent communication between the asynchronous sensor network and the synchronous EnergyPlus simulation engine. The incoming data are subsequently translated into simulation control commands through EnergyPlus API Actuators, allowing the Digital Twin to dynamically update its boundary conditions and mirror the current operating state of the physical environment.

Following simulation execution, the framework enters the Simulation Feedback and Analysis Pipeline. EnergyPlus API Sensors continuously extract predictive outputs from the virtual asset, including HVAC energy consumption, indoor air temperature, relative humidity, and thermal comfort conditions represented by the Fanger Predicted Mean Vote (PMV) index. These outputs are processed through a Pandas-based data aggregation engine and synchronized with their corresponding measured values

obtained from the physical environment. The resulting dataset is stored as a continuous operational record and simultaneously streamed to an interactive Streamlit-based dashboard.

The operational dashboard provides real-time visualization of the Digital Twin by presenting measured and simulated HVAC energy consumption, indoor temperature, relative humidity, and PMV, together with a three-dimensional thermal representation of the conditioned space. Through this interface, the physical conference room and its virtual counterpart operate in parallel, enabling continuous comparison between observed and simulated performance. This virtual parallel assessment framework allows operators to evaluate both energy performance and thermal comfort conditions in real time while maintaining synchronization between the physical and digital domains.

By integrating real-time telemetry, dynamic EnergyPlus co-simulation, thermal-comfort assessment, and live visualization within a single operational framework, the developed Digital Twin extends beyond conventional building monitoring systems and functions as an active virtual representation of the target building. The continuous exchange of information between the physical asset and the simulation environment enables ongoing performance assessment, HVAC energy quantification, and comfort validation during live operation.

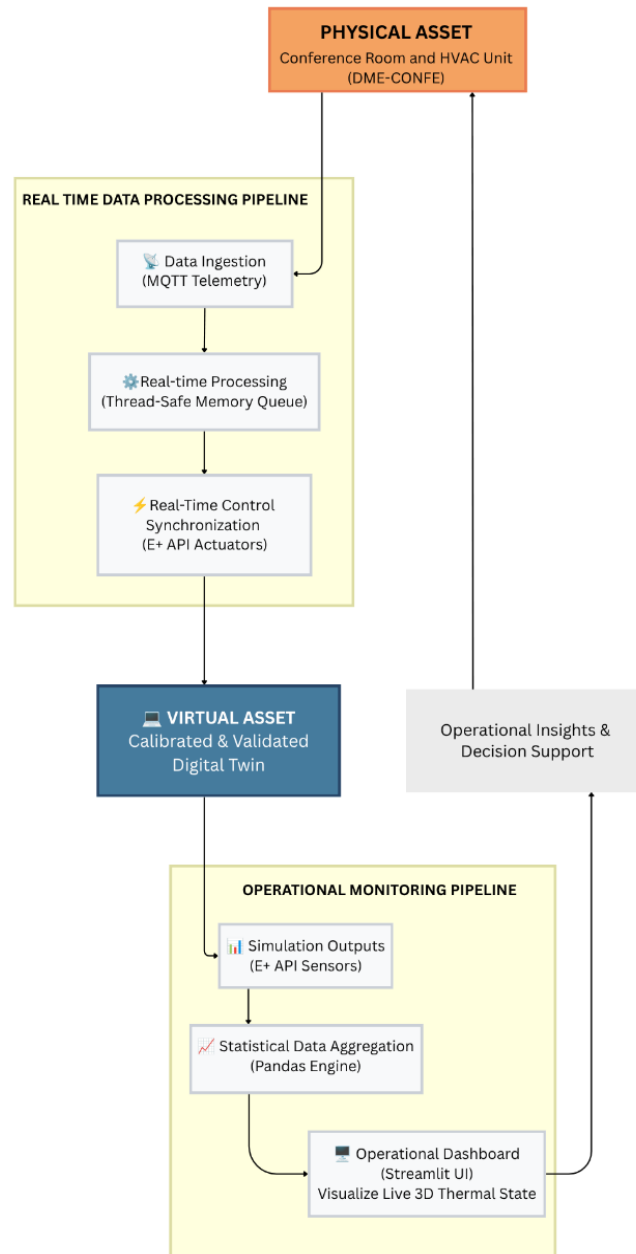


Figure 44. Operational Digital Twin for Real-Time Simulation Framework

Architecture of the Co-Simulation Framework and Session Archival

The transition from a static, historically calibrated building energy model to a live, operational digital twin marks the most critical phase of the methodology. To achieve real-time, synchronous simulation that identically mirrors the physical facility,

a dynamic co-simulation framework was engineered utilizing the EnergyPlus Python Application Programming Interface (`pyenergyplus.api`). Traditional EnergyPlus workflows rely on static batch processing, which inherently precludes real-time intervention. The implementation of the Python API fundamentally alters this architecture, converting the simulation engine into an interactive, state-driven computational environment.

A master orchestration script was developed to instantiate a localized memory state (`api.state_manager.new_state()`) and register dedicated callback functions (specifically `callback_begin_zone_timestep_after_init_heat_balance`). To guarantee strict data provenance and immutable record-keeping during live deployments, a dynamic session management architecture was engineered. Upon initialization, the script automatically captures the localized system time to generate a unique, chronologically indexed directory (`SESSION_ID`). This dedicated namespace functions as the centralized repository for all subsequent runtime memory states, high-frequency telemetry logs, and final automated diagnostic reports, ensuring every operational deployment is distinctly archived.

Asynchronous IoT Telemetry and Concurrent State Management

Feeding the computational twin with real-time environmental data required the deployment of a highly responsive Internet of Things (IoT) telemetry pipeline. This pipeline was established utilizing the MQTT messaging protocol (`paho.mqtt.client`). Operating as an active subscriber on the local network, the simulation framework continuously monitors an expanded array of high-frequency data streams originating

from distributed microcontroller nodes within the physical facility. The captured variables encompass the unconditioned workshop environment, bi-nodal internal spatial temperatures and relative humidity, real-time occupancy headcounts, dynamic thermostat setpoints, and live electrical power draws from the physical HVAC units.

The asynchronous nature of physical sensor data arriving continuously and unpredictably via MQTT callbacks presents a fundamental structural conflict with the synchronous, discrete-timestep calculation engine of EnergyPlus. To resolve this computational bottleneck, a rigorous concurrent state management protocol was deployed.

A centralized memory dictionary (`live_sensor_data`) was instantiated alongside a strict multi-threading lock (`threading.Lock()`). This concurrency control mechanism safely isolates the shared memory space, preventing race conditions or data corruption when the asynchronous MQTT receiver thread writes incoming data at the exact millisecond the synchronous EnergyPlus thread attempts to read it.

Pre-Run State Synchronization and the Eradication of Data Starvation

An inherent vulnerability in real-time co-simulation occurs when the mathematical engine initializes prior to the IoT network successfully delivering the primary empirical boundary conditions. To eliminate this risk of "data starvation," a rigorous Pre-Run State Synchronization algorithm was implemented.

Upon execution, the script enters a suspended polling state, actively pausing all EnergyPlus initialization protocols until the multi-threading dictionary mathematically

verifies the receipt of all requisite physical parameters. Once the physical baseline is established, the script utilizes the eppy Python library to physically parse the foundational Input Data File (IDF). It dynamically overwrites the default Schedule:Constant objects in specifically the initial workshop temperature, room temperature, and default equipment availability with the precise empirical values captured at physical hour zero. This algorithmic intervention ensures the digital twin is perfectly synchronized with the physical reality of the facility before the core calculation engine is invoked.

Thermodynamic Warmup Stabilization and Boundary Control

A profound challenge in live building performance simulation is managing the computational "warmup" initialization. Before generating valid real-time predictions, the EnergyPlus engine must simulate several preliminary calculation days to mathematically establish the deep thermal capacitance of the structural envelope.

Allowing the injection of live, real-time sensor data during this deep historical warmup phase would catastrophically skew the thermodynamic gradient of the virtual building mass. To circumvent this thermodynamic distortion, the methodology leverages the API's `warmup_flag` function to intelligently pre-condition the structural mass. While the engine actively processes the historical warmup algorithms, the master script intentionally suppresses external IoT telemetry. Instead, it extracts the exact, empirically measured internal spatial averages captured during the Pre-Run Sync phase and forces the simulated exterior weather boundaries to strictly mirror these internal states.

This mathematical manipulation locks the building mass into a state of thermodynamic equilibrium corresponding precisely to the actual physical room conditions.

Fractional Load Transformation and Dynamic HVAC Translation

During the active simulation phase, physical telemetry vectors are injected directly into the core calculations utilizing `get_actuator_handle` methods. To align with the inherent mathematical logic of the EnergyPlus calculation engine, raw empirical data must be algorithmically translated prior to injection. The translation logic executes the following discrete operations at every timestep:

- **Occupancy Normalization.** Physical occupancy is registered by field sensors as discrete integer headcounts. However, EnergyPlus requires internal loads to be defined fractionally. A normalization algorithm processes the raw headcount against the facility's verified design capacity of 24 individuals, bounding this quotient to a strict fractional spectrum between 0.0 and 1.0. This ensures anthropogenic heat gains scale accurately without mathematically exceeding established thermal parameters.
- **Deactivation Override.** If the physical MQTT sensor registers the AC state as "OFF," the script intervenes and assigns an artificial 40.0°C cooling setpoint. This mathematical maneuver thermodynamically forces the simulated cooling coils into a dormant state without requiring complex, mid-simulation structural alterations to the equipment availability schedules.

- **Latent Load Processing.** If the physical system operates in "DRY" mode, the algorithm computationally depresses the targeted setpoint by exactly 2.0°C. This forces the simulated cooling coil to artificially increase its run-time fraction, thereby successfully emulating the enhanced latent cooling and psychrometric dehumidification inherent to the physical unit's dry mode.

Strict Temporal Pacing and Bivariate Data Extraction

To function as a true operational digital twin, the mathematical engine must be strictly paced to advance synchronously with the physical progression of time. Left unconstrained, the EnergyPlus core processed an entire year of calculations in minutes, rapidly decoupling from the physical timeline.

A rigorous temporal throttling algorithm was implemented to anchor the 1-minute simulated timestep directly to the real-world atomic clock. At the conclusion of every calculation step, the script calculates the temporal delta between the scheduled simulation target and the current physical hardware time (`datetime.now()`). If the calculation finishes prematurely, the script triggers a programmatic suspension protocol (`time.sleep()`), physically halting the execution thread until the timeline aligns.

Concurrently, at the exact moment of temporal alignment, the API extracts the immediate predictive outputs: internal zone temperature, relative humidity, and combined electrical power draw. The physical AC power data, extracted natively in Watts, is mathematically integrated into Kilowatt-hours per minute (kWh/min) to exactly match the digital twin's output format. Both vector simulated predictions and

empirical reality are subsequently appended to a continuous time-series log (live_data.csv).

Automated Interruption and Diagnostic Performance Reporting

The operational framework is engineered for continuous deployment but incorporates a robust interrupt protocol to facilitate controlled termination. Upon the detection of a user-initiated system interrupt, the algorithm executes a graceful shutdown sequence that safely terminates the asynchronous MQTT loops and disconnects from the telemetry broker.

To definitively validate the real-time accuracy of the session, an automated diagnostic reporting phase is triggered (generate_accuracy_report). Utilizing the pandas and numpy libraries, the script aggregates the synchronized empirical and simulated arrays to compute critical industry-standard performance metrics, specifically the Root Mean Square Error (RMSE), the Coefficient of Variation of the Root Mean Square Error (CV(RMSE)), the Normalized Mean Bias Error (NMBE), and the Coefficient of Determination (R2). To augment this quantitative analysis, the algorithm applies a 15-minute moving average filter to the raw temporal arrays, mathematically smoothing transient sensor noise to construct high-resolution comparative time-series plots and bivariate scatter plots.

Development of the Real-Time Graphical User Interface and Visualization Dashboard

The operational digital twin dashboard was developed using Streamlit to enable real-time visualization and comparison of measured and simulated building performance. The platform continuously synchronizes with live simulation sessions, acquires sensor and digital twin data, and presents key operational indicators including occupancy, HVAC operation, indoor environmental conditions, energy consumption, and thermal comfort metrics.

A three-dimensional representation of the conference room was integrated to visually communicate the simulated thermal state of the space through dynamic color mapping. Combined with real-time comparative metrics and interactive time-series visualizations, the dashboard provides continuous monitoring of digital twin accuracy and supports operational assessment during live deployment.

Architectural Framework and Dynamic Session Synchronization

The successful deployment of the operational digital twin required a centralized, high-performance Graphical User Interface (GUI) to monitor, visualize, and statistically compare the high-frequency thermodynamic data streams. The Streamlit Python framework was utilized to construct this interactive, web-based dashboard. To ensure sustained visibility of the dynamic calculations without requiring manual browser intervention, the architecture employs an infinite while True polling loop encapsulated within a dynamic Streamlit placeholder object (`st.empty()`).

A critical engineering enhancement to this visualization framework is the implementation of an autonomous session-tracking algorithm (`get_latest_session`). Rather than reading from a static, hardcoded directory path, the script utilizes the Python (`os`) module to actively scan the primary localized data repository (`live_data`). It computationally evaluates the directory modification timestamps (`os.path.getmtime`) to isolate and programmatically bind to the most recently generated session folder. If the primary simulation engine has not yet initialized a session, the dashboard gracefully enters a suspended waiting state, averting systemic file-not-found errors and ensuring flawless computational synchronization the exact moment the EnergyPlus engine boots. At the conclusion of each rendering cycle, the execution thread is programmatically suspended for precisely two seconds (`time.sleep(2)`), effectively throttling the dashboard's refresh rate to optimize local central processing unit (CPU) utilization while aligning with the asynchronous updates of the primary simulation.

Asynchronous Data Ingestion and Operational Shift Aggregation

Maintaining strict comparative synchronization with the physical facility requires the visualization algorithm to continuously ingest both the immediate empirical inputs and the corresponding simulated outputs. The script programmatically interfaces with the active session directory via two custom, fault-tolerant fetching functions:

- **Instantaneous Memory State (`fetch_instant_memory`).** Extracts the immediate physical sensor states specifically the unconditioned workshop temperature, workshop relative humidity (RH), active HVAC setpoint, physical

occupancy headcount, and AC operational mode from the `live_memory.json` file.

- **Continuous Predictive Archive (`fetch_live_data`).** Loads the comprehensive high-resolution CSV log (`live_data.csv`) into a dynamic pandas DataFrame.

Upon successful ingestion of the CSV array, the algorithm executes rigorous time-series data manipulation. The raw temporal strings are mathematically cast into formally recognized datetime objects. To provide scientifically relevant context to the real-time metrics, the script employs boolean masking to isolate data generated exclusively during the facility's active standard operating shift (strictly defined as 08:00 and < 18:00). Within this active temporal mask, the algorithm computes the cumulative simulated electrical consumption (`shift_sim_kwh`) against the actual measured consumption (`shift_act_kwh`). If the dashboard is accessed outside of this operational window, an automated logic gate assigns the facility an "Off-Shift" status and zeroes the active calculations, neutralizing dormant background data to prevent the mathematical distortion of daily operational efficiency metrics.

Programmatic Geometry Parsing and 3D Colorimetric Thermal Mapping

A foundational element of the visualization framework is the dynamic, three-dimensional geometric representation of the physical conference room. To circumvent the severe computational latency typical of commercial 3D rendering libraries, a highly optimized custom geometry parser (`load_obj`) was engineered to natively interpret a localized Wavefront object file (`thesis_room.obj`). To further minimize computational overhead during the rapid dashboard refresh cycles, this extraction function is wrapped

in a framework caching decorator (`@st.cache_data`), ensuring the static spatial geometry is loaded into system memory only once upon initial execution.

The extraction algorithm systematically parses the raw text structure of the object file. It isolates Cartesian coordinate vertices (`v`) and polygonal faces (`f`), cross-referencing these arrays against embedded material tags (`usemtl`) to mathematically dissect the mesh into discrete physical categories: fenestration arrays (windows), ingress points (doors), and opaque envelope boundaries. Crucially, the script includes a mathematical triangulation sequence; it automatically decomposes complex polygons (faces defined by more than three vertices) into standard triangles, guaranteeing an unbroken, watertight 3D mesh compatible with the rendering engine.

Utilizing the `plotly.graph_objects` library, these coordinates are reconstructed into interactive 3D structures (`Mesh3d`). To provide intuitive spatial analytics, the opaque envelope surfaces are dynamically bound to a custom colorimetric mapping algorithm (`get_thermal_color`). This function evaluates the instantaneously simulated zone temperature (`sim_temp_C`) and translates it into a specific hexadecimal color code along a rigid thermodynamic gradient:

- Blue (`#0055FF`): Deep sub-cooling, temperatures $\leq 22.0^{\circ}\text{C}$.
- Cyan (`#00FFFF`): Standard cooling, temperatures $\leq 24.0^{\circ}\text{C}$.
- Green (`#00FF55`): Nominal setpoint conditions $\leq 26.0^{\circ}\text{C}$.
- Yellow (`#FFDD00`): Elevated thermal conditions $\leq 28.0^{\circ}\text{C}$.
- Red (`#FF0000`): Critical zone overheating $> 28.0^{\circ}\text{C}$.

To enhance visual fidelity, specific architectural elements are assigned distinct rendering parameters: window arrays are rendered with high transparency (opacity=0.3) and extreme specular reflection (specular=1.0), while solid boundaries maintain diffuse lighting and high opacity.

Comparative Interface Topology and Real-Time Delta Tracking

The graphical user interface was structured to provide continuous monitoring of both the physical conference room and its corresponding Digital Twin without inducing cognitive overload. The dashboard architecture was organized into hierarchical visualization layers consisting of operational boundary conditions, Digital Twin performance indicators, three-dimensional thermal visualization, and detailed comparative analytics.

The upper tier displays the real-time physical boundary conditions received through the MQTT telemetry network, including occupancy, HVAC operating mode, thermostat setpoint, workshop temperature, and workshop relative humidity. These variables represent the primary boundary conditions governing the operation of the Digital Twin simulation environment.

Immediately below, the dashboard presents side-by-side comparisons between measured and simulated performance indicators. These include HVAC energy consumption, indoor air temperature, indoor relative humidity, and Predicted Mean Vote (PMV). Each metric is displayed using Streamlit's real-time monitoring components, enabling continuous assessment of the agreement between physical measurements and Digital Twin predictions.

By presenting both measured and simulated values simultaneously, the dashboard provides immediate visibility into the operational accuracy of the virtual asset during deployment.

In addition to numerical performance indicators, the interface incorporates a three-dimensional thermal representation of the conference room generated from the virtual building geometry. The visualization dynamically reflects the current simulated thermal state of the Digital Twin, providing an intuitive representation of the operational condition of the virtual environment.

High-Fidelity Time-Series Analytics and Visual Aggregation

The lower tier features an interactive, multi-tab layout dedicated to granular time-series analytics, utilizing the Plotly rendering engine for high-fidelity visualization. The script algorithmically truncates the overarching DataFrame, isolating strictly the most recent 60 temporal intervals (representing exactly one hour of 1-minute resolution data). This truncated array drives three specific comparative charts:

- **Thermodynamic Trajectories (Temp & RH).** Bivariate line charts utilizing go.Scatter traces. These plots strictly map the simulated digital twin metrics (rendered as dashed lines to denote virtual origin) against the actual measured physical metrics (rendered as solid, heavily weighted lines to denote empirical reality).
- **Thermal Comfort Assessment (PMV).** Comparative visualization of measured and simulated Predicted Mean Vote (PMV) values. The measured PMV is obtained from the physical environment, while the simulated PMV is

extracted directly from the EnergyPlus thermal-comfort calculations utilizing the Fanger comfort model. This module enables real-time virtual parallel assessment of occupant thermal comfort conditions and extends the Digital Twin beyond energy-performance monitoring alone.

- **Hourly Energy Consumption (AC kWh).** The granular 1-minute electrical consumption arrays for both the actual physical units and the simulated virtual units are mathematically grouped by their truncated hour index (`hour_group`) and integrated into discrete hourly sums. These summed values populate a comparative line-and-marker chart.

This rigorous visual aggregation protocol ensures that the real-time energy visualization remains strictly comparable to the industry-standard hourly performance metrics utilized during the baseline ASHRAE model calibration and validation phases.

Distributed IoT Mechatronic Controller

An IoT-enabled hierarchical distributed mechatronic controller was developed using microcontroller-based node units equipped with temperature sensors, IR transmitters (for interfacing with IR-controlled air-conditioning units), and Wi-Fi connectivity.

Each node communicated through the MQTT protocol via a local Mosquitto broker, allowing for localized data processing while still permitting supervisory monitoring from a central host. This reflects a hierarchical distributed architecture wherein lower-level nodes handle real-time actuation and data collection while a

higher-level interface only performs oversight and data aggregation rather than direct control.

Since the entire architecture of the system is distributed in nature, the system can be broken down into several component groups, which would communicate with each other through the MQTT broker. These groups were labeled as: Main Hub, Dynamic Inside Sensor, Static Inside Sensor, Outside Sensor, ACU Power Monitoring Node 1, and ACU Monitoring Node 2. The corresponding schematic diagram and functions of each node were detailed in their subsection.

Main Hub

The main hub of the distributed system houses the main controller, which is the Raspberry Pi 5. It handles all major tasks: run the ML/AI algorithm, be the MQTT broker, handle image recognition from RPi Camera using Yolov8n, and collect/process system metrics. The Wi-Fi Access Point is connected wirelessly to Raspberry Pi 5 for portability. The RPi Camera Module is connected through the Pi's MIPI CSI port to minimize traffic in the Wi-Fi network since images can take more milliseconds to upload and read.

To monitor the power consumption of the brain, a python script will estimate the power draw based on the temperature of the CPU cores, calibrated using a USB Inline Power Meter. The main hub also used an external RTC (real time clock) DS3231 to sync time better, necessary for time-based analysis. This assembly is summarized in the schematic diagram below.

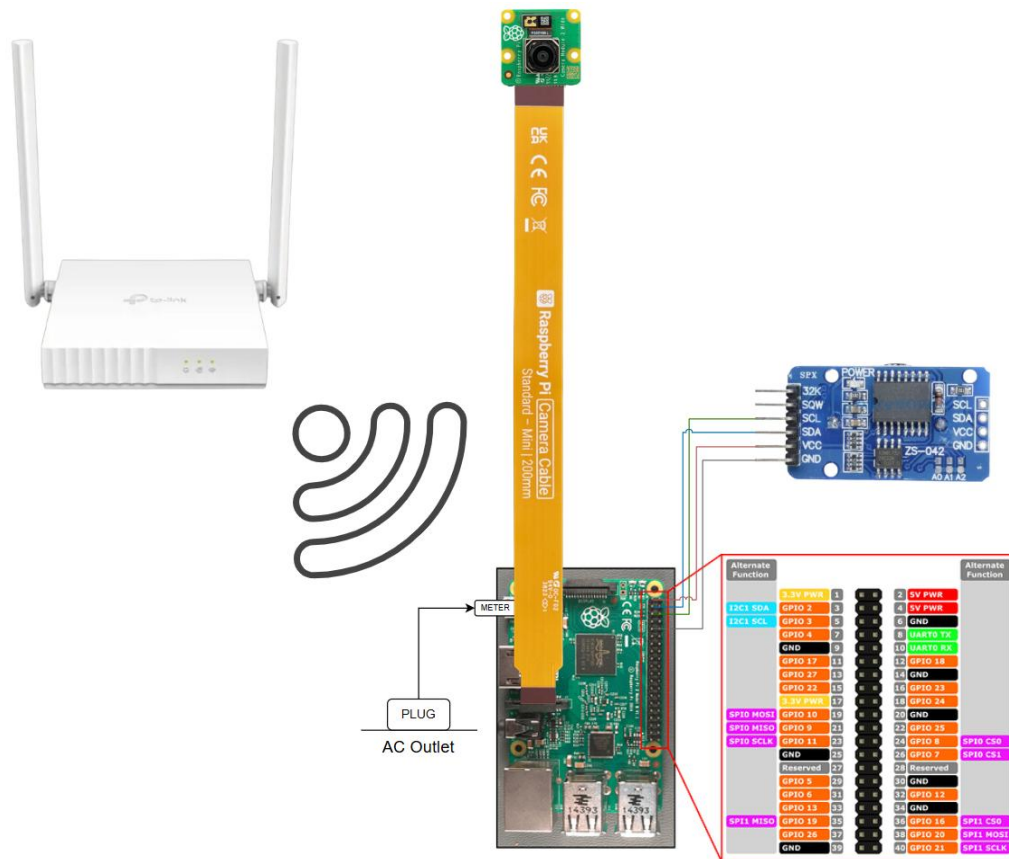


Figure 45. Main Hub Schematic Diagram

This assembly—with only the camera being visible from outside—is then protected by a 3D-printed enclosure. The enclosure is made of PLA (Polylactic Acid) Lite filament from Bambu Lab. The selected PLA Lite filament exhibits tensile strength values comparable to the lower range of common ABS materials used in electronic housings and protective enclosures (Jairaj, 2025). Considering that the developed system primarily supports lightweight electronic components and operates under low mechanical loading conditions, the material was determined to provide sufficient structural integrity for the intended application.

Additionally, PLA Lite offers advantages in cost efficiency, ease of fabrication, dimensional accuracy, and rapid prototyping capability, making it suitable for iterative

enclosure development using additive manufacturing (Smith3D, 2024). The same filament material was also used on other node's enclosures as well.

Per design choices, the enclosure is designed to be wall mounted, all the while making the camera adjustable with 3 degrees of freedom (DOF) to enable flexible positioning and orientation. The first DOF is provided by a linear sliding platform that allows front-back translational adjustment of the enclosure. The second DOF is a rotating base beneath the enclosure which enables horizontal yaw rotation. The third DOF is located at the camera module mount, allowing vertical pitch adjustment of the camera orientation. The assembly as shown in a 3D modeling software is shown in Figure 46, along with the actual node.

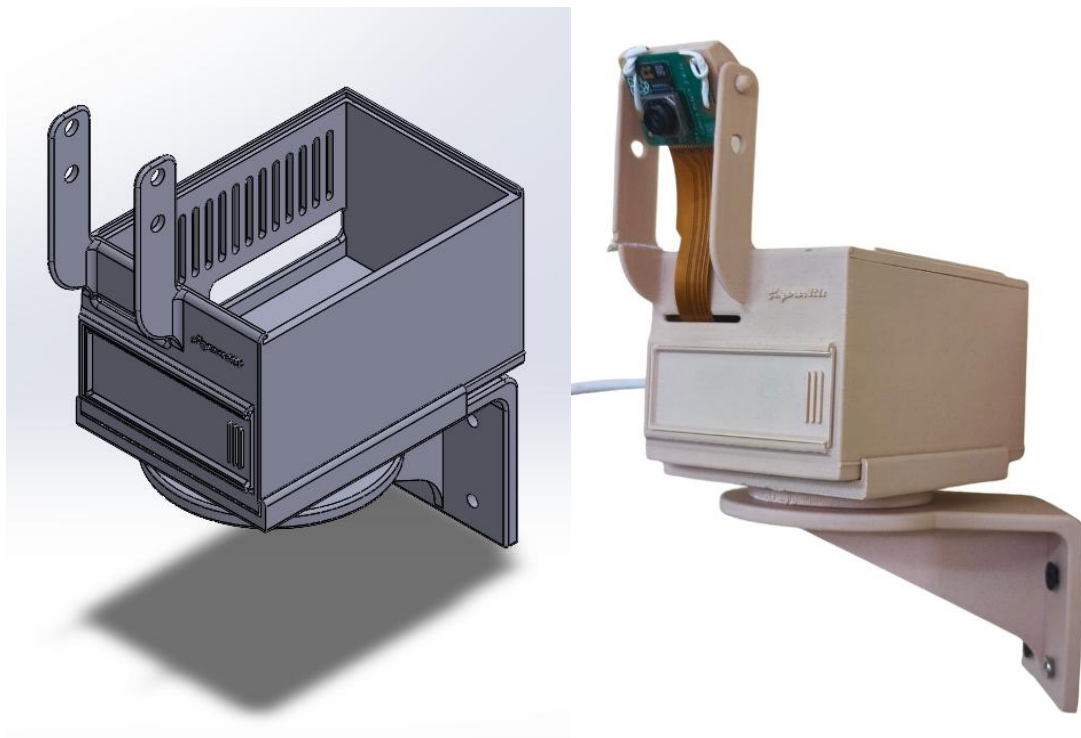


Figure 46. Main Hub Assembly, Digital (left) and Actual (right)

ESP32 Nodes

There are five (5) ESP32 managerial nodes tasked to gather regularly recorded data from their respective sensors. The nodes gathered readings from their sensors every 30 seconds and publish them to the MQTT broker as they are read, immediately. Three of the nodes will gather temperature and relative humidity data, and the other two will gather ACU power consumption data.

Each node also has their own power management system, composed of a battery, a supercapacitor, and switches. This subsystem is arranged so that the node will still run for a brief period during battery replacement. This is done by placing a battery parallel with a supercapacitor, putting one switch between the two to temporarily disconnect the battery to the circuit before replacement, and another switch after the supercapacitor which serves as the main power switch.

The capacity of the battery will dictate how long the node can run without battery swaps, while the supercapacitor's farads will dictate how much the node will last during battery swaps. The battery is rated at 3350 mAh, with a rated voltage of 3.7V (maximum of 4.2V), while the supercapacitor is rated at 5.5V maximum with a capacitance of 15F. These components were selected based on system compatibility and their availability during procurement, with the chosen capacities being among the highest readily available at the time of purchase.

Additionally, a DC power monitor is also present to monitor the battery health, the INA226 module. This module can monitor the bus supply voltage (VBus), and the voltage drop via a shunt resistor which is in turn used to measure the current draw and

power used by the node (Texas Instruments Inc., 2024). The local ESP32 then gathers the voltage readings every 30 seconds and processes them for a LED (light-emitting diode) to display.

The collected data are subjected to weighted moving averaging, which takes the last nine voltage readings, averages them, and gives it 90% of the weight, while the newest reading is given 10%. This way, the filtered average will transition smoothly from a value to the next, preventing the LED to flicker during state transitions.

The LED states are defined by filtered average ranges: green (fully charged) if between 4.2V and 3.9V, yellow (normal) if between 3.9V and 3.6V, orange (low battery) if between 3.6V and 3.4V, and red (shutdown soon) if below 3.4V.

For easier identification, each of the five ESP32's has their own tags and labels. ESP1 which handles gathering air quality data inside the room is named 'Dynamic Inside Sensor.' ESP2 does the same purpose as ESP1 but will also receive commands from the Main Hub to control the ACU via infrared signals, called 'Static Inside Sensor.' ESP3 also does air quality data gathering but unlike the previous sensors, it is positioned outside the room, garnering the label 'Outside Sensor.' ESP4 and ESP5 both monitor the power consumption of the air-conditioning units, with each assigned to one unit. Since ESP4 monitors the Samsung ACU, and the ESP5 monitors the General Royal ACU, the two are called 'ACU Power Monitoring Node (Samsung)' and 'ACU Power Monitoring Node (General Royal),' respectively.

Dynamic Inside Sensor and Outside Sensor

These two sensors are built similarly but will record two different room states. The former will record the humidity and temperature inside the room, and the latter will record the weather state just outside the room.

Both nodes handle real time data collected from the sensor SHT31. The SHT31 is a temperature and relative humidity sensor, needed for the algorithm to monitor air quality conditions (Sensirion, 2022).

An automatic boost-buck converter was used to make sure that ESP32 sees constant voltage despite the variable voltage from the battery, TPS63070 (Texas Instruments Inc., 2019). An RGB LED was also used, serving as a signal which will indicate the node status. The LED will alternate between blue and the battery status color to indicate something is wrong with the node. The entire schematic diagram is shown in Figure 47.

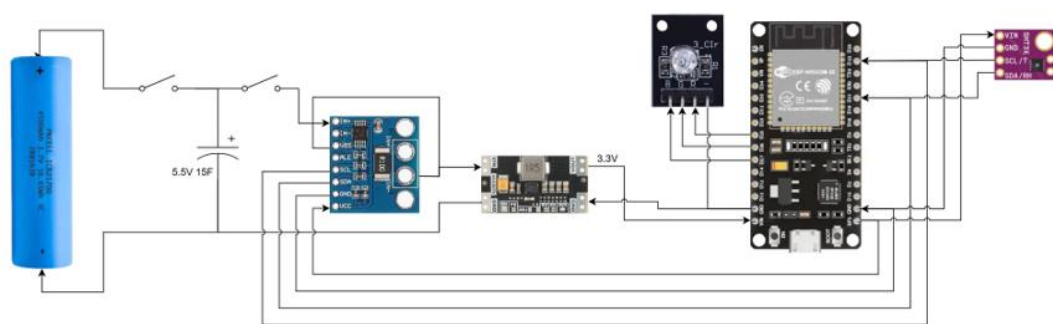


Figure 47. Temperature Sensors Schematic Diagram

To avoid loose wirings and to reduce voltage losses, the connections are reinforced using a printed circuit board (PCB), also shown in Figure 48. The full schematic of which is in Appendix A.

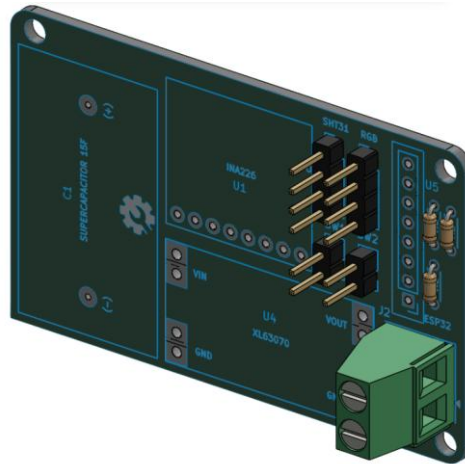


Figure 48. Temperature Sensors, PCB Design

From its name, ‘Dynamic Inside Sensor’ meant that it is stationary. To better reflect the actual sensation the occupants are feeling, this sensor should be near them. As such, this sensor will be the only one without a mount. But the node will still need an enclosure to protect the electronics inside. The enclosure is also made from PLA Lite, with the CAD model and actual node displayed below.

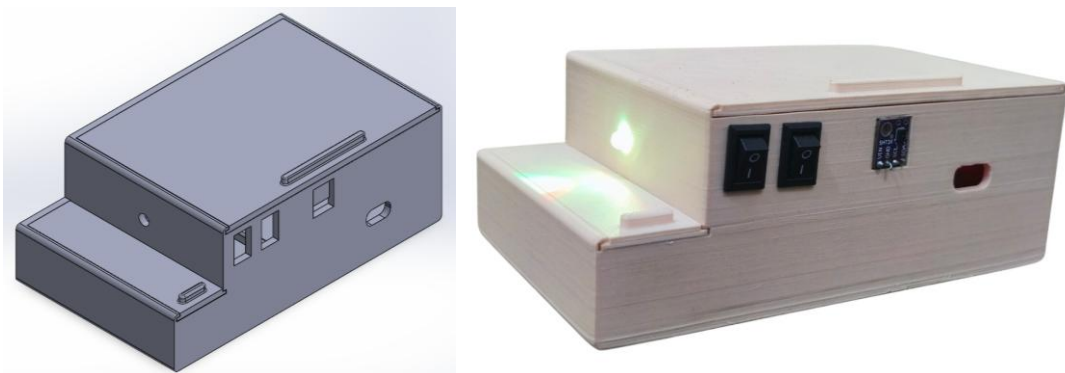


Figure 49. Dynamic Inside Sensor, Digital (left), Actual (right)

The ‘Outside Sensor,’ being of the same purpose as the ‘Dynamic Outside Sensor,’ only differing in deployment location, shares the same enclosure appearance

with the latter. However, since the node must be fixed in terms of location outside for consistency, the node needed to be wall mounted, thus requiring a mount. The CAD model and actual node is displayed below.

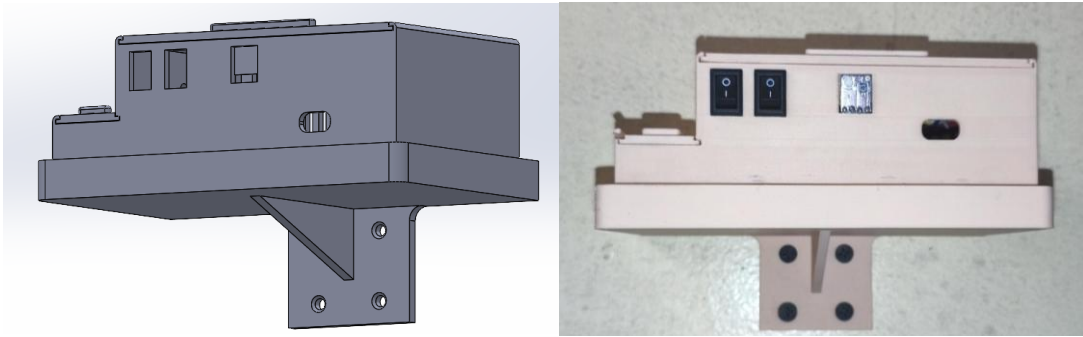


Figure 50. Outside Sensor, Digital (left), Actual (right)

Static Inside Sensor

This node serves two purposes, inside temperature and relative humidity sensing and as the air-conditioning unit actuator. Comparably, this node has the same components and arrangement as the previous node, but with additional components to accommodate the extra function, as an actuator.

The actuator side of the sensor utilizes two infrared (IR) LEDs to transmit infrared signals for controlling the two air-conditioning units. To drive the IR LEDs reliably, an NPN transistor (2N2222) is used as a switching device operating in saturation mode (onsemi, 2013). The base of the transistor is connected to a PWM output pin of the ESP32, while the emitter is connected to ground. The cathode of the IR LED is connected to the collector of the transistor, and its anode is connected to a 5V supply. The constant 5V supply is provided by a boost converter, TPS61023. As per

the datasheet from Texas Instruments (2020), the TPS61023 is built for boosting voltages as low as 0.5V, to 5V (minimum start-up voltage 1.8V). In this configuration, the circuit is completed only when the ESP32 outputs a 3.3V signal to the transistor base, causing the transistor to saturate and allowing current to flow through the IR LED. The full schematic diagram is in the diagram below.

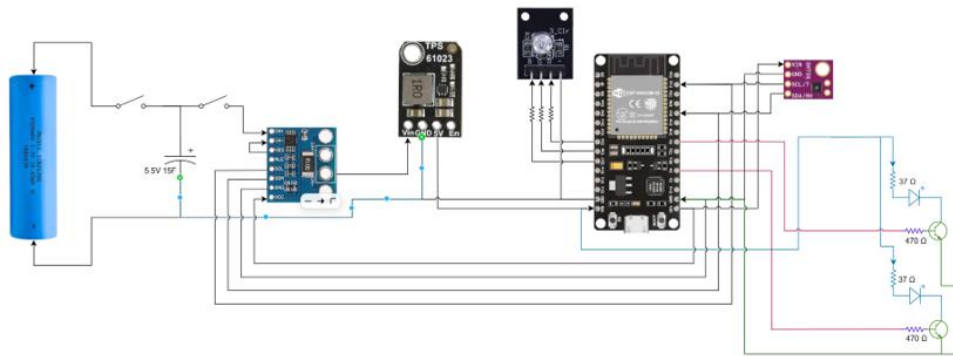


Figure 51. Static Inside Sensor, Schematic Diagram

To avoid loose wirings and to reduce voltage losses, the connections are reinforced using a printed circuit board (PCB), also shown in the Figure 52. The full schematic of which is in Appendix A.

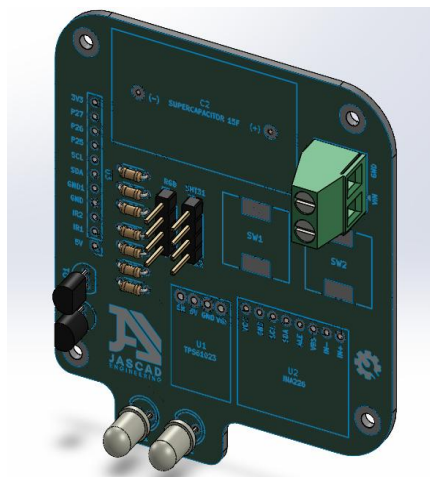


Figure 52. Static Inside Sensor, PCB Design

Being the node that will control the air-conditioning units, it must be positioned strategically, in such a way that the IR LED will point directly to the ACUs with minimum obstructions, while taking the SHT31's orientation into account. As such, the mount and enclosure are designed to take this into account. The CAD model and actual node is displayed below.

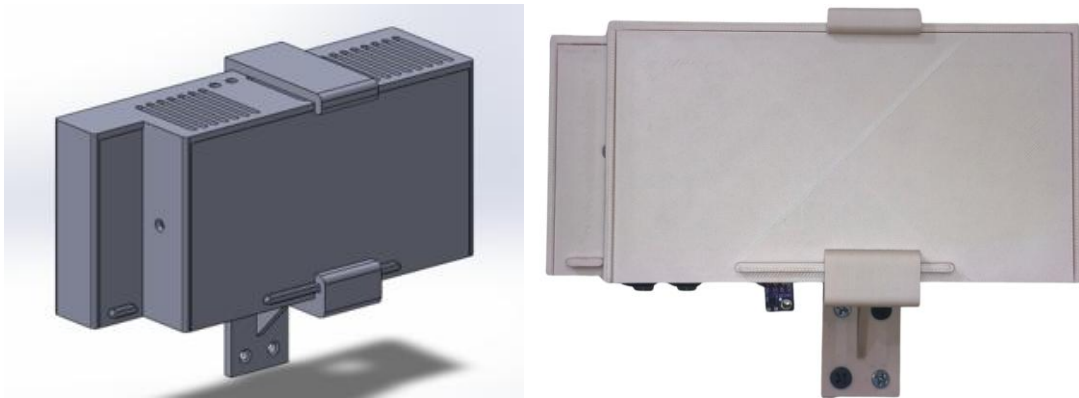


Figure 53. Static Inside Sensor, Digital (left), Actual (right)

ACU Power Monitoring Nodes

These nodes are used solely for measuring the power draw of the air-conditioning unit during the testing period. The module to be used is the PZEM-004T v4.0, which is built to measure AC voltages using a current transformer (CT), which leads to a safer, reversible, and non-invasive method of measuring an otherwise dangerous AC voltage (Components101, 2026).

Additionally, using UART protocol, which the ESP32 supports, makes wiring easier. Also, the PZEM-004T has a high accuracy fresh-out-of-the-box, with the datasheet claiming 0.5% error or deviation (Peacefair, 2024). This was supported by a study conducted by Sofwan et. al. (2025), wherein they tested the overall accuracy of

the AC monitoring module by comparing the values by comparing its values with an industrial grade AVOMeter, garnering a maximum average error of 0.41% across three testing devices.

A boost converter (TPS61023) will be used to step up the 3.3V from the ESP32 to the 5V requirement of the module. Alongside that, the converter's output was also used to make sure that ESP32 sees constant 5V despite the variable voltage from the battery. The module is to be connected to the AC L (live) wire and N (neutral) wire, while also powering the ACU.

The CT then must be clamped around the live wire connected to the ACU. This setup will allow the CT sensor to read the wattage through the live wire, like how a clamp ammeter works. The pinout connections of the components are detailed in the schematic diagram below.

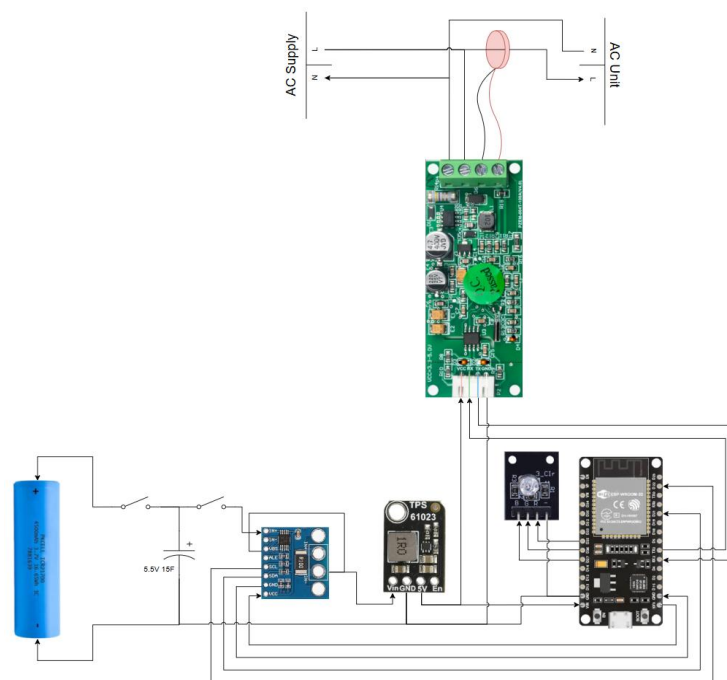


Figure 54. ACU Power Monitoring Nodes, Schematic Diagram

To avoid loose wirings and to reduce voltage losses, the connections are reinforced using a printed circuit board (PCB), also shown in the Figure 55. The full schematic of which is in Appendix A.

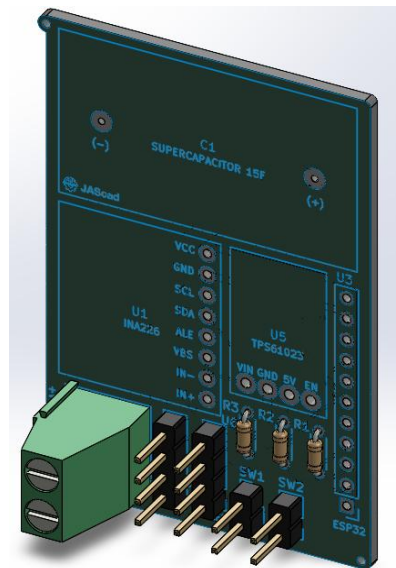


Figure 55. ACU Power Monitoring Nodes, PCB Design

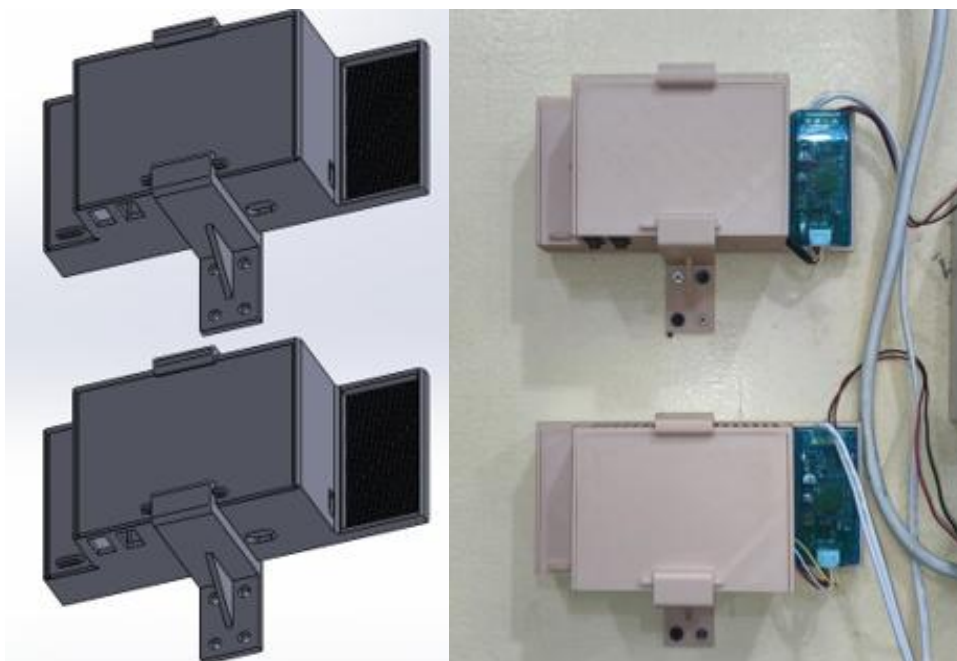


Figure 56. ACU Power Monitoring Nodes, Digital (left), Actual (right)

Since this node must be clamped to the live wire only and not including the neutral wire, it must be positioned so that the short CT wire can still reach the wire. It must also be near an outlet since the PZEM-004T requires a reference voltage, which an AC outlet can provide via a plug. With those taken into consideration, the CAD model and the actual node is as shown in Figure 56.

Since the placements of the sensor nodes are to be considered in terms of accessibility and function, the final node locations are as shown in Figure 18. These locations ensure that the camera of the Main Hub can see every person inside, Dynamic Inside Sensor will record the air quality near the occupants, the IR LED if the Static Inside Sensor signal reaches the ACU IR receivers reliably, the Outside Sensor represents the outside air quality well, and the ACU Power Monitoring Nodes are near the live wires of the air conditioning unit, all the while being non-invasive with the occupants, and being able to easy to conduct maintenance activities.

Prototype Testing for Technoeconomic Analysis

The preliminary prototype was deployed across a test zone (DME Conference Room) within a controlled indoor environment, with the zone representing an independently managed HVAC space. A standard Wi-Fi access point served as the common communication medium for all nodes. System data including temperature readings, actuation events, packet transmission data, and power consumption will be continuously collected. Power consumption per ESP32 node will be measured using an INA226 which is connected to the local ESP32, which sends the data separately through the broker. MQTT traffic statistics such as latency, packet loss, and network load will

be monitored using a python script, which will save the data in a CSV file for ease of processing. The monitoring and testing diagram is shown below.

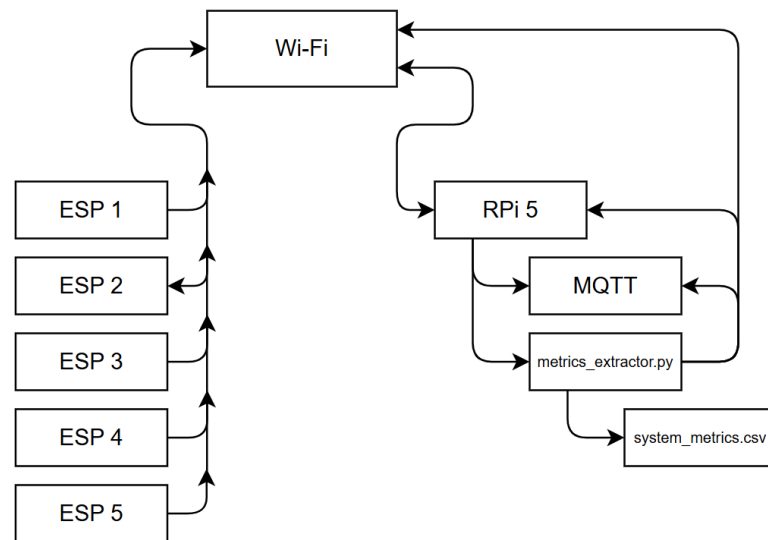


Figure 57. Metrics Extraction Layout

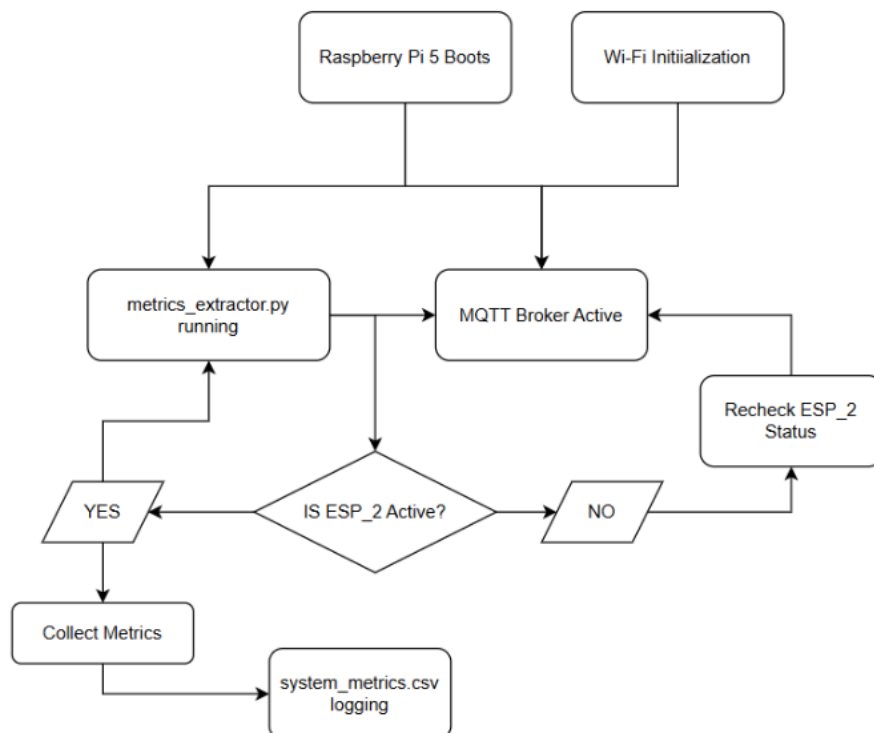


Figure 58. Metrics Extraction Flowchart.

Technical Evaluation

The entire system went through a technical performance evaluation according to four metric groups: (1) computational performance, (2) communication reliability, (3) system stability, and (4) energy consumption.

Computational performance. This measures the resource utilization of the Raspberry Pi 5, while the program and its peripherals are running. This includes metrics like CPU utilization percentage, CPU temperature (°C), RAM utilization percentage, and RAM usage (MB). These metrics, quantified through descriptive statistics such as mean, maximum, minimum, and standard deviation, are used to evaluate the computational demand, thermal behavior, and operational stability of the proposed framework. Furthermore, these values provide insight into the Raspberry Pi 5's available computational headroom and its potential capability to accommodate additional processes or sensor nodes during continuous operation.

Communication reliability. This includes latency, packet loss rate, and network load. These metrics, also quantified through descriptive statistics, evaluate the responsiveness and consistency of data transmission between the ESP32 sensor nodes and the Raspberry Pi 5 through the MQTT communication framework. Low latency and packet loss values indicate stable and reliable communication performance, while network load reflects the amount of communication traffic handled by the system during operation.

Energy consumption. This evaluates the operational power demand of both the distributed sensor nodes and the central Raspberry Pi 5 hub. ESP32 node energy

consumption is computed from logged current measurements obtained through INA226 monitoring modules, while Raspberry Pi 5 power consumption is estimated computationally using CPU utilization, and operating frequency. Additionally, the battery life of each node and the supercapacitor runtime are quantified as well. These metrics are used to evaluate the operational efficiency and technoeconomic feasibility of the proposed system.

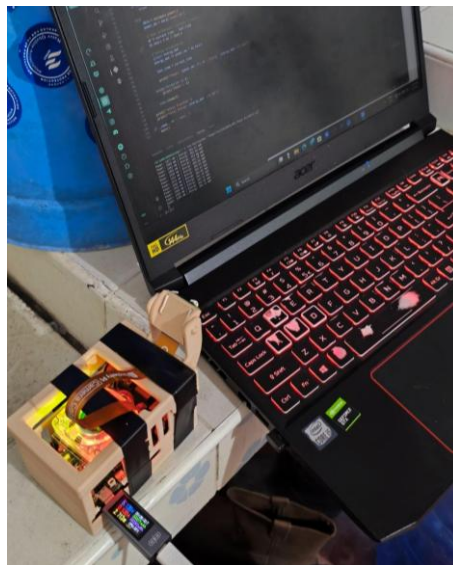


Figure 59. Manual Calibration of the Raspberry Pi 5 Power Estimator

The Raspberry Pi 5 power estimator was based on an article (Cadence PCB Solutions, 2023) where it is stated that total system power may be approximated as the combination of baseline system power and activity-dependent dynamic power. The baseline component represents the approximate idle power required to maintain continuous system operation, while the dynamic component accounts for additional power demand caused by processing activity. CPU utilization, and operating frequency were incorporated into the estimator following the principle that processor activity and frequency influence power consumption during operation (Beloglazov et al., 2011). An

inline USB power meter was used to calibrate the estimated power data against actual measured power consumption, as shown in Figure 59.

System stability. This refers to the evaluation of the capability of the proposed monitoring and control framework to sustain continuous and reliable operation throughout the deployment period. This study adopted the concept of operational continuity, defined as the ability of a system to continuously function under operational pressure (Soft&Solution, 2026). However, instead of directly quantifying uninterrupted operation, stability was evaluated through “Operational Monitoring Continuity,” defined as the proportion of complete operational records successfully generated relative to the total expected operational intervals during deployment. Periods affected by interrupted transmission, communication inconsistency, temporary node malfunction, or other operational interruptions resulted in incomplete operational records and were incorporated into the stability analysis. Through this metric, the persistence of communication, processing, logging, and control operations within the deployed framework can be quantitatively assessed.

Economic Assessment

For the economic component, capital expenditure (CapEx) will be computed from the total hardware and assembly costs of the controller modules, while operational expenditure (OpEx) will account for electricity usage and maintenance requirements. Projected annual savings will be derived from the difference in energy consumption between automated and baseline manual HVAC usage. Formulas to be used to quantify the economic metrics are given below.

$$CapEX = C_E + C_I \quad (19)$$

Where:

C_E = Equipment costs

C_I = Installation costs

$$OpEX = C_{EL} + C_M \quad (20)$$

Where:

C_{EL} = Electricity Costs

C_M = Maintenance Costs

Annual HVAC operating costs (regardless of mode of operation) were projected using an assumed operating schedule representative of classroom-based HVAC operation. The annual operating duration considered daily operational hours, weekly operating days, reduced operational weeks due to academic interruptions, declared holidays, and preventive maintenance downtime. Electricity costs were evaluated using the local residential electricity tariff rate of ₱10.5049/kWh (Leyeco IV, 2026).

The economic viability of the proposed Smart HVAC Controller was evaluated using consumer-side lifecycle cash flow analysis through Simple Payback Period (SPP), Return on Investment (ROI), and cumulative cash flow projection methods. The evaluation was conducted from the perspective of a consumer retrofitting an already existing HVAC system with the developed Smart HVAC Controller package.

The analysis incorporated experimentally measured energy savings obtained from the Comparative Performance Evaluation between BORF operation and conventional manual setpoint operation. The measured energy reduction percentage of 20.5% was applied to the projected annual HVAC energy consumption to estimate yearly electricity savings under smart controller implementation.

To account for future operational conditions, the economic model incorporated an annual electricity rate escalation of 9%, shown in Table 7 (Philippine Statistics Authority, 2026). This is included to represent the long-term effects of electricity price inflation, and its effect on consumer side cash flow. An annual HVAC energy consumption variation of 2% is also incorporated to acknowledge occupancy variation, operational schedule changes, and equipment aging throughout the analysis horizon, which is assumed equivalent to the controller's expected life.

Table 7. Year-on-Year Inflation Rates by Commodity Group: NCR

Commodity Group	Apr 2025	Mar 2026	Apr 2026
All Items	2.4	3.5r	5.5
Food and non-alcoholic beverages	2.5	1.8r	3.0
Alcoholic beverages and tobacco	2.5	1.7	1.8
Clothing and footwear	0.8	2.8	3.3
Housing, water, electricity, gas, and other fuels	5.1	5.9r	9.0
Furnishings, household equipment and routine household maintenance	0.3	2.4	3.0
Health	1.6	3.7	4.0
Transport	-2.5	11.1r	18.1
Information and communication	0.0	0.1	0.6
Recreation, sport and culture	0.6	3.1	3.1
Education services	2.5	2.0	2.0
Restaurants and accommodation services	1.9	0.5	1.5
Financial services	0.0	0.0	0.0
Personal care, and miscellaneous goods and services	2.4	2.5	2.8

Source: Philippine Statistics Authority, Retail Price Survey of Selected Commodities for the Generation of Consumer Price Index

Note: "r" revised.

The lifecycle cash flow model evaluated yearly operating expenses under both manual HVAC operation and Smart HVAC Controller operation. Net annual savings were determined from the difference between the projected manual-operation electricity cost and smart-controller electricity cost, while accounting for recurring annual maintenance allocation associated with the controller package. Formulas used for cash flow analysis are given below.

$$AES = (E_{\text{manual}} - E_{\text{BORF}}) \times R_{\text{EL}} \quad (21)$$

Where:

AES = Annual energy savings

E_{manual} = Manual operation energy use

E_{BORF} = AI controlled AC energy use

R_{EL} = Electricity rate

$$ES \% = \frac{(E_{\text{manual}} - E_{\text{BORF}})}{E_{\text{manual}}} \quad (22)$$

Where:

$ES \%$ = Energy Savings (%)

$$ERP = R_{\text{EL}} \times (1 + i_{\text{EL}})^t \quad (23)$$

Where:

ERP = Electricity rate projection

i_{EL} = Annual electricity rate escalation (%)

t = Time of ownership (years)

$$MHCP = E_{manual} \times (1 + ACU_{var})^t \quad (24)$$

Where:

$MHCP$ = Manual HVAC Consumption Projection

E_{manual} = Manual Operation Energy Use

ACU_{var} = ACU Consumption Variation (%)

$$SHCP = ES \% \times MHCP \quad (25)$$

$$Manual\ Bill = R_{EL} \times MHCP \quad (26)$$

$$Smart\ Bill = R_{EL} \times SHCP \quad (27)$$

$$S_N = (Manual\ Bill - Smart\ Bill) - OpEx \quad (28)$$

The Simple Payback Period was used to determine the estimated duration required for the cumulative savings to recover the initial controller investment cost. Meanwhile, Return on Investment (ROI) was used to evaluate the long-term profitability of the Smart HVAC Controller throughout its expected operational life. In addition, cumulative cash flow projection was utilized to visualize the progressive financial recovery and long-term economic benefit of the proposed system under escalating electricity costs. The formulas for which are given below.

$$PBP = \frac{P_S}{S_{N1}} \quad (29)$$

Where:

P_S = Controller Proposed Selling Price

S_{N1} = Year 1 Net Savings

$$ROI = \frac{S_{N1} \times L - P_S}{P_S} \quad (30)$$

Where:

L = Expected Life

To further assess economic sensitivity, conservative, base-case, and optimistic consumer adoption scenarios were also evaluated by varying controller package cost assumptions and electricity tariff conditions.

CHAPTER 4

RESULTS AND DISCUSSIONS

This chapter presents and discusses the findings obtained from the development, implementation, and evaluation of the proposed distributed IoT mechatronic HVAC optimization system. The performance of the hybrid AI controller, calibrated digital twin, and distributed IoT architecture is analyzed in terms of energy efficiency, thermal comfort, predictive accuracy, operational reliability, and technoeconomic feasibility. The results provide a comprehensive assessment of the system's effectiveness in achieving intelligent HVAC management under real-world operating conditions.

Machine Learning (B.O.R.F.) Training

The training results for the Bayesian Optimized Random Forest (*BORF*) predictive model are illustrated in Figure 60. The foundational development of this machine learning algorithm relied on an extensive empirical dataset containing approximately 3,000 individual data points. This information was systematically aggregated over several weeks of continuous environmental logging.

The empirical data acquisition process yielded two distinct datasets essential for model development and validation. The primary training corpus consists of approximately 2,500 discrete samples, capturing minute-by-minute endogenous and exogenous environmental conditions, comprehensive HVAC telemetry, and localized occupancy densities.

1	inside air temp	inside air humidity	occupancy	outside air temp	outside air humidity	time of the day	samsung setpoint	samsung mode	samsung fan speed	samsung power (W)	coolix setp
1212	27.4,65.9,4,35.31	68.42,11:25	27.0,dry	auto,1431.8	27.0,dry	auto,619.3	2851.2,0.4	2026-05-13	11:25:46		
1213	27.4,65.7,4,35.15	68.47,11:26	27.0,dry	auto,1544.9	27.0,dry	auto,617.1	2162.0,0.39	2026-05-13	11:26:46		
1214	27.3,65.7,4,35.18	68.42,11:27	27.0,dry	auto,1489.5	27.0,dry	auto,623.5	2113.1,0.39	2026-05-13	11:27:46		
1215	27.3,65.7,4,35.73	68.5,11:28	27.0,dry	auto,1337.1	27.0,dry	auto,613.7	1950.8,0.38	2026-05-13	11:28:46		
1216	27.4,65.9,3,35.64	68.48,11:29	27.0,dry	auto,1386.6	27.0,dry	auto,621.5	2088.1,0.4	2026-05-13	11:29:46		
1217	27.4,65.9,4,35.54	68.42,11:30	27.0,dry	auto,1526.7	27.0,dry	auto,623.6	2150.3,0.42	2026-05-13	11:30:46		
1218	27.5,65.6,4,35.26	68.35,11:31	27.0,dry	auto,1342.6	27.0,dry	auto,614.3	1956.9,0.43	2026-05-13	11:31:46		
1219	27.5,65.6,3,34.99	68.28,11:32	27.0,dry	auto,1334.2	27.0,dry	auto,617.4	1951.6,0.42	2026-05-13	11:32:46		
1220	27.5,65.5,4,34.88	68.19,11:33	27.0,dry	auto,1500.0	27.0,dry	auto,611.8	2111.9,0.42	2026-05-13	11:33:46		
1221	27.4,65.4,4,34.98	68.18,11:34	27.0,dry	auto,1541.9	27.0,dry	auto,620.9	2162.0,0.41	2026-05-13	11:34:46		
1222	27.5,65.3,3,34.92	68.15,11:35	27.0,dry	auto,1548.0	27.0,dry	auto,619.3	2167.3,0.42	2026-05-13	11:35:46		
1223	27.5,65.0,3,35.04	68.13,11:36	27.0,dry	auto,1549.5	27.0,dry	auto,620.8	2168.5,0.42	2026-05-13	11:36:46		
1224	27.4,65.1,4,35.04	68.11,11:37	27.0,dry	auto,1484.4	27.0,dry	auto,624.2	2078.6,0.41	2026-05-13	11:37:46		
1225	27.4,65.2,4,35.14	68.09,11:38	27.0,dry	auto,1341.6	27.0,dry	auto,619.3	1960.0,0.42	2026-05-13	11:38:46		
1226	27.5,65.3,5,35.23	68.11,11:39	27.0,dry	auto,1378.0	27.0,dry	auto,618.7	1997.5,0.45	2026-05-13	11:39:46		
1227	27.6,65.4,5,35.18	68.13,11:40	27.0,dry	auto,1531.8	27.0,dry	auto,616.4	2148.2,0.46	2026-05-13	11:40:46		
1228	27.6,65.1,5,35.33	68.08,11:41	27.0,dry	auto,1546.2	27.0,dry	auto,620.3	2166.6,0.47	2026-05-13	11:41:46		
1229	27.6,65.0,4,35.5	68.02,11:42	27.0,dry	auto,1351.2	27.0,dry	auto,618.7	1969.8,0.48	2026-05-13	11:42:46		
1230	27.6,66.2,4,35.4	67.95,11:43	27.0,dry	auto,1311.6	27.0,dry	auto,620.4	1952.8,0.49	2026-05-13	11:43:46		
1231	27.6,66.2,4,35.33	67.91,11:44	27.0,dry	auto,1477.6	27.0,dry	auto,617.7	2095.3,0.49	2026-05-13	11:44:46		
1232	27.6,66.0,4,35.29	67.88,11:45	27.0,dry	auto,1529.6	27.0,dry	auto,619.8	2149.4,0.49	2026-05-13	11:45:46		
1233	27.6,65.7,4,35.25	67.88,11:46	27.0,dry	auto,1543.4	27.0,dry	auto,620.1	2163.5,0.48	2026-05-13	11:46:46		
1234	27.6,65.5,4,35.33	67.79,11:47	27.0,dry	auto,1470.8	27.0,dry	auto,622.0	2093.4,0.48	2026-05-13	11:47:46		
1235	27.7,65.4,4,35.04	67.75,11:48	27.0,dry	auto,1327.0	27.0,dry	auto,618.8	1945.8,0.49	2026-05-13	11:48:46		
1236	27.7,65.9,4,35.09	67.73,11:49	27.0,dry	auto,1375.9	27.0,dry	auto,616.8	1992.5,0.5	2026-05-13	11:49:46		
1237	27.7,65.1,4,35.35	67.79,11:50	27.0,dry	auto,1529.6	27.0,dry	auto,623.3	2152.8,0.49	2026-05-13	11:50:46		
1238	27.6,65.3,4,35.4	67.74,11:51	27.0,dry	auto,1530.7	27.0,dry	auto,621.3	2152.0,0.48	2026-05-13	11:51:46		
1239	27.6,65.2,4,35.29	67.7,11:52	27.0,dry	auto,1539.5	27.0,dry	auto,619.3	2158.8,0.46	2026-05-13	11:52:46		
1240	27.5,65.4,4,35.28	67.71,11:53	27.0,dry	auto,1538.2	27.0,dry	auto,619.2	2157.4,0.45	2026-05-13	11:53:46		
1241	27.6,65.3,4,35.47	67.7,11:54	27.0,dry	auto,1546.3	27.0,dry	auto,629.9	2176.2,0.46	2026-05-13	11:54:46		
1242	27.6,65.6,4,35.8	67.74,11:55	27.0,dry	auto,1412.4	27.0,dry	auto,624.5	2036.9,0.46	2026-05-13	11:55:46		
1243	27.6,65.8,5,35.8	67.66,11:56	27.0,dry	auto,1330.4	27.0,dry	auto,623.5	1953.9,0.46	2026-05-13	11:56:46		
1244	27.6,65.8,5,35.88	67.63,11:57	27.0,dry	auto,1456.1	27.0,dry	auto,630.2	2086.2,0.47	2026-05-13	11:57:46		
1245	27.6,65.4,5,36.05	67.63,11:58	27.0,dry	auto,1358.0	27.0,dry	auto,629.9	2067.9,0.46	2026-05-13	11:58:46		
1246	27.6,66.0,5,36.04	67.57,11:59	27.0,dry	auto,1531.8	27.0,dry	auto,628.3	2152.6,0.48	2026-05-13	11:59:46		
1247	27.6,65.9,5,35.67	67.46,12:00	27.0,dry	auto,1336.0	27.0,dry	auto,612.8	2138.8,0.49	2026-05-13	12:00:46		

Figure 60. Training Dataset Structure

Crucially, this primary dataset was rigorously filtered to include exclusively steady-state operational data. Transient thermal phases periods where the room had not yet achieved thermodynamic equilibrium, were isolated into a separate repository.

This filtration ensures the machine learning algorithm trains strictly on stable, deterministic input-output mappings. To empirically validate the resulting BOREF model, a secondary, independent dataset comprising approximately 500 steady-state samples was gathered during a separate temporal period.

This holdout validation set was utilized post-training to evaluate the algorithm's predictive generalization. By subjecting the model to entirely unseen, real-world boundary conditions collected on a divergent date, the system's real-world applicability was rigorously quantified without risk of data leakage.

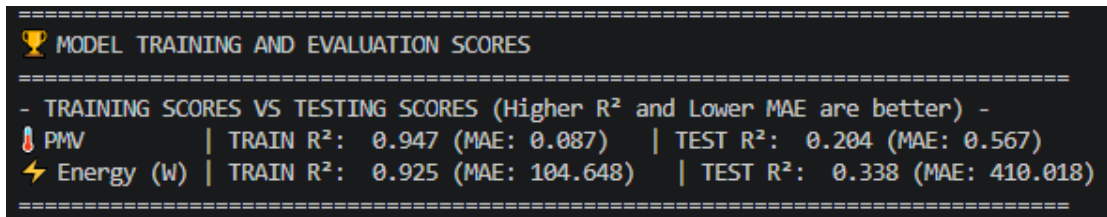


Figure 61. BORF Training and Evaluation Results

Figure 61 presents the results of the Bayesian Optimized Random Forest model development and training on 2,500 datapoints, as well as its performance on 500 unseen real-world datapoints. The training score represented by R-Squared for both PMV and Energy shows 94.7% and 92.5% along with corresponding MAE scores. In comparison, to the left, shows the brutal honest scores of the real-world deployment of the model which achieve a score of 20.4% for PMV and 33.8% for the Energy along with its corresponding MAE scores. The Table 8 also shows the Bayesian Optimized Values for the parameters of the random forest.

Table 8. Bayesian Optimized Random Forest model highest score parameters

Parameters	BO Values
<i>N_estimators</i>	65
<i>Max_depth</i>	6
<i>Min_samples_split</i>	20
<i>Min_samples_leaf</i>	1
<i>Random_state</i>	42

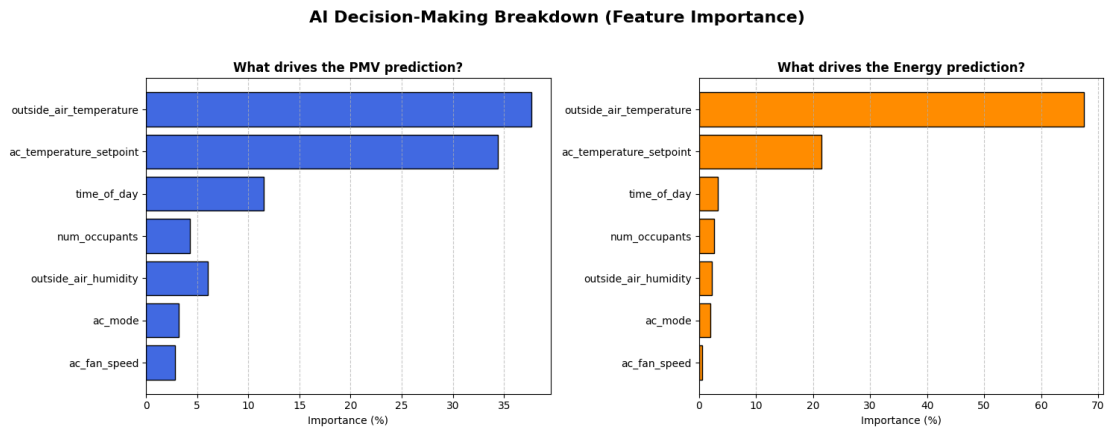


Figure 62. Model Features Importance Breakdown

Figure 62 shows the feature importance of the model, showing what factors greatly affect the decision making of the model and its corresponding weight.

The large gap between the high training score and the low testing score shows that the model "overfitted." This means the model memorized the specific training data instead of learning the actual physics of the room. To fix this, the researchers investigated if there was any "data leakage" which happens when a model accidentally uses information it should not have access to

After reviewing the input features and the predicted outputs, we found no connection that would cause the model to cheat. The input data is completely separated from the outcomes, and there is no way for the model to use the inputs to look into the future or guess the results.

To address the poor performance, the focus was shifted to parameter tuning. While Bayesian Optimization was used to identify the best hyperparameters, the researchers did not rely on it exclusively due to the persistently low-test scores. It was

doubted the optimizer might be converging on local minima that favored training accuracy over generalization. Subsequently, the researchers conducted an exhaustive manual parameter search, running multiple trials and recording the resulting test scores to verify the limits of the model.

The examples of these manual tuning attempts and their corresponding test scores are shown in Table 9.

Table 9. Sample scores hunted by Bayesian Optimizer across different parameters

Trial	PMV R²	Energy R²	n_estimators	max_depth	Min Samples Split	Min Samples Leaf
1	-0.078	0.333	727	13	16	6
2	-0.068	0.335	429	9	10	4
3	0.162	0.338	848	3	20	7
4	0.162	0.338	50	3	20	1
5	0.163	0.338	900	3	20	10
6	0.168	0.338	50	3	2	10
7	0.168	0.338	50	3	9	10
8	0.168	0.338	50	3	13	10
9	0.174	0.334	606	7	2	1
10	0.179	0.334	823	6	4	1
11	0.178	0.336	900	6	2	1
12	0.179	0.336	806	6	2	1
13	0.180	0.336	841	6	2	1
14	0.180	0.336	772	6	2	1
15	0.180	0.336	827	6	2	1
16	0.180	0.337	874	6	2	1
17	0.180	0.337	876	6	2	1
18	0.190	0.334	73	6	17	1
19	0.193	0.334	119	6	20	1

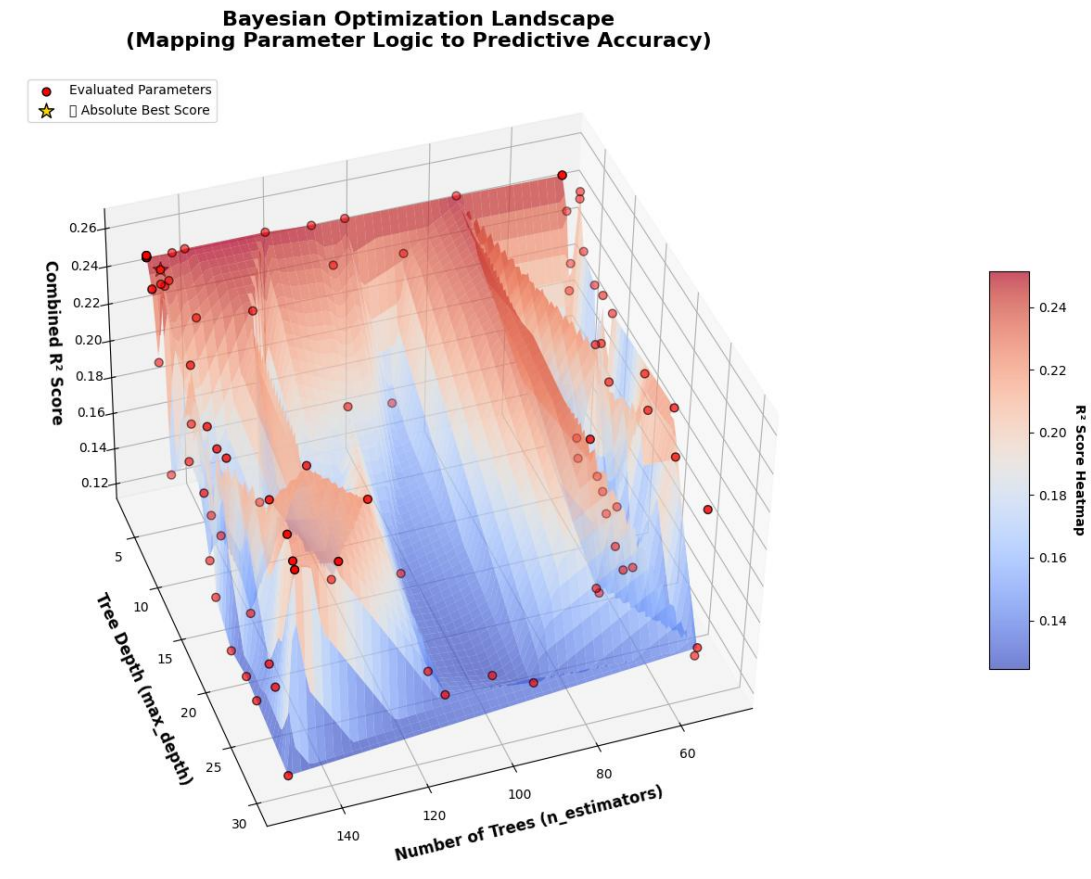


Figure 63. 3D Graph of Model Parameters vs Model Performance

Despite the exhaustive Bayesian Optimization and manual attempts, even after testing extreme parameter values such as *n_estimators* exceeding 10,000 and *max_depth* surpassing 300, the model failed to improve upon deployment. In most cases, these complex settings caused the performance to drop further, often resulting in negative scores. Even at its best, the model could not exceed a ceiling of 20.4% for PMV and 33.8% for Energy on unseen data.

This confirms that the issue is not with the hyperparameter tuning. Instead, the problem lies in the size and quality of the training data. The 2,500 data points collected over three weeks did not provide a strong enough foundation for the model to learn the room's complex thermodynamic behavior.

Furthermore, the BORF model is poor at extrapolation; it cannot predict values outside of its training range, and it essentially starts guessing when exposed to data that falls outside the boundaries of what it has already seen.

The model lacks the necessary information to form generalized rules that work in noisy, dynamic, real-world conditions. This proves that a larger, more diverse dataset covering a wider variety of weather and occupancy scenarios is required for this application.

However, collecting this data is costly and time-consuming. Gathering a wide range of weather conditions, temperature ranges, and occupancy levels is difficult to achieve in a short period. Similar studies often require six months to one year of data collection to be effective. The Random Forest model was chosen for this project precisely because among various machine learning algorithms, it is known for its ability to train quickly and learn patterns effectively even when working with smaller datasets compared to deep learning approaches.

Furthermore, external factors created additional challenges. The energy crisis of the country, combined with strict compliance to energy-saving mandates and the shift from face-to-face to online classes, significantly restricted the researchers' ability to access the facility.

Additionally, sudden weather shifts caused by the Super El Niño event introduced unpredictable environmental variables, placing immense strain on the model's predictive capabilities. These combined factors made it exceptionally difficult to collect a consistent, stable dataset during this period.

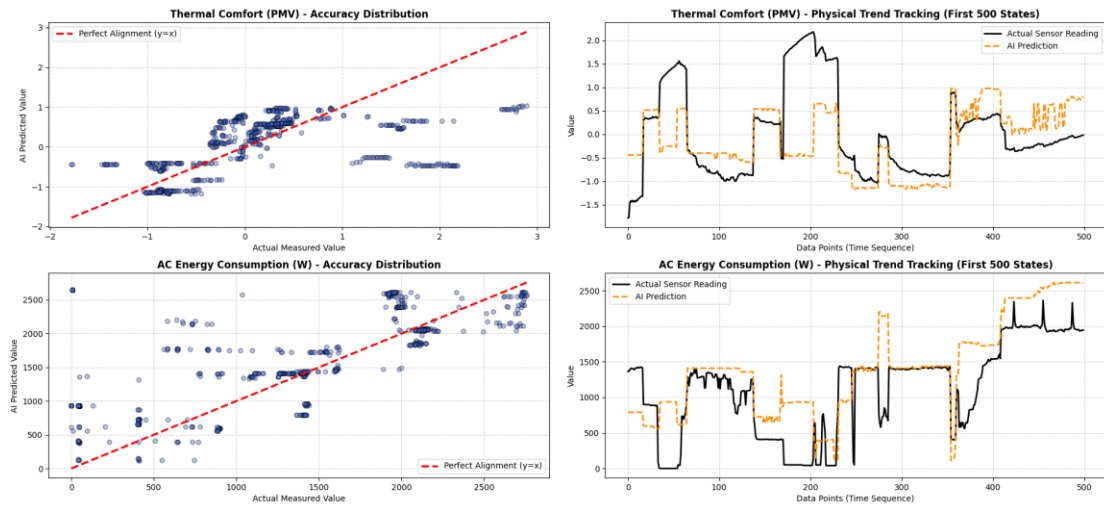


Figure 64. True Value vs. Model Predictions

Despite the low statistical validation scores, Figure 64 illustrates that the BORF model successfully identified the underlying physical trends within the system. The scatter plots demonstrate that, while the environment introduced significant stochastic noise, most of the predictive outputs (represented by blue circular markers) cluster along the ideal parity line (indicated by the red dashed line). This clustering confirms that the model correctly learned the primary thermodynamic correlations for both PMV and energy consumption.

Furthermore, the time-series plots on the right provide visual evidence of the model's predictive logic; specifically, the system correctly recognizes that reductions in the PMV index necessitate a corresponding increase in AC power consumption. While the model achieved R^2 scores of 0.204 (PMV) and 0.338 (Energy) during external validation, these metrics represent a partial capture of the room's complex thermodynamic behavior rather than a complete absence of learning.

The divergence between the predicted (orange dashed lines) and actual (solid black lines) values is not an indication of algorithmic failure, but rather a reflection of the extreme environmental volatility and data sparsity encountered during the deployment phase.

The model successfully learned the basic physical relationships, yet its capacity to fully map the room's response was limited by the highly dynamic, noisy nature of real-world conditions.

While the machine learning model B.O.R.F managed to identify the underlying trends among complex, non-linear variables, its standalone performance was limited by real-world environmental noise. To address this, the study implements a hybrid control architecture.

This approach integrates the predictive BORF model with a physics-informed, rule-based supervisory logic. This hard-coded logic acts as a system safeguard, designed to override mathematically anomalous or physically impractical decisions generated by the AI. By applying necessary adjustments to the model's preliminary outputs based on established empirical limits, the hybrid system ensures reliable occupant thermal comfort even in volatile real-world conditions.

As detailed in the methodology, a series of conditional 'if-then' protocols were established through extensive stress testing and failure mitigation analysis to guide the model during physical deployment. The operational success of this hybrid framework is evaluated in the subsequent section.

Comparative Performance Evaluation: BORF vs. Manual Operation

To quantify the performance differential between standard manual Air Conditioning operation (Human-Controlled) and the automated machine learning architecture (BORF-Controlled), a rigorous comparative evaluation was conducted. The primary objective of this testing phase was to empirically validate whether the BORF system successfully maintained ASHRAE thermal comfort standards while maximizing energy conservation, and to mathematically quantify the exact margin of those energy savings.

To ensure statistical validity and prevent exogenous variables from skewing the comparative energy data, the physical testbed was subjected to strict experimental controls. The baseline conditions for both operational scenarios were equalized using the following parameters:

- **Occupant Heat Load.** The occupant density within the target zone was strictly regulated. The number of individuals present was equalized across both testing phases to ensure a consistent biological thermal load (metabolic heat).
- **Internal Equipment Loads.** Transient sensible heat gains generated by electronics were actively managed. The number of active computer terminals, overhead lighting fixtures, and mobile devices was standardized to guarantee identical baseline electrical heat dissipation.
- **Weather Normalization.** Because external microclimates inevitably fluctuate between testing days, outdoor thermal loads were mathematically neutralized using the Cooling Degree Hour (CDH) metric, in strict accordance with

ASHRAE Guideline 14. This allowed the AC energy consumption to be evaluated against an equalized weather baseline.

- **Temporal Consistency.** Both operational setups were deployed and logged over identical timeframes and durations to eliminate diurnal temperature biases.

The resulting comparative telemetry tracking both weather-normalized energy consumption and dynamic thermal comfort (PMV) across both control methodologies is presented in the figures below.

Test A

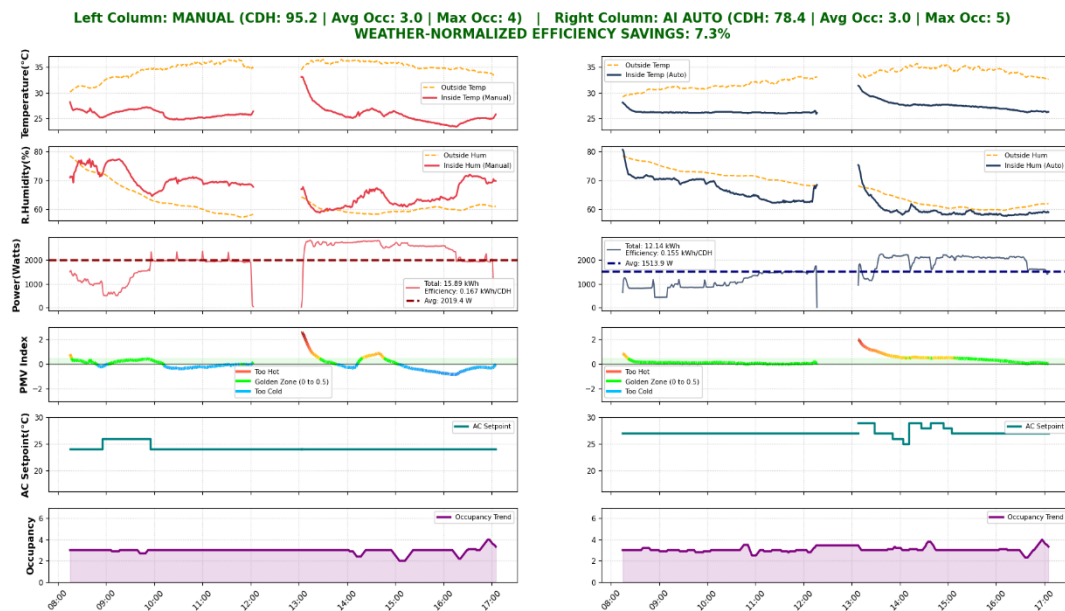


Figure 65. Performance Test A

Figure 65 illustrates a comparative performance analysis between the manually operated AC system (left panel) and the automated, AI-driven BORF system (right

panel). The visual telemetry is composed of six distinct temporal plots: ambient temperature (°C), relative humidity (%), AC power consumption (Watts), thermal comfort index (PMV), AC setpoint, and localized occupancy density. Within the temperature and humidity plots, exogenous boundary conditions (outdoor weather) are denoted by yellow dashed lines, while the internal room conditions are represented by solid trendlines, red for the manual control scenario and blue for the automated BORE system. The thermal comfort index, empirically derived from direct occupant surveys, is mapped onto a color-coded psychophysical scale to visualize physiological thermal sensation:

- Red (≥ 2.0): Severe thermal discomfort; the environment is uncomfortably hot. Intense Sweating will occur.
- Orange (1.0 to 2.0): Moderate thermal discomfort; the environment is distinctly warm.
- Yellow (0.8 to 1.0): Marginal thermal discomfort; the environment is slightly warm.
- Light Yellow (0.6 to 0.8): Upper boundary of thermal acceptability. The space remains neutral for occupants with low metabolic activity, though newly entered individuals may experience transient warmth during initial acclimatization.
- Green (-0.1 to 0.5): Optimal thermal neutrality (ASHRAE target zone). The space is perfectly balanced, with absolute physiological equilibrium occurring at 0.

- Light Blue (-0.1 to -0.5): Lower boundary of thermal acceptability. The environment exhibits mild overcooling, potentially registering as slightly cold for acclimating occupants or during prolonged exposure.
- Blue (≤ -0.5): Severe overcooling; the environment is uncomfortably cold, capable of inducing localized physiological discomfort, such as extremity numbness, during extended stays.

Furthermore, the telemetry array tracks the temporal progression of the AC temperature setpoints. While discrete HVAC states such as specific operational modes and fan speeds are excluded from this visual representation and will be detailed in subsequent sections, the dynamic modulation of the temperature thresholds is clearly showed.

The lowest panel illustrates the localized occupancy density over time. To mitigate transient data noise caused by brief foot traffic such as rapid ingress and egress of occupants in the room, a moving average filter was applied to smooth the occupancy curve and provide a more accurate representation of the sustained biological heat load. The third panel details the aggregate active power consumption (measured in Watts) of the dual split-type AC units Samsung and General Royal.

This combined power draw is mapped continuously via solid trendlines, red representing the manual baseline and blue representing the BORF automated system, while the horizontal dashed lines indicate the mean active consumption across the operational period.

Finally, quantitative performance metrics are embedded within this plot, explicitly detailing both the cumulative electrical consumption (kWh) and the critical weather-normalized energy efficiency ratio (kWh/CDH).

Since the experimental testbed consisted of a single conditioned zone equipped with a singular primary HVAC system, simultaneous parallel evaluation of the manual and automated control methodologies was physically impossible. Consequently, the operational tests were conducted longitudinally across separate days. To account for the inherent variance in outdoor meteorological conditions between these sequential evaluation periods, ASHRAE Guideline 14 was implemented. Utilizing the Cooling Degree Hour (CDH) metric, the external weather data was mathematically normalized to establish an equivalent thermodynamic baseline. During the “Test A” evaluation phase, the manual control scenario was subjected to a significantly higher exogenous thermal load (CDH = 95.2), whereas the AI-driven BORF system operated under milder external conditions (CDH = 78.4). Despite this temporal weather variance, both testing environments were successfully regulated to maintain a consistent internal biological load, averaging three occupants throughout the duration. Based on this normalized baseline telemetry, the following key observations are derived:

Table 10. Test A Results Summary

Parameter	Manual Control	AI Control
CDH	95.2	78.4
Avg. Occupancy	3	3
Peak Occupancy	4	5
Total Energy Consumed	15.89 kWh	12.14 kWh
Norm. Efficiency	0.167 kWh/CDH	0.155 kWh/CDH

Parameter	Manual Control	AI Control
PMV	Overcools from 10 to 12, and 15 to 17	Neutral at 8 to 12, border 0.5 at 13 to 15, neutral at 15 to 17
AC Modes Utilized	Cool	Cool, dry, eco
BORF System Energy Saving Percentage		7.3%

The automated BORF system achieved a weather-normalized energy savings of 7.3% while demonstrating superior thermal regulation compared to the manual baseline. As evidenced by the PMV telemetry, the machine learning controller executed a smooth and controlled transition into optimal thermal neutrality, maintaining steady-state comfort throughout the deployment.

In contrast, the manual control strategy exhibited severe transient fluctuations and persistent overcooling. Furthermore, the BORF system's superior performance was driven by its dynamic actuation of HVAC modes. The algorithm initially deployed 'Cool' mode to rapidly depress the sensible heat load (dry-bulb temperature), subsequently transitioning to 'Dry' mode to actively extract the remaining latent heat. This intelligent psychrometric management is empirically validated by the humidity data: the BORF system successfully maintained the indoor relative humidity strictly below the ambient outdoor levels.

Conversely, the rigid reliance of the manual operator on a static 'Cool' mode failed to provide adequate dehumidification, ultimately resulting in an indoor relative humidity that inefficiently exceeded that of the unconditioned outdoor air.

Equivalent energy savings in kWh is calculated below:

$$\text{Equivalent Manual System Consumption} = \text{Efficiency} \times \text{CDH} \quad (31)$$

$$\text{Equivalent Manual System Consumption} = 0.167 \frac{\text{kWh}}{\text{CDH}} \times 78.4 \text{ CDH}$$

$$\text{Equivalent Manual System Consumption} = 13.0928 \text{ kWh}$$

$$\begin{aligned} \text{Energy Saving in kWh} = & \text{Manual Equivalent Consumption at same CDH} - \\ & \text{BORF System Consumption at same CDH} \end{aligned} \quad (32)$$

$$\text{Energy Saving in kWh} = 13.0928 \text{ kWh} - 12.14 \text{ kWh}$$

$$\text{Energy Saving in kWh} = 0.9528 \text{ kWh}$$

The aggregate energy conservation achieved by the BORF system is driven by four synergistic operational strategies. Primarily, the machine learning algorithm strictly enforces ASHRAE 55 thermal neutrality, effectively eradicating the massive energy waste associated with systemic overcooling.

Secondly, the system strategically deploys the AC unit's native 'Dry' mode to prioritize latent heat extraction. By dehumidifying the space, the system exponentially increases the air's capacity to absorb anthropogenic latent heat, achieving physiological comfort without the high energy expenditure required for aggressive sensible cooling (dry-bulb temperature reduction).

Then, upon establishing optimal thermal equilibrium ($\text{PMV} \approx 0$), the algorithm transitions into 'Eco' mode, locking the compressor into its peak thermodynamic efficiency state for steady-state maintenance.

Finally, the hybrid architecture, combining the predictive BORF model with deterministic supervisory logic, actively mitigates erratic compressor throttling. Rather than demanding maximum rotational frequency (RPM) for rapid thermal '*pull-downs*,' the system induces compressor partial-loading by issuing gradually stepped temperature setpoints, gently guiding the space into the comfort zone over a dynamic 30-minute evaluation window.

Crucially, this comprehensive multi-objective optimization is executed entirely exogenously; the AI manipulates high-level setpoints via infrared transmission, maximizing efficiency without directly interfering with the inverter's native hardware or internal PID control loops.

Test B

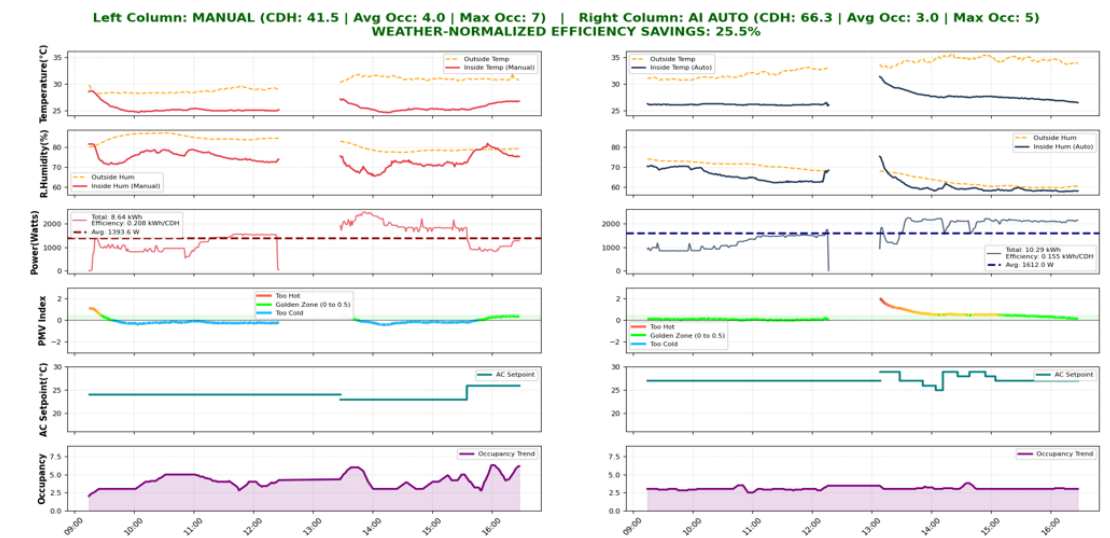


Figure 66. Performance Test B

Figure 66 (Test B) presents another comparative performance analysis, contrasting the manually operated AC baseline (left panel) against the automated BORF

architecture (right panel). Unlike the boundary conditions observed in prior evaluations, this specific test subjected the AI-driven system to a significantly more severe exogenous thermal load.

Environmental telemetry indicates that the automated controller was forced to mitigate a notably hotter outdoor climate, recording 66.3 Cooling Degree Hours (CDH), whereas the manual baseline was evaluated under a relatively mild thermal burden of just 41.5 CDH. Despite operating under this thermodynamic disadvantage, establishing this normalized baseline allows for the derivation of the following key observations:

Table 11. Test B Results Summary

Parameter	Manual Control	AI Control
CDH	41.5	66.3
Avg. Occupancy	4	3
Peak Occupancy	7	5
Total Energy Consumed	8.64 kWh	10.29 kWh
Norm. Efficiency	0.208 kWh/CDH	0.155 kWh/CDH
PMV	Overcools from 10 to 12, and 13 to 16	Neutral at 8 to 12, border 0.5 at 13 to 15, neutral at 15 to 17
AC Modes Utilized	Cool	Cool, dry, eco
BORF System Energy Saving Percentage		25.5%

Equivalent energy savings in kWh is calculated below:

$$\text{Equivalent Manual System Consumption} = \text{Efficiency} \times \text{CDH}$$

$$\text{Equivalent Manual System Consumption} = 0.208 \frac{\text{kWh}}{\text{CDH}} \times 66.3 \text{ CDH}$$

$$\text{Equivalent Manual System Consumption} = 13.7904 \text{ kWh}$$

Energy Saving in kWh

= Manual Equivalent Consumption at same CDH

– BORF System Consumption at same CDH

Energy Saving in kWh = 13.7904 kWh – 10.29 kWh

Energy Saving in kWh = 3.5004 kWh

Despite operating under a significantly higher exogenous thermal load, the BORF architecture successfully sustained the environment strictly within optimal ASHRAE comfort boundaries with negligible deviation.

The automated system achieved a substantial 25.5% weather-normalized energy reduction, conserving 3.5004 kWh relative to the manual baseline. This efficiency gain is fundamentally attributed to the eradication of systemic overcooling; while the manual system continuously overcompensated, the BORF controller achieved and maintained a steady-state thermodynamic equilibrium.

Furthermore, the psychrometric telemetry indicates that both operational scenarios utilized 'Dry' mode, as evidenced by the indoor relative humidity remaining consistently below outdoor ambient levels in both tests. Since latent heat management was effectively identical across both scenarios, the drastic 25.5% efficiency divergence isolates the value of dynamic setpoint modulation.

The AI's capacity to actively and continuously adjust the temperature setpoint prevented the massive energy waste inherent to the human operator's static control strategy.

Test C

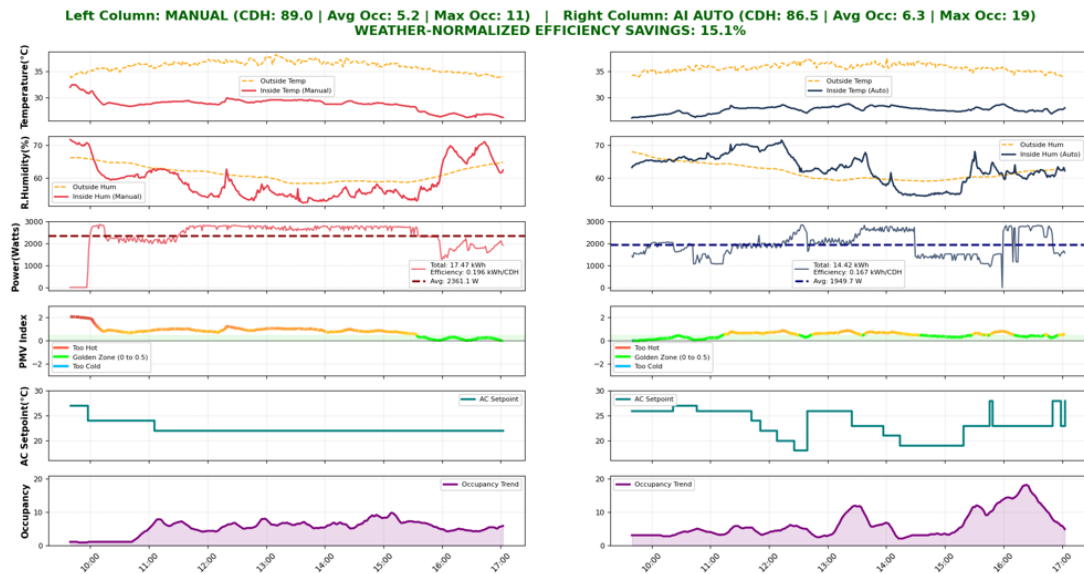


Figure 67. Performance Test C

Another comparative performance analysis (TEST C) between the manually operated AC system (left panel) and the automated, AI-driven BORF system (right panel). During this evaluation phase, both the manual and AI-controlled setups were subjected to nearly identical external weather conditions.

Unlike previous tests, occupant behavior during this period was unconstrained, as individuals freely entered and exited the space to attend to their regular academic or professional tasks. Despite this dynamic and uncontrolled biological heat load, the telemetry indicates that the automated AI system successfully managed the environment while being exposed to a slightly higher overall occupancy level than the manually controlled system. Based on this baseline telemetry, the following key observations are derived:

Table 12. Test C Results Summary

Parameter	Manual Control	AI Control
CDH	89.0	86.5
Avg. Occupancy	5.2	9.3
Peak Occupancy	11	19
Total Energy Consumed	17.47 kWh	14.42 kWh
Norm. Efficiency	0.196 kWh/CDH	0.167 kWh/CDH
PMV	Hot to slightly hot from 10 to 15:30	Maintained within PMV range
AC Modes Utilized	Cool	Cool, dry, eco
BORF System Energy Saving Percentage		15.1%

Equivalent energy savings in kWh is calculated below:

$$\text{Equivalent Manual System Consumption} = \text{Efficiency} \times \text{CDH}$$

$$\text{Equivalent Manual System Consumption} = 0.196 \frac{\text{kWh}}{\text{CDH}} \times 86.5 \text{ CDH}$$

$$\text{Equivalent Manual System Consumption} = 16.954 \text{ kWh}$$

$$\text{Energy Saving in kWh}$$

$$= \text{Manual Equivalent Consumption at same CDH}$$

$$- \text{BORF System Consumption at same CDH}$$

$$\text{Energy Saving in kWh} = 16.954 \text{ kWh} - 14.42 \text{ kWh}$$

$$\text{Energy Saving in kWh} = 2.354 \text{ kWh}$$

The comparative telemetry from Test C demonstrates a pronounced operational superiority of the BORF architecture, yielding a 15.1% weather-normalized energy savings (equivalent to an absolute reduction of 2.534 kWh) over the manual baseline. This efficiency was achieved despite the AI controller operating under a severe, asymmetric anthropogenic load.

As illustrated in the occupancy plot, the BORF system was subjected to a highly erratic occupancy profile peaking at 19 individuals, compared to a peak of only 11 during the manual test. This transient high-density foot traffic necessitated high-frequency door actuations, introducing massive influxes of unconditioned ambient air and severely elevating both sensible and latent heat infiltration.

Despite these extreme boundary conditions, the BORF system rapidly established and maintained optimal ASHRAE thermal neutrality with minimal transient deviation.

In contrast, the manual control scenario was governed by a static 24°C–26°C setpoint policy, strictly adhering to the University’s Energy Conservation Memorandum.

Despite facing a significantly lighter occupancy burden, this rigid, policy-driven approach failed to achieve thermal neutrality. The space predominantly hovered in a state of moderate thermal discomfort (≥ 1.0), leaving occupants in a distinctly warm environment susceptible to physiological strain or sweating, and only achieved marginal comfort late in the diurnal cycle as external ambient temperatures dropped.

Ultimately, the BORF system successfully managed a harsher thermodynamic environment and secured a 15.1% energy reduction strictly through the exogenous, dynamic modulation of operational setpoints.

Furthermore, this evaluation empirically proves that human thermal comfort cannot be universally defined by a static thermostatic setpoint. As demonstrated in prior testing phases, the BORF system successfully established optimal thermal neutrality utilizing a setpoint range of 26°C to 27°C. However, applying these identical parameters under divergent meteorological conditions fails to produce an equivalent thermodynamic response. The thermal behavior of the conditioned space is inherently dynamic, heavily influenced by fluctuating external boundary conditions. Consequently, the BORF architecture sustains thermal equilibrium strictly through the continuous, dynamic modulation of AC parameters to actively reject these environmental disturbances. Crucially, this evaluation highlights that the ultimate energy-saving potential of the BORF system is bounded by the mechanical limitations of the primary HVAC unit. Because the AI functions exclusively as an exogenous supervisory controller (modulating signals without physical hardware modification), its efficacy relies on the baseline mechanical viability of the AC unit. Provided the HVAC system is correctly sized for the maximum thermal load and not under-designed, the BORF algorithm will successfully optimize its operation.

Test N

To empirically demonstrate that the BORF architecture's multi-objective optimization hierarchy strictly prioritizes occupant thermal comfort over absolute energy minimization, the comparative telemetry is presented in (TEST N) Figure 68.

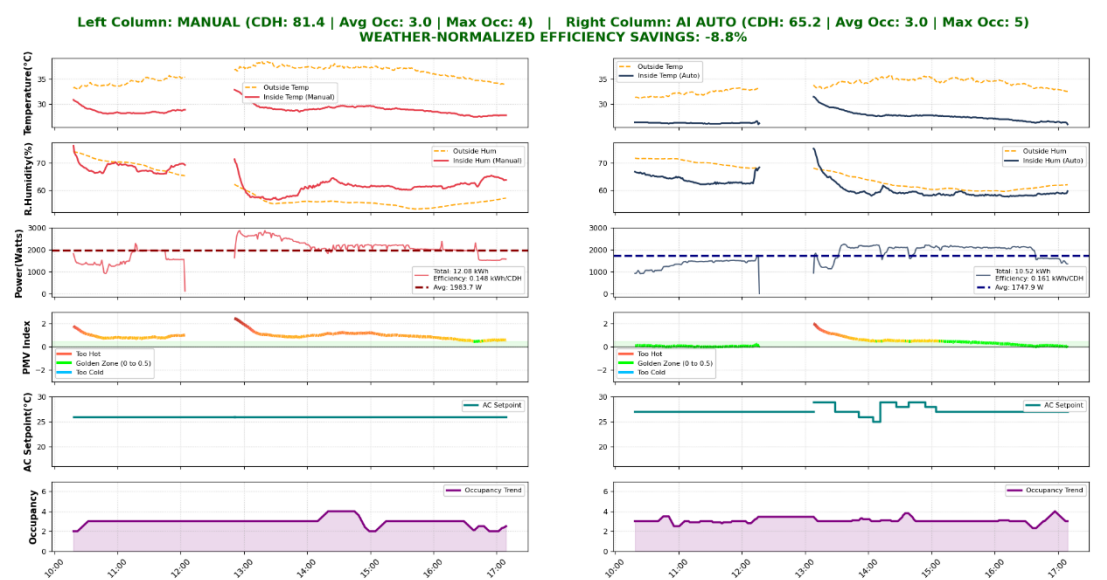


Figure 68. Performance Test N

Table 13. Test N Results Summary

Parameter	Manual Control	AI Control
CDH	81.4	65.2
Avg. Occupancy	3	3
Peak Occupancy	4	5
Total Energy Consumed	12.08 kWh	10.52 kWh
Norm. Efficiency	0.148 kWh/CDH	0.161 kWh/CDH
PMV	Undercool, slightly warm to warm	Maintained within PMV range
AC Modes Utilized	Cool	Cool, dry, eco
BORF System Energy Saving Percentage		-8.8%

In contrast to prior evaluations, this test phase illustrates a scenario in which the BORF architecture incurred an 8.8% energy penalty relative to the manually controlled baseline. Environmental telemetry indicates that the manual system operated under a more severe exogenous heat load ($CDH = 81.4$) compared to the BORF system ($CDH = 65.2$), while both environments maintained equivalent occupancy densities. As evidenced by the relative humidity data, the manual operator rigidly maintained a static 'Cool' mode, whereas the BORF algorithm dynamically actuated a combination of Cool, Dry, and Eco modes. While the manual system recorded lower overall power consumption, the PMV telemetry reveals a critical failure in its primary operational objective: the space remained in a continuous state of moderate-to-marginal thermal discomfort. Conversely, the BORF system deliberately expended additional compressor energy to successfully drive the environment into the strict ASHRAE thermal comfort zone.

This operational divergence empirically validates the algorithm's multi-objective optimization hierarchy: the AI strictly prioritizes occupant physiological comfort over unconditional energy reduction. Consequently, the manual baseline's lower power draw represents a false efficiency, as it was achieved entirely by sacrificing the thermal well-being of the occupants.

Test D

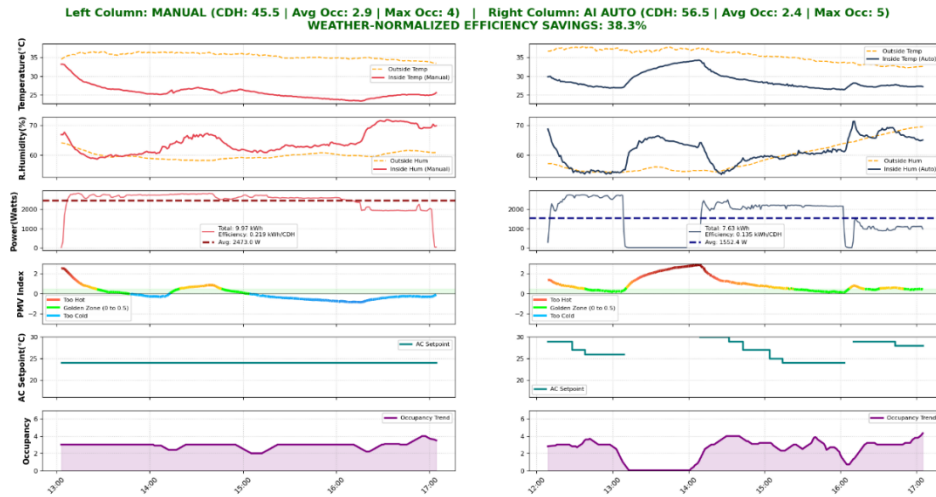


Figure 69. Performance Test D

Table 14. Test D Results Summary

Parameter	Manual Control	AI Control
CDH	45.5	56.5
Avg. Occupancy	2.9	2.4
Peak Occupancy	4	5
Total Energy Consumed	9.97 kWh	7.63 kWh
Norm. Efficiency	0.219 kWh/CDH	0.135 kWh/CDH
AC Modes Utilized	Cool	Cool, dry, eco
BORF System Energy Saving Percentage		38.3%

Equivalent energy savings in kWh is calculated below:

$$\text{Equivalent Manual System Consumption} = \text{Efficiency} \times \text{CDH}$$

$$\text{Equivalent Manual System Consumption} = 0.219 \frac{\text{kWh}}{\text{CDH}} \times 56.5 \text{ CDH}$$

$$\text{Equivalent Manual System Consumption} = 12.3735 \text{ kWh}$$

Energy Saving in kWh

= Manual Equivalent Consumption at same CDH

– BORF System Consumption at same CDH

Energy Saving in kWh = 12.3735 kWh – 7.63 kWh

Energy Saving in kWh = 4.7435 kWh

Figure 69 (Test D) demonstrates the BORF architecture's capacity for autonomous, demand-responsive state modulation. The comparative telemetry illustrates that the AI dynamically dictates HVAC operation based on real-time spatial demand, notably deactivating the AC unit entirely during zero-occupancy periods to eliminate parasitic power draw. Upon occupant re-entry, the controller actively modulates parameters to counteract the renewed thermal load.

The temporal data reveals a critical event between 14:00 and 15:00: a sharp influx of occupants coincided with the system's effort to overcome the room's accumulated thermal inertia from its prior dormant state. The setpoint telemetry illustrates the controller's precise response, executing a gradual, responsive decrease in the temperature setpoint to steadily drive the Predicted Mean Vote (PMV) into optimal thermal neutrality. Subsequently, as occupancy decreased at 16:00, the algorithm autonomously elevated the setpoint to proactively prevent systemic overcooling.

Furthermore, the BORF system seamlessly alternated between 'Cool' and 'Dry' modes to independently manage sensible and latent heat loads. In stark contrast, the manual baseline exhibited conventional, rigid human operation, relying on a static setpoint and a continuous 'Cool' mode, which resulted in persistent overcooling and

poor latent heat management. Ultimately, the integration of real-time presence detection, localized occupant density tracking, exogenous weather awareness, and ASHRAE 55 PMV optimization enabled the BORF machine learning model to achieve an exceptional 38.3% energy savings.

The following figures provide further comparative analyses detailing the performance of the manually controlled baseline against the AI-driven BORF system. These additional test cases were conducted under a variety of distinct operational scenarios, exposing the control strategies to unique outdoor climates and internal occupancy densities.

Across all evaluated conditions, the data unequivocally demonstrates a consistent outcome: the BORF architecture successfully achieved notable energy savings in every instance, proving its adaptability and overall efficiency.

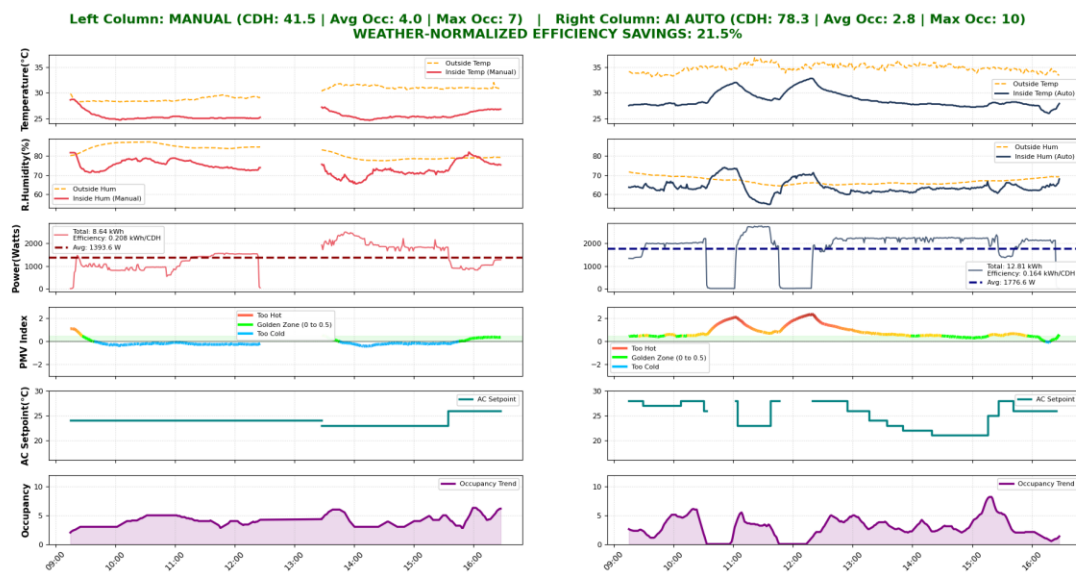


Figure 70. Performance Test E

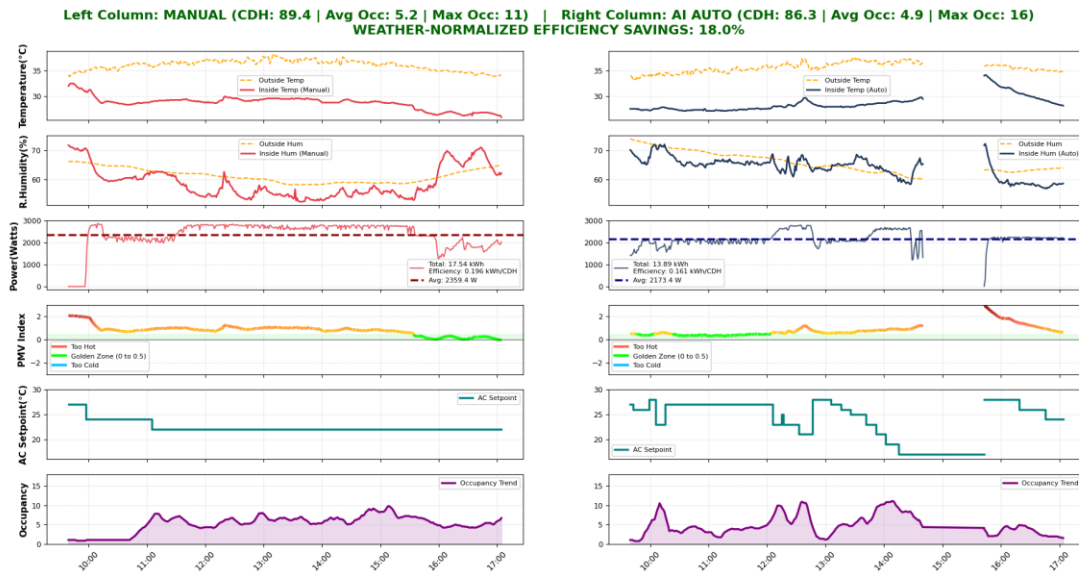


Figure 71. Performance Test F

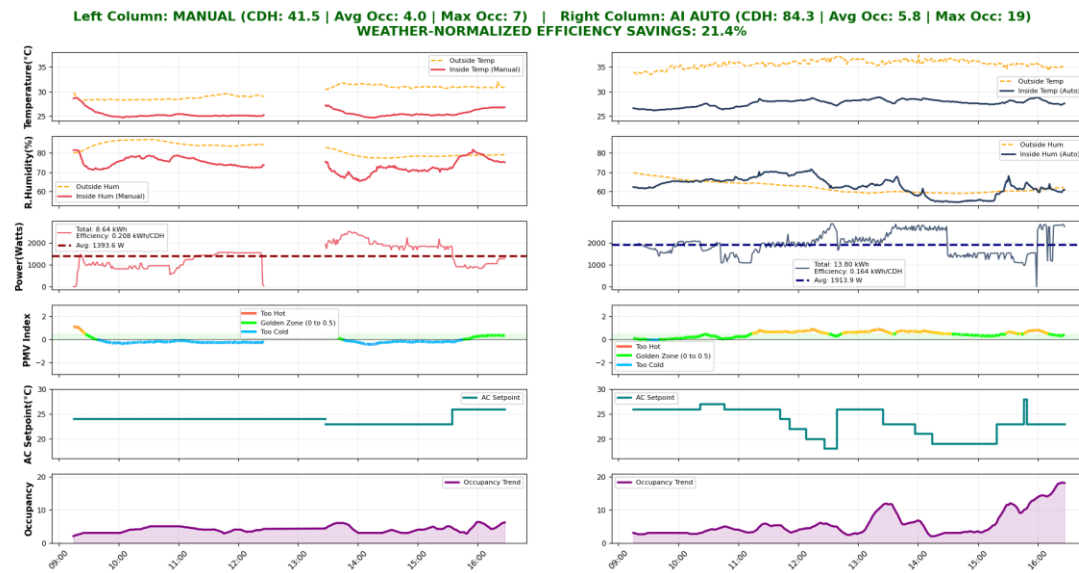


Figure 72. Performance Test G

Table 15 presents a systematic consolidation of the aggregated performance metrics for the BORF architecture across all deployment phases.

Table 15. Consolidation of hybrid system's energy saving performance

Test	BORF Energy Saving (%)	BORF Consumption (kWh)	BORF Cooling Degree Hour	Manual Baseline Efficiency (kWh/CDH)
Test A	7.3 %	12.14	78.4	0.167
Test B	25.5 %	10.29	66.3	0.208
Test C	15.1 %	14.42	86.5	0.196
Test D	38.3 %	7.63	56.5	0.219
Test E	21.5 %	12.81	78.3	0.208
Test F	18.0 %	13.89	86.3	0.196
Test G	21.4 %	13.80	84.3	0.208

- $Test A = 0.167 \frac{kWh}{CDH} \times 78.4 CDH = 13.0928 kWh$
- $Test B = 0.208 \frac{kWh}{CDH} \times 66.3 CDH = 13.7904 kWh$
- $Test C = 0.196 \frac{kWh}{CDH} \times 86.5 CDH = 16.954 kWh$
- $Test D = 0.219 \frac{kWh}{CDH} \times 56.5 CDH = 12.3735 kWh$
- $Test E = 0.208 \frac{kWh}{CDH} \times 78.3 CDH = 16.2864 kWh$
- $Test F = 0.196 \frac{kWh}{CDH} \times 86.3 CDH = 16.9148 kWh$
- $Test G = 0.208 \frac{kWh}{CDH} \times 84.3 CDH = 17.5344 kWh$

Total Manual Baseline Energy Consumption = 106.9463 kWh

Total B.O.R.F. Consumption = 84.98 kWh

Average BORF Energy Savings = $\frac{106.9463 kWh - 84.98 kWh}{106.9463 kWh} \times 100$

Average BORF Energy Savings = 20.5395%

To establish a mathematically rigorous comparison across all seven deployment phases (Test A through Test G), a weather-normalized baseline was synthesized. Because the BORF architecture and the manual baseline were evaluated under divergent meteorological conditions, direct energy comparisons are inherently flawed.

To isolate the algorithmic efficiency, the manual baseline's operational efficiency quantified in kilowatt-hours per Cooling Degree Hour (kWh/CDH) was multiplied by the specific CDH load experienced by the BORF system during each respective test. This normalization calculates the projected energy the manual system would have consumed had it been subjected to identical external boundary conditions. Aggregating these normalized values yields a total projected baseline consumption of 106.9463 kWh.

In contrast, the actual total energy expended by the BORF system across all testing phases was 84.98 kWh. By calculating the differential between the projected baseline and the actual automated consumption, the empirical data conclusively demonstrates that the BORF architecture achieves an aggregate, weather-normalized energy savings of 20.54%.

Drivers of Energy Savings

The observed energy savings were achieved by the Hybrid control system through external operational mode control and optimization strategies that balanced thermal comfort and energy consumption. The behavior and performance of the system were found to be consistent with established HVAC theories and principles which are discussed below.

Dynamic Actuation of Native HVAC Modes

The fundamental energy savings achieved by the BORF system stem from its strategic utilization of the AC unit's diverse operational modes such as Cool, Dry, Fan, and Eco. Because the machine learning architecture was trained across various matrices of these states, it algorithmically selects the exact mode optimized for the real-time thermal boundary conditions.

This is critical when addressing high-latent-load environments. For instance, in a humid scenario (e.g., $T_a = 32^\circ\text{C}$, $\text{RH} = 87\%$), forcefully applying the standard Cool mode lowers the dry-bulb temperature but drastically alters the psychrometric balance of the space. As dry-bulb temperature decreases, the volumetric capacity of the air to hold water vapor drops, causing the relative humidity to spike exponentially.

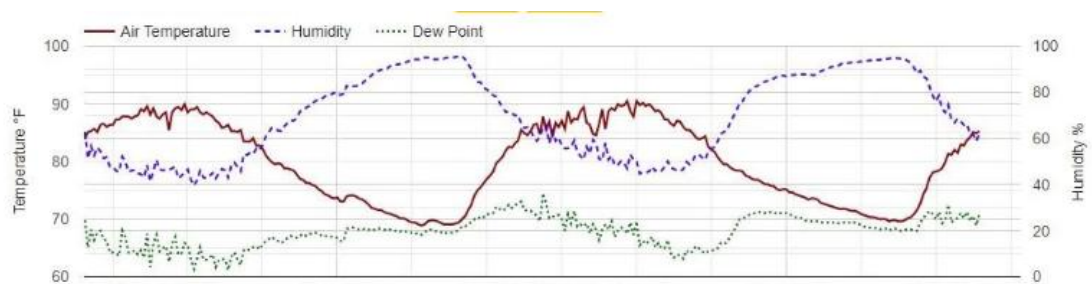


Figure 73. Relationship of Temperature and Relative Humidity

While the ambient air becomes physically colder (sensible cooling), the saturated air mass fundamentally inhibits the occupants' primary physiological cooling mechanism: the evaporation of perspiration. Because the saturated environment cannot absorb additional latent heat, occupants will continue to feel damp and uncomfortably warm, prompting human operators to lower the thermostat further and waste immense energy.

The BORF system recognizes this psychrometric limitation. By autonomously shifting to Dry mode, the system prioritizes latent heat removal (dehumidification) over sensible cooling, successfully achieving thermal equilibrium using significantly less compressor energy.

At an identical temperature setpoints, Dry mode inherently consumes less electrical power than standard Cool mode due to the native PID logic governing the inverter system. When dehumidification is triggered, the controller restricts the compressor's operational frequency (RPM) to a partial-load state, lowering the evaporator coil temperature just marginally below the ambient dew point. Simultaneously, the indoor blower fan is modulated to a minimum velocity. This reduced airflow increases the residence time of the air mass across the heat exchanger, maximizing the condensation rate of water vapor. This mechanical process aggressively extracts moisture (latent heat removal) while yielding a moderate drop in dry-bulb temperature. By significantly reducing the humidity ratio of the room, the system exponentially increases the air's volumetric capacity to absorb subsequent biological latent heat, achieving physiological comfort at a fraction of the energy cost.

The comparative telemetry in Figure 74 visually confirms the systemic differences between 'Cool' and 'Dry' operational modes. In the left panel (red trendline), the AC unit is locked in standard Cool mode.

The graph clearly illustrates that as the system drives the indoor dry-bulb temperature down, the indoor relative humidity sharply increases, surpassing even the unconditioned outdoor humidity (yellow dashed line). This perfectly demonstrates how

cooling without dehumidification creates a saturated, damp environment. In contrast, the right panel (blue trendline) demonstrates the BORF system utilizing Dry mode.

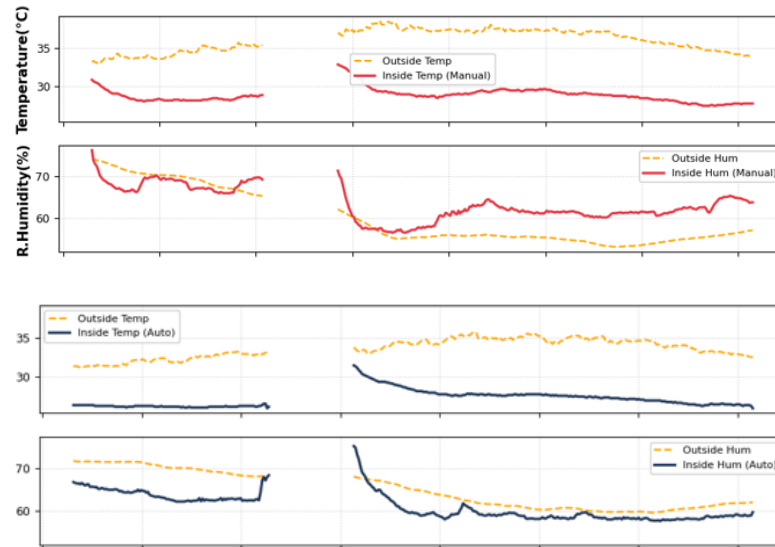


Figure 74. Effects of Cool and Dry Modes on Humidity

Under these settings, the system successfully manages the latent heat load, maintaining it just below the outside relative humidity, cooling the space without inducing the severe humidity spikes observed during manual operation.

Conversely, deploying Dry mode under boundary conditions characterized by high ambient temperatures, dense occupancy, and low-to-moderate relative humidity (e.g., $T_a = 34^{\circ}\text{C}$, $\text{RH} = 62\%$) yields diminishing thermodynamic returns. Because Dry mode is fundamentally engineered to prioritize latent heat removal, attempting to mechanically extract moisture from an already unsaturated air mass requires a disproportionate expenditure of compressor energy. Under such high-load scenarios, Cool mode becomes the optimal actuation strategy.

Cool mode strictly prioritizes sensible cooling, designed to rapidly depress the dry-bulb temperature. Crucially, unlike the partial-load restrictions inherent to dehumidification, Cool mode imposes no artificial limitations on the inverter's rotational frequency. The native PID controller commands the compressor to operate at peak RPM, executing a rapid thermal 'pull-down' of the space. As the ambient temperature approaches the target setpoint, the PID logic actively modulates the frequency downward to prevent thermal overshoot. Therefore, when confronting a massive sensible heat load driven by extreme outdoor temperatures and high occupant density, the BORF system intelligently selects Cool mode to maximize the sensible cooling capacity and swiftly restore thermal equilibrium.

Another critical feature of modern inverter HVAC architecture is the 'Eco' (Economy) mode. In this operational state, the internal PID controller imposes a strict governor on the compressor's rotational frequency, capping it at approximately 50% to 60% of its maximum capacity, while simultaneously maximizing the indoor blower fan speed to optimize convective heat transfer and spatial air distribution. This represents the unit's peak thermodynamic efficiency state. Unlike 'Cool' mode, which aggressively expends maximum power to execute rapid sensible temperature pull-downs, or 'Dry' mode, which targets latent heat removal, Eco mode is engineered strictly for steady-state thermal maintenance. It does not forcefully pursue setpoint deviations; rather, it provides a stabilized, low-capacity sensible cooling output.

Consequently, this mode is thermodynamically ideal for scenarios where the conditioned space has already achieved thermal equilibrium. The BORF architecture strategically prioritizes Eco mode precisely when the room stabilizes at optimal thermal

neutrality ($PMV \approx 0$). By deploying this mode at equilibrium, the AI restricts the AC unit to producing only the exact cooling capacity required to offset transient heat infiltration. This prevents inefficient overcooling and mitigates parasitic power draw, effectively leveraging the room's established thermal inertia to sustain occupant comfort at an absolute minimum energy cost.

Partial Loading of Air-Conditioning System

To further substantiate the energy-saving mechanisms of the Hybrid BORE System, it is critical to examine the thermodynamic behavior of modern air conditioning units under varying load conditions.

As demonstrated in the referenced study by Tukura et al., the Coefficient of Performance (COP) for split-type inverter systems does not scale linearly with operational load. Specifically, the empirical data indicates that peak thermodynamic efficiency is achieved at a partial-load ratio of approximately 50%. Deviating significantly from this optimal mid-range threshold yields a marked degradation in the system's COP. Significantly, the literature highlights a highly non-linear relationship between load ratio and active power consumption; operating an inverter unit at 50% capacity demands only roughly 30% to 35% of the electrical power required at near-maximum (91%) capacity. Consequently, forcing an AC unit to operate at full power not only severely diminishes its cooling efficiency but also accelerates mechanical wear on the compressor, prematurely reducing the hardware's lifespan.

By algorithmically restricting the AC to operate near this optimal partial-load state (such as through the actuation of Eco mode and the gradual temperature

adjustments), the BORF system inherently capitalizes on this non-linear efficiency curve to maximize COP and eliminate wasteful power spikes.

Table 1. Power consumption of the two AC

Part load ratio	T _{outside} , °C	P _{inverter} (1), W	P _{non-inverter} (2), W	(1)-(2)	((1)-(2)) / (2)
25%	28	150	311	161	52%
	30	170	321	151	47%
	32	190	332	142	43%
	34	210	342	132	39%
50%	28	312	570	258	45%
	30	343	592	250	42%
	32	373	614	241	39%
	34	403	636	233	37%
75%	28	703	837	134	16%
	30	757	887	129	15%
	32	812	936	124	13%
	34	866	986	120	12%
91%	28	1004	1115	111	10%
	30	1018	1148	130	11%
	32	1033	1181	148	13%
	34	1047	1214	167	14%

Table 2. COP of the two AC

Part load ratio	T _{outside} (°C)	COP _{inverter} (1)	COP _{non-inverter} (2)	(1)-(2)	((1)-(2)) / (2)
25%	28	4.78	2.34	2.44	104%
	30	4.72	2.50	2.22	89%
	32	4.67	2.67	2.01	75%
	34	4.62	2.83	1.79	63%
50%	28	5.17	2.85	2.32	82%
	30	4.94	2.85	2.09	73%
	32	4.71	2.86	1.85	65%
	34	4.48	2.87	1.61	56%
75%	28	3.54	2.95	0.60	20%
	30	3.43	2.89	0.55	19%
	32	3.32	2.83	0.49	17%
	34	3.22	2.77	0.44	16%
91%	28	3.10	2.85	0.25	9%
	30	3.08	2.81	0.27	10%
	32	3.07	2.78	0.29	11%
	34	3.05	2.74	0.31	11%

Figure 75. COP of ACU Across Various Compressor Loads

Consequently, the operational efficacy of the BORF architecture is inextricably linked to the mechanical capacity of the native HVAC hardware. Since the system functions exclusively as an exogenous supervisory controller transmitting high-level setpoints via infrared (IR) signals without direct hardware integration, it cannot transcend the physical limitations of an inherently under-designed AC unit. If the primary cooling infrastructure lacks the volumetric capacity to manage the space's maximum thermal load, the BORF algorithm's dual objectives of energy conservation

and thermal comfort are rendered unachievable. Therefore, this evaluation concludes that optimal system performance necessitates a meticulously sized AC unit. Avoiding both under-designed (capacity-deficient) and over-designed (excessive capacity) configurations allows the machine learning algorithm to strategically maintain the compressor near its peak thermodynamic efficiency point of 50% partial load, thereby maximizing verifiable energy savings.

Performance Assessment of the Digital Twin

The performance assessment of the Digital Twin was conducted through a comprehensive framework consisting of baseline evaluation, Bayesian calibration, independent validation, and real-time operational deployment. The baseline model demonstrated a reasonable ability to reproduce indoor temperature and relative humidity trends but exhibited substantial discrepancies in HVAC energy consumption, highlighting the need for calibration.

To improve the predictive accuracy, key building and HVAC parameters were optimized using a Bayesian calibration approach constrained by physically realistic boundaries and evaluated using ASHRAE Guideline 14 statistical metrics. The calibrated model was subsequently validated against an independent dataset not used during optimization, where it successfully met the prescribed performance criteria, demonstrating its ability to generalize beyond the calibration period and accurately represent the thermodynamic behavior of the physical conference room.

Finally, the validated model was integrated into a real-time Digital Twin platform that continuously synchronized measured and simulated data, enabling live

monitoring of energy consumption, indoor environmental conditions, and thermal comfort. Collectively, these assessments confirmed that the Digital Twin provides a reliable representation of the conference room's thermal and energy performance, supporting both virtual parallel assessment and intelligent HVAC operation.

Baseline Model Assessment Prior to Bayesian Calibration

The baseline EnergyPlus model was evaluated prior to calibration to establish a reference for assessing the effectiveness of the Bayesian optimization process. Results showed that the model could reasonably reproduce the indoor temperature and relative humidity trends of the conference room; however, significant discrepancies were observed in HVAC energy consumption, indicating that the initial assumptions regarding HVAC performance, airflow, and internal loads did not adequately represent actual operating conditions.

Statistical analysis revealed that HVAC energy consumption failed to satisfy the ASHRAE Guideline 14 acceptance criteria, with simulated energy consistently underpredicting measured consumption. Correlation analysis further confirmed weak agreement between measured and simulated energy use, while temperature and relative humidity exhibited stronger but still imperfect correspondence.

These findings demonstrate that, although the baseline model captured the general thermal and moisture behavior of the space, calibration was necessary to improve its ability to accurately quantify HVAC energy consumption and support reliable digital twin deployment.

Times Series Assessment of Baseline Model

The baseline EnergyPlus model was initially evaluated to determine its ability to reproduce the measured HVAC energy consumption and indoor environmental conditions of the conference room prior to calibration. Establishing baseline performance is an essential step in Building Energy Modeling because discrepancies between simulated and measured performance commonly arise from uncertainties associated with HVAC operation, internal loads, occupancy behavior, and other model assumptions. The resulting performance therefore provides a reference against which the effectiveness of the Bayesian calibration process can be assessed.

These findings indicate that the baseline model provided an acceptable representation of the conference room's thermal behavior but remained unsuitable for reliable HVAC energy quantification as shown in Table 16. Consequently, calibration became necessary to reduce uncertainty in key HVAC and internal-load parameters and improve agreement between the physical conference room and its virtual counterpart while maintaining the environmental conditions required for virtual parallel assessment.

Table 16. Baseline Performance Metrics

Metrics	CV(RMSE)	NMBE
Hourly AC Energy (kWh)	54.125357	40.461157
Indoor Temperature (°C)	16.821629	-9.770116
Indoor Relative Humidity (%)	12.793794	-0.865266

The greatest discrepancy was observed in HVAC energy consumption, which served as the primary calibration objective of this study. The baseline model produced

a CV(RMSE) of 54.13% and an NMBE of 40.46%, substantially exceeding the ASHRAE Guideline 14 acceptance limits for hourly calibration.

As illustrated in Figure 76, the simulated HVAC energy consumption consistently remained below the measured values throughout most of the monitoring period. Although the model was able to reproduce the general pattern of cooling demand, it failed to accurately capture the magnitude of electricity consumption required by the physical system.

The observed underprediction suggests that the baseline assumptions regarding cooling-system performance, equipment efficiency, airflow characteristics, and internal heat gains did not adequately represent the actual operating conditions of the conference room.

This finding is consistent with the observations of Coakley et al. (2014), who identified specification uncertainty and operational uncertainty as major contributors to discrepancies between measured and simulated building performance. The results therefore confirm the existence of a performance gap between the initial Building Energy Model and the physical building, a condition commonly reported in Digital Twin and Building Energy Modeling studies.

Indoor environmental conditions exhibited comparatively better agreement with measured observations. Indoor air temperature achieved a CV(RMSE) of 16.82% and an NMBE of -9.77% , while indoor relative humidity produced a CV(RMSE) of 12.79% and an NMBE of -0.87% . As shown in Figure 77, the simulated temperature profile generally followed the measured thermal trend throughout the monitoring

period. However, simulated temperatures remained consistently higher than measured values, indicating that the baseline model tended to overpredict indoor thermal conditions despite reproducing the overall thermal response of the space.

A similar behavior was observed for indoor relative humidity. Figure 78 shows that the simulated humidity profile reproduced the general variation in moisture conditions and remained within a realistic operating range. The relatively low NMBE value indicates minimal overall bias; however, noticeable deviations remained during several operating periods, suggesting that the baseline model did not fully capture transient moisture dynamics occurring within the conference room.

The differing performance between HVAC energy consumption and indoor environmental conditions is an important observation in the context of Digital Twin development. While the baseline model demonstrated a reasonable capability to represent the thermal and moisture behavior of the space, it was unable to accurately quantify the energy required to maintain those conditions. This distinction supports the findings of Ramírez et al. (2025), who emphasized that successful reproduction of indoor environmental variables should not automatically be interpreted as evidence of accurate energy-consumption prediction. Their work highlighted that additional HVAC-focused calibration is necessary when energy consumption becomes the primary performance objective of a Digital Twin.

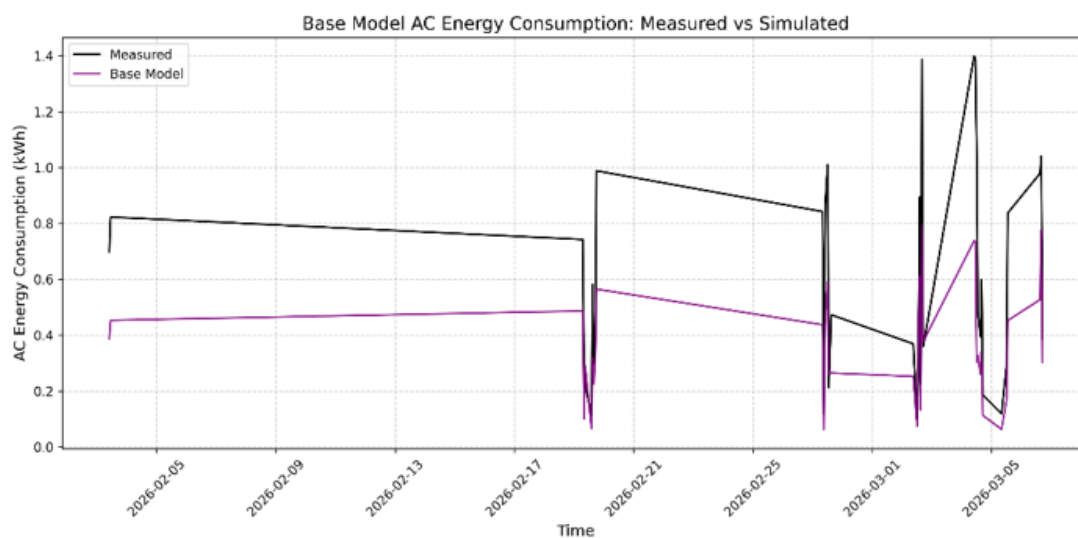


Figure 76. Simulated vs Measured of Baseline AC Energy Consumption

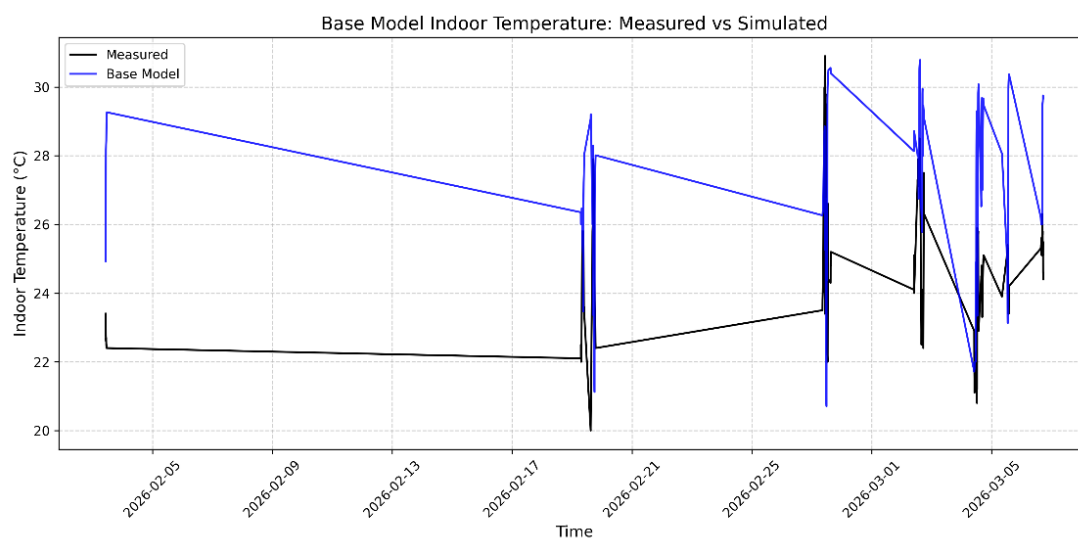


Figure 77. Simulated vs Measured of Baseline Indoor RH

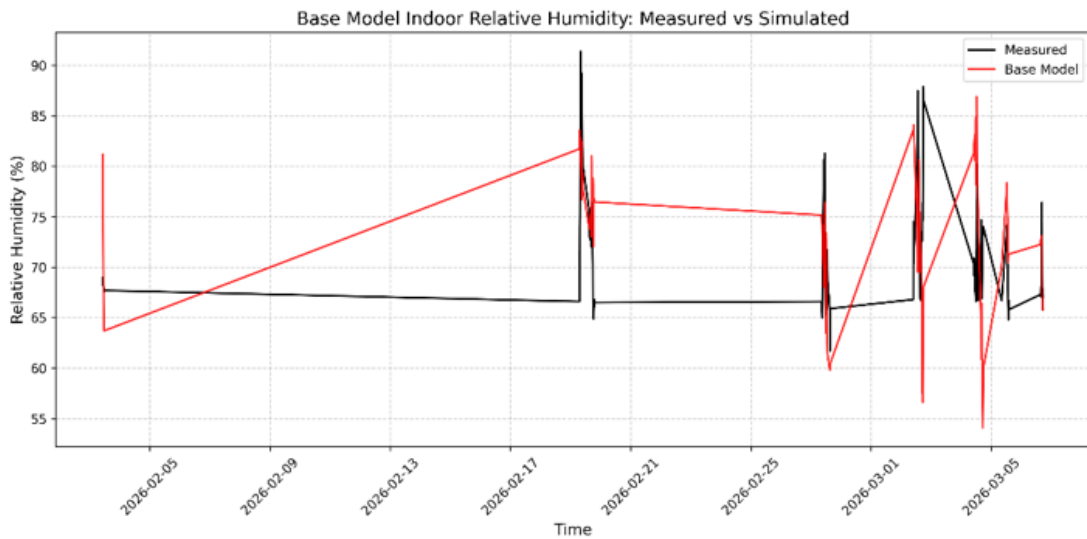


Figure 78. Simulated vs Measured of Baseline Indoor Temperature

Correlation Analysis of Baseline Model

The correlation analysis provides additional evidence supporting the observations obtained from the preceding time-series assessment of the baseline model. While the time-series plots illustrate the temporal agreement between measured and simulated variables, the scatter plots evaluate the strength of the relationship between the two datasets and the distribution of prediction errors across the monitoring period.

As shown in Figure 79, HVAC energy consumption exhibited the weakest agreement among the evaluated variables. The baseline model produced a coefficient of determination of $R^2 = 0.3950$, indicating a relatively weak correlation between measured and simulated cooling-energy consumption. Although a positive linear relationship is evident, the observations remain widely dispersed from the 1:1 reference line, particularly at higher cooling-energy levels. This distribution further confirms the

inability of the baseline model to accurately reproduce the measured HVAC energy behavior identified in the time-series assessment.

The temperature correlation plot presented in Figure 80 similarly reflects the trend observed in the previous assessment. Most observations are distributed above the 1:1 reference line, indicating that the simulated temperatures were generally higher than the measured values. While the overall distribution demonstrates that the model captured the general thermal response of the conference room, the displacement from the parity line confirms the temperature offset observed in the time-series comparison.

A comparable pattern is evident for indoor relative humidity, as illustrated in Figure 81. The observations remain centered around the 1:1 reference line but exhibit noticeable dispersion throughout the measured humidity range. This scatter distribution is consistent with the time-series results, where the simulated humidity profile generally followed the measured trend while exhibiting localized deviations during certain operating periods. Despite these differences, the overall distribution indicates that the baseline model preserved the dominant moisture behavior of the space.

The correlation results therefore reinforce the conclusions drawn from the baseline time-series assessment. While the model demonstrated a reasonable ability to represent the general thermal and moisture conditions within the conference room, the weak correlation observed for HVAC energy consumption confirms that the baseline model was not sufficiently representative of the actual cooling-energy behavior of the physical system. These findings further support the need for Bayesian calibration to improve the predictive capability of the Digital Twin prior to validation and operational deployment.

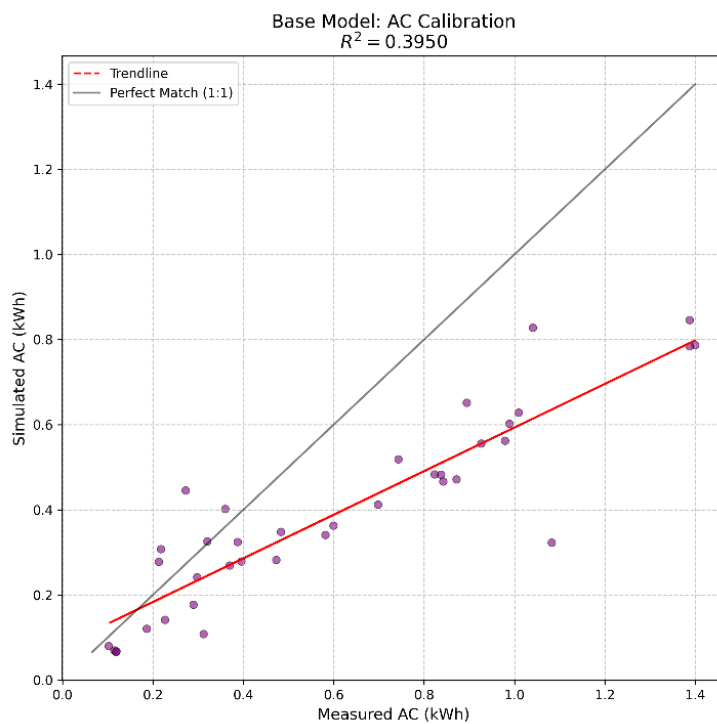


Figure 79. Correlation Analysis of Baseline AC Energy Consumption

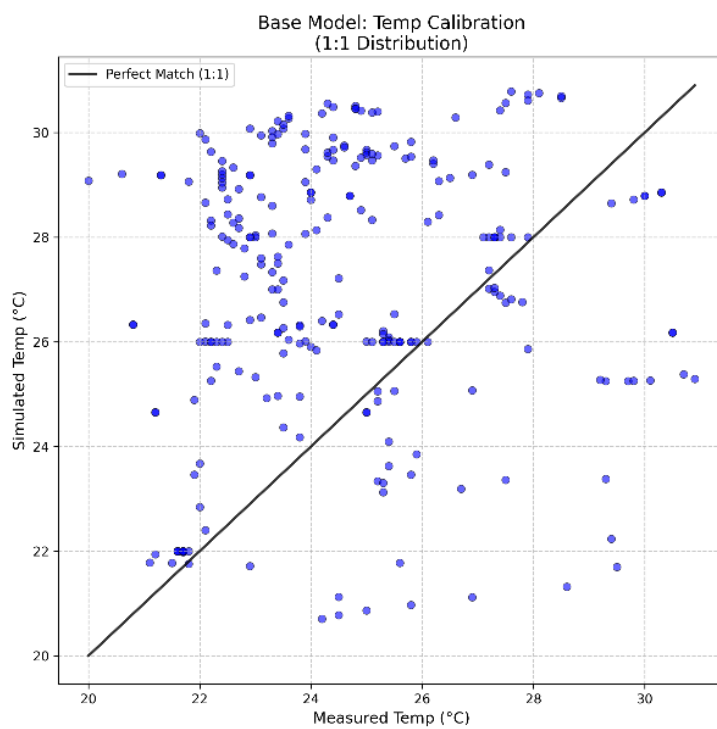


Figure 80. Correlation Analysis of Baseline Indoor Temperature

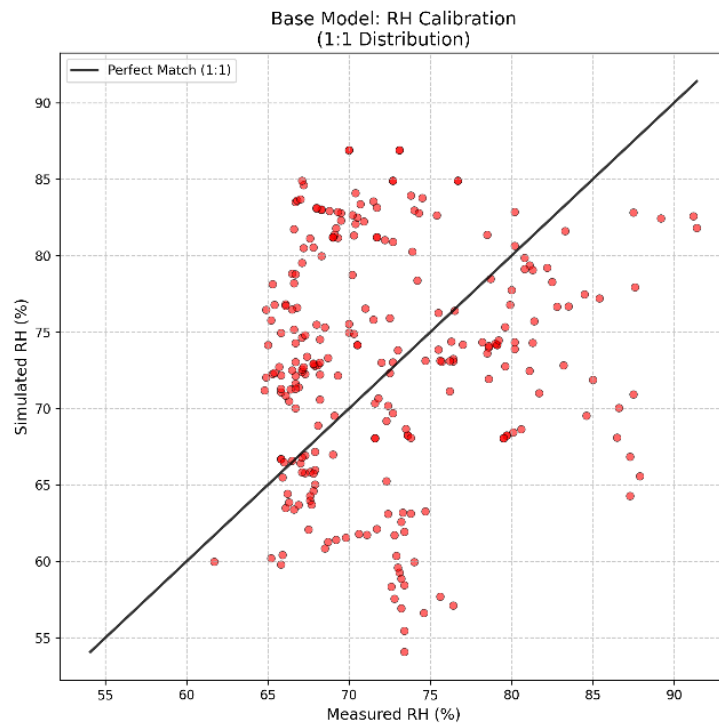


Figure 81. Correlation Analysis of Baseline Indoor RH

Performance of the Calibrated Model

Following the baseline-model assessment discussed in the previous section, Bayesian optimization was implemented to reduce the discrepancy between measured and simulated HVAC energy consumption while preserving the thermal and moisture behavior of the conference room. Table 17 presents the ten highest-performing parameter combinations identified during the calibration process. The results indicate a high degree of consistency among the top-ranked solutions, with only minor variations observed across both the optimized parameter values and the resulting performance metrics.

Table 17. Ten Highest-Performing Parameter Combination

Iteration	Lighting_ Multiplier	Equipment_ Multiplier	COP	Occupancy_ Multiplier	Cooling_ Capacity_ Multiplier	Fan_ Flow_ Multiplier
14	0.880286324	0.800840098	2.700098279	0.999173656	1.406931355	0.572203977
12	0.88	0.801	2.7	0.999	1.407	0.563294365
15	0.08	0.08	2.7	1.0	1.407	0.555656004
34	0.88	0.8	2.7	0.999	1.407	0.552505953
33	0.88	0.8	2.7	1.0	1.407	0.551803492
22	0.08	0.08	2.7	0.999348412	1.407	0.551452637
22	0.88	0.8	2.7	0.999875718	1.407	0.551206529
38	0.88	0.8	2.7	1.0	1.407	0.550664312
39	0.88	0.8	2.7	1.0	1.407	0.550623175
20	0.88	0.8	2.7	0.999	1.407	0.550613149

A notable characteristic of the optimization results is the convergence of the top-performing solutions toward a narrow region of the parameter space. The coefficient of performance (COP) remained approximately 2.70 across all top-ranked models, while the cooling-capacity multiplier converged near 1.407. Similarly, the occupancy multiplier remained essentially equal to unity, and the fan-flow multiplier converged within a relatively narrow range of approximately 0.55–0.57. The lighting and equipment multipliers likewise stabilized near 0.88 and 0.80, respectively. The limited variation observed among the top ten solutions indicates that the optimization process repeatedly identified similar parameter combinations capable of reproducing the measured building response. This convergence behavior suggests that the Bayesian calibration framework successfully reduced parameter uncertainty and consistently identified a stable solution space, which is a commonly reported outcome in Bayesian calibration studies (Heo et al., 2012; Coakley et al., 2014).

The improvement achieved through calibration was most evident in HVAC energy-consumption performance, which served as the primary calibration objective of the study. The highest-ranked solution, obtained during Iteration 14, achieved an hourly AC CV(RMSE) of 29.96% and an NMBE of 9.96% as shown in Table 18, satisfying the hourly acceptance criteria specified in ASHRAE Guideline 14. Compared with the baseline model discussed in Section 4.1, which produced a CV(RMSE) of 54.13% and an NMBE of 40.46%, the calibrated model substantially reduced prediction error and improved agreement between measured and simulated cooling-energy consumption. The remaining top-ranked solutions produced nearly identical performance metrics, further demonstrating the stability of the calibrated parameter region.

This substantial reduction in error is consistent with the purpose of model calibration, which seeks to improve agreement between measured and simulated performance by refining uncertain model parameters (Fabrizio & Monetti, 2015; Coakley et al., 2014).

The supporting indoor environmental indicators remained relatively consistent throughout the optimization process. Temperature CV(RMSE) values ranged from 17.11% to 17.12%, while relative humidity CV(RMSE) values varied only between 12.73% and 12.74%. Similarly, temperature NMBE values ranged from -8.55% to -8.52%, whereas relative humidity NMBE values remained between -0.36% and -0.31%. The limited variation among these metrics indicates that the optimization process primarily improved HVAC energy-consumption performance while maintaining a stable representation of the thermal and moisture conditions within the conference room. This outcome is desirable because improvements in energy prediction

were achieved without significantly altering the indoor environmental behavior represented by the model, supporting the requirements of virtual parallel assessment and Digital Twin development (Ramírez et al., 2025).

Table 18. Ten Highest-Performing Metrics

Iteration	Temp_ CVRMSE	Temp_ NMBE	RH_ CVRMSE	RH_ NMBE	AC_ CVRMSE	AC_ NMBE	Score
14	17.123189	-8.515185	12.734114	-0.360153	29.964546	9.956472	19.878327
12	17.118400	-8.527817	12.735483	-0.338994	29.964915	9.932100	19.876625
15	17.114136	-8.538364	12.737102	-0.322230	29.967832	9.918336	19.875987
34	17.112468	-8.542776	12.737962	-0.315395	29.969387	9.912658	19.875917
33	17.112098	-8.543755	12.738167	-0.313896	29.969754	9.911421	19.875912
22	17.111914	-8.544244	12.738271	-0.313149	29.969941	9.910807	19.875911
22	17.111785	-8.544586	12.738344	-0.312626	29.970072	9.910378	19.875911
38	17.111502	-8.545346	12.738512	-0.311470	29.970365	9.909437	19.875914
39	17.111481	-8.545404	12.738525	-0.311383	29.970388	9.909366	19.875914
20	17.111476	-8.545417	12.738528	-0.311362	29.970393	9.909348	19.875914

The time-series comparison between measured observations, the baseline model, and the calibrated model further illustrates the improvement achieved through Bayesian calibration.

As shown in Figures 82–84, the calibrated model exhibited noticeably closer agreement with the measured HVAC energy-consumption profile than the baseline model throughout the monitoring period. The calibrated energy profile successfully reproduced the general magnitude and variation of cooling demand, reducing the systematic underprediction observed in the baseline model. The improvement is particularly evident during periods of elevated cooling demand, where the calibrated model more closely followed the measured energy-consumption trends.

In contrast, the temperature and relative humidity profiles exhibited comparatively smaller changes following calibration. The calibrated temperature response remained similar to the baseline model and continued to overpredict measured indoor temperatures, while the calibrated relative humidity profile showed only minor deviations from the baseline results.

These observations are consistent with the adopted calibration strategy, which prioritized HVAC energy-consumption performance as the primary optimization objective. Consequently, the most substantial improvement was achieved in HVAC energy prediction, while the overall thermal and moisture behavior of the conference room was preserved. The time-series results therefore provide visual confirmation that the calibrated parameter set improved the Digital Twin's ability to represent the operational energy behavior of the physical conference room while maintaining realistic indoor environmental conditions.

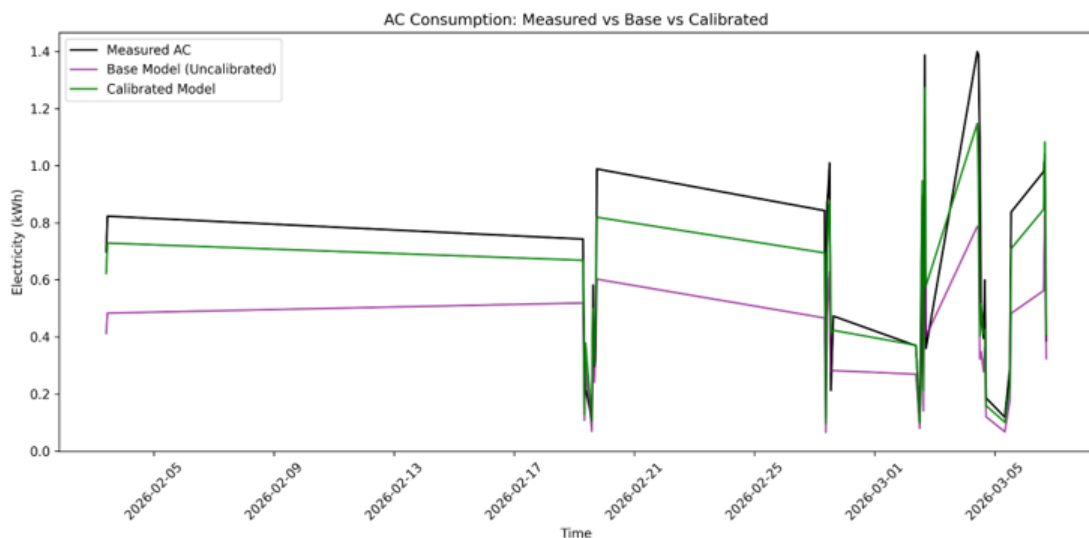


Figure 82. Simulated vs Measured of Calibrated AC Energy Consumption

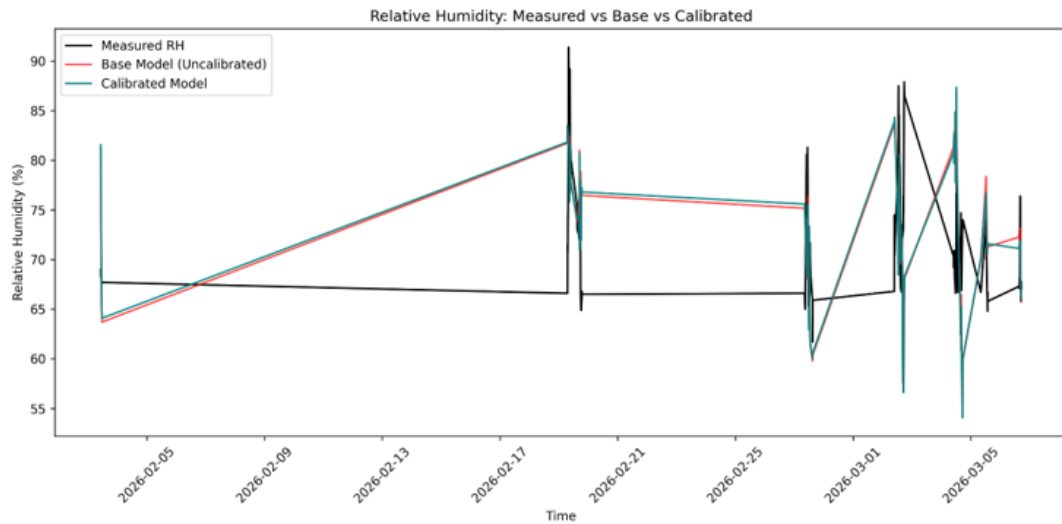


Figure 83. Simulated vs Measured of Calibrated Indoor RH

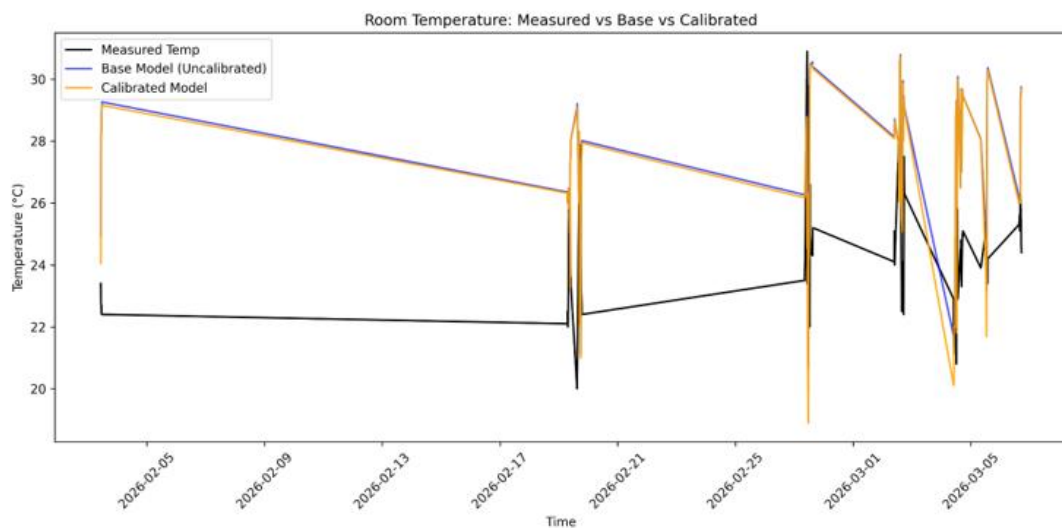


Figure 84. Simulated vs Measured of Calibrated Indoor Temperature

To better understand the physical implications of the calibration process, Table 19 compares the baseline assumptions with the final parameter values obtained from the champion model. The optimized parameter set provides insight into the adjustments required for the EnergyPlus model to reproduce the measured operational behavior of the conference room.

Table 19. Interpretation Baseline Assumptions and Final Parameter Values

Calibration Parameter	Baseline Value	Champion Model Value	Physical Interpretation
Lighting Multiplier	1.00	0.88	Lower lighting heat gains
Equipment Multiplier	1.00	0.80	Lower plug-load heat gains
Occupancy	1	0.999173656	Occupancy nearly unchanged
COP	3.20	2.70	Lower HVAC efficiency
Cooling Capacity Multiplier	1.00	1.407	Higher effective cooling capacity
Fan Flow Multiplier	1.00	0.572	Reduced operational airflow

The lighting multiplier decreased from 1.00 to approximately 0.88, while the equipment multiplier decreased from 1.00 to approximately 0.80 as shown in Table 19. These adjustments correspond to a reduction in effective lighting power density from 3.00 W/m² to approximately 2.64 W/m² and a reduction in equipment load from 200 W to approximately 160 W as shown in Table 20.

The optimized values indicate that lower internal heat gains provided improved agreement between measured and simulated performance. Because lighting and plug loads contribute directly to the sensible cooling demand of the conditioned space, adjustments to these parameters influenced the cooling load predicted by the simulation and improved correspondence with the measured HVAC energy-consumption profile. Similar findings have been reported by Raftery et al. (2011) and Coakley et al. (2014), who identified internal loads as among the most influential parameters affecting cooling-energy calibration.

The calibrated coefficient of performance decreased from the baseline value of 3.20 to approximately 2.70. This adjustment indicates that the original HVAC efficiency assumption did not adequately reproduce the measured cooling-energy behavior and that a lower effective cooling-system efficiency improved agreement with the observed operational data. Consequently, the calibrated model required greater electrical energy input to produce the cooling output necessary to match the measured HVAC energy consumption. Comparable reductions in effective HVAC performance have been reported in calibration studies where actual operational conditions differ from nominal equipment specifications (Heo et al., 2012; O'Neill et al., 2013).

In contrast, the cooling-capacity multiplier increased from 1.00 to approximately 1.407, corresponding to an effective cooling capacity of approximately 3518 W compared with the baseline value of 2500 W. The increase indicates that a greater effective cooling capacity was required for the simulation to reproduce the measured cooling response of the conference room. The optimized value improved the model's ability to represent variations in cooling demand and indoor-conditioning behavior throughout the calibration period. Similar adjustments have been observed in calibrated building energy models where design assumptions differ from actual operating conditions (Raftery et al., 2011).

Among the optimized parameters, occupancy exhibited the smallest adjustment. The occupancy multiplier remained essentially unchanged, converging to approximately 0.999. This result indicates that the measured occupancy schedules incorporated into the simulation were already representative of the actual room utilization and therefore required minimal modification during calibration. The result

further supports the reliability of the measured occupancy data used as an input to the Digital Twin framework.

Similarly, the fan-flow multiplier converged to approximately 0.572. Because the EnergyPlus PTAC system was configured using auto-sized airflow rates, the calibrated fan-flow multiplier modified the auto-sized cooling supply airflow calculated by the simulation rather than directly prescribing a fixed airflow value. The resulting effective cooling supply airflow rate was approximately 0.0915 m³/s. This adjustment improved agreement between measured and simulated HVAC performance by modifying the cooling-air delivery characteristics of the system. Since airflow directly affects heat transfer, sensible cooling delivery, and electrical energy consumption, the optimized airflow conditions contributed to reproducing the measured operational behavior of the conference room.

The Iteration 14 solution achieved the lowest objective-function value among all evaluated parameter combinations and was therefore selected as the champion model for the subsequent calibration assessment, independent validation, and operational deployment stages of the Digital Twin framework.

Table 20. Physical Value

Resulting Model Characteristic	Baseline Model	Calibrated Model
Lighting Power Density	3.00 W/m ²	2.64 W/m ²
Equipment Load	200 W	160 W
COP	3.20	2.70
Design Occupancy	24 persons	~24 persons
Cooling Capacity	2500 W	~3517 W
Effective Cooling Supply Air Flow Rate*	Auto-sized	0.09152 m ³ /s

Correlation Analysis of Calibrated Model

The improvements identified in the calibration assessment are further examined through the correlation plots shown in Figures 85-87. As a continuation of the preceding analysis, the scatter plots provide additional evidence of the agreement achieved between measured and simulated performance using the champion parameter set and illustrate the extent to which the calibration process improved the representation of the physical conference room.

The most evident improvement can be observed in the HVAC energy-consumption correlation plot shown in Figure 85. Relative to the baseline model discussed in Section 4.1, the calibrated model exhibits substantially closer clustering of observations around the 1:1 reference line.

The coefficient of determination increased from $R^2 = 0.3950$ to $R^2 = 0.7843$, indicating a considerably stronger relationship between measured and simulated cooling-energy consumption. The reduced scatter and improved alignment with the parity line confirm the improvement in HVAC energy prediction previously identified through the calibration metrics. Similar improvements in correlation following calibration have been reported in Building Energy Model calibration studies, where refinement of uncertain HVAC and operational parameters resulted in closer agreement between measured and simulated energy performance (Coakley et al., 2014; Heo et al., 2012).

The indoor air temperature correlation plot presented in Figure 86 exhibits a distribution generally consistent with the thermal behavior discussed in the previous

subsection. Although the observations remain concentrated around the 1:1 reference line, a noticeable proportion of points continues to lie above the parity line, indicating that the calibrated model retained a tendency to predict slightly warmer indoor conditions than those measured in the physical space. Compared with the baseline results, the distribution exhibits a modest reduction in deviation from the reference line; however, temperature prediction uncertainty remained evident throughout the calibration period.

A similar trend is observed in the indoor relative humidity correlation plot shown in Figure 87. The scatter distribution remains centered around the 1:1 reference line and preserves the general moisture-response characteristics observed in the measured data. While variability remains visible across portions of the humidity range, the overall distribution exhibits minimal directional bias and remains consistent with the relative-humidity metrics discussed in the calibration assessment. The results indicate that the improvements achieved in HVAC energy prediction did not compromise the model's ability to represent indoor moisture behavior.

Taken together, the correlation plots support the conclusions drawn from the calibration assessment. The most substantial improvement occurred in HVAC energy consumption, which served as the primary calibration objective, while indoor temperature and relative humidity maintained reasonable agreement with measured observations. The resulting distributions indicate that the calibrated model achieved a closer representation of the measured building performance and provided a suitable foundation for the independent validation stage presented in the following section.

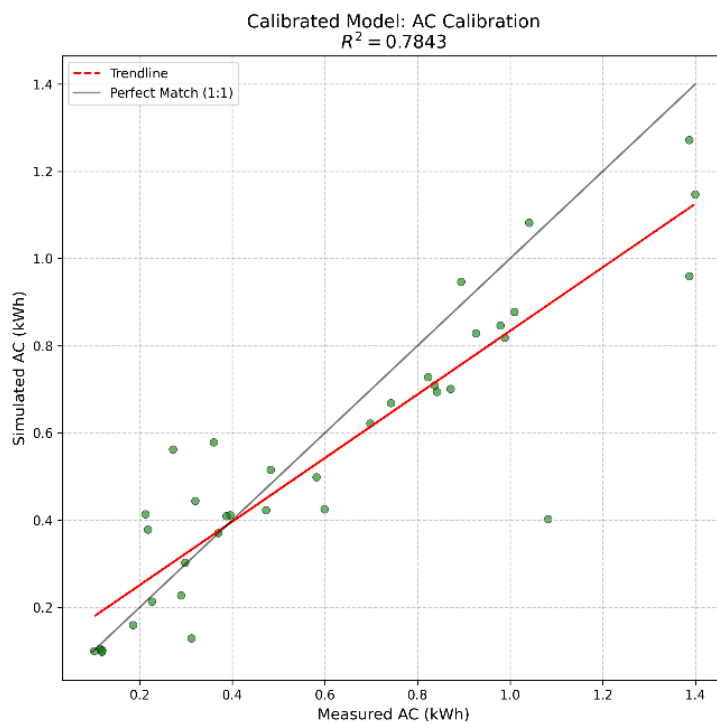


Figure 85. Correlation Analysis of Calibrated AC Energy Consumptions

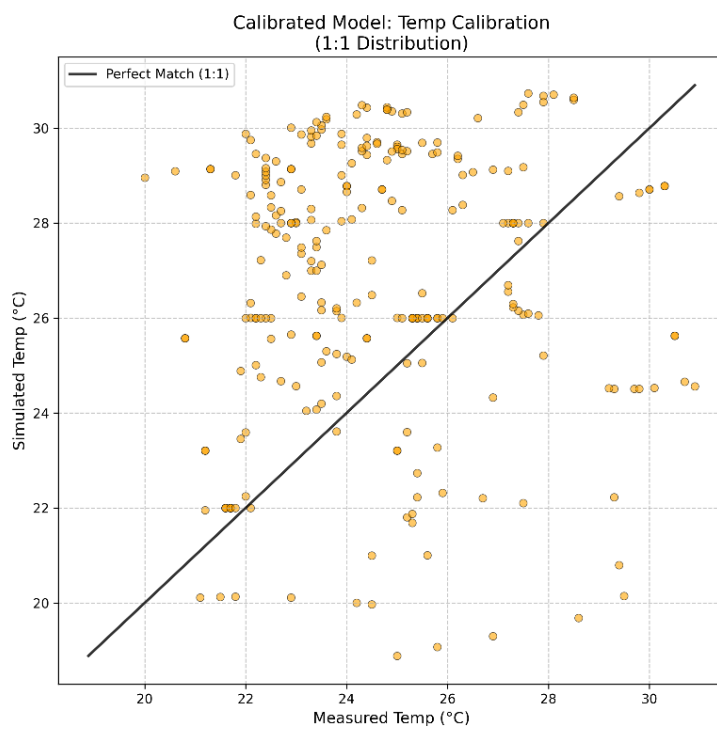


Figure 86. Correlation Analysis of Calibrated Indoor Temperature

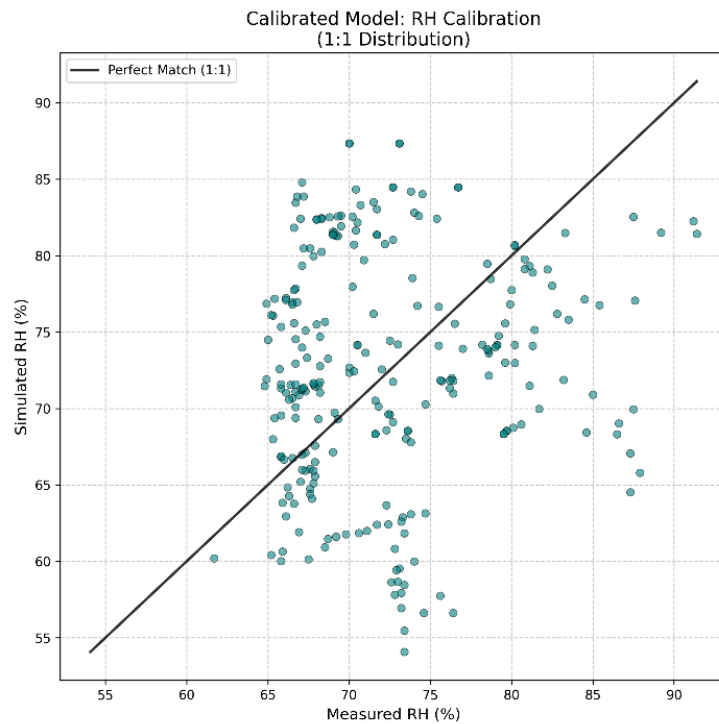


Figure 87. Correlation Analysis of Calibrated Indoor RH

Residual Error Analysis of the Calibrated Model

The residual error distributions shown in Figure 88 provide a complementary assessment of the calibrated model and further support the findings obtained from the calibration metrics and correlation analyses discussed in the preceding subsections. By examining the distribution of the remaining prediction errors, the residual plots provide additional insight into the consistency of the champion parameter set across the calibration dataset.

Consistent with the improvements previously observed in HVAC energy-consumption performance, the residual distribution for cooling energy remained concentrated near the zero-error reference line. The mean residual was approximately

–0.06 kWh, and the residuals were distributed across both positive and negative values without a pronounced directional trend. The relatively balanced distribution indicates that the remaining prediction errors were not dominated by systematic overprediction or underprediction, further supporting the improved agreement between measured and simulated HVAC energy consumption achieved through calibration. Similar residual characteristics have been reported in calibrated Building Energy Models where parameter optimization successfully reduced systematic prediction bias (Coakley et al., 2014; Heo et al., 2012).

A comparable pattern can be observed for indoor relative humidity. As shown in Figure 89, the residual distribution remains centered near the zero-error reference line, with a mean residual of approximately 0.26% RH. Although variability remains present throughout the humidity range, the distribution exhibits minimal directional bias and is consistent with the humidity behavior discussed in the previous calibration assessments. The results indicate that the calibrated model maintained a stable representation of indoor moisture conditions while improvements were achieved in HVAC energy prediction.

The residual distribution for indoor air temperature, presented in Figure 90, exhibits a different pattern. The mean residual of approximately 2.09°C indicates that the calibrated model retained a tendency to predict slightly warmer indoor conditions than those measured within the conference room. This behavior is consistent with the temperature correlation analysis, where a noticeable proportion of observations remained above the 1:1 reference line. Although a systematic offset remains evident, the residuals remain bounded within a relatively consistent range throughout the

calibration period, indicating that the overall thermal response of the space continued to be represented by the model.

The residual analysis therefore reinforces the conclusions obtained from the calibration metrics and correlation plots. The most substantial improvement was observed in HVAC energy consumption, which served as the primary calibration objective, while indoor temperature and relative humidity continued to exhibit reasonable agreement with measured observations. Together, these results provide additional evidence that the calibrated model achieved a closer representation of the physical conference room and was suitable for the independent validation stage.

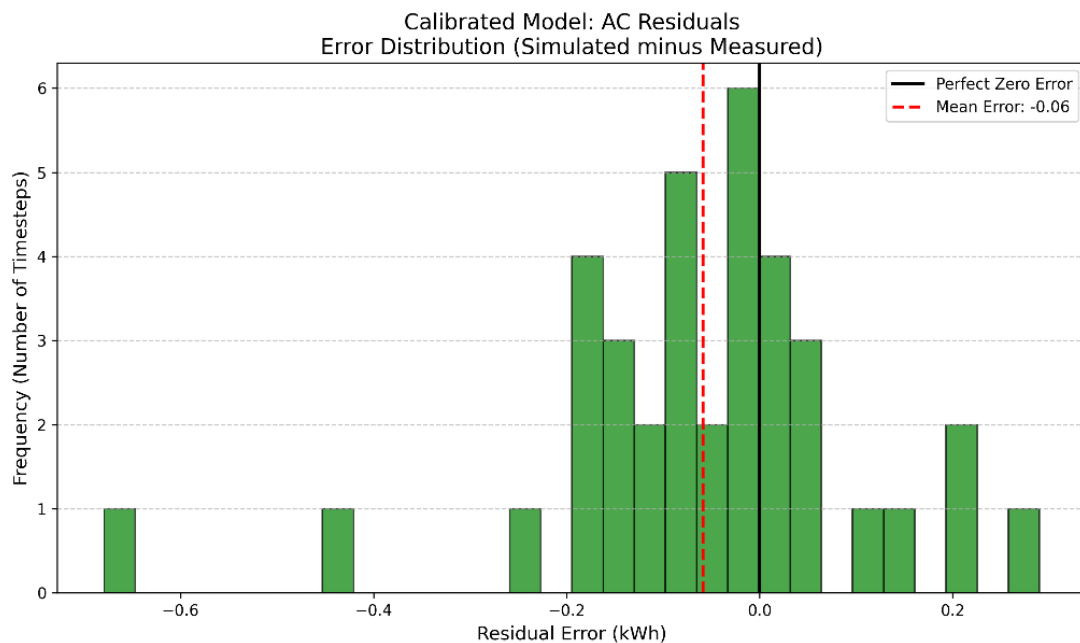


Figure 88. Residual Analysis of Calibrated AC Energy Consumptions

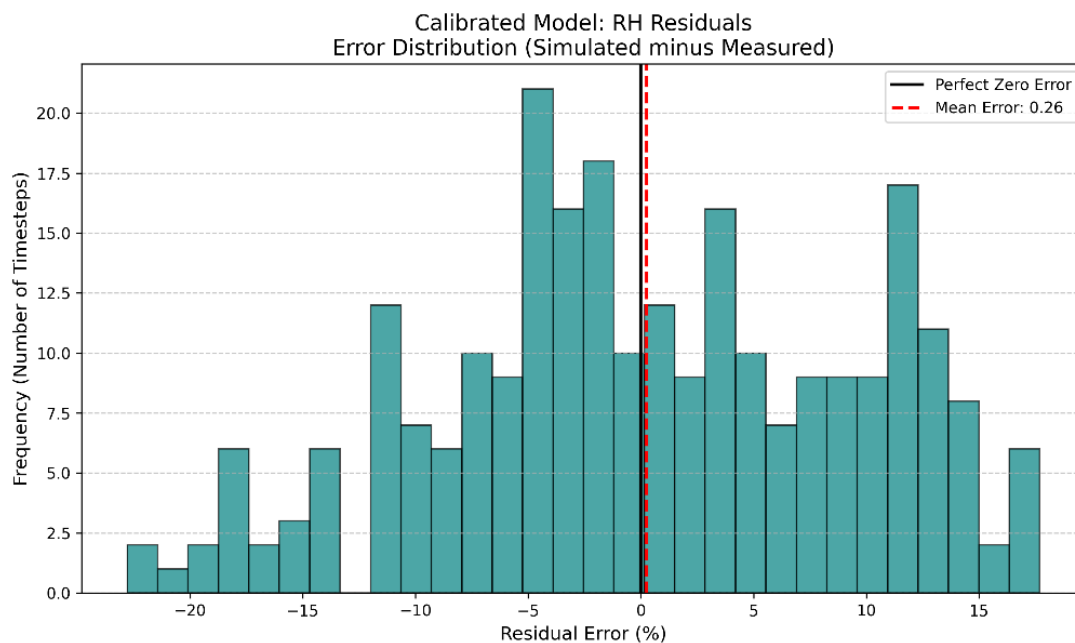


Figure 89. Residual Analysis of Calibrated Indoor RH

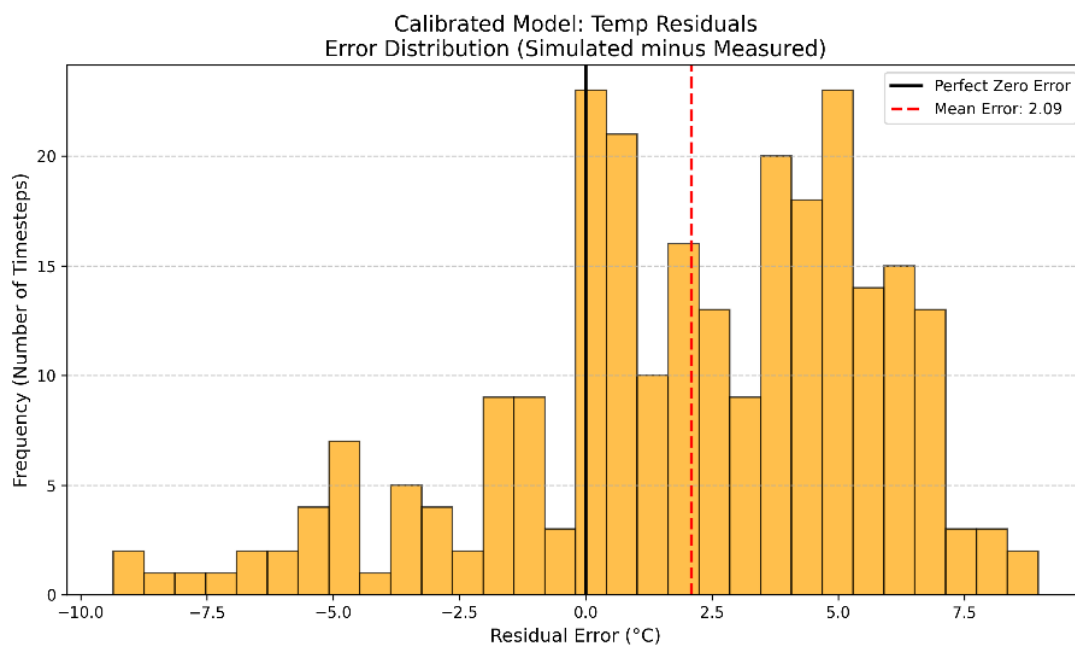


Figure 90. Residual Analysis of Calibrated Indoor Temperature

Validation Performance Metrics Using Unseen Operational Data

The predictive capability of the calibrated Digital Twin was further evaluated using an independent validation dataset consisting of previously unseen operational measurements. Unlike the calibration dataset, these observations were not utilized during the Bayesian optimization process and therefore provide an independent assessment of the model's ability to reproduce building performance under new operating conditions. The corresponding validation metrics are summarized in Table 21.

Table 21. Validation Performance Metrics

Metric	RMSE	CV(RMSE)	NMBE
Temperature (°C)	2.7626284493	10.4217675491	-4.4947783043
Relative Humidity (%)	15.3270226311	26.1449327780	-18.1058066840
Hourly AC Energy (kWh)	0.1633383719	28.2274458355	7.6393825148

The strongest validation performance was observed for HVAC energy consumption, which served as the primary calibration objective throughout the study. As shown in Figure 91, the simulated AC energy profile closely followed the measured cooling-energy behavior and successfully reproduced the major fluctuations in cooling demand throughout the validation period. Although localized deviations remained evident during several high-load intervals, the overall agreement between measured and simulated energy consumption was maintained.

The model achieved a CV(RMSE) of 28.23% and an NMBE of 7.64%, satisfying the hourly acceptance criteria specified in ASHRAE Guideline 14. More importantly, the validation performance remained comparable to the calibration results

obtained in Section 4.2, indicating that the optimized parameter set retained its predictive capability when exposed to operational conditions that were not included during calibration. Similar consistency between calibration and validation performance has been identified as an important indicator of model robustness and successful generalization in calibrated Building Energy Models (Raftery et al., 2011; Coakley et al., 2014).

The indoor air temperature results shown in Figure 92 further demonstrate that the calibrated model preserved the general thermal behavior of the conference room during validation. The simulated temperature profile reproduced the dominant temperature variations observed throughout the monitoring period and maintained reasonable agreement with the measured thermal response. Although differences in magnitude remained present at certain intervals, the model achieved a CV(RMSE) of 10.42% and an NMBE of -4.49%, indicating that the calibrated HVAC and internal-load parameters remained representative of the thermal response of the space beyond the calibration dataset. These results suggest that the improvements achieved during calibration did not compromise the model's ability to reproduce indoor environmental behavior under independent operating conditions.

Among the evaluated variables, indoor relative humidity exhibited the largest prediction errors during validation. As illustrated in Figure 93, the simulated humidity profile reproduced the general moisture-response pattern of the conference room but exhibited greater deviations from the measured observations than those observed for HVAC energy consumption and temperature.

The validation metrics yielded a CV(RMSE) of 26.14% and an NMBE of -18.11% , indicating increased uncertainty in humidity prediction. This behavior is consistent with the calibration strategy adopted in the present study, where optimization focused primarily on HVAC operational parameters associated with cooling-energy consumption. Parameters governing detailed moisture transport, infiltration effects, latent loads, and envelope-related humidity behavior were not explicitly included among the calibrated variables and therefore remained potential sources of prediction uncertainty. Similar limitations have been reported in previous calibration studies where energy-focused optimization produced stronger energy-prediction performance than humidity prediction accuracy (Heo et al., 2012; Coakley et al., 2014).

The validation results provide evidence that the calibrated Digital Twin generalized beyond the calibration dataset and retained its predictive capability under previously unseen operating conditions. The model has maintained compliance with the ASHRAE Guideline 14 acceptance criteria for HVAC energy consumption while preserving reasonable agreement with the measured indoor temperature and relative humidity profiles. This consistency between calibration and validation performance indicates that the optimized parameter set captured the underlying operational characteristics of the conference room rather than merely fitting the calibration dataset. Such behavior is considered a key requirement for Digital Twins intended for virtual parallel assessment and operational energy-performance evaluation (Ramírez et al., 2025).

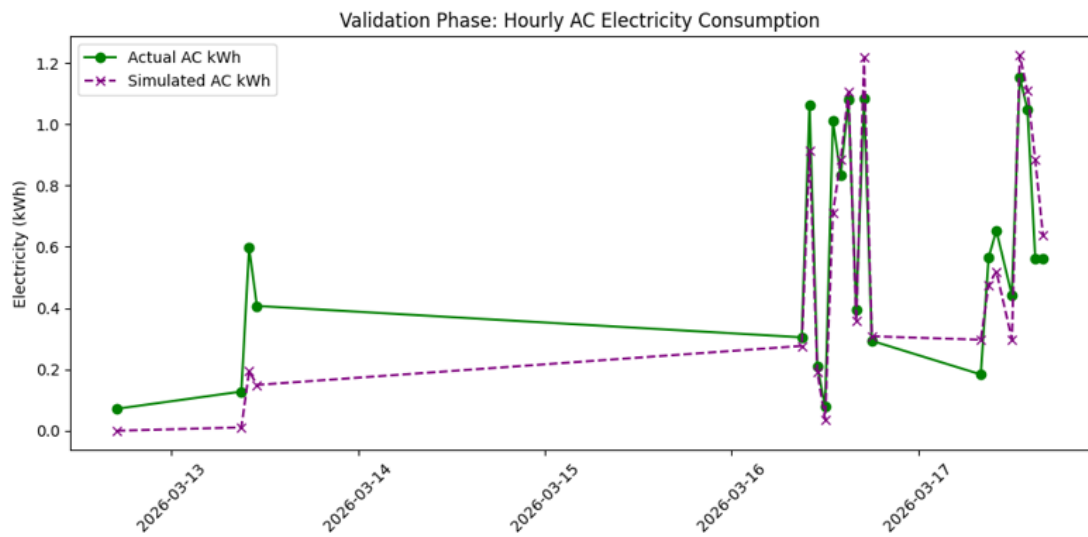


Figure 91. Simulate vs. Measured of Validated AC Energy Consumption

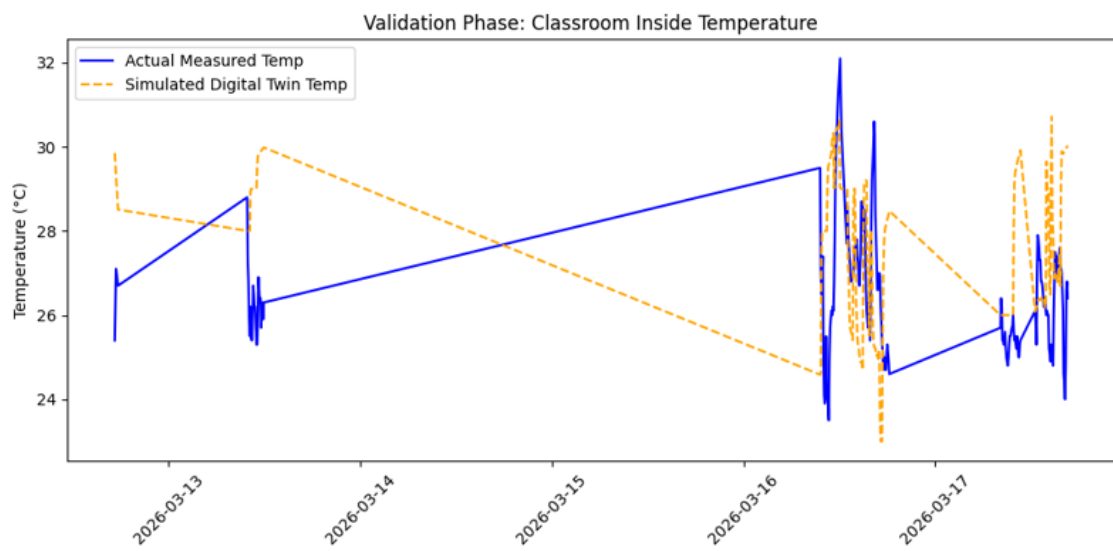


Figure 92. Simulate vs. Measured of Validated Indoor Temperature

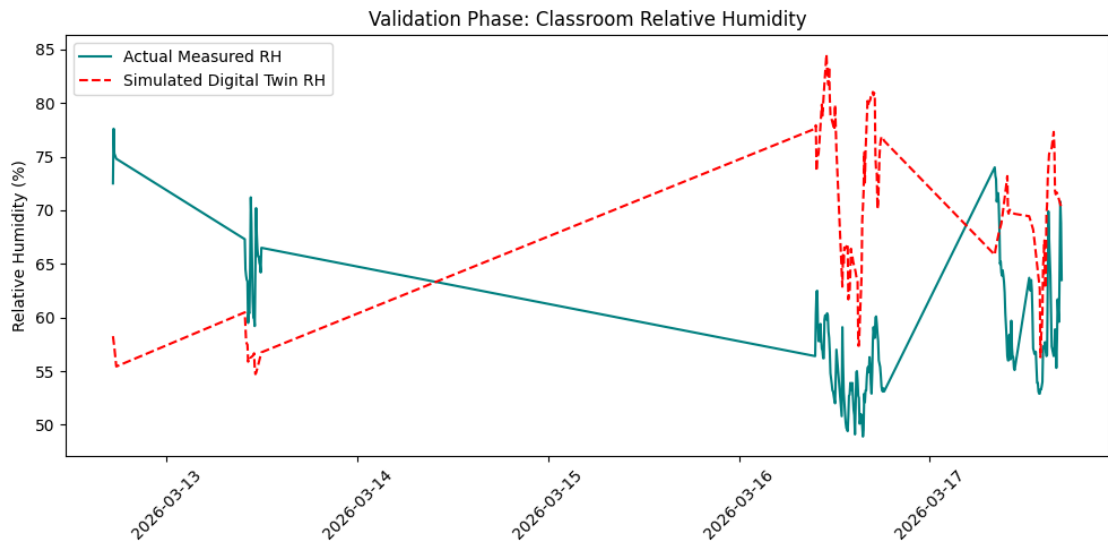


Figure 93. Simulate vs Measured of Validated Indoor RH

Validation Correlation Analysis

The predictive performance observed in the validation assessment is further examined through the correlation plots presented in Figures 94–96. As a continuation of the preceding validation analysis, the scatter plots provide additional evidence of the agreement between measured and simulated performance when the calibrated Digital Twin was evaluated using previously unseen operational data.

The strongest agreement was again observed for HVAC energy consumption. As shown in Figure 94, the validation scatter plot produced a coefficient of determination of $R^2 = 0.7993$, with most of observations clustered near the 1:1 reference line. Compared with the baseline model presented in Section 4.1, the validation distribution exhibits substantially improved alignment with the parity line, indicating that the calibrated model retained its ability to reproduce both the magnitude and variation of cooling-energy consumption beyond the calibration period. The close

agreement observed in the scatter distribution further supports the consistency between the calibration and validation results and indicates that the optimized parameter set remained representative of the operational behavior of the conference room. Similar validation characteristics have been reported in calibrated Building Energy Models where successful parameter optimization resulted in improved predictive capability under independent operating conditions (Raftery et al., 2011; Coakley et al., 2014).

The indoor air temperature correlation plot shown in Figure 95 exhibits a distribution generally concentrated around the 1:1 reference line. Although a degree of dispersion remains evident throughout the measured temperature range, the overall pattern is consistent with the validation time-series results and indicates that the calibrated model preserved the general thermal response of the conference room under unseen operating conditions. The distribution also suggests that the thermal behavior achieved during calibration remained stable when applied to an independent dataset.

A broader degree of scatter can be observed in the indoor relative humidity correlation plot presented in Figure 96. While the observations generally follow the measured humidity range, deviations from the parity line remain visible across portions of the dataset. This behavior is consistent with the humidity trends identified in the validation assessment and indicates that greater uncertainty remained in the prediction of moisture-related behavior compared with HVAC energy consumption and indoor temperature. Nevertheless, the overall distribution demonstrates that the calibrated model continued to reproduce the dominant humidity-response characteristics of the conference room.

The validation correlation analysis therefore reinforces the findings obtained from the validation metrics and time-series assessment. The strongest agreement remained associated with HVAC energy consumption, while indoor temperature and relative humidity continued to exhibit reasonable correspondence with the measured environmental conditions. Together, these results provide further evidence that the calibrated parameter set maintained its predictive capability when applied to operational data that were not included during the Bayesian calibration process and support the suitability of the developed Digital Twin for virtual parallel assessment.

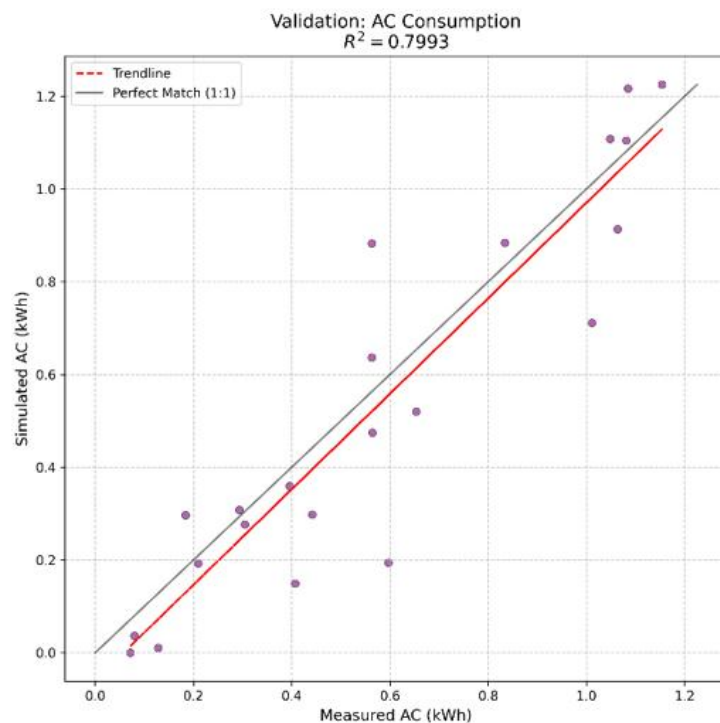


Figure 94. Correlation Analysis of Validated AC Energy Consumption

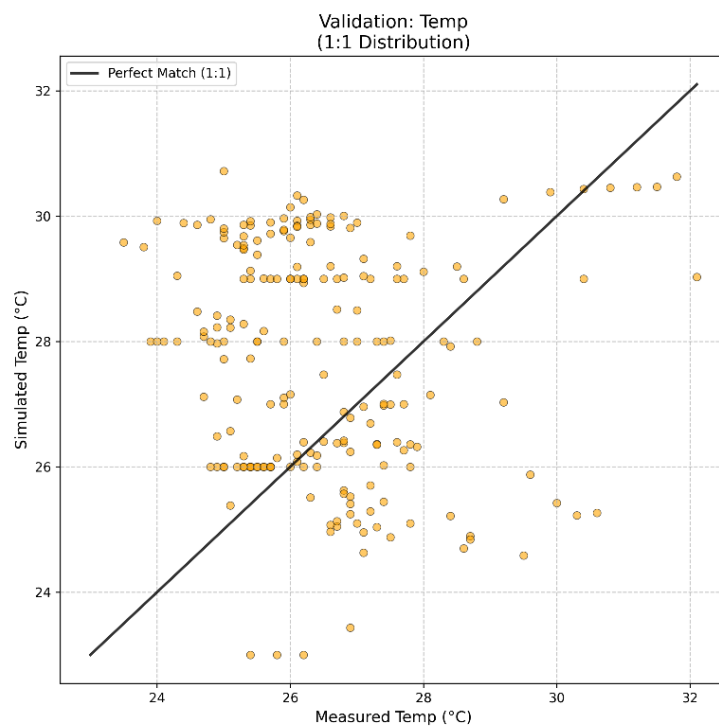


Figure 95. Correlation Analysis of Validated Indoor RH

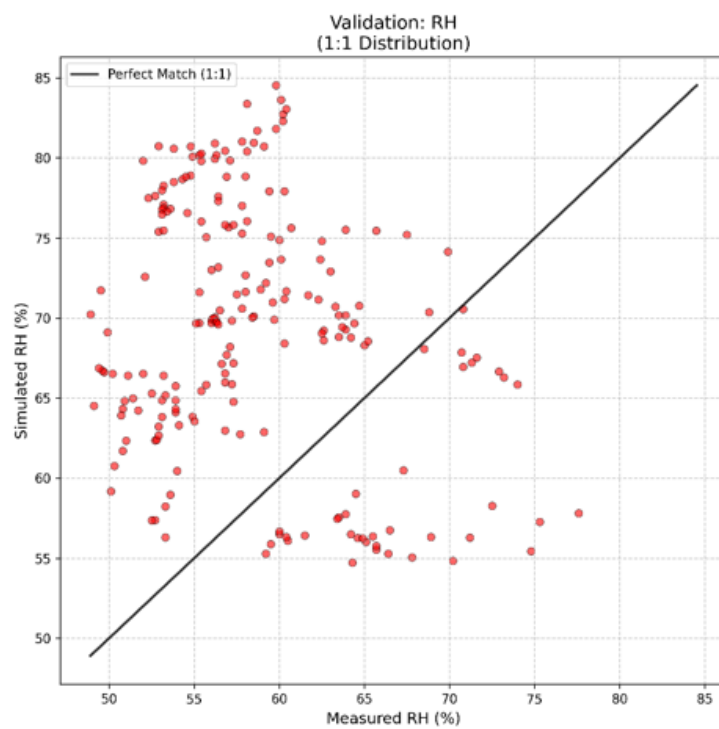


Figure 96. Correlation Analysis of Validated Indoor Temperature

Operational Digital Twin Dashboard

The final stage of the developed framework involved the deployment of the calibrated Digital Twin within an operational environment. As shown in Figure 97, the developed dashboard enabled real-time synchronization between the physical conference room and its EnergyPlus-based virtual counterpart through the MQTT communication framework. Live operational inputs, including indoor temperature, relative humidity, occupancy, AC setpoint, and AC operating mode, were continuously transmitted to the Digital Twin, allowing the simulation model to update according to the current operating state of the physical system.

The dashboard provided a unified interface for monitoring both measured and simulated building performance. Real-time HVAC energy consumption, indoor environmental conditions, and system operating parameters were displayed alongside the corresponding Digital Twin predictions, enabling direct comparison between the physical asset and its virtual representation.

In addition to numerical and graphical outputs, the platform incorporated a three-dimensional representation of the conference room, providing a visual interpretation of the simulated thermal state within the conditioned space.

A key feature of the developed platform was its ability to perform continuous virtual parallel assessment. By simultaneously displaying measured and simulated HVAC energy consumption, indoor temperature, and relative humidity, the dashboard enabled ongoing evaluation of the agreement between the physical conference room and the Digital Twin under the same operating conditions. This capability aligns with

the fundamental concept of Digital Twins, where real-time data exchange is used to maintain synchronization between physical and virtual assets and support operational decision-making (Fuller et al., 2020; Aheleroff et al., 2021).

The successful deployment of the dashboard demonstrates that the developed framework extended beyond offline simulation and calibration activities to support real-time operational implementation. The continuous integration of live building data with EnergyPlus simulation outputs confirms the capability of the Digital Twin to maintain an updated representation of the physical conference room, consistent with Digital Twin architectures reported in recent building-performance studies (Deng et al., 2024; Wang et al., 2024; Ramírez et al., 2025).

The operational dashboard therefore provides the foundation for real-time HVAC energy monitoring, performance assessment, and virtual parallel assessment within the proposed Digital Twin framework.



Figure 97. Graphical User Interface Live Dashboard

Operational Performance Assessment

The operational performance of the Digital Twin was evaluated using real-time data received through the MQTT communication framework during live deployment. The resulting time-series comparisons and performance metrics are summarized in Table 22 and Figures 98–100.

As a continuation of the validation stage, the operational assessment examines whether the calibrated Digital Twin could maintain its predictive capability while continuously interacting with the physical conference room under actual operating conditions.

Table 22. Operational Performance Metrics

Metric	RMSE	CV(RMSE)	NMBE
Temperature (°C)	2.7626284493	10.4217675491	-4.4947783043
Relative Humidity (%)	15.3270226311	26.1449327780	-18.1058066840
Hourly AC Energy (kWh)	0.1633383719	28.2274458355	7.6393825148

The strongest operational performance was again observed for HVAC energy consumption. As shown in Figure 98, the simulated AC energy profile closely followed the measured cooling-energy trend throughout the deployment period and successfully reproduced the major fluctuations in cooling demand. Although localized deviations remained visible during several operating intervals, the overall agreement between measured and simulated energy consumption was maintained.

The model achieved an RMSE of 0.277 kWh, a CV(RMSE) of 15.60%, and an NMBE of 14.24%, indicating that the Digital Twin retained its capability to quantify

HVAC energy consumption during live operation. The consistency between the calibration, validation, and deployment results suggests that the calibrated parameter set remained representative of the operational behavior of the conference room and continued to provide reliable energy-performance predictions under real-time conditions. Similar behavior has been identified as a key characteristic of operational Digital Twins, where the virtual asset must remain synchronized with the physical system while preserving predictive accuracy during deployment (Fuller et al., 2020; Ramírez et al., 2025).

The operational indoor temperature results presented in Figure 99 indicate that the Digital Twin remained responsive to changes in the thermal conditions of the physical environment. The simulated temperature profile reproduced the general thermal trends observed throughout the deployment period and responded to variations in occupancy, HVAC operation, and environmental conditions. Although differences in magnitude remained evident, with simulated temperatures generally lower than the measured observations, the model continued to capture the dominant thermal behavior of the conference room. The resulting RMSE of 4.53°C and CV(RMSE) of 16.18% are consistent with the temperature behavior previously observed during calibration and validation, indicating that the thermal-response characteristics of the model remained stable during deployment.

A similar observation can be made for indoor relative humidity. As illustrated in Figure 100, the simulated humidity profile reproduced the dominant moisture variations observed within the conference room and remained within a comparable operating range throughout the monitoring period. While deviations in magnitude

occurred at several intervals, the overall pattern remained consistent with the measured humidity behavior. The operational metrics yielded an RMSE of 3.92% RH, a CV(RMSE) of 6.26%, and an NMBE of -2.69% , indicating that the Digital Twin maintained a realistic representation of indoor moisture conditions during live operation. The relatively stable humidity performance further demonstrates that the virtual model remained physically linked to the environmental conditions of the conditioned space throughout deployment.

The operational assessment confirms that the Digital Twin remained synchronized with the physical conference room while continuously receiving and processing real-time operational inputs. Consistent with the calibration and validation stages, HVAC energy consumption continued to exhibit the strongest predictive performance and remained the most accurately represented variable during deployment. At the same time, the indoor temperature and relative humidity results indicate that the Digital Twin preserved a reasonable representation of the thermal and moisture behavior of the conference room. These findings demonstrate the capability of the developed framework to support real-time HVAC energy monitoring and virtual parallel assessment under actual building operating conditions (Deng et al., 2024; Wang et al., 2024).

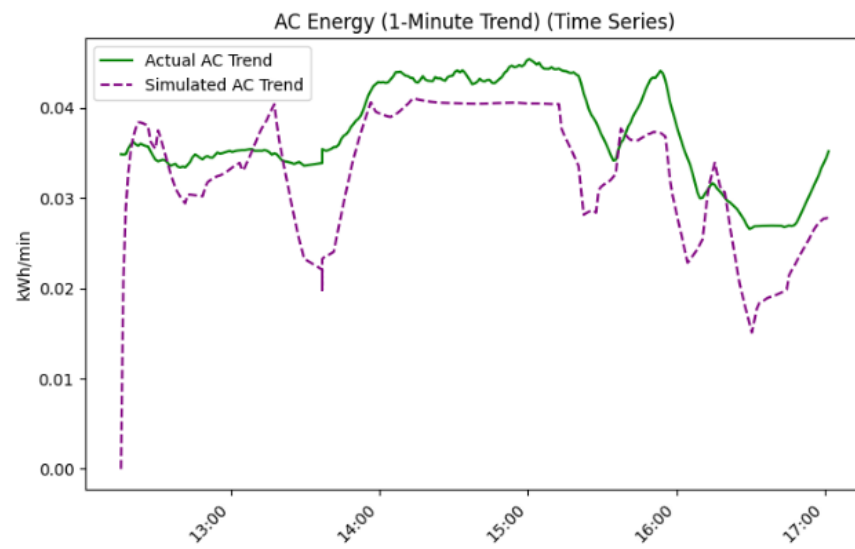


Figure 98. Simulated vs. Measured of Operational Digital Twin of AC Energy Consumption

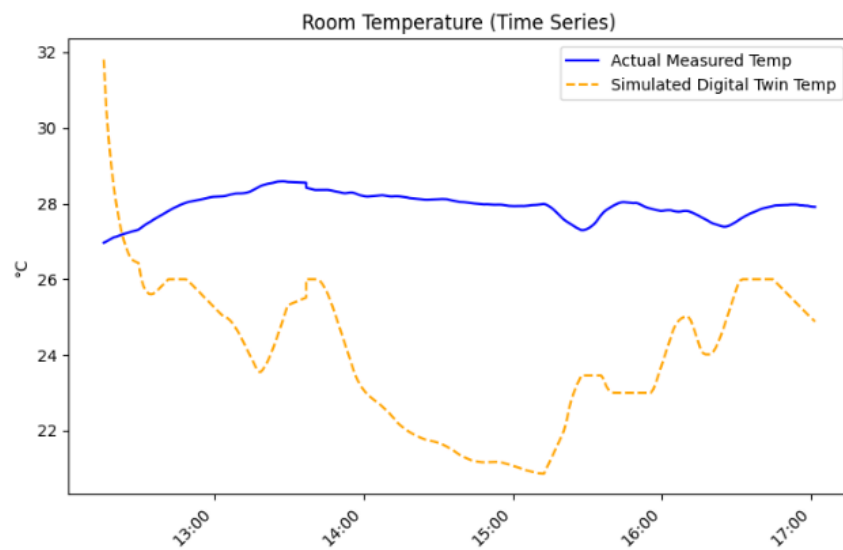


Figure 99. Simulated vs. Measured of Operational Digital Twin of Indoor Temperature

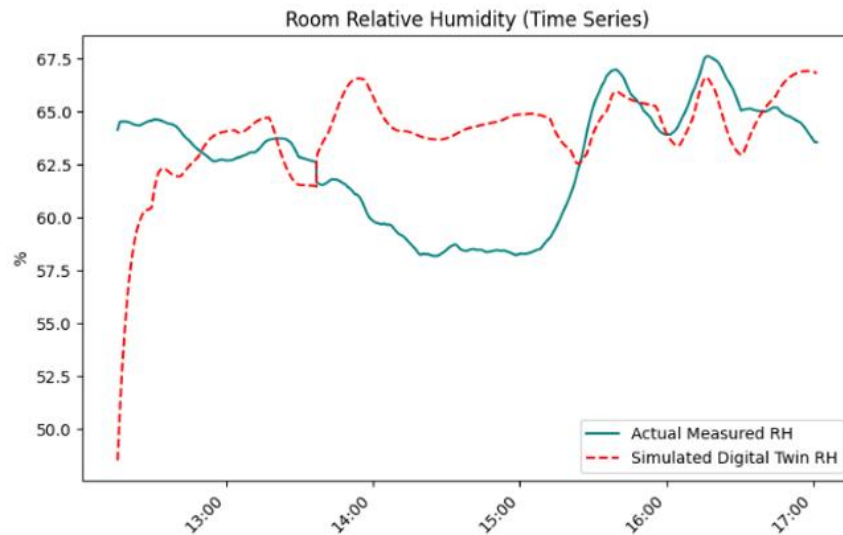


Figure 100. Simulated vs. Measured of Operational Digital Twin of Indoor RH

Technoeconomic Analysis

Technoeconomic analysis is a method of evaluating the effectiveness of a system by combining technical performance metrics with economic assessments. It consists of two main components: the technical side, which measures factors such as power consumption, response time, and control accuracy to establish the operational value of the controller, and the economic side, which focuses on cost-savings, investment requirements, and payback periods to assess feasibility.

Technical Metrics

Important system metrics were gathered to assess the performance of the system. These system data are grouped into four metric groups. Such groups and the individual data analysis are detailed in the next sections.

Computational Performance

This metric group evaluates the computational demand imposed on the Raspberry Pi 5 during continuous operation of the proposed framework. Such demand includes, CSV logging, MQTT broker operation, execution of the YOLOv8n-based occupancy detection, and the HVAC-control machine learning processes. To ensure representative evaluation of normal operating conditions. Only datasets obtained during stable deployment periods were included in the computational performance analysis. Datasets collected during debugging stages, unstable voltage conditions, and power interruption events were excluded from the treated data. In total, 1,743 minutes of operational data, recorded at 1-minute intervals across four deployment days, were included in the analysis.

CPU Utilization Percentage. Shown in Table 23 is an excerpt of CPU utilization dataset containing 1,743 timestamped entries, which represents how much of the central processing unit's capacity is utilized. The whole version of the dataset and the corresponding data analysis sheet is uploaded in the GitHub repository available in Appendix A.

Table 23. Excerpt of CPU Utilization % Dataset

Timestamp	CPU Utilization (%)
5/6/2026 12:06	5
5/6/2026 12:07	4.7
5/6/2026 12:08	4.8
5/6/2026 12:09	8.1
5/6/2026 12:10	5.3
5/6/2026 12:11	4.9

Timestamp	CPU Utilization (%)
5/6/2026 12:12	5.1
5/6/2026 12:13	5.2
5/6/2026 12:14	5.1
5/6/2026 12:15	5.2
5/18/2026 16:55	5.4
5/18/2026 16:56	6.3
5/18/2026 16:57	5.3
5/18/2026 16:58	5.5
5/18/2026 16:59	5.8
5/18/2026 17:00	7
5/18/2026 17:01	10.8
5/18/2026 17:02	5.5
5/18/2026 17:03	6.2
5/18/2026 17:04	5.8

The entire set is then subjected into descriptive statistics to quantify its implications, summarized in Table 24.

Table 24. CPU Utilization Data Summary

Parameter	Value (%)	Timestamp
Average CPU Utilization	4.58	—
Maximum CPU Utilization	36.50	5/12/2026 8:56 AM
Minimum CPU Utilization	2.10	5/12/2026 8:47 AM
Sample Standard Deviation	1.46	—

Results show an average CPU utilization of 4.58%, indicating that the proposed monitoring and control framework imposed relatively low computational demand on the central processing unit despite simultaneously performing data processing, MQTT broker operation, CSV logging, YOLOv8n execution, alongside the main ML program.

The maximum recorded CPU utilization reached 36.50% on May 12, 2026, at 8:56 AM, while the minimum recorded utilization was 2.10% on the same day at 8:47 AM. Furthermore, the computed sample standard deviation of 1.46 indicates relatively stable CPU utilization behavior throughout the treated operational periods, suggesting that the Raspberry Pi 5 maintained sufficient computational headroom during continuous system operation.

CPU Temperature. Shown in Table 25 is an excerpt of CPU temperature dataset containing 1,743 timestamped entries.

Table 25. Excerpt of CPU Temperature (°C) Dataset

Timestamp	CPU Temperature (°C)
5/6/2026 12:06	50.7
5/6/2026 12:07	50.1
5/6/2026 12:08	53.5
5/6/2026 12:09	51.2
5/6/2026 12:10	52.4
5/6/2026 12:11	51.2
5/6/2026 12:12	53.5
5/6/2026 12:13	52.9
5/6/2026 12:14	52.9
5/6/2026 12:15	54
5/18/2026 16:55	51.8
5/18/2026 16:56	52.9
5/18/2026 16:57	52.4
5/18/2026 16:58	52.4
5/18/2026 16:59	51.8
5/18/2026 17:00	52.9
5/18/2026 17:01	53.5
5/18/2026 17:02	51.8

Timestamp	CPU Temperature (°C)
5/18/2026 17:03	52.4
5/18/2026 17:04	51.8

The entire set is then subjected into descriptive statistics to quantify its implications, summarized in Table 26.

Table 26. CPU Temperature Data Summary

Parameter	Value (°C)	Timestamp
Average CPU Temperature	51.70	—
Maximum CPU Temperature	58.40	5/06/2026 2:07 PM
Minimum CPU Temperature	31.40	5/12/2026 8:47 AM
Sample Standard Deviation	2.65	—

Results show an average CPU temperature of 51.70°C during continuous system operation, with a maximum recorded temperature of 58.40°C and a minimum recorded temperature of 31.40°C. Furthermore, the computed sample standard deviation of 2.65°C indicates relatively stable thermal behavior throughout the treated operational periods, suggesting minimal temperature fluctuation during continuous deployment.

Operational observations during data gathering suggested that the implemented 52Pi Low-Profile ICE Tower with heatsink contributed to maintaining acceptable operating temperatures and reducing thermal buildup during sustained system operation. This is consistent with a benchmark test conducted by 52 PI (2024), on the same model of cooler, and it maintained a temperature between 44°C and 50 °C, way

below the throttling temperature of 80 °C (Alasdair, 2023). The exact results of the 52PI benchmarking are in Figure 101.

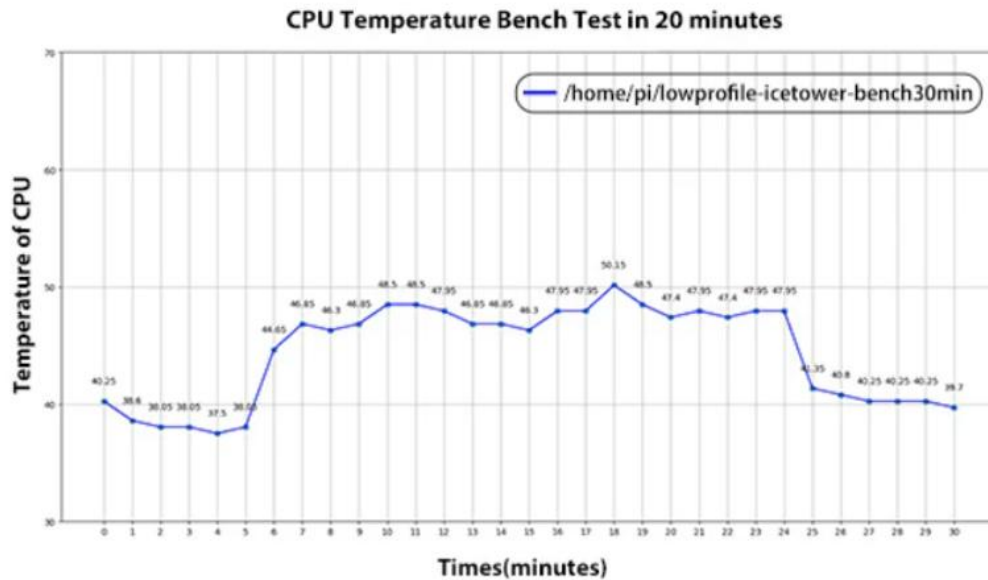


Figure 101. 52PI Low-Profile ICE Tower w/ Heatsink 30-min Bench Test Results

RAM Percentage and Usage. Shown in Table 27 is an excerpt of RAM dataset containing 1,743 timestamped entries, which represents how much of the available random-access memory, in this case 8GB, is utilized by the program and its peripherals.

Table 27. Excerpt of CPU Utilization % Dataset

Timestamp	RAM Utilization (%)	RAM Usage (MB)
5/6/2026 12:06	25.9	2086.53
5/6/2026 12:07	32.6	2625.69
5/6/2026 12:08	32.8	2642.47
5/6/2026 12:09	32.8	2648.31
5/6/2026 12:10	33	2658.94
5/6/2026 12:11	32.5	2622.08
5/6/2026 12:12	32.8	2640.91
5/6/2026 12:13	33.1	2671.16

Timestamp	RAM Utilization (%)	RAM Usage (MB)
5/6/2026 12:14	32.7	2635.08
5/6/2026 12:15	32.8	2644.19
5/18/2026 16:55	27.4	2213.16
5/18/2026 16:56	27	2180.7
5/18/2026 16:57	27.6	2223.7
5/18/2026 16:58	27.6	2228.34
5/18/2026 16:59	27.6	2222.98
5/18/2026 17:00	28.4	2292.94
5/18/2026 17:01	28.1	2265.86
5/18/2026 17:02	28.3	2284.64
5/18/2026 17:03	27.7	2229.47
5/18/2026 17:04	27.7	2232.38

The entire set is then subjected into descriptive statistics to quantify its implications, summarized in Table 28.

Table 28. RAM Percentage and Usage Data Summary

Parameter	Value (%)	Timestamp
Average RAM Percentage	26.19	—
Maximum RAM Percentage	44.10	5/12/2026 8:56 AM
Minimum RAM Percentage	9.30	5/15/2026 12:36 PM
Sample Standard Deviation (RAM %)	6.99	—
Parameter	Value (MB)	Timestamp
Average RAM Usage	2,111.49	—
Maximum RAM Usage	3,553.53	5/12/2026 8:56 AM
Minimum RAM Usage	745.88	5/15/2026 12:36 PM
Sample Standard Deviation (RAM Usage)	563.94	—

Results show an average RAM utilization of 26.19%, corresponding to an average RAM usage of 2,111.49 MB during continuous operation of the proposed

framework. The maximum recorded RAM utilization reached 44.10%, while the minimum recorded utilization was 9.30%.

Furthermore, the computed sample standard deviations of 6.99% and 563.94 MB indicate moderate variation in memory allocation behavior throughout the treated operational periods. This behavior is consistent with Linux-based systems, wherein dynamic memory allocation and caching mechanisms commonly produce observable fluctuations in RAM usage during multitasking operation.

Even so, these mechanisms make Linux-based systems efficient in handling RAM, under heavy memory needs (Yogesh, 2024). Despite such variation, the recorded values remained well below the available 8 GB system memory of the Raspberry Pi 5, indicating that sufficient memory headroom was maintained during continuous operation.

Overall, results indicate that the Raspberry Pi 5 maintained sufficient computational headroom throughout continuous operation of the proposed framework. This suggests potential capability for accommodating additional processing tasks or future system expansions, such as implementation of more advanced occupancy detection algorithms or enhanced data processing methods.

Communication Reliability

This metric group evaluates the system's reliability in communication between the Raspberry Pi 5 and the five ESP32 nodes, during continuous operation of the proposed framework.

To ensure the representative evaluation of normal operating conditions, only datasets obtained during stable deployment periods were included in the computational performance analysis. Datasets collected during debugging stages, unstable voltage conditions, and power interruption events were excluded from the treated data. In total, 1,743 minutes of operational data, recorded at 1-minute intervals across four deployment days, were included in the analysis.

Network Load. Shown in Table 29 is an excerpt of the network load dataset containing 1,743 timestamped entries, representing the amount of data in kbps being transmitted and received by the Raspberry Pi 5 through the wireless network during continuous system operation.

Table 29. Excerpt of Network Load Dataset

Timestamp	Net Sent (kbps)	Net Received (kbps)
5/6/2026 12:06	993.3	152.6
5/6/2026 12:07	29.17	27.23
5/6/2026 12:08	29.03	27.41
5/6/2026 12:09	325.65	51
5/6/2026 12:10	70.13	32.72
5/6/2026 12:11	64.07	30.4
5/6/2026 12:12	62.48	30.47
5/6/2026 12:13	266.1	33.63
5/6/2026 12:14	202.34	32.15
5/6/2026 12:15	250.17	31.54
5/18/2026 16:55	163.78	28.04
5/18/2026 16:56	168.64	27.99
5/18/2026 16:57	128.22	27.39
5/18/2026 16:58	130.23	27.42
5/18/2026 16:59	182.51	27.85

Timestamp	Net Sent (kBps)	Net Received (kBps)
5/18/2026 17:00	630.66	32.79
5/18/2026 17:01	3878.94	57.07
5/18/2026 17:02	339.6	28.54
5/18/2026 17:03	304.09	28.08
5/18/2026 17:04	822.09	32.66

The entire set is then subjected into descriptive statistics to quantify its implications, summarized in Table 30.

Table 30. Network Load Data Summary

Parameter	Value (%)	Timestamp
Average Sent Network Load	159.69	—
Maximum Sent Network Load	11,048.34	5/06/2026 2:15 PM
Minimum Sent Network Load	6.90	5/18/2026 9:40 AM
Sample Standard Deviation (Net Sent)	641.82	—
Parameter	Value (MB)	Timestamp
Average Received Network Load	28.44	—
Maximum Received Network Load	152.60	5/06/2026 12:09 PM
Minimum Received Network Load	6.86	5/18/2026 09:40 AM
Sample Standard Deviation (Net Received)	7.47	—

Results show an average outgoing network load of 159.69 kBps and an average incoming network load of 28.44 kBps during continuous system operation. While the computed sample standard deviation for outgoing traffic reached 641.82 kBps, inspection of the treated dataset showed that only 61 out of 1,743 recorded entries exceeded 1,000 kBps, with only 8 exceeding 5,000 kBps, indicating that high network load events occurred infrequently and were likely associated with transient burst transmissions rather than sustained communication congestion.

Furthermore, the incoming traffic standard deviation of 7.47 kbps suggests relatively stable sensor node reporting behavior throughout deployment. Overall, the recorded network load values indicate that the proposed MQTT-based communication framework imposed manageable demand on the wireless network infrastructure during continuous operation.

Latency. Shown in Table 31 is an excerpt of the recorded latency or transmission delay dataset containing 1,743 timestamped entries. Latency measurements were obtained by transmitting five ICMP (Internet Control Message Protocol) ping packets from the Raspberry Pi 5 to the wireless access point per minute, allowing the collected values to represent the responsiveness and reliability of the deployed local wireless communication infrastructure.

Table 31. Excerpt of Latency (ms) Dataset

Timestamp	Latency (ms)
5/6/2026 12:06	3.503
5/6/2026 12:07	4.333
5/6/2026 12:08	4.732
5/6/2026 12:09	4.238
5/6/2026 12:10	4.014
5/6/2026 12:11	5.573
5/6/2026 12:12	4.903
5/6/2026 12:13	5.835
5/6/2026 12:14	3.72
5/6/2026 12:15	3.857
5/18/2026 16:55	4.194
5/18/2026 16:56	3.58
5/18/2026 16:57	3.381
5/18/2026 16:58	5.164

Timestamp	Latency (ms)
5/18/2026 16:59	3.484
5/18/2026 17:00	3.822
5/18/2026 17:01	5.034
5/18/2026 17:02	5.946
5/18/2026 17:03	3.799
5/18/2026 17:04	4.353

The entire set is then subjected into descriptive statistics to quantify its implications, summarized in Table 32.

Table 32. Latency Data Summary

Parameter	Value (ms)	Timestamp
Average Latency	5.60	–
Maximum Latency	394.54	5/15/2026 3:20 PM
Minimum Latency	1.04	5/18/2026 12:36 PM
Sample Standard Deviation (Latency)	11.10	–

Results show an average communication latency of 5.60 ms during continuous operation of the proposed framework, indicating relatively low transmission delay throughout deployment. Although a maximum latency of 394.54 ms was recorded, the computed sample standard deviation of 11.10 ms suggests that elevated latency events occurred infrequently and that overall communication behavior remained relatively stable throughout the treated operational periods. The low average latency further indicates that the proposed MQTT-based communication framework imposed manageable demand on the wireless communication infrastructure during continuous operation.

Packet Loss (Ping at Access Point). Shown in Table 33 is an excerpt of the packet loss, or the percentage of transmitted data packets that fail to successfully reach their intended destination, from the dataset containing 1,743 timestamped entries

Table 33. Excerpt of Rate of Packet Loss (%) Dataset

Timestamp	Packet Loss (%)
5/6/2026 12:06	0
5/6/2026 12:07	0
5/6/2026 12:08	0
5/6/2026 12:09	0
5/6/2026 12:10	0
5/6/2026 12:11	0
5/6/2026 12:12	0
5/6/2026 12:13	0
5/6/2026 12:14	0
5/6/2026 12:15	0
5/18/2026 16:55	0
5/18/2026 16:56	0
5/18/2026 16:57	0
5/18/2026 16:58	0
5/18/2026 16:59	0
5/18/2026 17:00	0
5/18/2026 17:01	0
5/18/2026 17:02	0
5/18/2026 17:03	0
5/18/2026 17:04	0

The entire set is then subjected into descriptive statistics to quantify its implications, summarized in Table 34.

Table 34. Packet Loss Data Summary

Parameter	Value (%)	Timestamp
Average Packet Loss	0.0115	–
Maximum Packet Loss	20	5/15/2026 4:18 PM
Minimum Packet Loss	0	–
Sample Standard Deviation (PLR)	0.48	–

Results show an average packet loss rate of 0.0115% throughout continuous system operation, indicating highly reliable communication performance within the deployed wireless network infrastructure. Out of the 1,743 recorded data points, only one instance exhibited a nonzero packet loss value. Since each logging interval transmitted five ICMP ping packets from the Raspberry Pi 5 to the wireless access point, the recorded 20% packet loss corresponds to a single failed packet transmission out of five transmitted packets during that isolated event. Furthermore, the computed sample standard deviation of 0.48% indicates that packet loss occurrences were minimal and infrequent throughout deployment. Overall, the near-zero average packet loss suggests that the proposed MQTT-based monitoring and control framework maintained stable and reliable wireless communication during continuous operation.

Overall, the communication reliability results indicate that the proposed MQTT-based monitoring and control framework maintained stable and dependable wireless communication throughout continuous deployment. The recorded low latency, near-zero packet loss rate, and manageable network load suggest that the system imposed relatively light demand on the deployed wireless network.

Additionally, when considered alongside the computational performance results of the Raspberry Pi 5, the collected data indicate that the system still maintained

sufficient processing and communication headroom during operation. These observations suggest the potential capability of the framework to support future expansions, such as additional sensor nodes, more advanced occupancy detection methods, or enhanced system functionality.

Energy Consumption

This metric group evaluates the energy consumption behavior of the proposed distributed mechatronic controller. Energy consumption data for each ESP32 node were obtained through the implemented INA226 monitoring modules, while Raspberry Pi 5 power consumption was estimated through the developed power estimation model. Additionally, battery life and supercapacitor runtime duration of each ESP32 node were evaluated to assess operational endurance and uninterrupted deployment capability.

Raspberry Pi 5 Power Consumption. Shown in Table 35 is an excerpt of the estimated power usage, recorded per minute dataset containing 1,743 timestamped entries. Aside from the power usage in milliwatts (mW), the table also contains the equivalent energy use in milliwatt-minute (mW-min).

Table 35. Excerpt of Raspberry Pi 5 Power Dataset

Timestamp	Pi Power (mW)	Current (mA)	Interval (min)	Energy (mw-min)
5/6/2026 12:06	2792	558.4	1	2792
5/6/2026 12:07	2917	583.4	1	2917
5/6/2026 12:08	2971	594.2	1	2971
5/6/2026 12:09	3219	643.8	1	3219
5/6/2026 12:10	2998	599.6	1	2998
5/6/2026 12:11	2794	558.8	1	2794

Timestamp	Pi Power (mW)	Current (mA)	Interval (min)	Energy (mw-min)
5/6/2026 12:12	2997	599.4	1	2997
5/6/2026 12:13	2998	599.6	1	2998
5/6/2026 12:14	2989	597.8	1	2989
5/6/2026 12:15	3013	602.6	1	3013
5/18/2026 16:55	2990	598	1	2990
5/18/2026 16:56	3041	608.2	1	3041
5/18/2026 16:57	2828	565.6	1	2828
5/18/2026 16:58	3091	618.2	1	3091
5/18/2026 16:59	2998	599.6	1	2998
5/18/2026 17:00	2840	568	1	2840
5/18/2026 17:01	3033	606.6	1	3033
5/18/2026 17:02	3151	630.2	1	3151
5/18/2026 17:03	3484	696.8	1	3484
5/18/2026 17:04	2852	570.4	1	2852

The entire set is then subjected into power related calculations to quantify the total consumption, hourly average consumption, and average current draw, summarized in Table 36.

Table 36. Raspberry Pi 5 Power Metrics Summary

Parameter	Value and Unit
Total Energy Consumption Over Testing Period	84.17 Wh
Total Testing Time	29.05 hours
Hourly Averaged Energy Consumption	2.90 W
Average Current Draw	579.48 mA

Results show an average Raspberry Pi 5 power consumption of 2.90W during the deployment duration, corresponding to an estimated average current draw of 579.48 mA at 5V operation. Throughout the treated deployment periods, the Raspberry Pi 5

consumed a total estimated energy of 84.17 Wh across accumulated 29.05 hours of stable operation. The resulting power consumption values were subsequently used in the computation of system operating expenditures (OpEx).

Power Consumption and Battery Life of Five ESP32 Nodes. Due to the distributed nature of the system, each ESP32 node operates independently from one another and from the Raspberry Pi 5 in terms of processing workload, only requiring the MQTT broker for communication and data transmission. Each ESP32 node was programmed to publish sensor data to the MQTT broker every 30 seconds regardless of the active control mode. Therefore, all available INA226 power monitoring data were considered usable for analysis. Furthermore, retaining the complete dataset allows more representative computation of node-level power consumption and battery life during deployment. Shown in Table 37 is the excerpt of the said dataset.

Table 37. Excerpt of ESP Nodes Power Dataset

Timestamp	ESP ID	Voltage (V)	Power (mW)	Status
5/5/2026 8:13	ESP_1	3.82	248.72	OK
5/5/2026 8:13	ESP_2	3.83	411.99	OK
5/5/2026 8:13	ESP_3	3.34	227.36	OK
5/5/2026 8:13	ESP_4	3.62	533.75	OK
5/5/2026 8:13	ESP_5	2.75	473.75	OK
5/5/2026 8:14	ESP_1	3.81	283.81	OK
5/5/2026 8:14	ESP_2	3.83	411.99	OK
5/5/2026 8:14	ESP_3	3.23	548.55	OK
5/5/2026 8:14	ESP_4	3.62	534.37	OK
5/5/2026 8:14	ESP_5	2.73	566.88	OK
5/18/2026 17:04	ESP_1	3.49	292.21	OK
5/18/2026 17:04	ESP_2	3.42	469.21	OK
5/18/2026 17:04	ESP_3	3.43	305.94	OK

Timestamp	ESP ID	Voltage (V)	Power (mW)	Status
5/18/2026 17:04	ESP_4	3.25	535.63	OK
5/18/2026 17:04	ESP_5	3.59	454.38	OK
5/18/2026 17:05	ESP_1	3.49	289.15	OK
5/18/2026 17:05	ESP_2	3.42	469.21	OK
5/18/2026 17:05	ESP_3	3.43	305.94	OK
5/18/2026 17:05	ESP_4	3.25	535.63	OK
5/18/2026 17:05	ESP_5	3.59	454.38	OK

The entire set is then subjected into power related calculations to quantify the hourly average consumption, and average current draw, and the estimated battery life, summarized in Table 38. It should be noted that the listed supply voltage corresponds to the regulated output voltage supplied to the ESP32 module after the boost-buck converter stage, and not the bus voltage directly measured by the INA226 monitoring module at the battery side of the circuit.

Table 38. ESP Nodes Power Metrics Summary

Parameter	ESP_1	ESP_2	ESP_3	ESP_4	ESP_5
Supply Voltage (V)	3.3	5	3.3	5	5
Power Consumption (mW)	288.32	525.27	297.52	566.72	539.36
Average Current Draw (mA)	79.35	145.41	82.40	163.05	157.66
Battery Capacity (mAh)	3,350	3,350	3,350	3,350	3,350
Estimate Battery Life (hrs)	42.22	23.04	40.62	20.55	21.25

Results show that the combined average power consumption of all five nodes is at 2,217.19 mW. Differences in power consumption between nodes can be attributed to variations in node functionality, operating voltage, and connected peripherals. Nodes operating at 5 V generally exhibited higher average power consumption and current

draw compared to nodes operating at 3.3 V, resulting in shorter estimated battery life. In contrast, the 3.3 V nodes demonstrated relatively lower power demand and longer operational endurance. Despite these variations, all nodes maintained estimated battery runtimes exceeding 20 hours under assumed continuous deployment conditions, indicating that the proposed distributed framework is capable of prolonged battery-powered operation.

Supercapacitor Runtime of Five ESP32 Nodes. Supercapacitor runtime testing was conducted to evaluate the temporary backup operation capability of each ESP32 node during power interruption or battery replacement. For each trial, the supercapacitor was first fully charged before disconnecting the main battery supply, after which the elapsed time until ESP32 shutdown was manually recorded using a timer. Five trials were conducted for each node, and the resulting runtimes were averaged to quantify the available uninterrupted operating duration provided solely by the supercapacitor. The results are shown in Table 39.

Table 39. ESP Nodes Power Metrics Summary

Trial	ESP_1	ESP_2	ESP_3	ESP_4	ESP_5
Trial A	4:45	3:16	3:57	4:37	3:07
Trial B	4:50	3:29	4:05	4:54	2:52
Trial C	4:54	3:29	4:00	4:31	3:00
Trial D	4:51	3:00	3:49	4:56	2:54
Trial E	4:58	3:30	4:03	4:55	2:43
Average Runtime (min)	4:51	3:20	3:58	4:46	2:55

The recorded runtimes demonstrate the capability of the supercapacitor-assisted design to temporarily sustain node operation during brief power interruptions or battery replacement procedures.

During the testing, it was observed that once the LED blinks orange, battery replacement should be finished in maximum 90 seconds, and once it flashes RED, shutdown is imminent, only giving a leeway of 30 seconds, which is plenty of time. These visual cues assist the user in making sure the node doesn't just shut down, making sure sufficient data reaches the main hub.

Overall, the measured supercapacitor runtimes generally aligned with the computed average node power consumption. Nodes with lower average power demand exhibited longer uninterrupted runtimes, while nodes with higher average power consumption demonstrated shorter backup durations. For instance, ESP_4, which recorded the highest average power consumption among all nodes, also exhibited the shortest average supercapacitor runtime, while the opposite is true for ESP_1. Such observations indicate agreement between the experimentally measured runtimes and the computed power consumption data, further supporting the consistency and reliability of the gathered deployment metrics.

System Stability

System stability pertains to the capability of the system to sustain uninterrupted and consistent operation throughout deployment. In this metric group, operational continuity and data completeness was used as the stability indicators. Both of those sub metrics will be based on the main data logger of the system, which includes 3,367

expected data sets. An excerpt is shown in Figure 102, while the entire data set is given in the GitHub repository, link in Appendix A.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	inside air t	inside air h	occupancy	outside air	outside air	time of the	samsung si	samsung ri	samsung fi	samsung p	coolix setp	coolix mcr	coolix fan	coolix pow	total com	thermal cc	timestamp
2	28	80.5	1	29.15	78.52	8:13				0				7.4	7.4	0.86	5/5/2026 8:13
3	28.1	80.6	1	29.22	78.47	8:14	27 dry	auto		294.6	27 dry	auto		342.1	636.6	0.87	5/5/2026 8:14
4	28	80.1	1	29.29	78.36	8:15	27 dry	auto		699.8	27 dry	auto		456.9	1156.7	0.84	5/5/2026 8:15
5	27.8	78.4	1	29.38	78.3	8:16	27 dry	auto		795.2	27 dry	auto		456.7	1251.9	0.76	5/5/2026 8:16
6	27.7	76.8	1	29.46	78.23	8:17	27 dry	auto		787.7	27 dry	auto		458.7	1246.4	0.69	5/5/2026 8:17
7	27.5	75.5	1	29.53	78.11	8:18	27 dry	auto		728.4	27 dry	auto		457	1185.4	0.64	5/5/2026 8:18
8	27.3	74.2	1	29.59	77.99	8:19	27 dry	auto		623.3	27 dry	auto		458	1081.4	0.56	5/5/2026 8:19
9	27.2	73.2	1	29.55	77.89	8:20	27 dry	auto		489.3	27 dry	auto		457.5	946.8	0.51	5/5/2026 8:20
10	27.1	72.1	1	29.52	77.73	8:21	27 dry	auto		423	27 dry	auto		424.6	847.6	0.45	5/5/2026 8:21
11	26.9	72	1	29.56	77.64	8:22	27 dry	auto		419.3	27 dry	auto		390	809.3	0.39	5/5/2026 8:22
3359	26.6	63.3	5	33.89	64.53	16:56	18 cool		1	1065.9	18 cool		1	895.2	1961	0.11	5/18/2026 16:56
3360	26.6	62.6	5	33.86	64.55	16:57	18 cool		1	1063.6	18 cool		1	954.4	2018.1	0.09	5/18/2026 16:57
3361	26.5	62	4	33.92	64.57	16:58	18 cool		1	1061	18 cool		1	960.5	2021.5	0.07	5/18/2026 16:58
3362	26.4	61.6	5	33.95	64.66	16:59	18 cool		1	1068.2	18 cool		1	1038.1	2106.3	0.03	5/18/2026 16:59
3363	26.4	61.6	5	33.85	64.71	17:00	26 eco	auto		1064.8	26 eco	auto		1061.5	2126.4	0.02	5/18/2026 17:00
3364	26.3	61.7	8	33.89	64.77	17:01	26 eco	auto		945.8	26 eco	auto		1030.2	1976	0	5/18/2026 17:01
3365	26.3	62.4	8	34.05	64.82	17:02	26 eco	auto		930.7	26 eco	auto		993.5	1924.2	0	5/18/2026 17:02
3366	26.3	61.7	7	34.13	64.89	17:03	26 eco	auto		929.5	26 eco	auto		993.5	1923	0	5/18/2026 17:03
3367	26	62.2	7	34.15	64.95	17:04	26 eco	auto		1048.4	26 eco	auto		996	2044.3	-0.09	5/18/2026 17:04
3368	25.4	64.3	7			17:05	26 eco	auto		0	26 eco	auto		0	0	-0.27	5/18/2026 17:05

Figure 102. Excerpt of the System's Collected Dataset

The above data set will then be analyzed to get the system stability. Analysis will be based on the number of data present in each row. Out of 17 columns, the amount of lost data per row is counted and is summarized in Table 40. Unrecorded intervals pertain to the number of intervals with no data at all, while the other rows represented the total amount of intervals with only a certain number of data lost. Most of the intervals however did have a record for everything, representing the complete logging row. This table (Table 40) was used to quantify both operational continuity and data completeness.

Table 40. Logging Completeness Summary

Logging Status	Number of Intervals	Percentage
Unrecorded Intervals	237	7.04%
Eight Lost Data	58	1.72%
Seven Lost Data	0	0.00%
Six Lost Data	21	0.62%

Logging Status	Number of Intervals	Percentage
Five Lost Data	0	0.00%
Four Lost Data	0	0.00%
Three Lost Data	0	0.00%
Two Lost Data	19	0.56%
One Lost Data	3	0.09 %
Complete Logging	3029	89.96 %

Results indicate that the system maintained strong operational continuity throughout deployment, with 92.96% of the expected operational intervals successfully generating records. Among these recorded intervals, 89.96% contained complete operational data, demonstrating a high level of data completeness during system operation. Only a relatively small portion of the dataset consisted of incomplete records, with the largest occurrence involving intervals containing eight missing parameters at 1.72%, followed by intervals with six missing parameters at 0.62%. Minor instances of partial logging involving one to two missing parameters were also observed. Unrecorded intervals accounted for 7.04% of the expected operational duration, reflecting periods where operational records were not generated.

Overall, the system stability results indicate that the proposed distributed monitoring and control framework could maintain generally stable operation throughout deployment. The recorded operational continuity demonstrates that the most of the expected monitoring intervals were successfully sustained, while the high operational record completeness further indicates that most generated records contained complete operational data. Although occasional interruptions and incomplete logging events were observed, these were relatively limited compared to the overall deployment

duration and were primarily associated with temporary communication inconsistencies, node interruptions, and deployment-related operational conditions. Collectively, the stability results suggest that the framework was able to maintain persistent communication, processing, logging, and control operations under actual deployment conditions.

Economic Analysis

The second phase of the technoeconomic analysis focuses on the economic impact of the mechatronic controller. This section quantifies the system's payback period, and return of investment, based on the energy savings achieved by the algorithm, and the total expenditures (CapEX and OpEX).

CapEx Calculation

CapEX or capital expenditures, represents the one-time purchases or payments needed to progress in the fabrication of the system. This is mainly composed of two component costs, the Total Equipment Costs, and the Total Deployment Costs, as such the formula for CapEX, is the sum of the two.

Total Equipment Costs is the sum of all components costs and prices. This includes all electronic, mechanical, and miscellaneous materials. Specified in Table 41 are the models of such components and their corresponding prices, grouped per node, necessary for CapEX calculations. However, since some components are only included for prototyping and data gathering purposes, such components will be removed for marketability evaluation.

Table 41. Total Equipment Costs

COMPONENT		UNIT	PRICE	TOTAL
<i>Main Hub</i>				
Raspberry Pi 5	8Gb (RedWhite Case Kit - UK Plug)	1	₱ 9,101.93	₱ 9,101.93
Rpi Camera Module 3	12 MP w/ Auto-Focus and Wide-Angle Lens	1	₱ 2,271.50	₱ 2,271.50
Rpi 5 Active Cooler	52Pi Low-Profile ICE Tower w/ Heatsink	1	₱ 688.00	₱ 688.00
Wi-Fi Access Point	TP-Link WR820N	1	₱ 599.00	₱ 599.00
RTC Module	DS3231 w/ AT24C32 EEPROM	1	₱ 140.00	₱ 140.00
Filament Used	Bambu Lab Filament PLA Lite 1.75mm	0.1569 4	₱ 759.00	₱ 119.12
Screws	Wood Screw	4	₱ 0.75	₱ 3.00
<i>ESP 1 - Dynamic Inside Sensor</i>				
NodeMCU ESP32	Microcontroller with Expansion Board	1	₱ 477.90	₱ 477.90
Li-ion Battery	PKCell 3.7V NMC 18650 3350 mAh	0	₱ 205.00	₱ 00.00
Battery Holder	Series 18650 Single Cell	0	₱ 33.00	₱ 00.00
Switch	Single Pole Single Throw KCD11	1	₱ 7.80	₱ 7.80
Supercapacitor	5.5V 15F	0	₱ 128.00	₱ 00.00
DC Power Monitor	INA226 I2C Interface	0	₱ 72.00	₱ 00.00
Boost-Buck Converter	TPS63070 3.3V Regulator	0	₱ 105.00	₱ 00.00
Air Quality Sensor	SHT31	1	₱ 139.00	₱ 139.00
RGB LED Module	SMD KY-016	1	₱ 40.00	₱ 40.00
Resistor	1/4W 1% 230Ω	3	₱ 0.37	₱ 1.10
Screw Terminal	KF301-2P 5.0mm Straight Pin	1	₱ 5.50	₱ 5.50
Filament Used	Bambu Lab Filament PLA Lite 1.75mm	0.1146 1	₱ 759.00	₱ 86.99
PCB Costs	JASCad	1	₱ 400.00	₱ 400.00
Screws	Wood Screw	4	₱ 0.75	₱ 3.00
<i>ESP 2 - Static Inside Sensor</i>				
NodeMCU ESP32	Microcontroller with Expansion Board	1	₱ 477.90	₱ 477.90
Li-ion Battery	PKCell 3.7V NMC 18650 3350 mAh	0	₱ 205.00	₱ 00.00
Battery Holder	Series 18650 Single Cell	0	₱ 33.00	₱ 00.00
Switch	Single Pole Single Throw KCD11	1	₱ 7.80	₱ 7.80

COMPONENT		UNIT	PRICE	TOTAL
Supercapacitor	5.5V 15F	0	₱ 128.00	₱ 00.00
DC Power Monitor	INA226 I2C Interface	0	₱ 72.00	₱ 00.00
Boost Converter	TPS61023 5V Regulator	0	₱ 195.00	₱ 00.00
Air Quality Sensor	SHT31	1	₱ 139.00	₱ 139.00
RGB LED Module	SMD KY-016	1	₱ 40.00	₱ 40.00
Resistor	1/4W 1% 230Ω	3	₱ 0.37	₱ 1.10
Screw Terminal	KF301-2P 5.0mm Straight Pin	1	₱ 5.50	₱ 5.50
Transistor	2N2222	2	₱ 5.00	₱ 10.00
IR LED	5mm 940nm Infrared Light Emitter Diode	2	₱ 4.70	₱ 9.40
Resistor	1/4W 1% 37Ω	2	₱ 0.37	₱ 0.73
Resistor	1/4W 1% 470Ω	2	₱ 0.37	₱ 0.73
Filament Used	Bambu Lab Filament PLA Lite 1.75mm	0.17335	₱ 759.00	₱ 131.57
PCB Costs	JASCad	1	₱ 400.00	₱ 400.00
Screws	Wood Screw	4	₱ 0.75	₱ 3.00

ESP 3 - Outside Sensor

NodeMCU ESP32	Microcontroller with Expansion Board	1	₱ 477.90	₱ 477.90
Li-ion Battery	PKCell 3.7V NMC 18650 3350 mAh	0	₱ 205.00	₱ 00.00
Battery Holder	Series 18650 Single Cell	0	₱ 33.00	₱ 00.00
Switch	Single Pole Single Throw KCD11	1	₱ 7.80	₱ 7.80
Supercapacitor	5.5V 15F	0	₱ 128.00	₱ 00.00
DC Power Monitor	INA226 I2C Interface	0	₱ 72.00	₱ 00.00
Boost-Buck Converter	TPS63070 3.3V Regulator	0	₱ 105.00	₱ 00.00
Air Quality Sensor	SHT31	1	₱ 139.00	₱ 139.00
RGB LED Module	SMD KY-016	1	₱ 40.00	₱ 40.00
Resistor	1/4W 1% 230Ω	3	₱ 0.37	₱ 1.10
Screw Terminal	KF301-2P 5.0mm Straight Pin	1	₱ 5.50	₱ 5.50
Filament Used	Bambu Lab Filament PLA Lite 1.75mm	0.20753	₱ 759.00	₱ 157.52
PCB Costs	JASCad	1	₱ 400.00	₱ 400.00
Screws	Wood Screw	4	₱ 0.75	₱ 3.00

ESP 4 - ACU Power Monitor 1

NodeMCU ESP32	Microcontroller with Expansion Board	1	₱ 477.90	₱ 477.90
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COMPONENT		UNIT	PRICE	TOTAL
Li-ion Battery	PKCell 3.7V NMC 18650 3350 mAh	0	₱ 205.00	₱ 00.00
Battery Holder	Series 18650 Single Cell	0	₱ 33.00	₱ 00.00
Switch	Single Pole Single Throw KCD11	1	₱ 7.80	₱ 7.80
Supercapacitor	5.5V 15F	0	₱ 128.00	₱ 00.00
DC Power Monitor	INA226 I2C Interface	0	₱ 72.00	₱ 00.00
Boost Converter	TPS61023 5V Regulator	0	₱ 195.00	₱ 00.00
AC Power Monitor	PZEM-004T v3.0 w/ Open CT	1	₱ 670.00	₱ 670.00
RGB LED Module	SMD KY-016	1	₱ 40.00	₱ 40.00
Resistor	1/4W 1% 230Ω	3	₱ 0.37	₱ 1.10
Screw Terminal	KF301-2P 5.0mm Straight Pin	1	₱ 5.50	₱ 5.50
Stranded Wire		1	₱ 25.00	₱ 25.00
Plug		1	₱ 45.00	₱ 45.00
Filament Used	Bambu Lab Filament PLA Lite 1.75mm	0.1567 9	₱ 759.00	₱ 119.00
PCB Costs	JASCad	1	₱ 400.00	₱ 400.00
Screws	Wood Screw	4	₱ 0.75	₱ 3.00

ESP 5 - ACU Power Monitor 2

NodeMCU ESP32	Microcontroller with Expansion Board	1	₱ 477.90	₱ 477.90
Li-ion Battery	PKCell 3.7V NMC 18650 3350 mAh	0	₱ 205.00	₱ 00.00
Battery Holder	Series 18650 Single Cell	0	₱ 33.00	₱ 00.00
Switch	Single Pole Single Throw KCD11	1	₱ 7.80	₱ 7.80
Supercapacitor	5.5V 15F	0	₱ 128.00	₱ 00.00
DC Power Monitor	INA226 I2C Interface	0	₱ 72.00	₱ 00.00
Boost Converter	TPS61023 5V Regulator	0	₱ 195.00	₱ 00.00
AC Power Monitor	PZEM-004T v3.0 w/ Open CT	1	₱ 670.00	₱ 670.00
RGB LED Module	SMD KY-016	1	₱ 40.00	₱ 40.00
Resistor	1/4W 1% 230Ω	3	₱ 0.37	₱ 1.10
Screw Terminal	KF301-2P 5.0mm Straight Pin	1	₱ 5.50	₱ 5.50
Stranded Wire		1	₱ 25.00	₱ 25.00
Plug		1	₱ 45.00	₱ 45.00
Filament Used	Bambu Lab Filament PLA Lite 1.75mm	0.1567 9	₱ 759.00	₱ 119.00
PCB Costs	JASCad	1	₱ 400.00	₱ 400.00
Screws	Wood Screw	4	₱ 0.75	₱ 3.00

COMPONENT		UNIT	PRICE	TOTAL
<i>Other General Components</i>				
Li-ion Battery Charger	CMOCO Lii-402 Universal Li-ion Charger	1	₱ 446.00	₱ 446.00
Heat Shrink Tubes	LESU PE Insulation Materials Kit (127 pcs)	1	₱ 56.00	₱ 56.00
Jumper Wires	Dupont Wires 2.54mm Assorted	1	₱ 265.00	₱ 265.00
Cable Wires	JST-XH 2.54mm Assorted	1	₱ 380.00	₱ 380.00
Electrical Tape	Medium Size	2	₱ 45.00	₱ 90.00
Soldering Lead		1	₱ 45.00	₱ 45.00
Pin Header	10 pcs 1x40 Single Row Male 2.54mm Strip	3	₱ 5.60	₱ 16.80
USB-C Power Meter	POROMETISTO UT02C Max 12A	0	₱ 389.00	₱ 00.00
Breadboard	ENGLAB MB102 400 Tie Points	0	₱ 45.00	₱ 00.00
TOTAL EQUIPMENT COSTS				₱ 21,429.80

Another consideration for CapEX is the installation costs or deployment costs, which include costs that are necessary to deploy the system. This also includes the additional fees that are included in purchasing some equipment. Table 42 shows the details of the deployment cost.

Table 42. Total Deployment Costs

TYPE	DESCRIPTION	UNIT	RATE	TOTAL
Labor Costs	2 personnels (hardware)	1 day	₱ 500.00	₱ 1,000.00
	1 personnel (software)	2 days	₱ 500.00	₱ 1,000.00
Engineering Costs	PCB Designing	—	₱ 1,000.00	₱ 1,000.00
	Bambu Lab A1 Mini Power Consumption (57W Average)	28.87 hours	₱ 10.5049	₱ 17.28
Contingency Costs	5% of Total Material Costs	₱ 21,429.80	5%	₱ 1,071.49
TOTAL DEPLOYMENT COSTS				₱ 4,088.77

From the CapEX formula, the total capital is:

$$CapEX = Equipment\ Costs + Deployment\ Costs$$

$$CapEX = Php\ 21,429.80 + Php\ 4,088.77$$

$$CapEX = \textbf{Php 25,518.57}$$

Since the product will be sold as a set, with instructions for maintenance for the user to follow, the selling price can be based only on CapEX. The proposed selling price will be Php 26,000.00.

OpEX Calculation

OpEX or operational expenditures refer to the cost required by the system to function. This includes expenses from electricity and maintenance of the system. Electricity expenditures are calculated per node, using the average power consumption previously calculated in Technical Metrics, in this chapter. As of May 2026, the electricity rate of Baybay City, Leyte is at Php 10.5049/kWh (Leyeco IV, 2026). The calculations are detailed in Table 43.

Table 43. Electricity Costs

Node	Consumption	Hours per Day	Days per Week	Weeks per Year*	Hours of No Operation**	Annual kWh
RPi 5	2.90 W	8	5	48	256	4.82
ESP_1	0.29 W	8	5	48	256	0.48
ESP_2	0.53 W	8	5	48	256	0.87
ESP_3	0.30 W	8	5	48	256	0.50
ESP_4	0.57 W	8	5	48	256	0.94
ESP_5	0.54 W	8	5	48	256	0.90

ANNUAL SYSTEM ENERGY CONSUMPTION	8.51 kWh
TOTAL ELECTRICITY COSTS	₱ 89.40
*Weeks per year is at 48 instead of 52 to consider unforeseen events (4-week downtime)	
**256 hours of no operation came from 20 holidays and 12 monthly maintenance activities.	

An annual preventive maintenance allowance equivalent to 5% of the selling price (Php 1,300) was included in the operational expenditure analysis to account for periodic inspection, recalibration, cleaning, firmware maintenance, and minor component servicing required for sustained system reliability. This value will be added in the calculation of the OpEX.

From the OpEX formula, the total operational cost is:

$$OpEX = Electricity\ Costs + Maintenance\ Costs$$

$$OpEX = \text{Php } 89.40 + \text{Php } 1,300.00$$

$$OpEX = \text{Php } 1,389.40$$

Annual Energy Savings

From the results of the Comparative Performance Evaluation: BORF vs. Manual Operation, also found in this chapter, the annual energy savings can be calculated. The results of BORF vs Manual Operation and as to how they were treated is detailed in Table 44.

Table 44. Annual Energy Savings

Results of BORF vs Manual Operation		Testing consumption (kWh)	Hourly kWh			
Total Manual Energy Consumption		106.95 kWh	1.91 kWh			
Total BORF Energy Consumption		84.98 kWh	1.52 kWh			
Energy Savings		21.97 kWh	0.39 kWh			
Mode	Hourly kWh	Hours per Day	Days per Week	Weeks per Year	Hours of No Operation	Annual kWh
Manual	1.91 kWh	6	5	48	256	2,261.15
BORF	1.52 kWh	6	5	48	256	1,796.72
Savings	0.39 kWh	6	5	48	256	464.43
6 hours of utilization per day was assumed to mimic classroom settings, or when the room is occupied.						
Weeks per year is at 48 instead of 52 to consider unforeseen events (4-week downtime)						
256 hours of no operation came from 20 holidays and 12 monthly maintenance activities.						

As shown in Table 44, ‘Testing Consumption’ corresponds to the total amount of power consumed by both ACU units, across seven trials conducted, spanning 8 hours each. Therefore, ‘Hourly kWh’ is the hourly energy consumption, averaged across 56 cumulative testing hours conducted during the comparative evaluation. The resulting hourly energy consumption values were then projected into annual operation using assumed operating schedules, including hours per day, operating days per week, operational weeks per year, and estimated downtime periods.

Payback Period (PBP) and Return of Investment (ROI)

In determining the marketability of this system, the proposed Smart HVAC Controller shall be able to offer a reasonable ROI in the customer’s perspective. To do

that, a consumer-side lifecycle cash flow analysis is performed. Figure 103 shows the assumptions and parameters used in the analysis.

Smart HVAC Controller Consumer-Side Assumptions			
Input	Value	Unit	Purpose / Meaning
Electricity Rate	P10.5049	Peso/kWh	Consumer tariff paid for HVAC electricity consumption in Year 1.
Electricity Rate Escalation	9.0%	%/year	Expected yearly increase in electricity price.
Existing HVAC Annual Energy Consumption (Before Smart Controller)	2261.15	kWh/year	Measured or estimated Year 1 annual HVAC energy consumption under manual setpoint operation. The HVAC unit is not purchased; it already exists.
Annual HVAC Energy Consumption Variation	2.0%	%/year	Expected yearly change in existing HVAC energy use due to weather, occupancy, operating hours, and equipment aging.
Measured Energy Savings Achieved by Smart HVAC Controller	20.5%	%	Experimentally observed savings delivered by the controller; applied to each year's projected baseline HVAC consumption.
Smart HVAC Controller Package Cost	P26,000	Peso	One-time consumer cost for the controller package only. This does NOT include the HVAC unit.
Controller Installation & Setup Cost	P0	Peso	One-time installation, configuration, and commissioning cost for the controller.
Smart Controller Annual Maintenance	P1,389	Peso/year	Recurring consumer-side operating cost. Enter 0 if none.
Smart Controller Useful Life	12	years	Expected service life of the Smart HVAC controller package.
Smart Controller Salvage Value	0	Peso	Residual value at the end of the controller useful life for depreciation calculation.

Figure 103. Smart HVAC Controller Consumer-Side Assumptions

Those assumptions are set to ascertain the long-term economic feasibility of implementing the proposed Smart HVAC Controller under realistic consumer-side operational conditions. Using the experimentally obtained annual energy savings of 20.5%, the lifecycle cash flow projection was conducted to determine the projected payback period, annual savings progression, and overall return on investment throughout the system's expected operational life.

Figure 104 presents the consumer adoption scenarios evaluated in the study. The scenarios considered variations in controller package cost and electricity tariff assumptions to represent conservative, base-case, and optimistic market conditions. Among these, the base-case scenario was used as the primary basis for the economic evaluation discussion since it reflects the nominal projected operating condition of the proposed Smart HVAC Controller.

Consumer Adoption Scenarios							
Scenario	Controller Cost Factor	Savings %	Electricity Rate	Year 1 Net Savings	Total CAPEX	Simple Payback	12-Year ROI
Conservative	120.0%	20.5%	P9.50	P3,023	P31,200	10.32	16.3%
Base	100.0%	20.5%	P10.50	P3,489	P26,000	7.45	61.0%
Optimistic	90.0%	20.5%	P12.00	P4,184	P23,400	5.59	114.6%

Figure 104. Consumer Adoption Scenarios

Under the base-case scenario, the proposed Smart HVAC Controller achieved a projected Simple Payback Period (PBP) of approximately 7.45 years with an estimated end-of-life Return on Investment (ROI) of 61.0% throughout the 12-year analysis horizon. The obtained payback period indicates that the cumulative operational savings generated by the Smart HVAC Controller could recover the initial controller investment cost within the expected operational life of the system. Meanwhile, the positive ROI value indicates that the proposed system remained economically profitable beyond investment recovery.

Following the scenario-based economic indicators, the lifecycle consumer cash flow projection was further analyzed to observe how the financial performance evolved throughout the operational life of the system. Figure 105 presents the projected yearly cash flow comparison between conventional manual HVAC operation and Smart HVAC Controller operation under the base-case scenario.

Consumer Cash Flow: Existing HVAC vs Smart HVAC Controller Purchase									
Year	Electricity Rate	Manual HVAC kWh	Smart HVAC kWh	kWh Saved	Manual Bill	Smart Bill	Recurring Smart Cost	Net Annual Savings	Cumulative Net Cash Flow
0	P0.0000	-	-	-	-	-	-	(P26,000)	(P26,000)
1	P10.5049	2,261	1,797	464	P23,753	P18,874	P1,389	P3,489	(P22,511)
2	P11.4503	2,306	1,833	474	P26,409	P20,985	P1,389	P4,035	(P18,476)
3	P12.4809	2,353	1,869	483	P29,361	P23,331	P1,389	P4,641	(P13,834)
4	P13.6042	2,400	1,907	493	P32,644	P25,939	P1,389	P5,316	(P8,519)
5	P14.8285	2,448	1,945	503	P36,293	P28,839	P1,389	P6,065	(P2,454)
6	P16.1631	2,496	1,984	513	P40,351	P32,063	P1,389	P6,899	P4,445
7	P17.6178	2,546	2,023	523	P44,862	P35,648	P1,389	P7,825	P12,270
8	P19.2034	2,597	2,064	533	P49,878	P39,633	P1,389	P8,855	P21,125
9	P20.9317	2,649	2,105	544	P55,454	P44,064	P1,389	P10,001	P31,126
10	P22.8155	2,702	2,147	555	P61,654	P48,991	P1,389	P11,274	P42,400
11	P24.8689	2,756	2,190	566	P68,547	P54,468	P1,389	P12,690	P55,090
12	P27.1071	2,811	2,234	577	P76,210	P60,557	P1,389	P14,264	P69,354

Figure 105. Consumer Cash Flow: Existing HVAC vs Smart HVAC Control

Based on the lifecycle cash flow projection, the Smart HVAC Controller generated increasing annual savings throughout succeeding operational years. During Year 1, the projected net annual savings amounted to approximately ₱3,489. However, by Year 12, the projected annual savings increased to approximately ₱14,264. This progressive increase in yearly savings was primarily caused by the incorporated annual electricity tariff escalation and annual HVAC energy consumption variation assumptions within the economic model.

Although the experimentally measured energy savings percentage remained constant at 20.5%, the monetary value of the avoided energy consumption continuously increased over time due to rising electricity rates and projected increases in baseline HVAC energy consumption. As a result, the later operational years contributed significantly larger annual savings compared to the earlier years of operation.

The cumulative cash flow projection further demonstrated that the system initially operated under negative cash flow conditions due to the upfront controller investment cost. However, as yearly savings accumulated and progressively increased, the cumulative cash flow eventually transitioned into positive recovery within the projected payback period. Beyond the payback period, succeeding years primarily contributed to net positive savings accumulation, thereby strengthening the overall lifecycle profitability of the proposed Smart HVAC Controller.

These results indicate that the proposed Smart HVAC Controller becomes increasingly economically advantageous under long-term operation, particularly under conditions of rising electricity costs and increasing HVAC operational demand.

CHAPTER 5

CONCLUSIONS AND RECOMMENDATIONS

Conclusions

This chapter presents the overall conclusions derived from the design, implementation, and evaluation of the proposed Distributed IoT Mechatronic System. The findings generated from the experimental investigations and system performance analyses provided valuable insights into the application of data-driven control strategies for energy conservation in conditioned spaces.

Hybrid Machine Learning Model System

The Bayesian Optimized Random Forest (BORF) model was developed using a filtered dataset of 3,000 steady-state data points, mapping air conditioning settings to environmental boundary conditions. This dataset was partitioned into 2,500 training samples and 500 validation samples. Despite extensive Bayesian hyperparameter optimization benchmarked against unseen data, the standalone machine learning model achieved maximum R^2 scores of 0.204 for PMV and 0.338 for energy consumption. These metrics confirm that a 3,000-point dataset is fundamentally insufficient to comprehensively map the high-variance thermodynamic behavior of the target room.

To resolve the predictive limitations of the machine learning model, this study successfully designed and implemented a hybrid control architecture. By integrating the predictive BORF model with a physics-informed, rule-based supervisory logic, the system established a deterministic safeguard to override anomalous or

thermodynamically invalid AI outputs. This architecture ensures that the system maintains occupant thermal comfort within strict ASHRAE standards, even when exposed to harsh, volatile real-world environments.

Upon real-world deployment, the Hybrid BORF system achieved an average energy savings of 20.54% compared to standard human-operated AC systems, with specific operational savings ranging from 7% to 38%. These savings were realized by actively preventing system overcooling and dynamically matching AC operations to real-time environmental demand. Ultimately, this research proves that even a data-constrained machine learning model, when intelligently paired with a deterministic physical override, can deliver highly effective, energy-efficient automation in real-world HVAC applications.

Digital Twin

This study successfully developed, calibrated, validated, and deployed an EnergyPlus-based Digital Twin of the selected conference room using an open-source building performance simulation platform integrated with real-time telemetry through an MQTT communication framework. The developed framework established continuous synchronization between the physical asset and its virtual counterpart, enabling virtual parallel assessment of building energy performance and indoor environmental conditions under both offline and live operational scenarios.

The baseline Building Energy Model exhibited substantial discrepancies between measured and simulated HVAC energy consumption, producing a CV(RMSE) of 54.13% and an NMBE of 40.46%, which exceeded the hourly calibration acceptance

criteria specified in ASHRAE Guideline 14. Although the baseline model could reproduce the general thermal and moisture behavior of the conference room, it was unable to accurately quantify HVAC energy consumption, demonstrating the need for calibration prior to Digital Twin deployment.

Bayesian calibration using a Gaussian Process Regression-based optimization framework successfully reduced model uncertainty and improved agreement between measured and simulated performance. The calibrated model achieved an hourly HVAC energy CV(RMSE) of 29.96% and an NMBE of 9.96%, satisfying the ASHRAE Guideline 14 acceptance criteria. The optimization process converged toward a stable parameter set that improved the representation of measured cooling-energy behavior while maintaining realistic indoor environmental conditions.

Validation using an independent dataset confirmed the generalizability of the calibrated Digital Twin beyond the calibration period. The model achieved a validation CV(RMSE) of 28.23% and an NMBE of 7.64% for HVAC energy consumption, demonstrating that the optimized parameter set retained its predictive capability under previously unseen operating conditions. The consistency between calibration and validation performance indicates that the Digital Twin captured the operational characteristics of the conference room rather than merely reproducing the calibration dataset.

The operational deployment phase further demonstrated the practical applicability of the developed framework. Through real-time integration of occupancy, temperature, humidity, thermostat settings, and HVAC operating conditions, the Digital Twin continuously updated its virtual representation and maintained synchronization

with the physical environment. During live deployment, the Digital Twin achieved an HVAC energy RMSE of 0.277 kWh, a CV(RMSE) of 15.60%, and an NMBE of 14.24%, confirming its capability for real-time HVAC energy monitoring and operational performance assessment.

In addition to energy-performance evaluation, the developed Digital Twin successfully enabled comfort validation through virtual parallel assessment. Measured and simulated indoor temperature, relative humidity, and Predicted Mean Vote (PMV) were continuously compared throughout deployment, allowing simultaneous assessment of both building energy performance and occupant thermal comfort. The incorporation of EnergyPlus-generated PMV within the operational Digital Twin extended the framework beyond energy monitoring and demonstrated its capability to represent comfort-related building behavior under actual operating conditions.

Collectively, the findings confirm that the study successfully achieved its objectives. The developed Digital Twin provided a calibrated and validated virtual representation of the target building capable of quantifying HVAC energy consumption, evaluating indoor environmental conditions, and supporting comfort assessment through virtual parallel operation. The integration of EnergyPlus simulation, Bayesian calibration, MQTT-based communication, and real-time visualization demonstrates the feasibility of developing practical and scalable Digital Twin frameworks using open-source technologies for building performance assessment, operational monitoring, and future intelligent building applications.

Distributed IoT-Mechatronic Controller

The results of the study demonstrate that the proposed distributed mechatronic monitoring and control framework achieved stable and efficient operational performance throughout the deployment period. Computational performance analysis showed that the Raspberry Pi 5 maintained low average CPU utilization of 4.58%, stable CPU temperature behavior averaging 51.70 °C, and moderate RAM utilization averaging 26.19%, indicating that the system possessed sufficient computational and thermal headroom during continuous operation.

Communication reliability metrics further revealed low average latency of 5.60 ms, manageable network load, and an average packet loss rate of only 0.0115%, confirming that the MQTT-based communication framework could maintain dependable wireless data transmission with minimal interruptions. These findings indicate that the developed framework can sustain real-time monitoring and control operations while remaining scalable for future system expansion.

The energy consumption analysis also confirmed that the proposed system operated within practical and sustainable power requirements. The Raspberry Pi 5 exhibited an average power consumption of 2.90 W during deployment, while the ESP32 nodes maintained estimated battery life durations ranging from approximately 20 to 42 hours depending on node configuration and operating voltage.

In addition, the implemented supercapacitor-assisted backup system successfully provided temporary uninterrupted operation during power interruptions and battery replacement procedures, demonstrating the effectiveness of the backup

mechanism in preserving operational continuity. Overall, the results suggest that the framework is suitable for prolonged deployment under continuous operating conditions with manageable energy requirements.

Lastly, the system stability evaluation showed that the framework maintained strong operational continuity and high data completeness during actual deployment conditions. Results indicated that 92.96% of the expected operational intervals successfully generated records, while 89.96% of the collected records contained complete operational data. Although isolated cases of incomplete logging and temporary interruptions were observed, these incidents remained limited relative to the overall deployment duration and were primarily associated with temporary communication inconsistencies and operational conditions.

Collectively, the findings confirm that the proposed distributed monitoring and control framework could maintain persistent communication, processing, logging, and control operations under real deployment scenarios. Therefore, the study concludes that the developed framework is technically viable, operationally reliable, and capable of supporting future enhancements and larger-scale deployment applications.

The economic and marketability analysis indicates that the proposed Smart HVAC Controller is financially feasible and potentially marketable for long-term implementation. The system obtained a total projected CapEx of ₱25,518.57 with a proposed selling price of ₱26,000.00, while maintaining relatively low annual operational expenditures amounting to ₱1,389.40. Based on the comparative evaluation between BORF and manual HVAC operation, the controller achieved an estimated

annual energy savings of 20.5%, equivalent to approximately 464.43 kWh of reduced energy consumption annually.

Under the base-case scenario, the system attained a projected simple payback period of approximately 7.45 years and an end-of-life return on investment (ROI) of 61.0% across a 12-year operational lifespan. Furthermore, lifecycle cash flow projections demonstrated progressively increasing annual savings due to rising electricity rates and HVAC operational demand, indicating that the proposed Smart HVAC Controller becomes more economically advantageous over time. These findings suggest that the developed system is not only technically reliable but also economically sustainable and commercially viable for practical HVAC energy management applications.

Recommendations

While the study successfully demonstrated the feasibility of the proposed Hybrid BORF System, opportunities remain for further improvement and validation. The following recommendations are presented to address the limitations encountered during the research and to provide direction for future investigations aimed at advancing AI-driven HVAC optimization and energy-efficient building operation.

Expansion of Data Acquisition Parameters. To develop a more robust and highly generalized predictive model, future iterations must prioritize expansive data acquisition. It is recommended to collect approximately 50,000 or a minimum of 6 months of diverse data points over a multi-month and various season period of continuous, uninterrupted logging. This expanded dataset is necessary to capture a

wider spectrum of outdoor meteorological conditions, varying occupancy densities, and stochastic occupant behaviors, ensuring the algorithm can map seasonal and daily environmental shifts.

Integration of Dynamic PMV Variables. A critical limitation of the current study was treating highly variable Predicted Mean Vote (PMV) parameters such as air velocity, metabolic rate, mean radiant temperature of the walls, and clothing insulation, as static constants. In real-world thermodynamic environments, these dynamic factors are integral to both energy optimization and occupant comfort. It is also equally important to properly validate PMV to the actual thermal sensation of occupants and must match both values. To resolve this, future research should integrate:

- **Dynamic Air Velocity Sensor Deployment:** Installing localized anemometers to measure real-time air velocity changes in the room. Air velocity massively affects thermal comfort of occupants due to rapid heat transfer through convection.
- **Computer Vision Improvement:** Implementing a dedicated computer vision model, utilizing object detection architectures, to actively classify and determine the clothing insulation factor and level of activity of the occupants in real time which also directly affects the rooms thermodynamic and occupants' comfort.
- **Integration of Thermal Camera for localized heat mapping:** One of the major factors that affects the occupancy's thermal comfort is the surface temperatures of their skin, clothing, walls, and the surrounding environment which creates a localized thermal climate that cannot be

easily seen by native temperature sensors but massively affects the occupant's thermal sensation as documented in ASHRAE 55. It is a massive upgrade to include visual AI learners such as YOLO models to train with thermal cameras to learn the invisible thermal buildup and thermal inertia of the room.

- **Integration of Local Language Model:** The actual thermal sensation of the occupants is a very complex problem, one may say that the PMV matches the thermal feelings of the occupants at the moment, but it may not be the same all through out. Thus, integration of LLMs to listen and hear out the voices of the occupants on what they feel about the room and dynamically adjusts its decision weights based on what it hears from the occupants. Through this, it is also possible to establish a simple communication to the control system, ask some questions related to the system, or make some requests.

Integration of an Adaptive Neuro-Fuzzy Inference System (ANFIS) into the Supervisory Framework: The current control architecture relies on static, rule-based Boolean logic to serve as a safety net against machine learning model failures. While this provides a foundational safeguard, the rigidity of hard-coded logic precludes the system from adapting to evolving environmental conditions or learning from historical performance data.

Future iterations of this study should prioritize the integration of an Adaptive Neuro-Fuzzy Inference System (ANFIS) to replace these static protocols. By combining the adaptive learning capabilities of neural networks with the rule-based

transparency of fuzzy logic, ANFIS can dynamically manage process errors in real time while the predictive supervisory AI model makes future predictions. Unlike the existing binary approach, ANFIS utilizes self-adjusting decision weights to model the non-linear thermal dynamics of the environment, enabling the system to robustly mitigate errors and maintain thermal equilibrium even amidst harsh, fluctuating environmental conditions.

Use other more powerful models capable of learning time sequence events and chain reaction. As documented in this study, the Random Forest model had its shortcomings when it comes to learning complex behavior such as thermodynamics. It is noteworthy to try other lighter models with more powerful interpreters such as XGBoost or Reinforce Learning models.

Tradeoff with massive AI models and lighter models. Lighter AI models are easy to train and deploy on edge devices but can't learn complex chain reaction events of thermodynamic problem, Larger AI models are more capable but cannot be deployed to edge devices unless retrofitted with upgrades such as AI Pi Hat. Thus, in this study, we recommend the use of AI Pi Hat to further boost the computational power of the Raspberry Pi, allowing the integration of Local Language Models, Thermal viewing by YOLO, and ultimately the deployment of much better Neural Networks to learn the complex dynamic system of the room.

Accounting User Feedback. In terms of control, the current study does not take the occupants' thermal sensation into account. Future studies should integrate user feedback into consideration when training the model. This can be gathered either via a User Interface or as a button in the Dynamic Inside Sensor Node.

Mechanical and Hardware Considerations Future models must account for the physical degradation of the AC unit over time, as the predictive accuracy and efficiency of the control system are inherently tied to the mechanical performance of the HVAC hardware. Furthermore, physical interventions are recommended to improve the spatial air circulation within the target environment.

Localized thermal transients frequently occur when concentrated heat sources generate thermal loads that exceed the cooling capacity of natural convection, resulting in isolated temperature spikes. Integrating forced convection systems, such as auxiliary fans, will accelerate air distribution. This physical intervention minimizes spatial overcooling and saving energy, enhances overall occupant comfort, and ensures a more homogenous thermal gradient across the room.

Evaluating scalability and adding more data nodes. Since this study has proved that the system is stable and imposes little computational requirement, future studies should focus on evaluating the scalability of the system, either on simulating additional network loads or by adding more nodes. This will further evaluate if the system can be evaluated on a larger scale, like in an industrial or commercial setting.

Improving the prediction of thermal-comfort-related variables through calibration of building-envelope characteristics. While the present study successfully achieved ASHRAE Guideline 14 compliance for HVAC energy consumption, calibration efforts were primarily directed toward HVAC operational performance. Since indoor temperature, relative humidity, and PMV were evaluated but not established as primary calibration objectives, incorporating building-envelope

calibration may further improve the Digital Twin's capability to represent indoor environmental conditions and occupant thermal comfort.

Extending the operational Digital Twin platform beyond local deployment by integrating cloud-based hosting and remote-access capabilities. The current Streamlit-based dashboard operates within a localized environment, limiting accessibility to the system administrator or developer. Deploying the platform through cloud infrastructure or web-hosted services would enable authorized users, facility managers, and building operators to access real-time Digital Twin information remotely and support broader operational adoption.

Extended operational deployment and validation. Validation should be conducted over longer monitoring periods and under a wider range of occupancy and environmental conditions. Evaluating the Digital Twin across different seasons, operating schedules, and usage patterns would provide additional evidence of long-term robustness, predictive reliability, and applicability for real-world building-performance assessment

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APPENDICES

Appendix A

Software Repository, Libraries, and Database Accessibility

Due to the comprehensive multi-disciplinary scope and interconnected subsystems required to operate this system, the complete codebase, system libraries, architecture details, and datasets have been archived externally to maintain manuscript readability and formatting integrity. Comprehensive documentation, deployment instructions, and hardware configuration requirements for system replication are detailed within the repository's primary documentation file

QR Code:

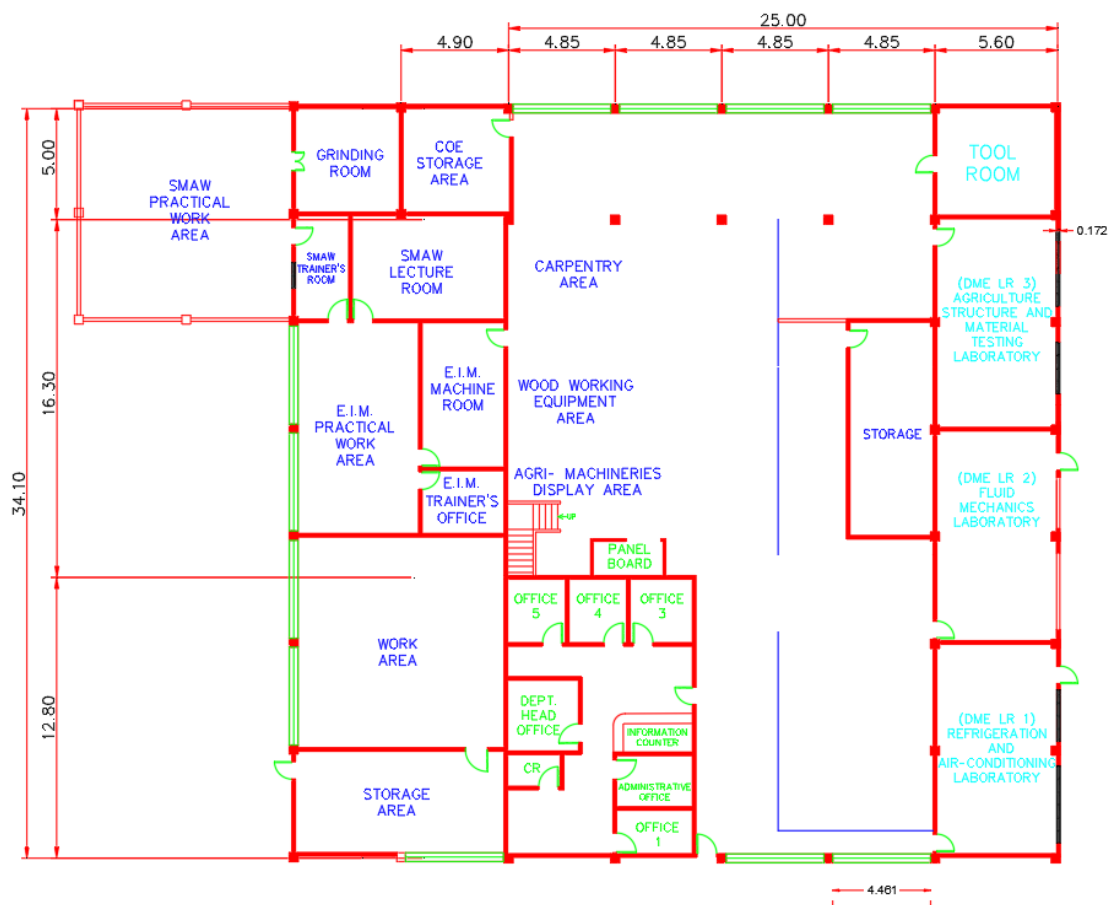


Repository Link:

<https://github.com/poor-charles/Distributed-IoT-Mechatronic-System-for-HVAC-Optimization-using-Hybrid-AI-Logic>

Appendix B

DME Floor Plan



Appendix C

General Royal ACU Specifications



**GENERAL[®]
Royal**
AIR CONDITIONER

1.0-3.0HP WALL SERIES

Premier Inverter Series



**PREMIER
iNVERTER**




3D AIR FLOW


FOLLOW ME


ECO MODE


SELF CLEAN


1 WATT STANDBY

SPECIFICATIONS

MODEL NUMBER

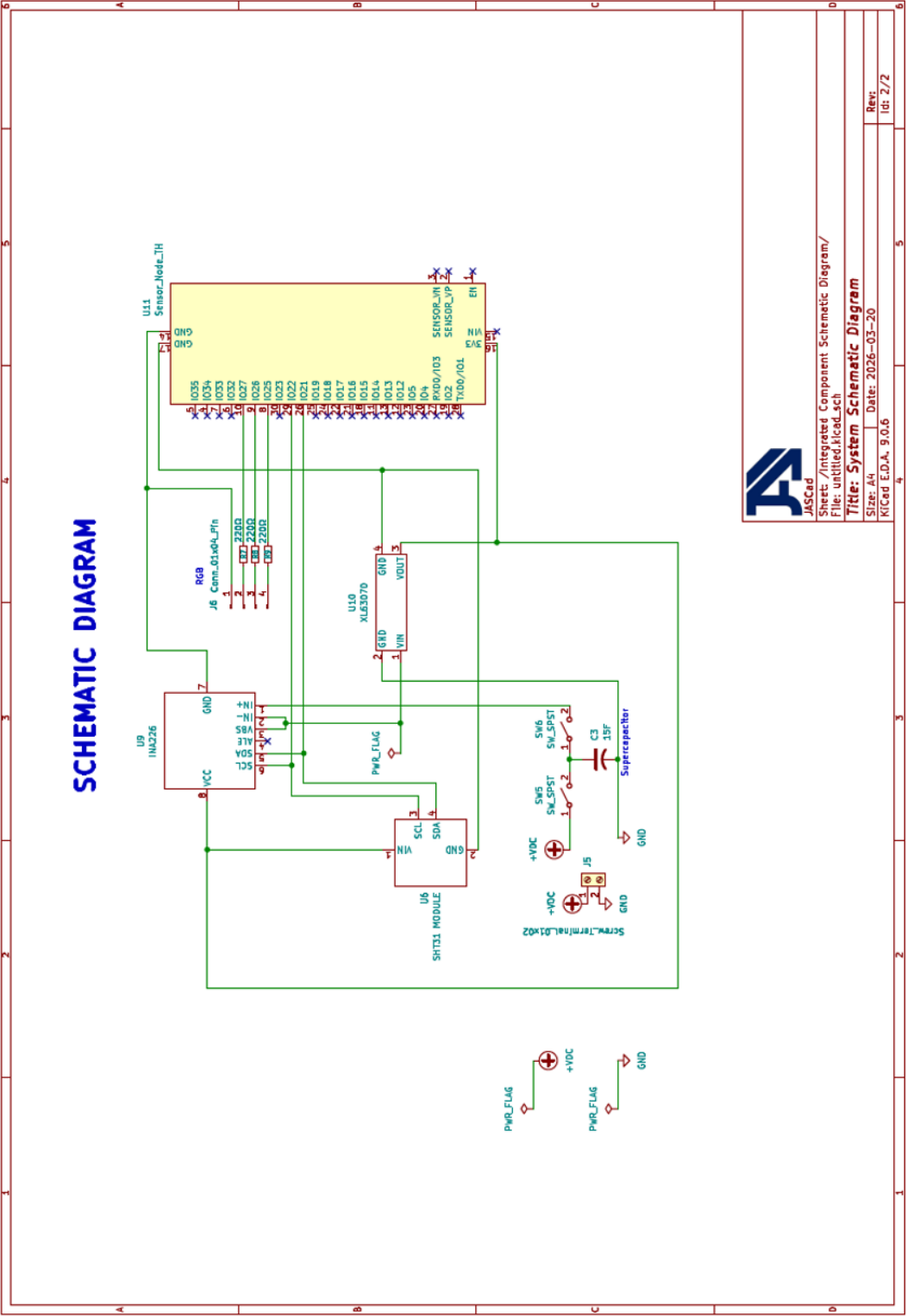
HP	UNIT	GRIU-10AWGIN3 GRIU-10AWGON3	GRIU-15AWGIN3 GRIU-15AWGON3	GRIU-20AWGIN3 GRIU-20AWGON3	GRIU-25AWGIN3 GRIU-25AWGON3	GRIU-30AWGIN3 GRIU-30AWGON3
TECHNICAL DATA						
Cooling Capacity	Btu/h	9,000	12,000	18,000	22,000	30,000
Power Supply	V,Hz,Ph	230,60,1	230,60,1	230,60,1	230,60,1	230,60,1
Rated Power Input	W	700	1,035	1,648	1,928	3,500
Rated Current	A	3.1	4.6	7.64	8.8	9.7
Pipe Size (L,G)	in	1/4, 3/8	1/4, 3/8	1/4, 1/2	1/4, 1/2	1/4, 5/8
Pipe Length / Difference	mm	25,10	25,10	30,20	30,20	50,25
Application Area	m ²	12-18	16-23	24-35	29-43	33-48
PHYSICAL DATA						
Indoor Dimension (WxHxD)	mm	722 x 290 x 187	802 x 297 x 189	965 x 319 x 215	1,080 x 335 x 226	1,259 x 362 x 282
Outdoor Dimension (WxHxD)	mm	720 x 495 x 270	720 x 495 x 270	765 x 555 x 303	805 x 554 x 330	946 x 810 x 410
Indoor Net/Gross Weight	Kg	8.1/10.4	8.9/11.4	11.7/14.9	14.3/17.9	16.7/22.6
Outdoor Net/Gross Weight	Kg	21.5/23.4	21.8/23.8	26.4/28.8	29.6/32.2	62.1/68.3

Save up to 60% Energy Consumption



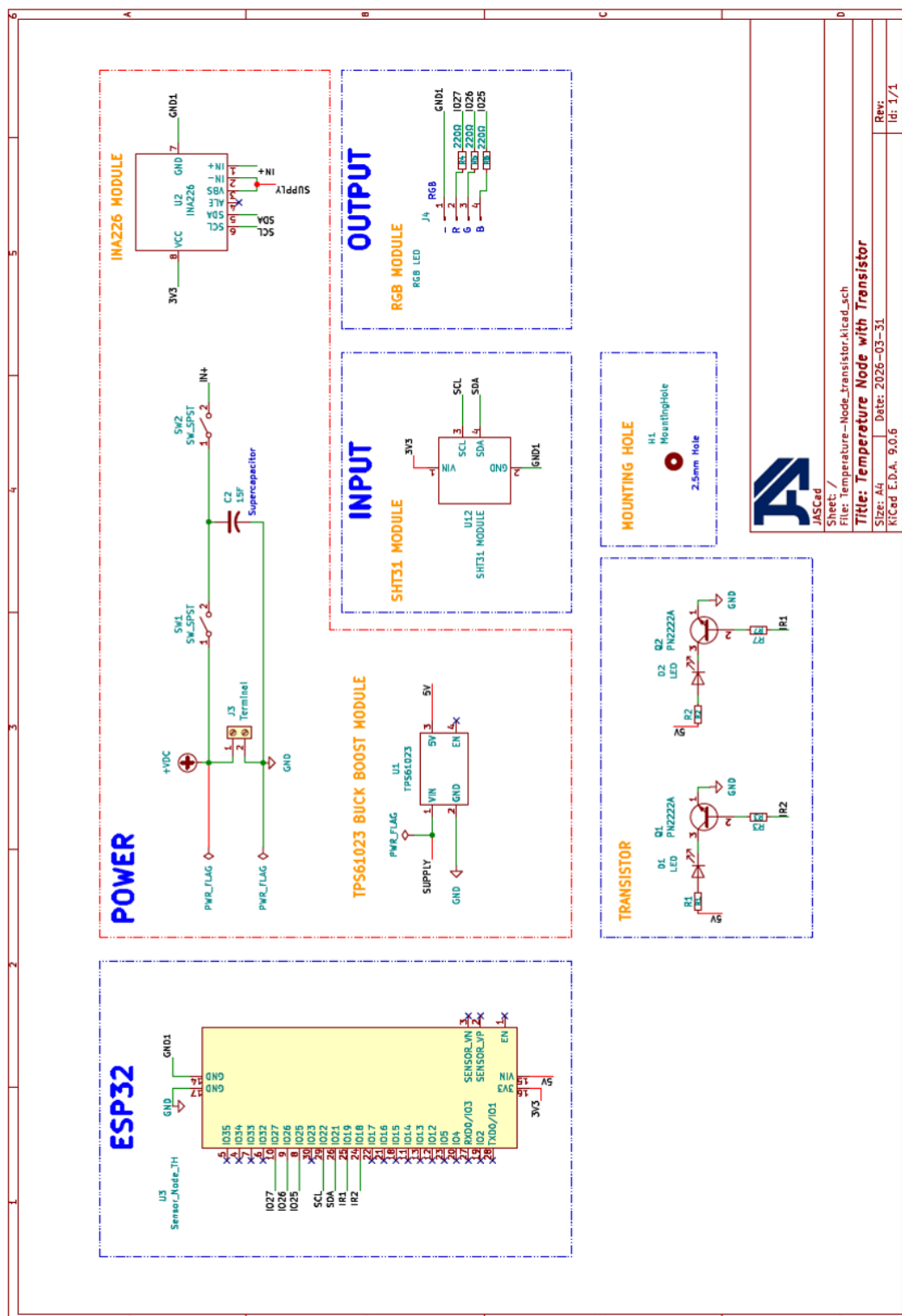
Appendix D

Temperature Nodes PCB Schematic Diagram



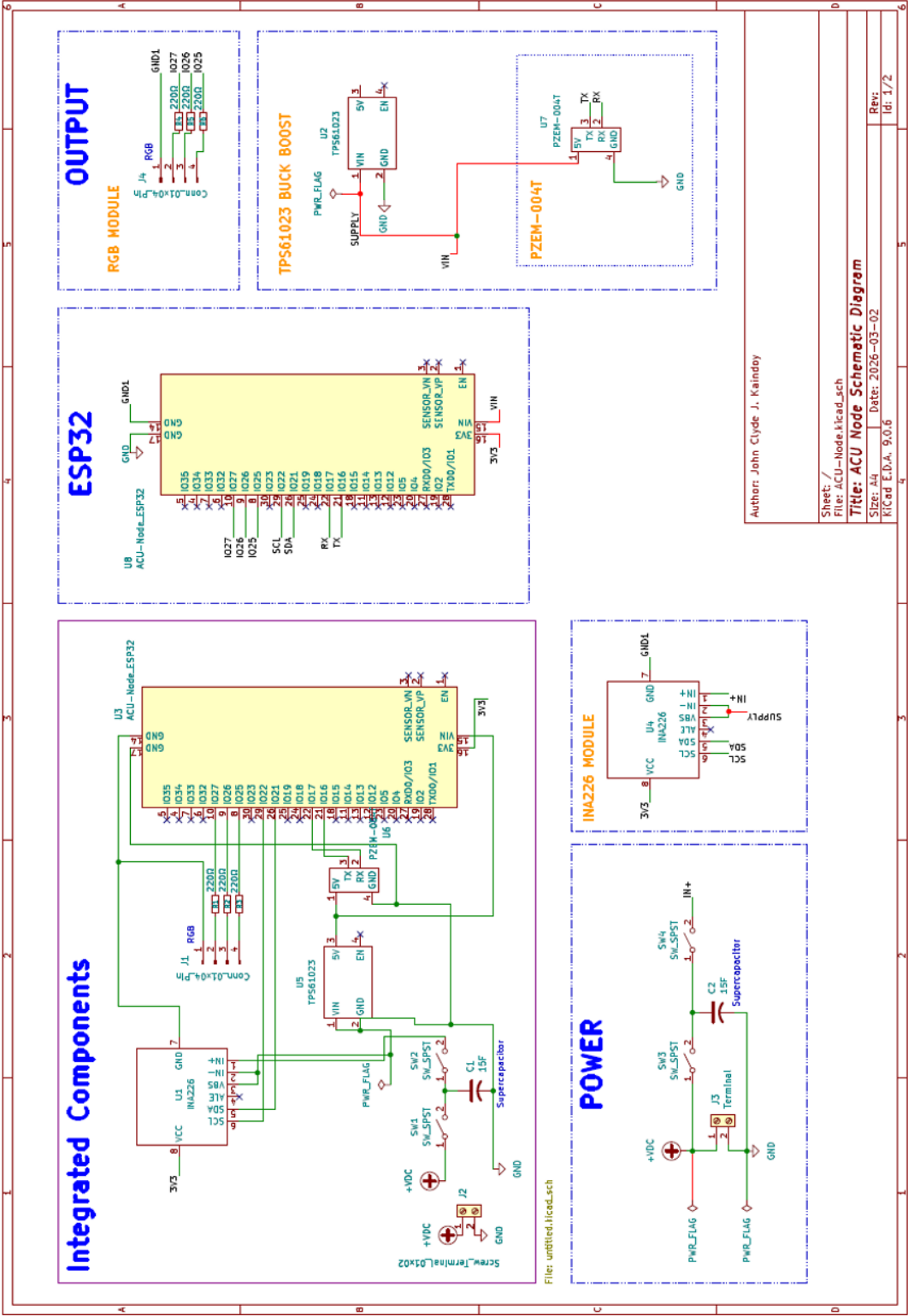
Appendix E

IR Transmitter Node PCB Schematic Diagram



Appendix F

ACU Power Monitoring Nodes PCB Schematic Diagram



Appendix G

DME Conference Room Usage Letter for Testing



VISAYAS
STATE UNIVERSITY

20 February 2026

To whom it may concern:

Subject: Request for Permission to Use Classroom for Thesis Deployment

Good day!

We, the undersigned students, are currently conducting our undergraduate thesis entitled "Design and Development of Bayesian Optimized Random Forest with Digital Twin Modeling and Hardware Implementation." As part of our research implementation, we respectfully request permission to utilize the DME Conference Room for the installation and testing of monitoring sensors required for data collection.

The sensors will be installed strictly for academic and research purposes and will not interfere with normal classroom activities or cause any damage to university property. However, during the course of the study, minor adjustments or calibration of the devices may be necessary, which may occasionally take place during lecture hours. We assure you that these activities will be conducted with permission from the instructor using the classroom and with minimal disruption to ongoing classes.

In this regard, we respectfully seek your approval and cooperation to allow the deployment and maintenance of our research equipment in the said classroom for the duration of our thesis implementation.

Thank you very much for your support and consideration.

Respectfully,

CHARLES VINCENT MAYAMAN
Researcher

PAOLO RENOMERON
Researcher

RISBBY ARNE CERNA
Researcher

Noted by:

ENGR. PHILIP CAESAR L. EBIT
Thesis Adviser

ENGR. EDGARDO C. OCHAVILLO
Department Head, DME

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Mission: To produce graduates equipped with advanced knowledge and lifelong learning skills with ethical standards through high quality instruction, innovative research, and impactful community engagements.

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Appendix H

DME Building Plan Request Letter



VISAYAS
STATE UNIVERSITY

05 March 2026

To whom it may concern:

Subject: Request for a Copy of the DME Building Plan

Good day!

We, the undersigned students, are currently conducting our undergraduate thesis entitled "Design and Development of Bayesian Optimized Random Forest with Digital Twin Modeling and Hardware Implementation." As part of our research implementation, we respectfully request a copy of the building plan of the Department of Mechanical Engineering, including the workshop, offices, and classrooms.

The building plan will be used solely for the development of a digital twin model, which is necessary to accurately simulate and predict the behavior of our selected test area, the DME Conference Room. Rest assured that all obtained documents will be handled with the utmost confidentiality and will be used strictly for academic and research purposes, unless otherwise permitted.

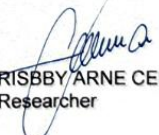
In this regard, we humbly seek your approval and support in granting access to the requested building plan, which will greatly contribute to the development of our 3D model and the overall progress of our research.

Thank you very much for your time, support, and consideration.

Respectfully,


CHARLES VINCENT MAYAMAN
Researcher


PAOLO RENOMERON
Researcher


RISBY ARNE CERNA
Researcher

Noted by:


ENGR. PHILIP CAESAR L. EBIT
Thesis Adviser


ENGR. EDGARDO C. OCHAVILLO
Department Head, DME

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No. FE, ME 26

Appendix I

DME Conference Room Usage Letter for Deployment



VISAYAS
STATE UNIVERSITY

28 February 2026

To whom it may concern:

Subject: Request for Permission to Use Conference Room for Thesis Deployment

Good day!

We, the undersigned students, are currently conducting our undergraduate thesis entitled "Design and Development of Bayesian Optimized Random Forest with Digital Twin Modeling and Hardware Implementation." In line with this, we respectfully request permission to utilize the DME Conference Room for the installation and testing of monitoring sensors required for data collection.

Each test requires four (4) hours of uninterrupted occupancy to ensure data accuracy. As the study requires a minimum of eight (8) tests, the researchers request access to the room for at least four (4) days. However, extensions to the overall testing period may be necessary, if time is insufficient. Either way, rest assured that system testing will only be conducted during periods when no classes or official activities are scheduled.

In this regard, we respectfully seek your approval to deploy and maintain our research equipment in the said room for the entire duration of the system evaluation and hardware testing phase.

Thank you very much for your time, support, and consideration.

Respectfully,


CHARLES VINCENT MAYAMAN
Researcher


PAOLO RENOMERON
Researcher


RISBBY ARNE CERNA
Researcher

Noted by:


ENGR. PHILIP CAESAR L. EBIT
Thesis Adviser

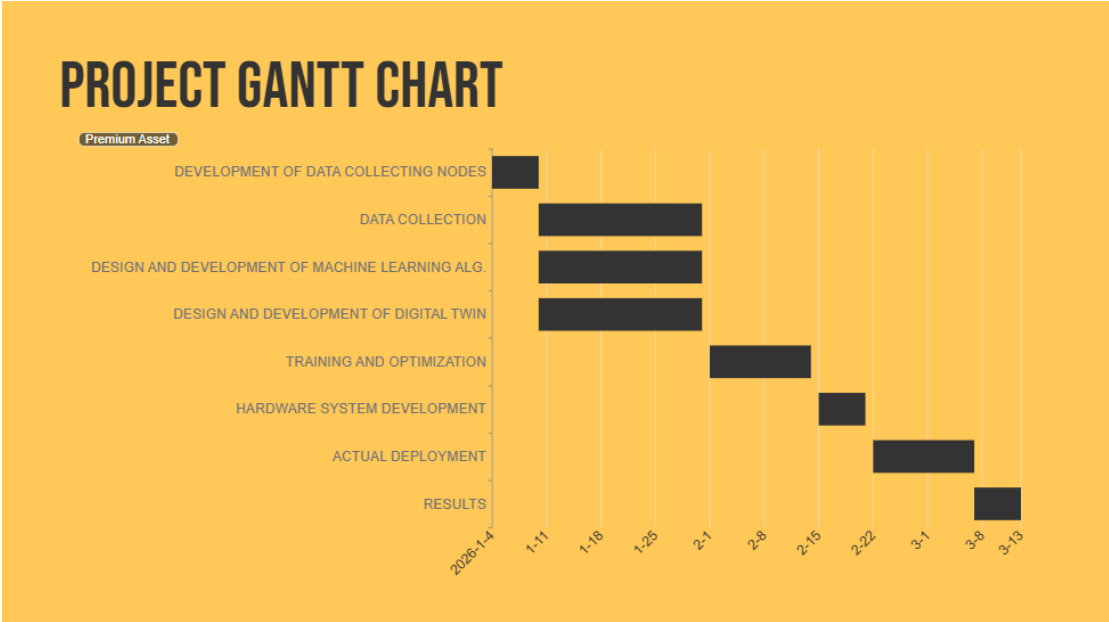

ENGR. EDGARDO C. OCHAVILLO
Department Head, DME

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Mission: To produce graduates equipped with advanced knowledge and lifelong learning skills with ethical standards through high quality instruction, innovative research, and impactful community engagements.

Appendix J

Gantt Chart of Activities



Appendix K

Thesis Proposal Approval Sheet



VISAYAS
STATE UNIVERSITY

DEPARTMENT OF
**MECHANICAL
ENGINEERING**

THESIS PROPOSAL
APPROVAL SHEET
1ST Semester AY 2025-2026

DESIGN AND DEVELOPMENT OF BAYESIAN OPTIMIZED RANDOM FOREST WITH DIGITAL TWIN MODELING AND HARDWARE IMPLEMENTATION
Title of thesis

NAME OF STUDENT : Risbby Arne P. Cerna, Charles Vincent Mayaman, Paolo L. Renomeron

Degree Program : Bachelor of Science in Mechanical Engineering

Major Field : N/A

APPROVED :  Adviser 01/24/24
Date Signed

REVIEWED : Student Research Committee:

 **PROF. MICHAEL ANTHONY JAY B. REGIS** Chairperson 01/24/24
Date Signed

 **ENGR. EDGARDO C. OCHAVILLO** Member 01/24/24
Date Signed

APPROVED:  **ENGR. EDGARDO C. OCHAVILLO** Department Head 01/26/24
Date Signed

 **DR. HANZEL N. MEJIA** College Dean 01/26/24
Date Signed



**DEPARTMENT OF MECHANICAL
ENGINEERING**
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Leyte
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Website: www.vsu.edu.ph
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Appendix L

Documentations



